



Course Guide



Automotive Electrical Diagnosis

Course Guide

AED.01

SAFETY FIRST

Appropriate service methods and proper repair procedures are essential for safe, reliable operation of all motor vehicles as well as the personal safety of the individual performing the repair. There are numerous variations in procedures, techniques, tools and parts for servicing vehicles, as well as in the skill of the individual performing the service. This module cannot possibly anticipate all such variations and provide advice or caution to each. Accordingly, anyone who departs from the instruction provided in this module must first establish that they compromise neither their personal safety nor the vehicle integrity by their choice of methods, tools or parts. The following list contains general warnings that should always be followed while working on a vehicle.

- Always wear safety glasses for eye protection.
- Use safety stands whenever a procedure requires underbody work.
- Be sure the ignition switch is always off unless otherwise specified by a procedure.
- Set the parking brake when working on the vehicle.
- Operate the engine only in a well-ventilated area.
- Keep clear of moving parts when the engine is running.
- To prevent serious burns, avoid contact with hot metal parts such as the radiator, exhaust manifold, tailpipe, catalytic converter and muffler.
- Do not smoke while working on a vehicle.

Within this module you will find Notes, Cautions and Warnings which provide critical information and help you do your job safely and efficiently. Below are the definitions of these terms.



NOTE

The purpose of a Note is to help you do your job more efficiently. A Note may provide additional information to help clarify a particular point or procedure.



CAUTION

A Caution alerts you to the possibility of damage to tools, equipment, or the vehicle. A Caution recommends that a procedure must be done in a certain way to avoid potential problems resulting from improper techniques or methods.



WARNING

A Warning alerts you to the highest level of risk. Warnings inform you that a procedure must be done in a particular way to minimize the chances of an accident that could result in personal injury or even loss of life.

When you see a Note, Caution or Warning, be certain you understand the message before you attempt to perform any part of a service procedure.

Automotive Electrical Diagnosis

Course Guide Module

COURSE OVERVIEW	The Automotive Electrical Diagnosis Course provides you with the opportunity to increase your ability and efficiency in testing, diagnosing and repairing moderately complex electrical/electronic systems and components. In this performance-based course, you will demonstrate your testing, diagnostic and repair proficiency through intermediate to advanced level hands-on, instructor supervised guided practices and diagnostic scenarios including: • Electrical circuit and component operation and diagnosis • Testing, diagnosis and repair techniques for Kia wiring harnesses • Battery, Charging, and Starting system operation & diagnosis
COURSE GOAL	Upon completion of this course, you will have the ability to diagnose, troubleshoot, and repair basic to moderately complex electrical systems and components the first time, every time.
TARGET AUDIENCE	Kia Dealership Technicians who have mastered the fundamentals of automotive electrical principles and diagnostic techniques.
PREREQUISITES	You must complete the following: • <i>Introduction to Kia Automotive Electrical Web Course</i> • <i>Introduction to Kia Circuit Diagram Analysis Web Course</i>

**ABOUT TRAINING
MODULES**

Today's complex automotive technology demands that you stay up-to-date with the latest service information, special tools and complex repair procedures. We have adopted a modular training delivery system that breaks-down the critical information into logical groupings. First you will be presented with system theory and operation, then given a chance to practice what you have learned. Finally, we will test what you have learned through a Performance Assessment.

REVIEW

A **Review Module** gives the technician the opportunity to demonstrate he/she has mastered the key objectives from a prerequisite course. This module is only used where appropriate.

THEORY

A **Theory Module** explains the subject from basic to complex. This allows you to obtain a working knowledge of a component or system, which is a prerequisite for successful diagnosis and repair.

GUIDED PRACTICE

The **Guided Practice Module** affords you the opportunity to familiarize yourself with a component or system through hands-on experience. The guided practices are to be instructor supervised and verified. These exercises may include: the use of service manuals or KSIS, accessing kdealer.net, lab disassembly and reassembly, live vehicle activates and much more.

**PERFORMANCE
ASSESSMENT**

The **Performance Assessment Module** provides the opportunity for you to prove that you can perform the subject matter related tasks and procedures. Each technician must successfully complete this module, which is designed to test your cognitive (knowledge) and motor (hands-on) abilities. The module must be completed individually, not as a team.

**GETTING THE MOST OUT
OF THIS COURSE**

These modules are designed to be part of a structured training plan consisting of lecture, interactive classroom discussion, and hands-on shop activities under the direction of a trained Kia Instructor. After completing the course modules your understanding of the material will be verified through our Performance Assessment Module, you must pass a written and hands-on evaluation.

COURSE MANAGEMENT

The course and its materials are here for you to learn. Use them and your time in a way that will benefit you when you return to your dealership.

TAKE NOTES

Make drawings, jot down notes, and highlight these materials to help you remember important details. Each module is designed with ample margins for your important notes.

ASK QUESTIONS

If you do not understand something in this course, **ask your instructor** for clarification. Asking questions is strongly encouraged to help you get the most out of this course.

TEAMWORK

During the hands-on activities, you will often be working as a team. By actively engaging in each activity, you will maximize your learning experience. While in the lab, feel free to ask the instructor questions at any time.

**LEARN AT EVERY
OPPORTUNITY**

This course is an opportunity for you to learn in a controlled environment under the guidance of a trained Kia Instructor. Through active participation you can build confidence in your abilities to diagnose customer concerns right the first time, every time!

COURSE MATERIALS**Day 1**

Module Number	Module Title	Course Guide	Guided Practice	Performance Assessment	Review	Theory
AED.01	Course Guide	X				
AED.02	WBT Prerequisite Review				X	
AED.03	Electrical Circuits 1					X
AED.04	Electrical Circuits 2					X
	Virtual Breadboard Sims	Computer Simulation				
AED.05	Electrical Components					X
AED.06	Wiring Harness					X
AED.07	Electrical Diagnosis		X			
	Vehicle Simulations	Computer Simulation				
AED.08	Battery, Starting, Charging					X
AED.09	Battery, Starting, Charging		X			
AED.10	Performance Assessment			X		

Day 2**Day 3**

ICONS

Throughout this course you will come across several icons designed to keep you on track.



NOTE: Indicates additional information to help clarify a particular point or procedure to help you perform your job more efficiently.



FEEDBACK: Indicates a progress check meant to provide you with feedback on your understanding of the course material. Based on this information, we recommend you review any areas where you have not mastered the material.



COMPUTER SIMULATION:



CAUTION: Indicates a procedure must be done in a certain way to avoid potential problems and possibility of damaged tools, equipment or the vehicle.



GUIDED PRACTICE EXERCISE: Indicates an activity that supports a critical learning objective. These activities are offered to help you master the material.



WARNING: Indicates a procedure must be done in a particular way to minimize the chances of an accident that could result in personal injury or even loss of life.



VIDEO: Indicates there is a video segment corresponding to the module information.



WAIT FOR FURTHER INSTRUCTIONS



**PERFORMANCE
ASSESSMENT SCORECARD**

The **Performance Assessment Scorecard** is used to track your classroom participation, performance on guided practices, performance on diagnosis & troubleshooting, and any written knowledge assessments.

COURSE ACHIEVEMENT

A final score of **80% or higher** is needed for completion credit of this course.

SCORECARD ROUTING

One copy of the **scorecard** is yours, and one copy is used to update your Kia technical training records. Should you not complete the course, the third copy is forwarded to your Kia District Parts and Service Manager (DPSM).

Automotive Electrical Diagnosis Review

MODULE GOAL Upon completion of this module you will demonstrate, through discussion, your knowledge of:

- Relationship of current to resistance
- Power flow in an ETM schematic
- Determining unwanted circuit resistance
- Types of electrical circuits
- DVOM measurements
- Number Conversions

MODULE OBJECTIVES Given this module, and information learned from successfully completing the prerequisite courses, you will be able to participate in class discussions and answer questions on the following:

- ETM Schematics
- DVOM Usage
- Circuit Operation
- DVOM Readings

MODULE INSTRUCTIONS During this review, participate in the discussion to get the most out of this module while answering the questions and recording your answers.

Questions are based upon the web courses that are required prerequisites.

EXPECTATIONS It is expected that you passed the prerequisite web course that supports this course, and will be able to complete this review module, and participate in class discussion on the information presented.

TIME TO COMPLETE Approximately: 60 minutes for review

**WRITE IN YOUR ANSWER
BASED UPON THE
PREREQUISITE WEB
COURSE DISCUSSION**

1. Which of the following is not a load device?
 - a. Lamp
 - b. Fuse
 - c. Solenoid
 - d. Motor
2. Which of the following is typically at the top of an ETM schematic?
 - a. Ground
 - b. Load
 - c. Signal
 - d. Power
3. Which of the following best describes wire size on an ETM schematic?
 - a. The wire size equals to the number of strands that make up the wire
 - b. The larger the number the larger the wire size
 - c. The larger the number the smaller the wire size
 - d. Each number doubles in size
4. Which of the following best describes hyperlinks on an ETM schematic?
 - a. They link to a photo of the component
 - b. They link to a view of the connector
 - c. They link to a view of the wiring harness
 - d. All the above
5. Which of the following is a characteristic of a series circuit?
 - a. When circuit resistance increases, circuit current flow decreases
 - b. When circuit voltage increases circuit resistance decreases
 - c. Current is reduced as it passes through each load
 - d. None of the above

6. Which method is the best for identifying unwanted resistance in a circuit?
 - a. Measure circuit resistance
 - b. Measure circuit voltage drop
 - c. Measure circuit amperage
 - d. None of the above

7. How is a DVOM connected in a circuit to measure amperage?
 - a. In series with the load being measured
 - b. In parallel with the load being measured
 - c. Across the source of power
 - d. None of the above

8. Convert the values listed in the chart below:

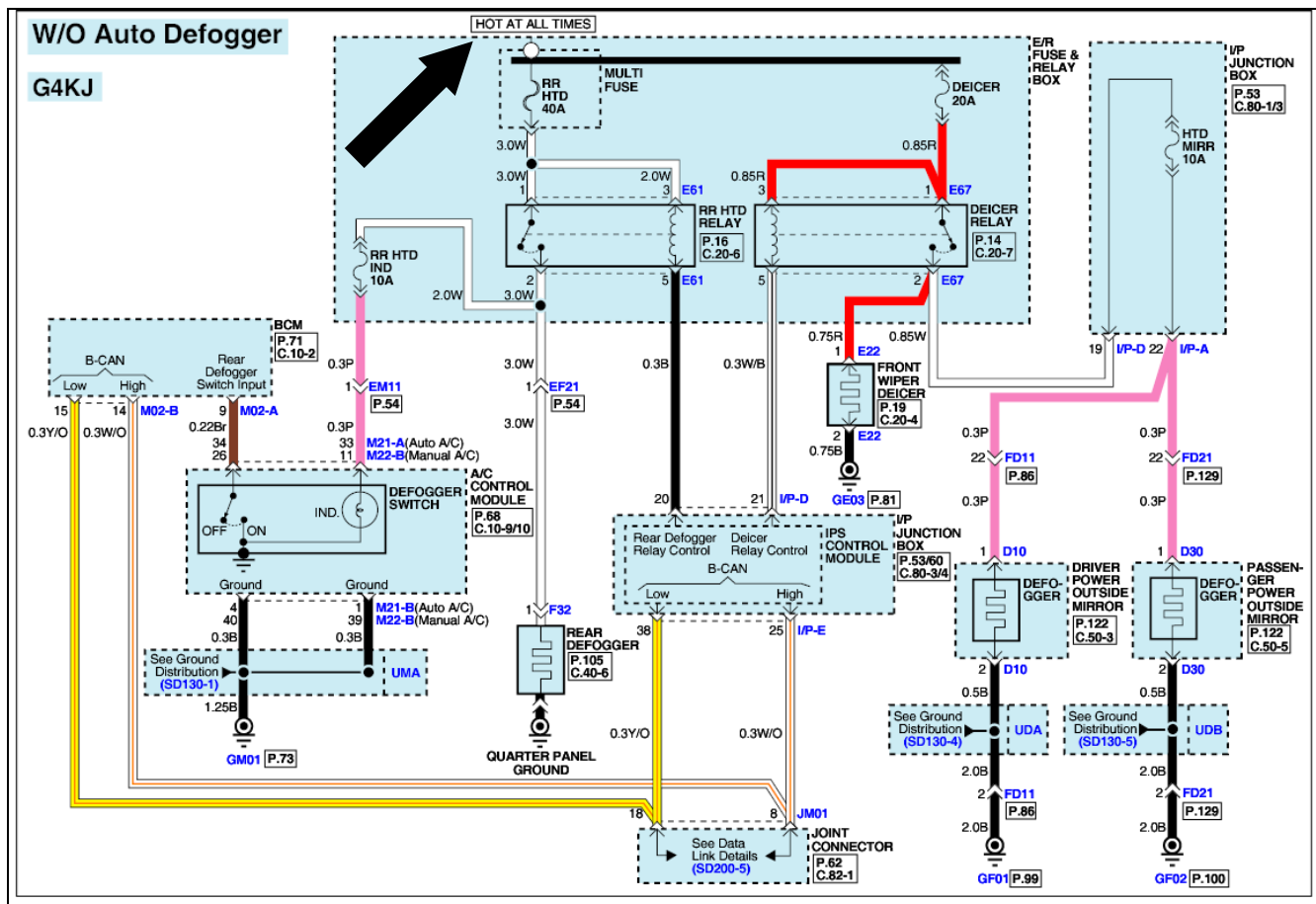
Mega	Kilo	Base Value	milli
	.869 K Ω		
	7.45 K Ω		
2.2M Ω			
			50 mA
		.208 A	
			1,086 mA
			186mV
		.450 V	
		3.3 V	

9. What does an O.L reading on a DVOM indicate when measuring resistance of a load device?
 - a. Shorted
 - b. Over Limit
 - c. Over Load
 - d. Continuity

10. Which of the following would cause a fuse in a circuit to blow?
 - a. Decreased Total Circuit Resistance
 - b. Decreased Total Circuit Amperage
 - c. Decreased Applied Circuit Voltage
 - d. None of the above

11. Which of the following would be considered a control device?
 - a. Switch
 - b. An ECU
 - c. Relay Contacts
 - d. All the above

12. What do you need to do to the DVOM to measure Amperage?
 - a. Move the selector switch to Amps position and leave the Red lead in the Volt Jack
 - b. Move the Red lead to the Amps Jack and leave the selector switch to Volt position
 - c. Move the Red lead to the Amps Jack and the selector switch to Amps position
 - d. Move the selector switch to Volts and the Red lead to Volts Jack



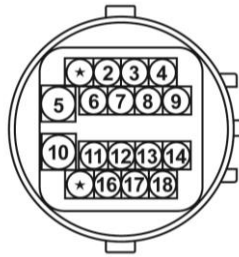
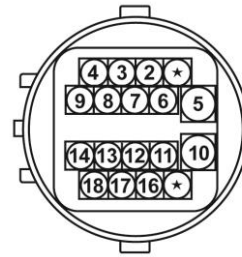
Use the schematic above for the next two (2) questions.

13. What is indicated in the box where the arrow is pointing to on the ETM schematic above?

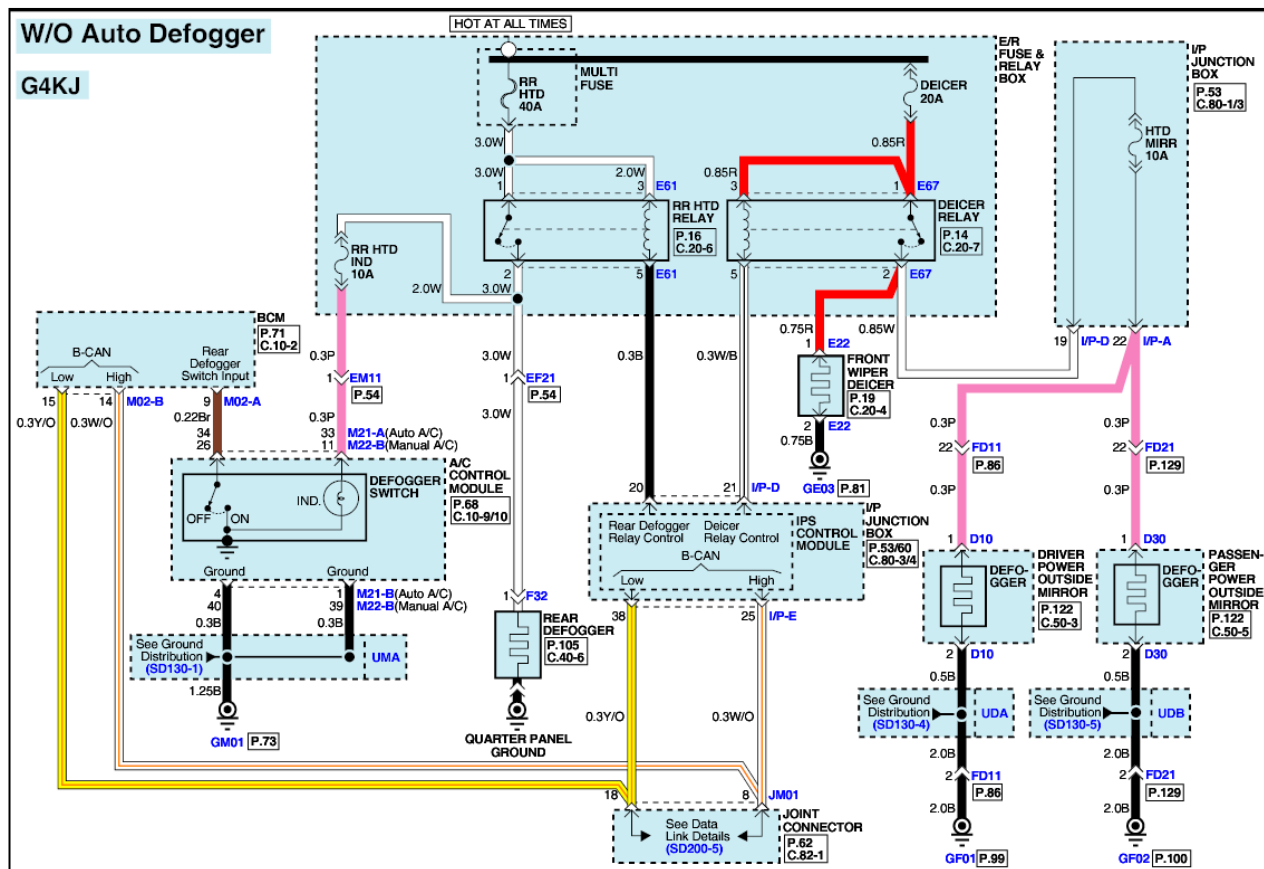
- a. Power is available only when the ignition is ON
- b. Power is available all the time
- c. Power is available only when the RR HTD Relay is ON
- d. Power is available only when the Defogger Switch is ON

14. What turns on the RR HTD Relay and Deicer Relay?

- B-CAN Low pulled High
- B-CAN High pulled Low
- IPS Control Module applying voltage to the relays
- IPS Control Module grounding the relays

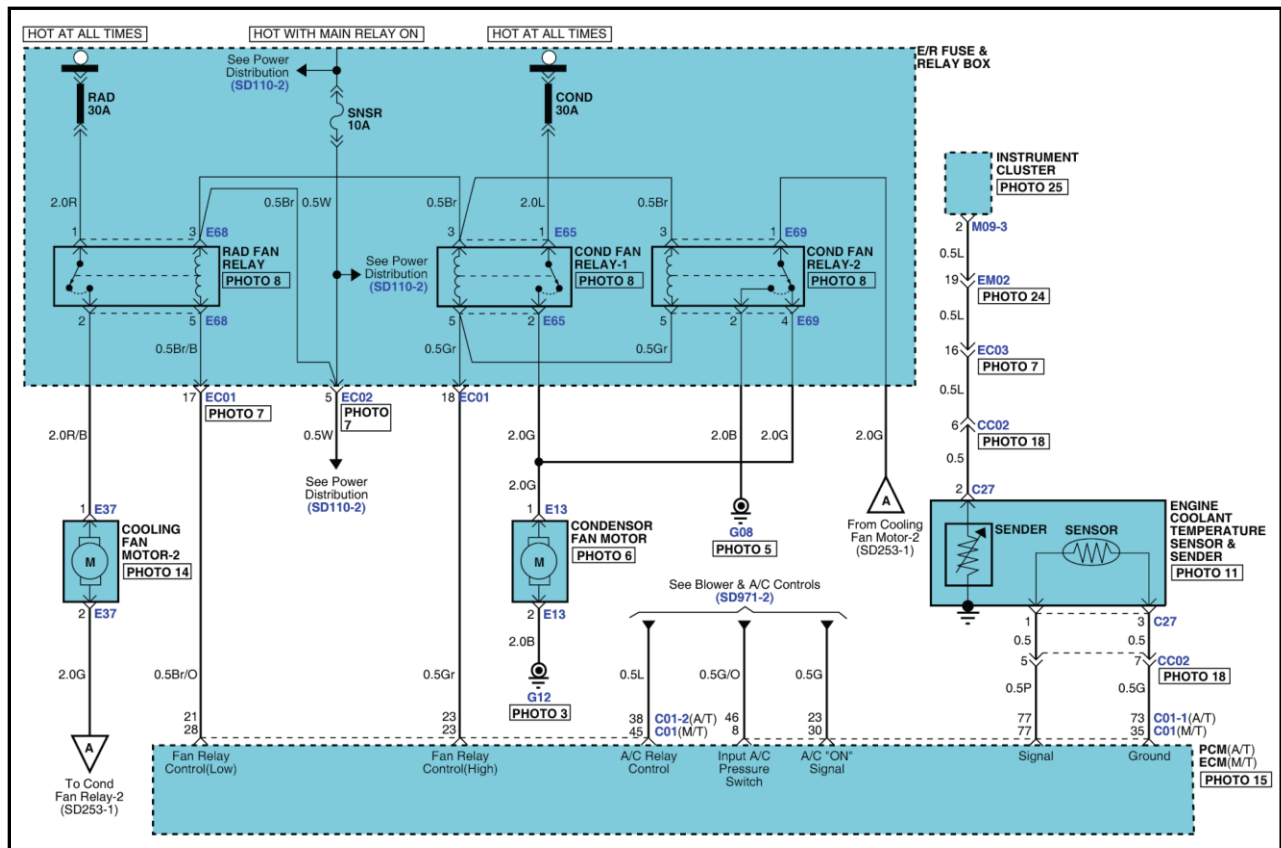
**Connector A****Connector B**

15. Which connector shown above is a female?
- a. Connector A
 - b. Connector B
 - c. Unable to determine from drawing
 - d. Varies depending on wiring harness configuration
16. What action is required when measuring the resistance of a component installed in the vehicle?
- a. Operate the circuit while testing
 - b. Turn Key ON Engine OFF while testing
 - c. Disconnect the battery negative cable while testing
 - d. Isolate the component from the rest of the circuit
17. What will cause the amperage in a circuit to decrease?
- a. Resistance decreases
 - b. Voltage decreases
 - c. Resistance increases
 - d. None of the above
18. What is the function of a load in a circuit?
- a. Produce heat
 - b. Create light
 - c. Perform movement
 - d. All the above



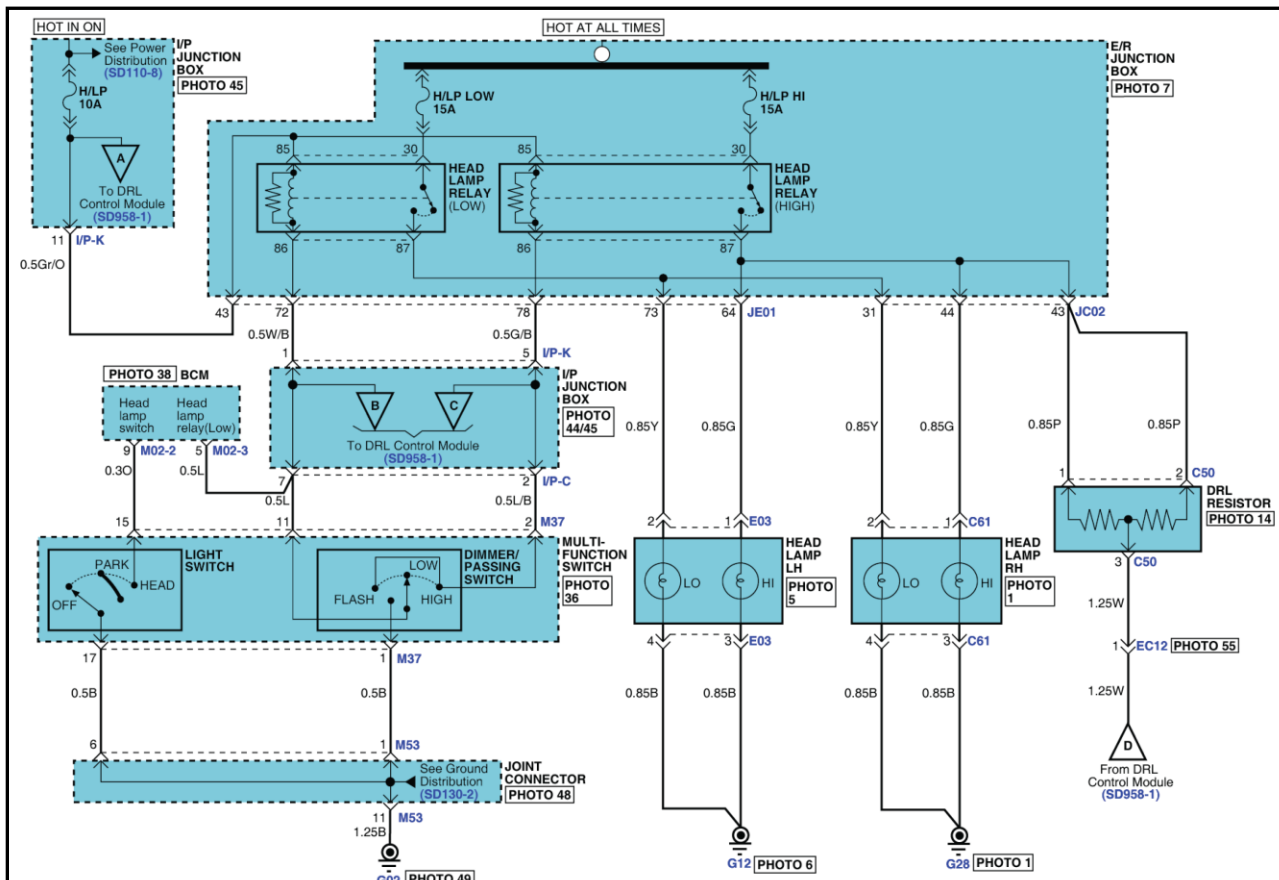
Use the schematic above for the next three (3) questions.

19. What is the purpose of the Defogger Switch
 - a. Turn on the Indicator Lamp in the switch
 - b. Supply ground to the Rear Defogger Element
 - c. Directly ground the RR HTD and Deicer relay coils
 - d. Indirectly controls the RR HTD and Deicer relay coils
20. Which fuse supplies voltage to the Power Outside Mirror Defoggers?
 - a. HTD MIRR
 - b. RR HTD IND
 - c. RR HTD
 - d. Both a. and b.
21. What is the function of the IPS Control Module?
 - a. Supply voltage to the RR HTD and DEICER relays
 - b. Control the B-CAN lines
 - c. Monitor the contact closing of the RR HTD and DEICER relays
 - d. Supply ground to the RR HTD and DEICER relays



Use the schematic above for the next two (2) questions.

22. Under what conditions does Cooling Fan Motor-2 receive ground for Low Speed operation?
 - a. Rad Fan Relay (OFF) Cond Fan Relay-1 (ON)
 - b. Rad Fan Relay (OFF) Cond Fan Relay-2 (ON)
 - c. Rad Fan Relay (ON) Cond Fan Relay-1 (OFF)
 - d. Rad Fan Relay (ON) Cond Fan Relay-2 (OFF)
23. Which relay has a Normally Open (NO) and a Normally Closed (NC) set of contacts
 - a. Cond Fan Relay-1
 - b. Cond Fan Relay-2
 - c. Rad Fan Relay
 - d. All the above



Use the schematic above for the next two (2) questions.

24. With the Headlamp Switch OFF & Daytime Running Lamps ON, how is the DRL RESISTOR wired with the HI BEAM HEAD LAMPS?

- Series
- Parallel
- Series – Parallel
- Parallel – Parallel

25. What is required to turn the LO BEAM HEADLIGHTS ON?

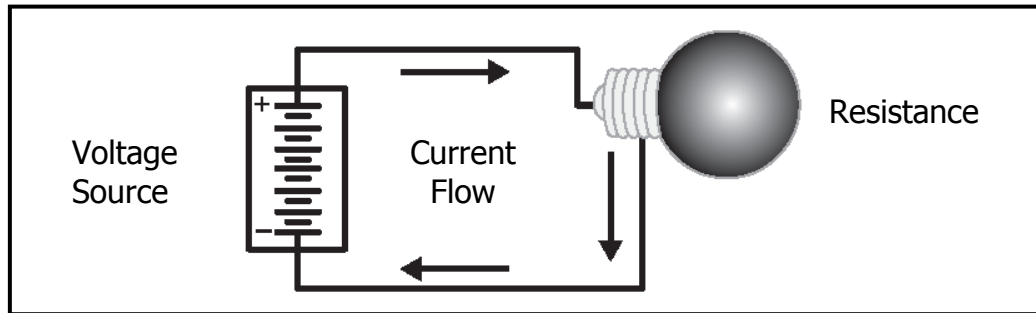
- Light Switch to HEAD, Dimmer Passing Switch to LOW, and Ignition to OFF position
- Light Switch to HEAD, Dimmer Passing Switch to LOW, and Ignition to RUN position
- Either a or b
- Neither a nor b

NOTES:

Automotive Electrical Diagnosis

Electrical Circuits

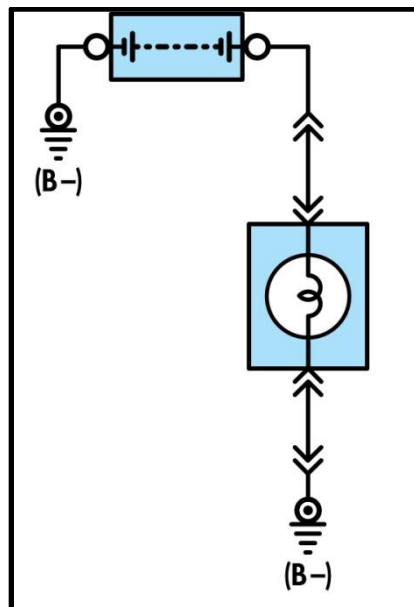
TARGET AUDIENCE	The target audience for this module consists of Kia service technicians who have basic understanding of automotive electricity.
MODULE GOAL	Upon completion of this module, you will be able to list expected measurements and measure voltage, resistance, and amperage in different types of electrical circuits.
MODULE OBJECTIVES	Following the instructions in this module, and given a Digital Volt-Ohm Meter (DVOM) and electrical “breadboard”, you will be able to: • Build series electrical circuits and accurately predict and measure the following with 100% accuracy: ○ Applied Voltage ○ Voltage Drop ○ Amperage ○ Resistance • Diagnose abnormal voltage readings in a series circuit and list the root cause with 100% accuracy.
MODULE INSTRUCTIONS	Carefully read through the material, take notes based on the classroom discussion and study each illustration.
REQUIRED MATERIALS	In order to complete this module, you will need the following items: • DVOM • Electrical “breadboard” • Colored highlighters (pink/red, yellow, green)
TIME TO COMPLETE	Approximately 1.75 hours

**WHAT IS ELECTRICITY?**

Electricity is a form of energy where electrons move from one atom to another. Electricity exists when a _____ creates a _____

by _____ with enough force to overcome the _____ of the circuit.

Electricity itself is invisible but its effects can be seen in the forms of light, heat, and motion, which is often referred to as "work".

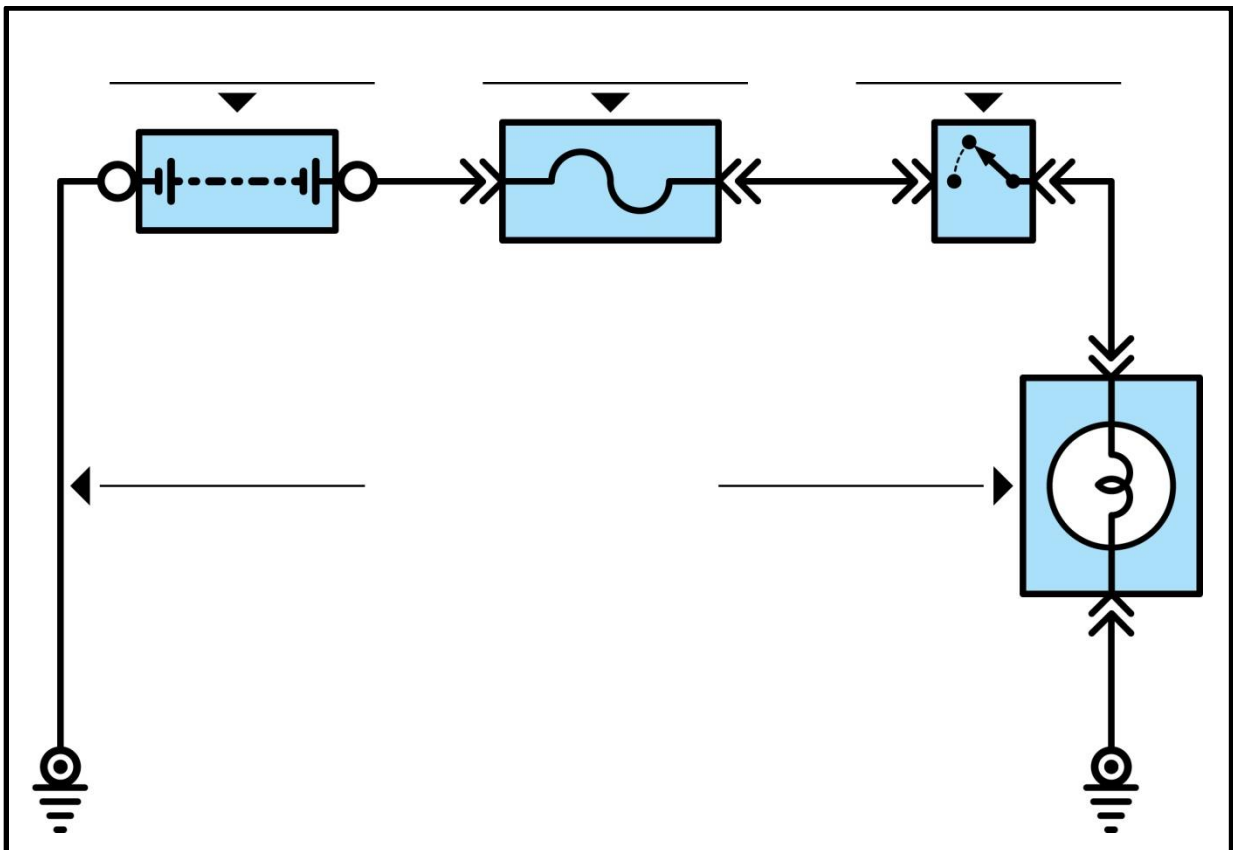
**CIRCUIT ELEMENTS**

In order to operate, all electrical circuits need three items:

1. _____
2. _____
3. _____

Fill in the blanks. Most automotive circuits are made up of a:

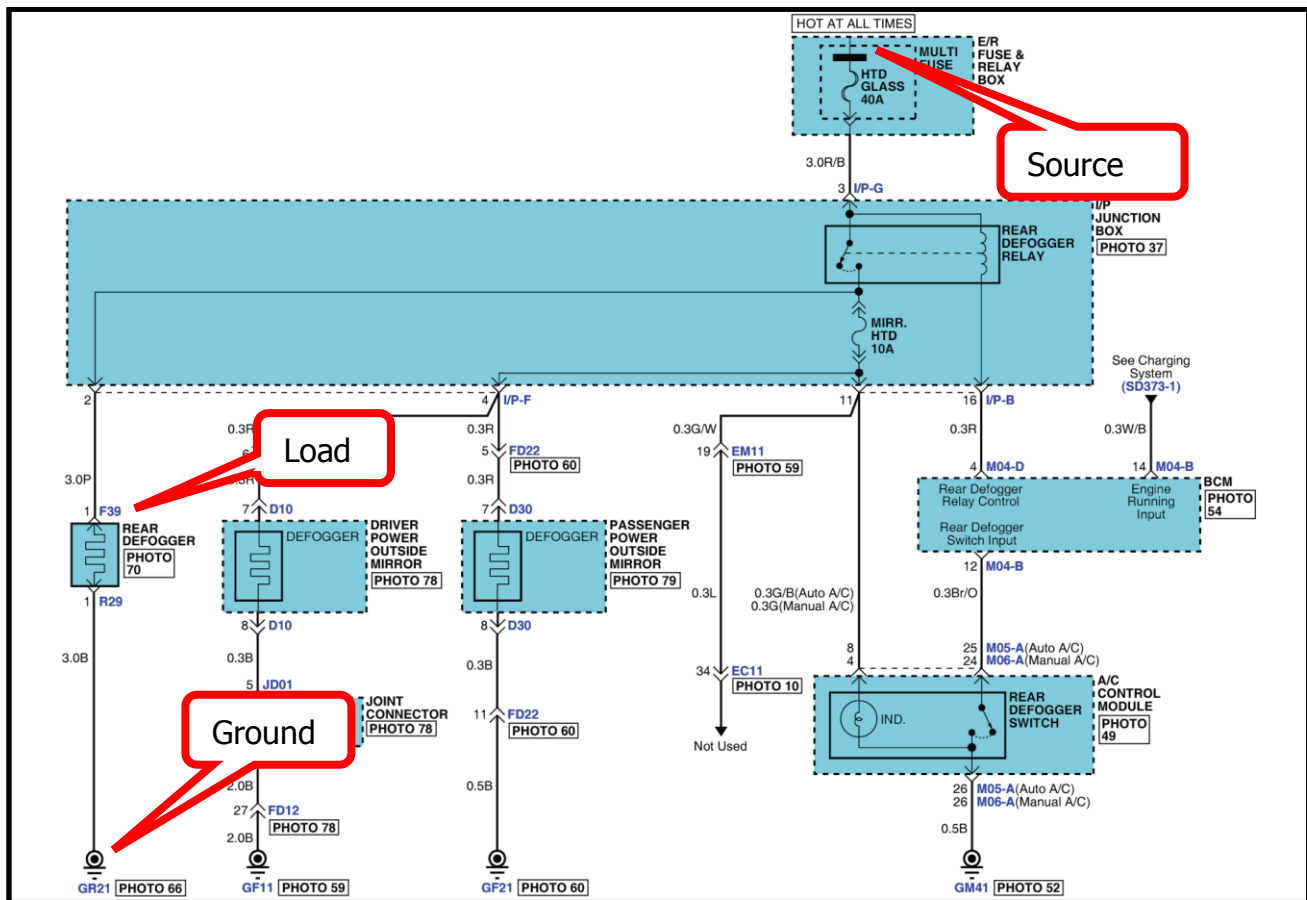
- Power Source _____
- Protection device to protect the wiring and components _____
- Control device to turn the circuit ON or OFF _____
- Load _____
- Conductors to allow current flow from one element to another _____
- Ground _____



The ground path is created by connecting the negative battery terminal to the engine, chassis and/or body. This allows circuits to have a ground point at any metal contact point on the vehicle.

An ECU can also provide a ground path by turning ON a transistor that is connected to ground

It is critical that ground points and ground related wiring are installed properly after vehicle service.

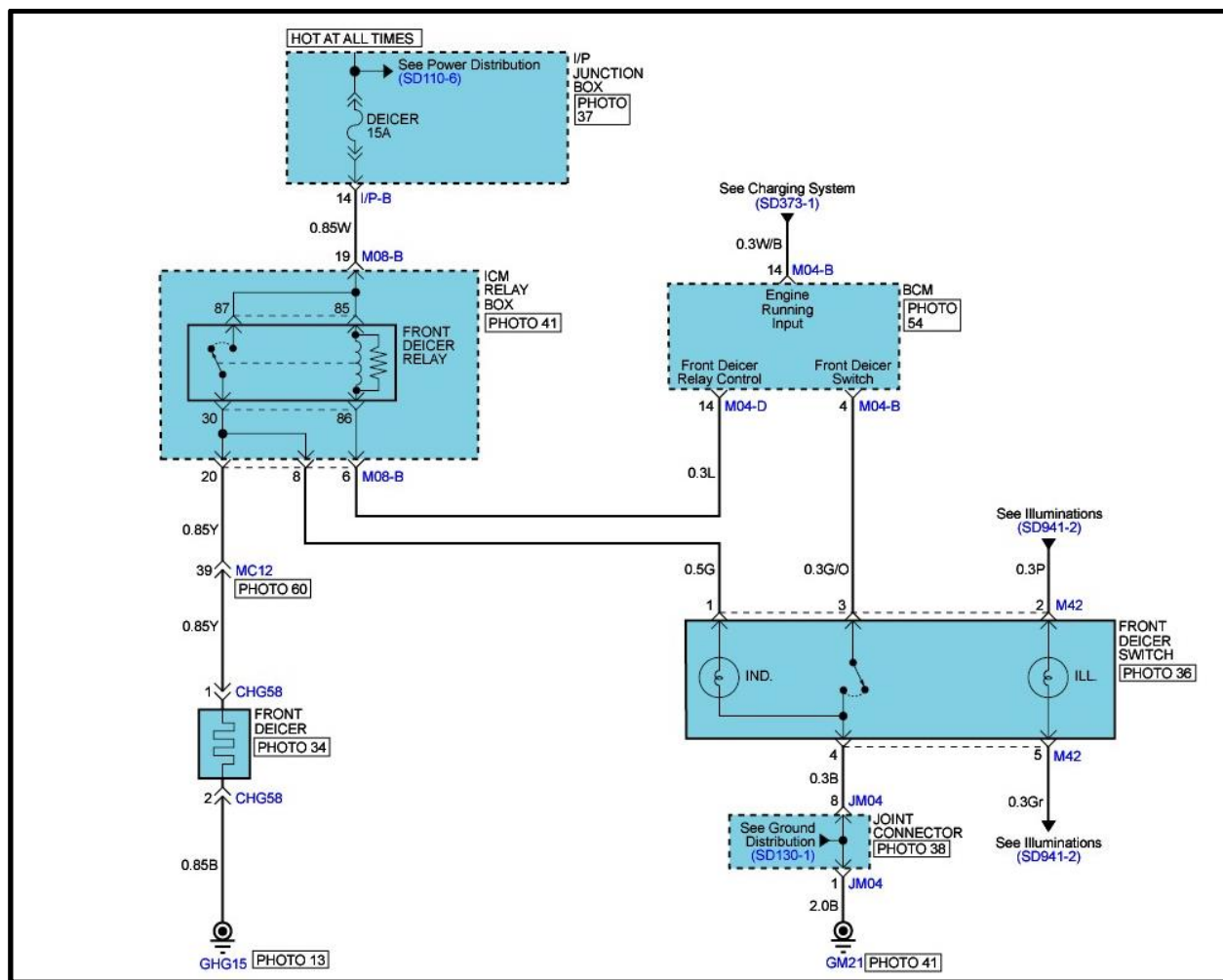


DIRECTION OF CURRENT FLOW

Conventional current flow is from positive to negative. In the automotive industry, we frequently "read" circuit diagrams from the top "positive" to the bottom "negative."

To support conventional current flow, Kia's wiring diagrams generally flow from top to bottom through three (3) sections:

1. The electrical source (positive) at the top (Power)
2. Circuit load is in the middle (Load)
3. The ground (negative) is shown at the bottom (Ground)



POWER FLOW DIAGRAMS

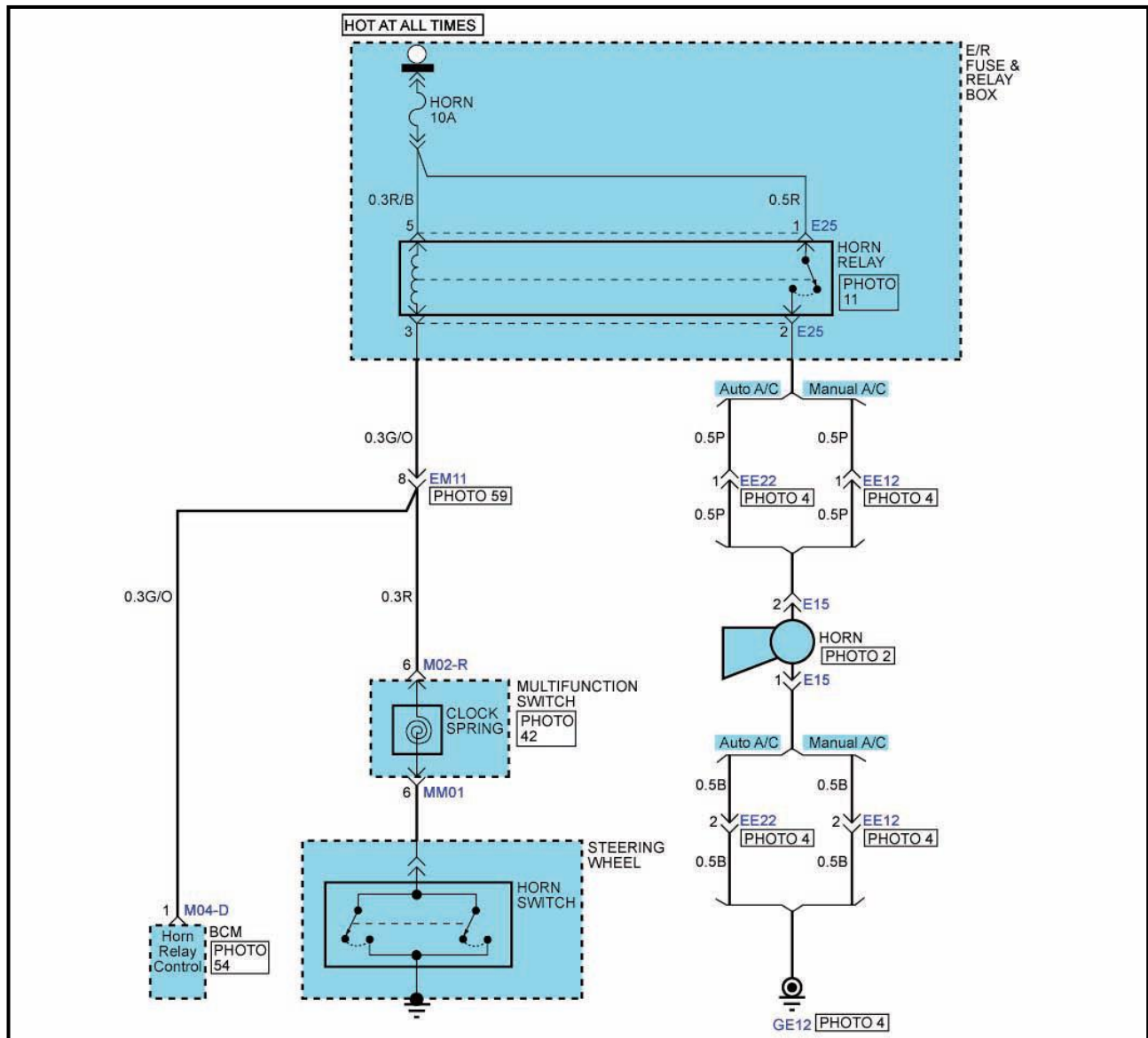
You need a plan when diagnosing electrical concerns.

Begin by printing out the wiring diagram located in the Electrical Troubleshooting Manual (ETM) in KGIS and identify the components.

Assume that the circuit is in operation.

Next, using colored highlighters, identify the three sections on the wiring diagram:

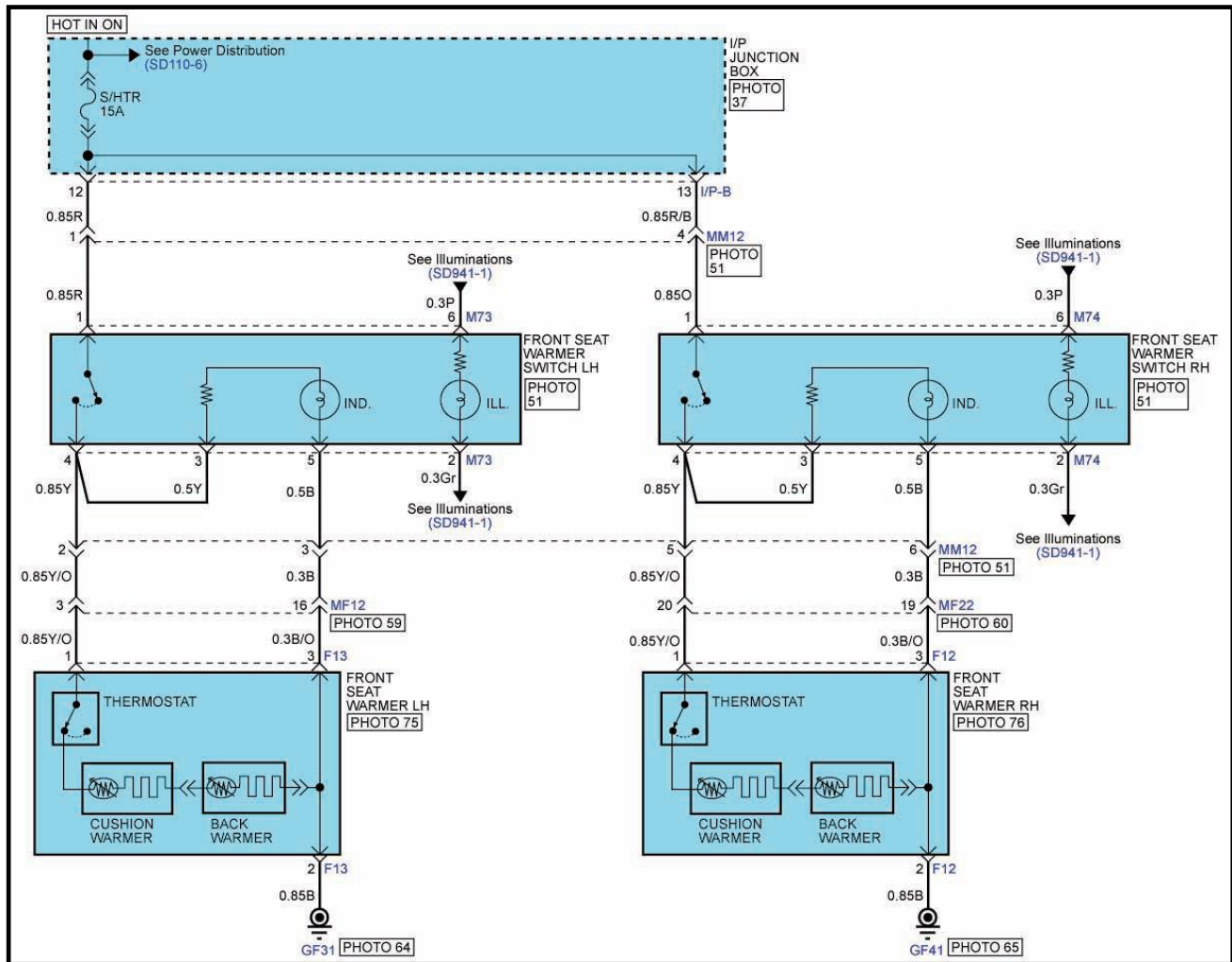
- Power in Red/Pink
- Load in Yellow
- Ground in Green



Using the wiring diagram above, color each circuit listed below:

- Power side in Pink / Red
- Load in Yellow
- Ground side in Green

Assume that the circuit is in operation.



Using the wiring diagram above, color each circuit listed below:

- Power side in Pink / Red
- Load in Yellow
- Ground side in Green

Assume that the circuit is in operation.

MEASUREMENT OF ELECTRICITY

Voltage is produced by chemical action in the battery or by magnetic induction in the alternator (generator).

Pressure _____ is the pressure in an electrical circuit and is measured in _____.

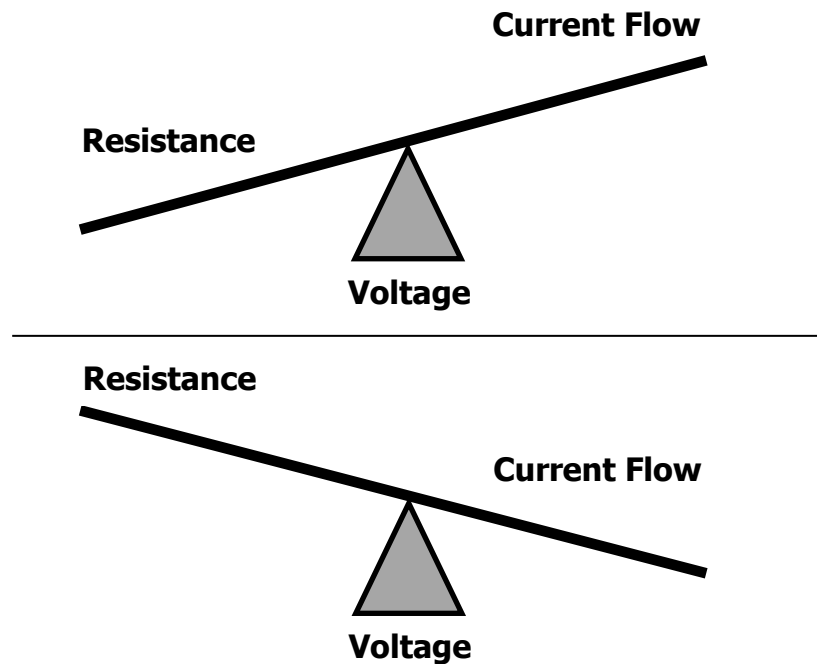
The symbol for voltage is V. You may see E used to represent electromotive force.

Current Flow The movement or “flow” of electrons is called _____ and is measured in _____.

The symbol for amperage is A. You may see I used to represent intensity.

Resistance Every electrical circuit or component has _____ which opposes current flow and is measured in _____.

Resistance is what creates heat, light, or motion, and does the work. The standard unit of resistance is the ohm. The symbol for resistance is Ω .

OHM'S LAW**Ohm's Law Relationship**

Understanding the relationship between volts, amps and ohms is important when performing automotive electrical diagnosis.

This relationship is referred to as "Ohm's Law".

According to Ohm's Law:

- When resistance _____,
current flow _____.
- When resistance _____,
current flow _____.

Ohm's Law Application

Complete the following statements for a better understanding of how Ohm's Law can help you in electrical diagnosis.

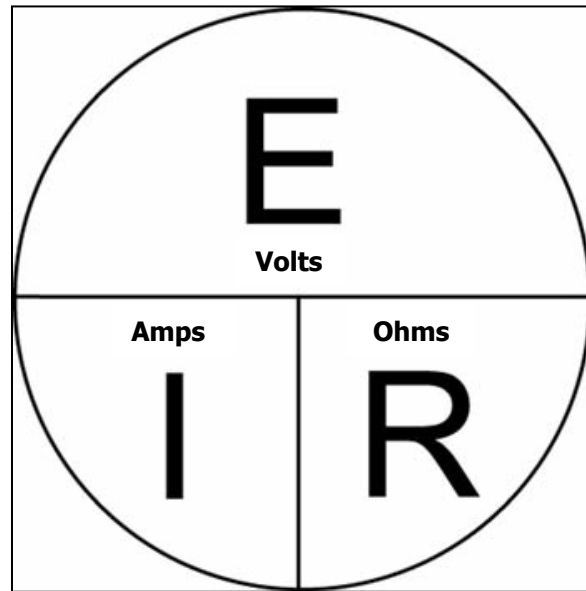
Fuses blow when _____ is too high.

According to Ohm's Law, _____ goes up,
when _____ goes down.

What causes a fuse to blow?

Ohm's Law Solving Circle

While you generally don't engineer electrical circuits, the divided circle can help you to remember the relationship between volts, amps and ohms.



$$\text{Volts} = \text{Amps} \times \text{Ohms}$$

$$\text{Amps} = \text{Volts} \div \text{Resistance}$$

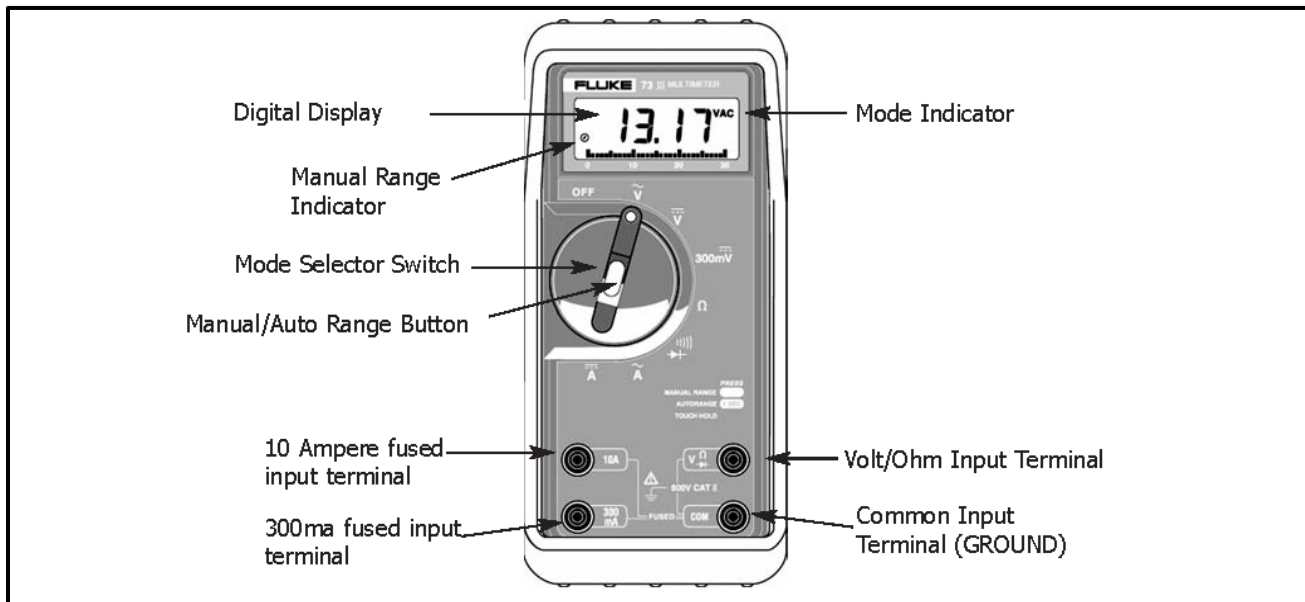
$$\text{Resistance} = \text{Volts} \div \text{Amps}$$

ELECTRICAL CIRCUIT CHARACTERISTICS

In addition to Ohm's Law, the following characteristics are true of electrical circuits.

- Voltage is always _____ when current flows through resistance
- Voltage is _____ used after the last load, or other resistance, in a circuit
- Voltage will always take the path of _____ resistance to ground

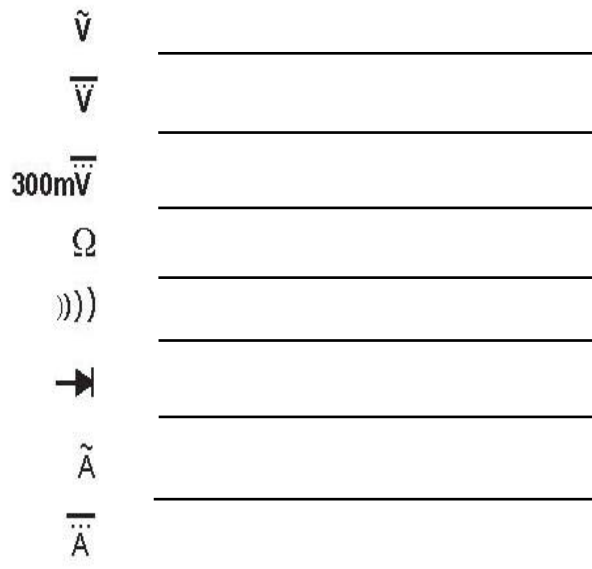
We'll talk more about this later when we make circuit measurements and look at electrical circuit diagnosis.



Multimeter Functions

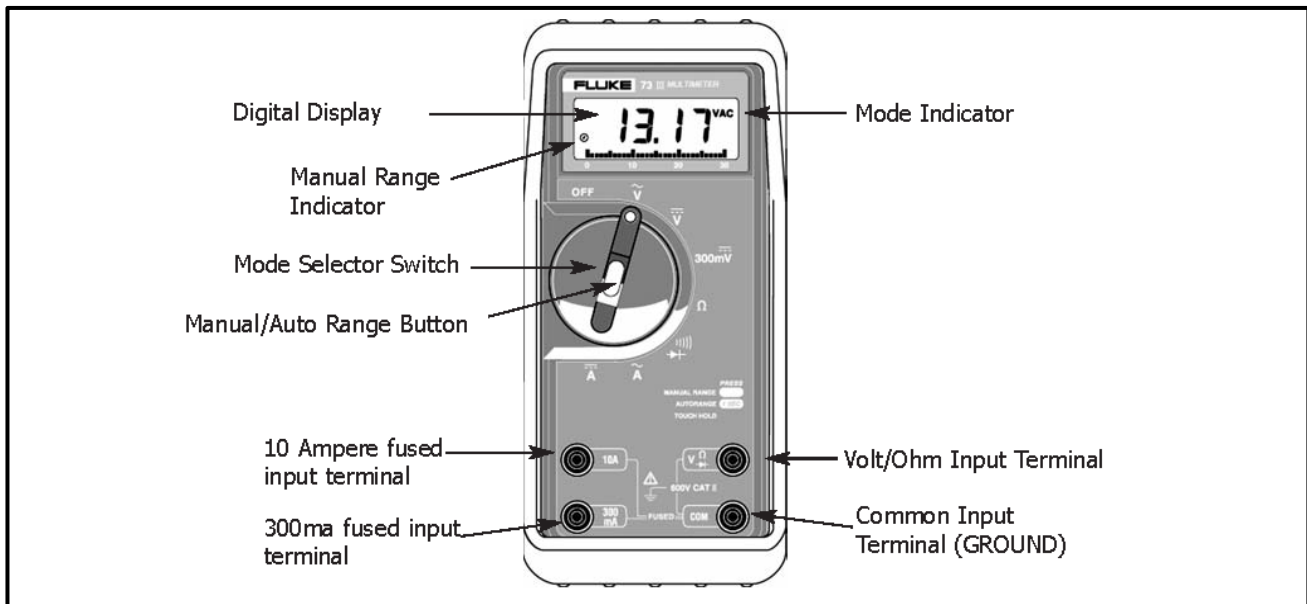
Most meters have a rotary selector switch that allows you to set the function and range for the type of reading that you wish to make.

Most meters will let you manually select the range, or will set the range automatically (auto-ranging).



NOTE: Always look at the meter display to determine if the range is set manually or auto-ranging.

Mistakes are often made by not reading the meter reading correctly.



Setting up the Multimeter

When setting up a multimeter:

- Set the mode selector switch to the measurement you want to make
- Insert the meter leads in the terminals for the measurement you want to make
- Select auto or manual range:
 - Auto range can be used when you are not sure what range your reading will be in, and to protect the meter
 - Manual range can be used when you know what the expected meter reading is, for a more accurate reading

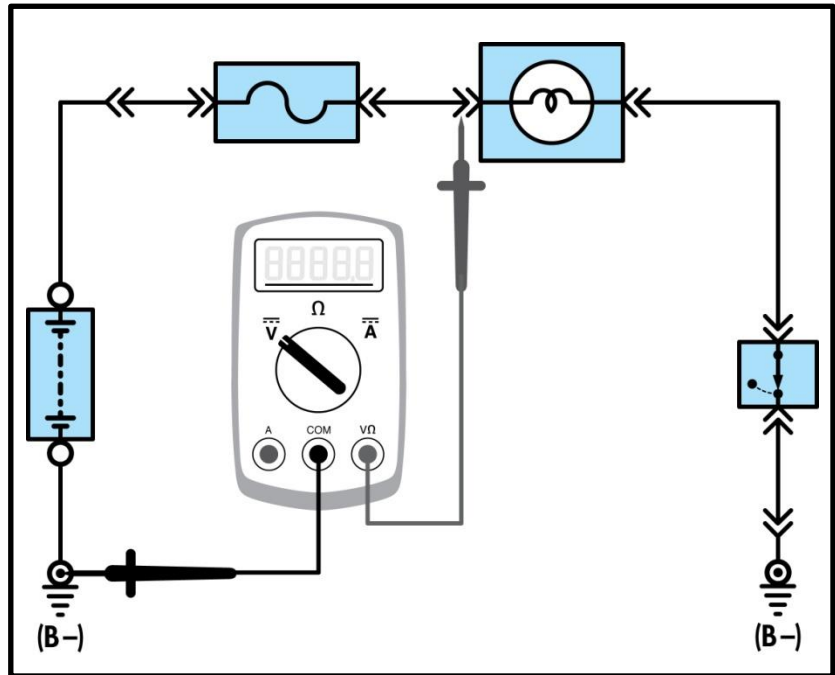
Using the Multimeter

Because digital multimeters are designed to measure voltage, current and resistance, you must properly set up the meter to avoid damaging the meter or the circuit, and to ensure that you are making an accurate measurement. Since there are a wide variety of meters available, always review the operator's manual before using a meter.



CAUTION: *To ensure that the meter is used safely, follow these instructions:*

- 1. Never use the meter if the meter or test leads look damaged.**
- 2. Be sure the test leads and rotary switch are in the correct positions for the desired measurement.**
- 3. Always insert the meter in the circuit in parallel for voltage and in series for current measurements.**
- 4. Never use the meter in a circuit that has higher voltage or amperage than the meter is capable of measuring safely.**
- 5. Always start measuring with the ammeter on the highest scale and work your way down.**
- 6. Never measure resistance in a circuit with power applied. Be certain capacitors are discharged before measuring.**
- 7. Keep your fingers behind the finger guards on the test probes when making measurements.**
- 8. Keep meter leads in good condition.**
- 9. Always test/check meter fuses.**
- 10. Check battery display - meters will not read accurately with a low battery.**
- 11. Calibrate meter leads resistance by zeroing the meter before doing resistance testing.**



VOLTMETER Applied Voltage

When measuring applied voltage:

- The voltmeter is connected in parallel
- Tests for voltage in a non-operating or operating circuit.

When to Use:

When verifying voltage values.

Readings:

- Approximately source voltage before the load = good power side
- Approximately 0 volts after the last load = good ground

Drawbacks:

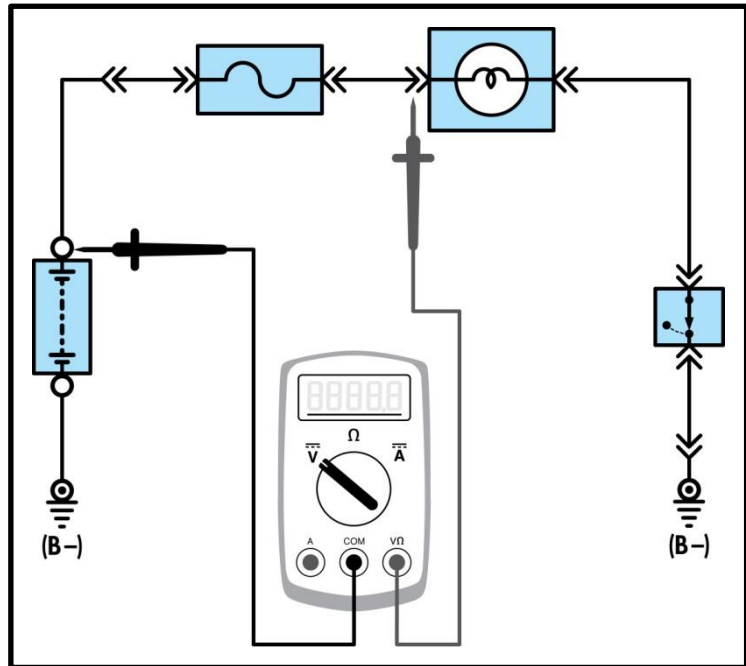
Voltage values may be misleading when the load is open in the circuit.

How to Measure Applied Voltage:

- Operate the circuit
- Connect the meter in parallel
- One meter lead to B(-) the other at key points along the circuit

Examples of When to Use:

- Testing for open
- Testing for short
- Testing control functions
- Testing for excessive resistance

**Voltage Drop**

When using a voltmeter for voltage drop

- The voltmeter is connected in parallel.
- Voltage drop test must be made when the circuit is operating.

When to Use:

When looking for excessive resistance.

Readings:

- Across power side should show very low voltage. If significant voltage, then there is resistance.
- Across ground side should show very low voltage. If significant voltage, then there is resistance.
- Most voltage drop should be across the load(s).

Drawbacks:

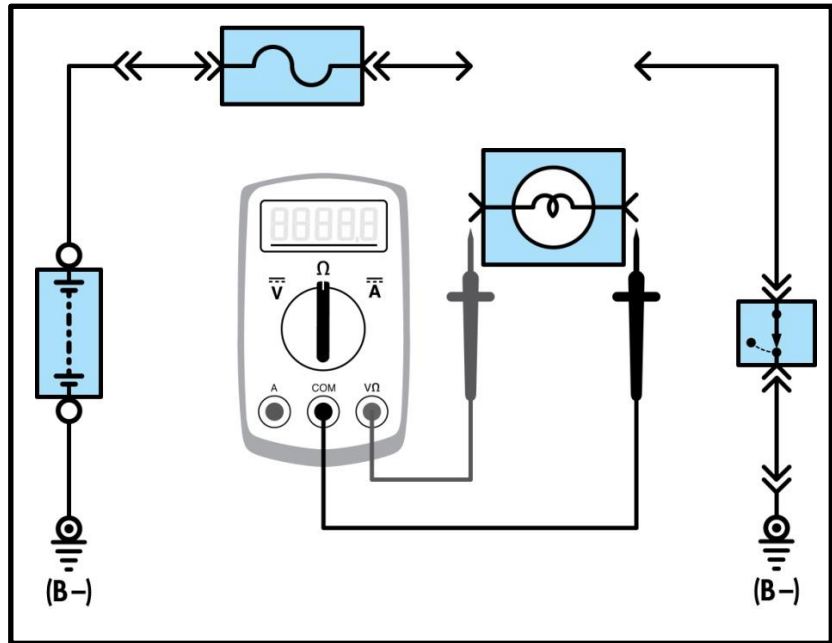
Not effective if the circuit does not operate.

How to Measure Voltage Drop:

- Connect the meter in parallel across the section of the circuit to be tested, usually source or ground
- Operate the circuit

Examples of When to Use:

- Lighting circuits are unusually dim
- Motors operate more slowly than normal
- Specific circuit is operating strangely
- Abnormal Current Data values
- Testing for excessive resistance

**OHMMETER**

When using an ohmmeter

- The circuit should be OFF
- The section being tested should be isolated from the power source otherwise the meter can be damaged.
- Connect the DVOM across the part of the circuit, or component, to be tested.

When to Use:

Testing for continuity, or when there is a resistance specification.

Readings:

- No continuity/Infinity/Over Limit = OPEN circuit
- 0 ohms = no resistance (good connection)
- An ohm reading = resistance

Drawbacks:

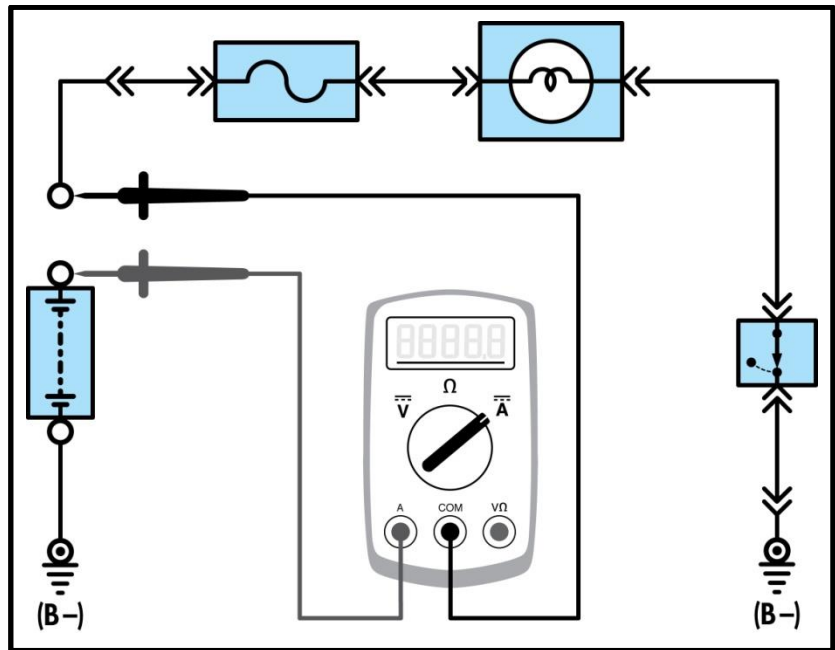
Cannot measure the effective resistance in a circuit under operating conditions.

How to Measure Ohms:

- Power down the circuit
- Isolate the circuit or component to be tested from the power source
- Place meter leads at each end of section

Examples of When to Use:

- Testing for broken wire (open)
- Testing for short
- Resistance of a component (relay, solenoid or other load)
- Continuity in switch or relay contacts

**AMMETER**

When using an ammeter:

- The ammeter is connected to the circuit in series.
- Never substitute the ammeter for the load.
- The ammeter should be used the same way you would use a fused jumper wire.
- The circuit must be ON in order to measure amperage.

When to Use:

When specified by the diagnostic process in the proper service publication.

Readings:

- Battery output
- Starting system draw
- Charging system output
- Parasitic draw

Drawbacks:

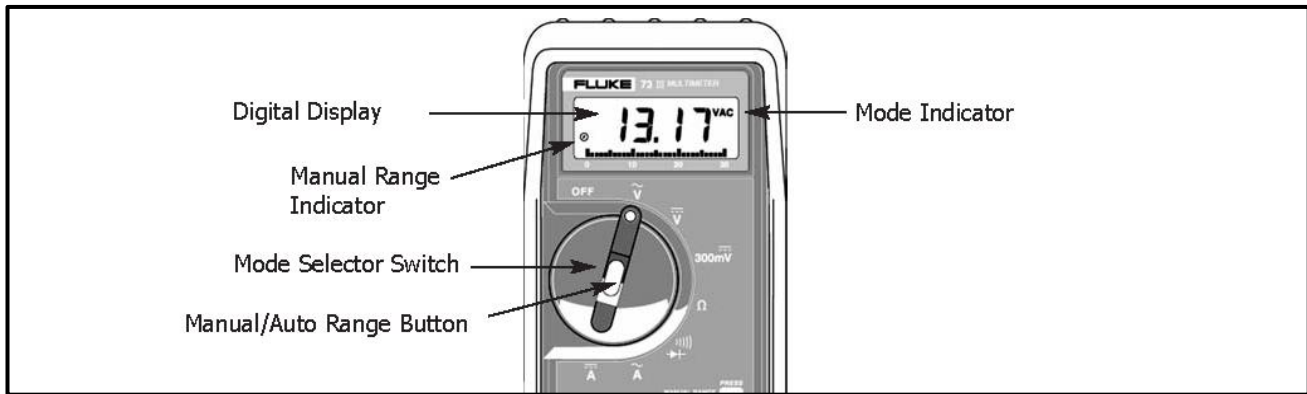
Not able to pinpoint location of open, low resistance, or high resistance.

How to Measure Amperage:

- Connect the meter in series
- Operate the circuit

Readings:

- No current = OPEN circuit
- Low current = high resistance
- High current = low resistance



INTERPRETING METER READINGS

Volts, amps and ohm readings are given in:

- Whole number (12 volts, 30 amps or 560 ohms)
- Milli = one thousandths of the whole number, x0.001 (12mV)
- Kilo = whole number X 1,000 (12KV)
- Mega = whole number X 1,000,000 (10MΩ)

Meter Reading Conversion Tables

Value	Meter Reading	Multiply by	Value	Practice
mV (milli-volts)	12mV	x 0.001	_____	213mV = _____ V
V Volts	12V	x 1	12V	
KV (kilo-volts)	1.2KV	x 1,000	_____	14.2KV = _____ V

Value	Meter Reading	Multiply by	Value	Practice
Ω Ohms	10 Ω	x 1	10 Ω	
Ω (Kilo-ohms)	10K Ω	x 1,000	_____	4.6k Ω = _____ Ω
M Ω (Mega-ohms)	10M Ω	x 1,000,000	_____	.63k Ω = _____ Ω

Value	Meter Reading	Multiply by	Value	Practice
mA (milli-amps)	125mA	x 0.001	_____	31mA = _____ A
A (amps)	125A	x 1	125A	

Multimeter Specification Exercise

Interpreting Resistance Specifications 1



This exercise requires using information from KGIS. The vehicle is the same for each question.

- ☐ Using KGIS, select the Component Inspection for the vehicle and DTC assigned by the Instructor:

Vehicle: _____

DTC: _____

- ☐ Using the Specification Table for checking ECTS resistance for the DTC, convert the $k\Omega$ reading into Ω . Write your answers below.

Temp. (°C)	Resistance $k\Omega$	Resistance Ω
-40		
-20		
0		
20		
40		
60		
80		

Interpreting Resistance Specifications 2



- ☐ Using KGIS, select the Component Inspection for the vehicle and DTC assigned by the Instructor:

Vehicle: _____

DTC: _____

- ☐ Using the Specification Table for checking the coil primary and secondary resistances for the DTC, answer the following questions.
- ☐ What is the range for the primary coil resistance?
_____ to _____
- ☐ What is the range for the secondary coil resistance?
_____ Ω to _____ Ω



NOTE: When measuring low resistance values it is necessary to zero your meter or take the resistance of the leads into consideration for accurate measurements.

Interpreting Voltage Specifications

- ☐ Using KGIS, select the Component Inspection for the vehicle and DTC assigned by the Instructor:

Vehicle: _____

DTC: _____

- ☐ Using the Inspection/Repair W/Harness Inspection Ground Circuit Inspection procedure for the DTC, answer the following question.

What is the specification _____mV _____V

If the voltage "A" measured 5.12V and voltage "B" measured 5.15V, what is the calculated value

_____V _____mV

Would the measurement be within specification?

Yes _____ No _____

End of Exercise

[illegible]

PREDICTING ELECTRICAL READINGS

Predicting voltage in a circuit is one of the most important diagnostic skills you develop. Analyzing the circuit and predicting the voltage at critical test points ensures the following:

- Your diagnosis is efficient
- Your testing is accurate
- Your understanding of proper circuit operation
- Confidence that you have isolated the problem based on the DVOM readings made during circuit testing

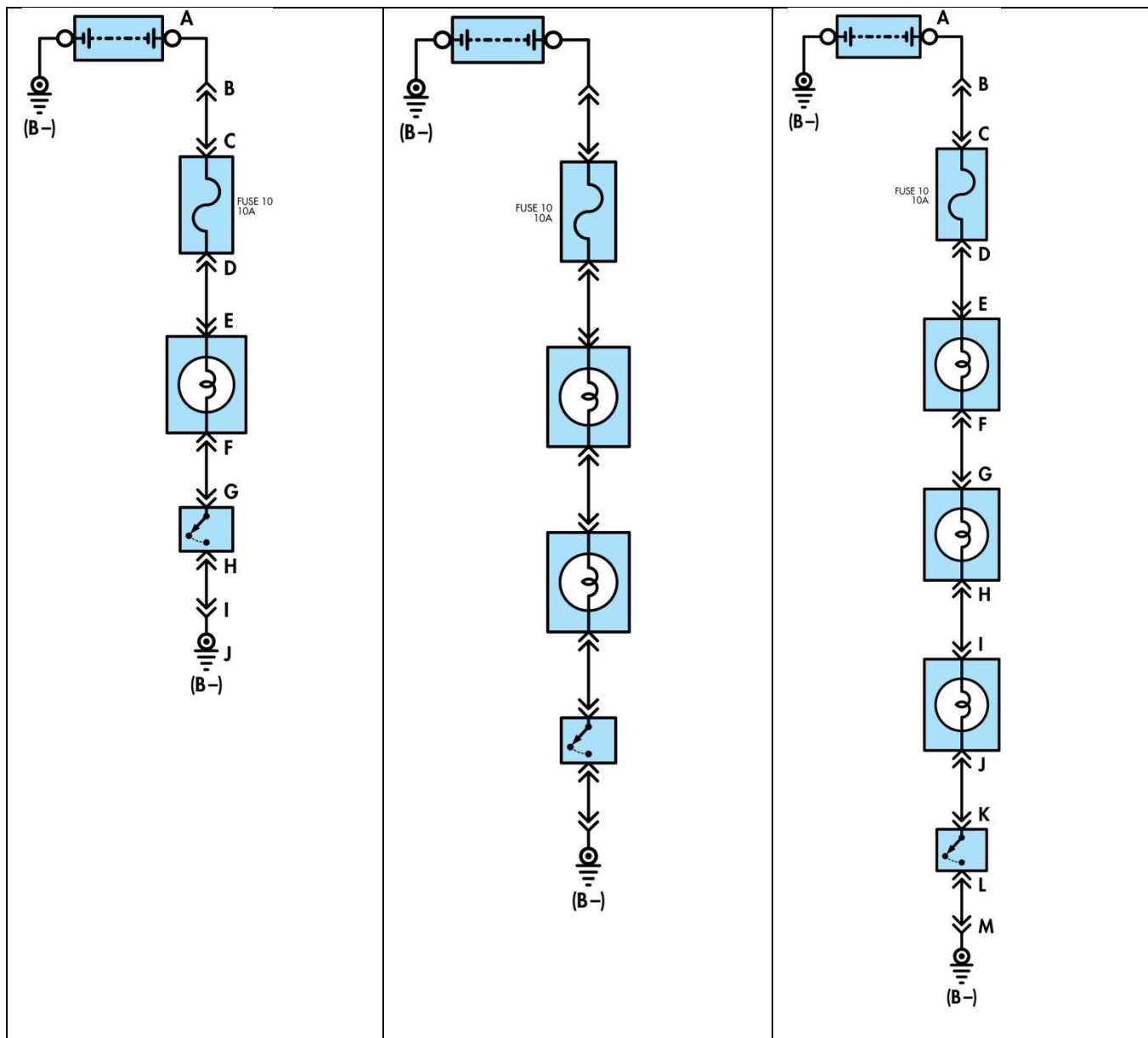
Whenever you make an electrical measurement, you should know what results to expect if the circuit is working properly.

SERIES CIRCUITS

A series circuit has only one path for current to flow. There are no branches to other loads. The components of a series circuit are assembled on the same path. A sample circuit is shown below.

Series Circuit Rules

1. **Voltage** – Voltage drops across each load, and the sum of the voltage drops equals the source voltage.
2. **Resistance** - The total resistance in the circuit is equal to the sum of each resistance in the circuit.
3. **Current** - There is only one current path in the series circuit. Current must be the same at every test point in the circuit.



Circuit A

Circuit B

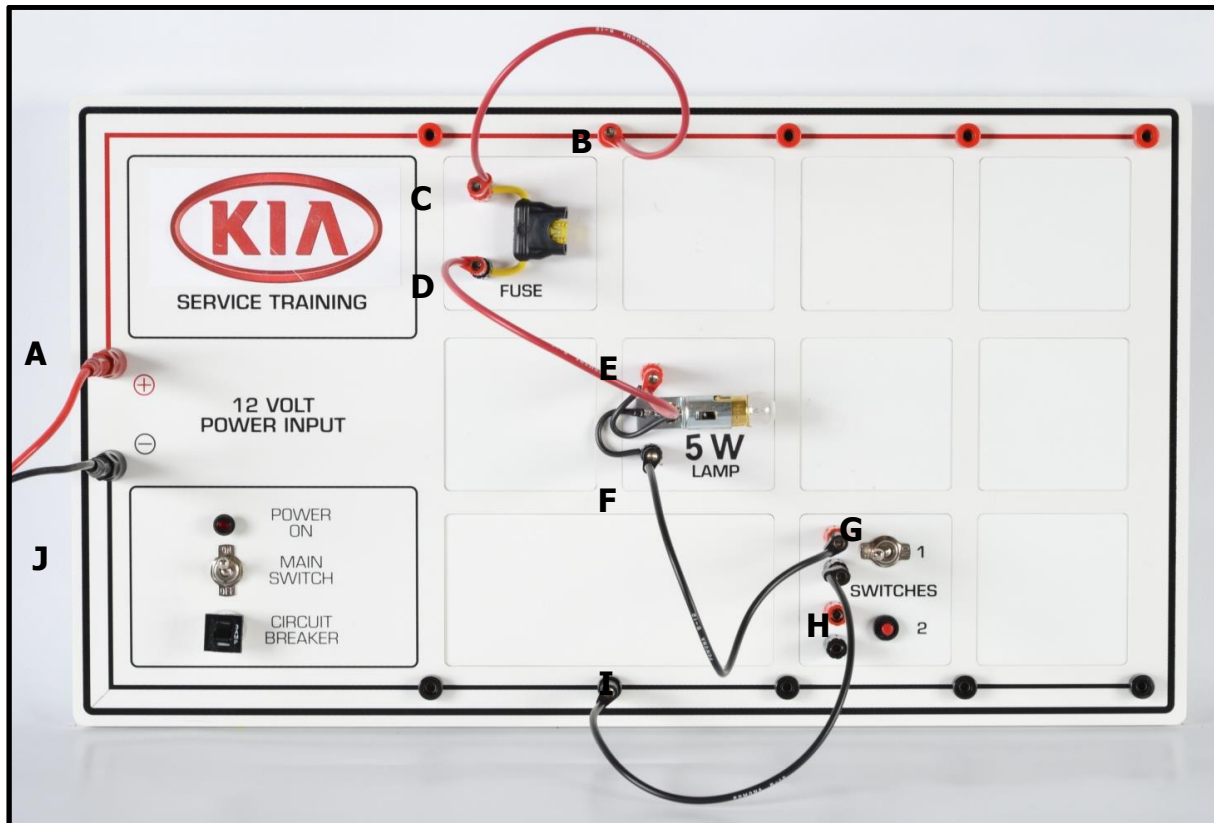
Circuit C

In the schematics shown above, record the following.

Circuit	A	B	C
Current Paths			
Loads			



NOTE: Control devices in wiring diagrams/schematics are shown in their normal state (open or closed) with the ignition in the OFF position. When predicting circuit measurements, you need to visualize the control device is in both the OFF and ON position.



SERIES CIRCUIT PREDICTING AND MEASURING

In the following exercises you will:

- Predict measurement results in a circuit
- Build the circuit using an electrical breadboard
- Confirm your predictions by making actual measurements on the electrical breadboard

APPLIED VOLTAGE EXERCISE 1

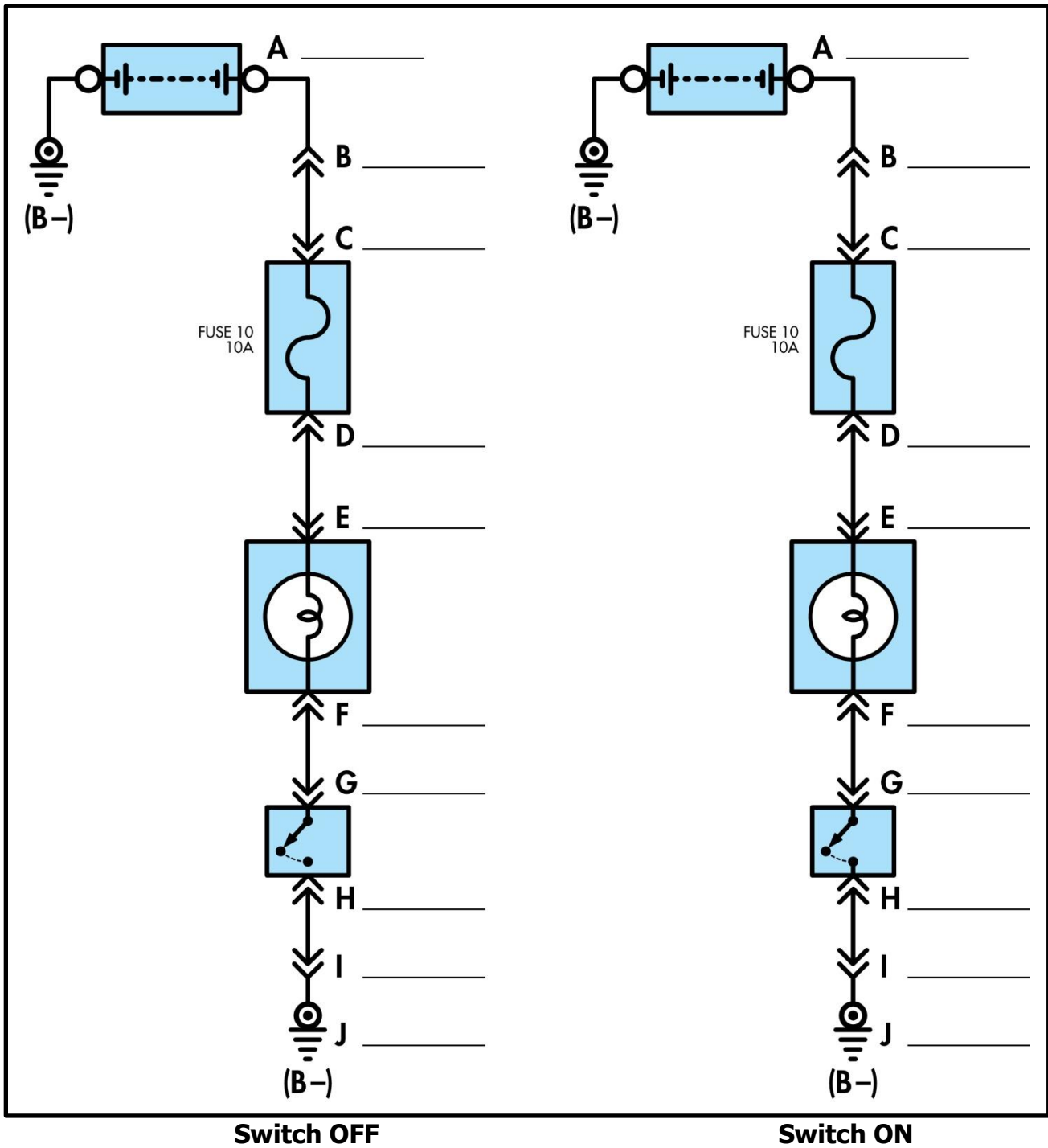


Set up your breadboard to match the circuit above. Work in teams as assigned by your instructor and build the circuit shown using the following components supplied in your Kia Electrical Workstation Kit:

- Power supply
- Jumper leads
- Fuse
- Lamp
- Switch (#1)



NOTE: The Kia Electrical Breadboard has built in wiring from the power supply connectors to the positive and ground connector terminals (view the bottom of the Workstation to see this wiring if desired).

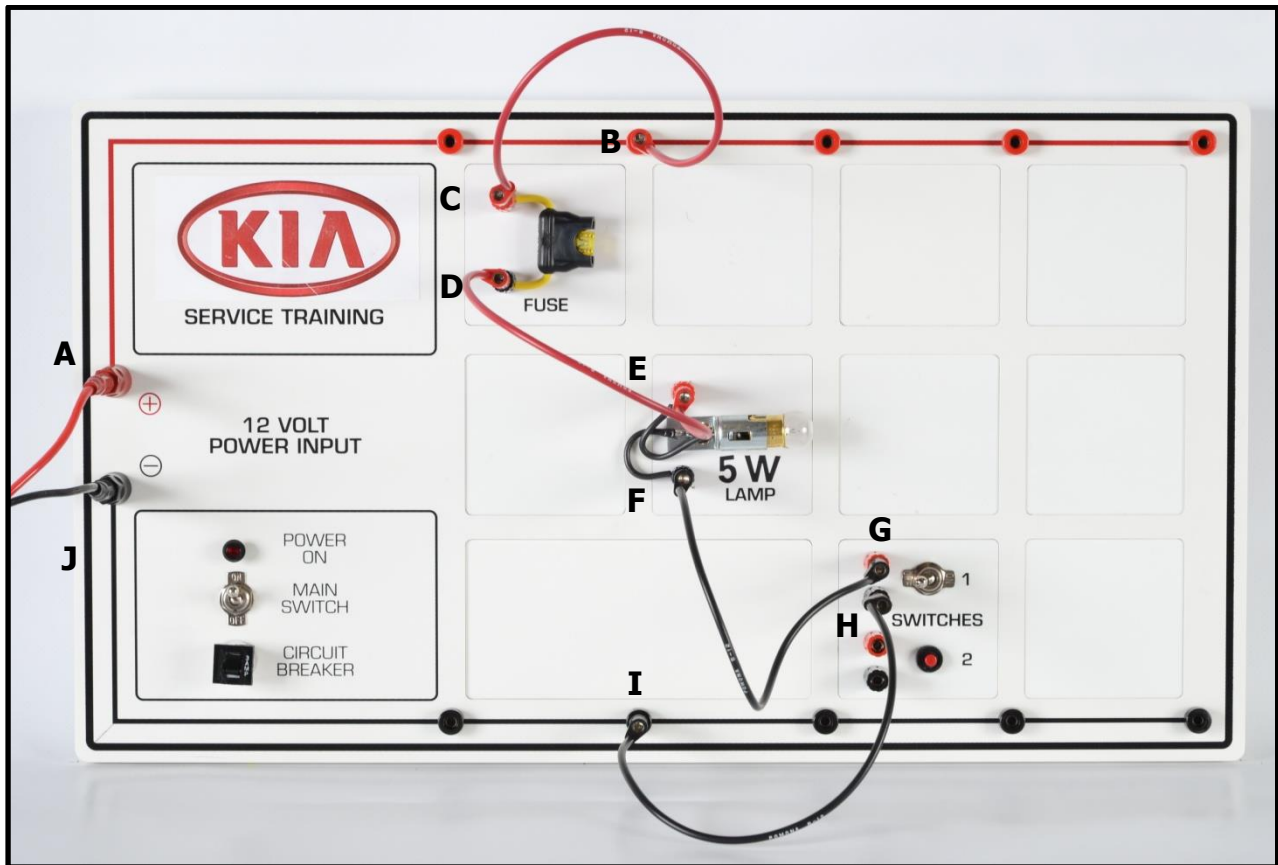


Predicting Applied Voltage

On the schematics above:



- ☐ Predict and record the voltage readings you expect to measure at each location (switch OFF and ON)



Measuring Applied Voltage

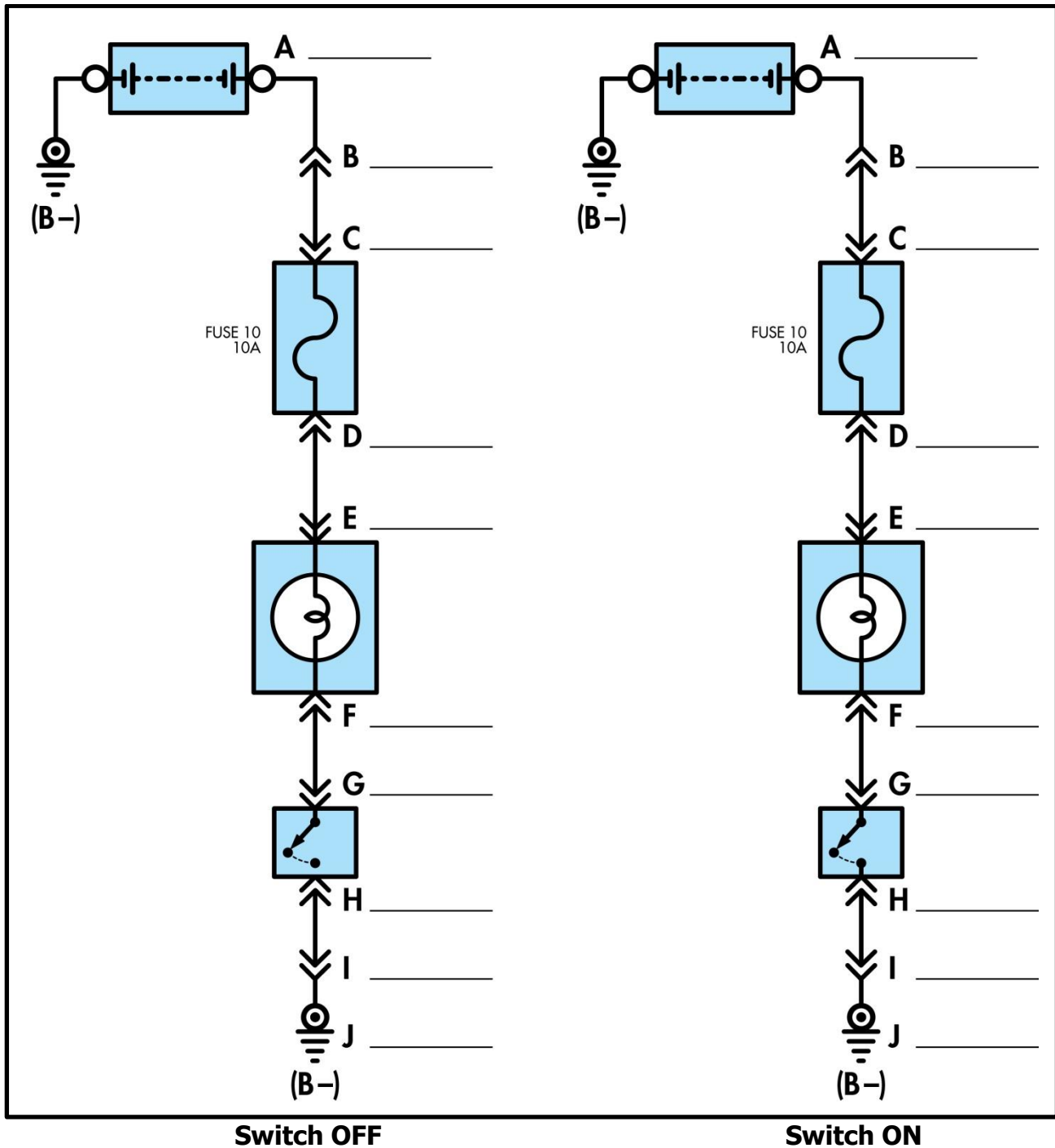
On the breadboard, build a series circuit as shown above:



- ☐ Turn the power supply on
- ☐ Turn the Main Switch ON
- ☐ Turn Switch #1 ON, the bulb should light!

If the bulb lights, continue on with the exercise.

If the bulb does not light, ask your instructor for assistance.



Measuring Applied Voltage

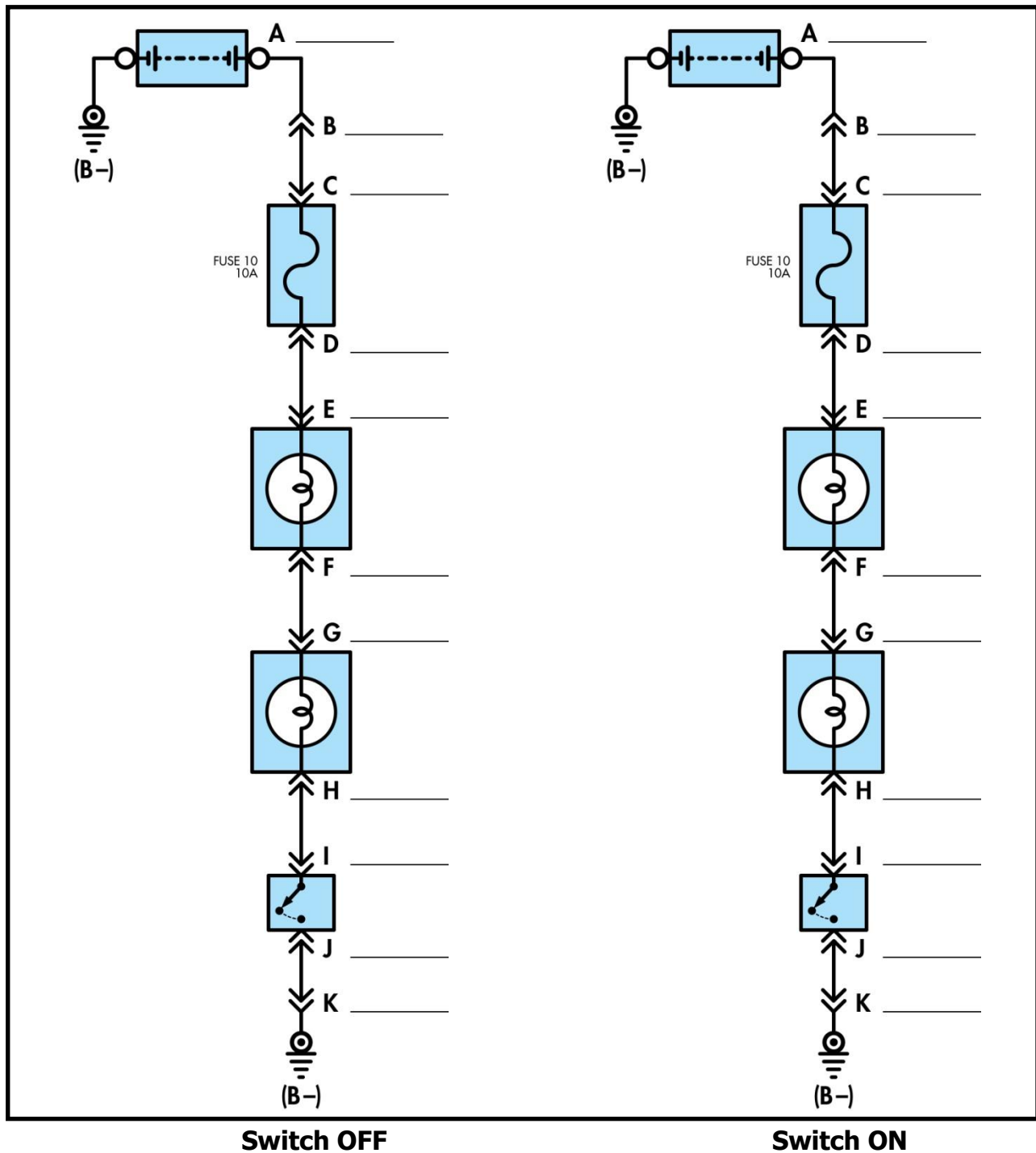
Measure the applied voltage as follows:

- ☐ Set the DVOM to measure Direct Current (DC) Voltage
- ☐ Place the DVOM ground (black) lead to the ground connection (J) at the bottom of the breadboard

- ☐ Place the DVOM positive (red) lead at each test point
- ☐ Record the Applied Voltage (switch OFF and ON) readings on the circuit schematics



Wait for further instruction.

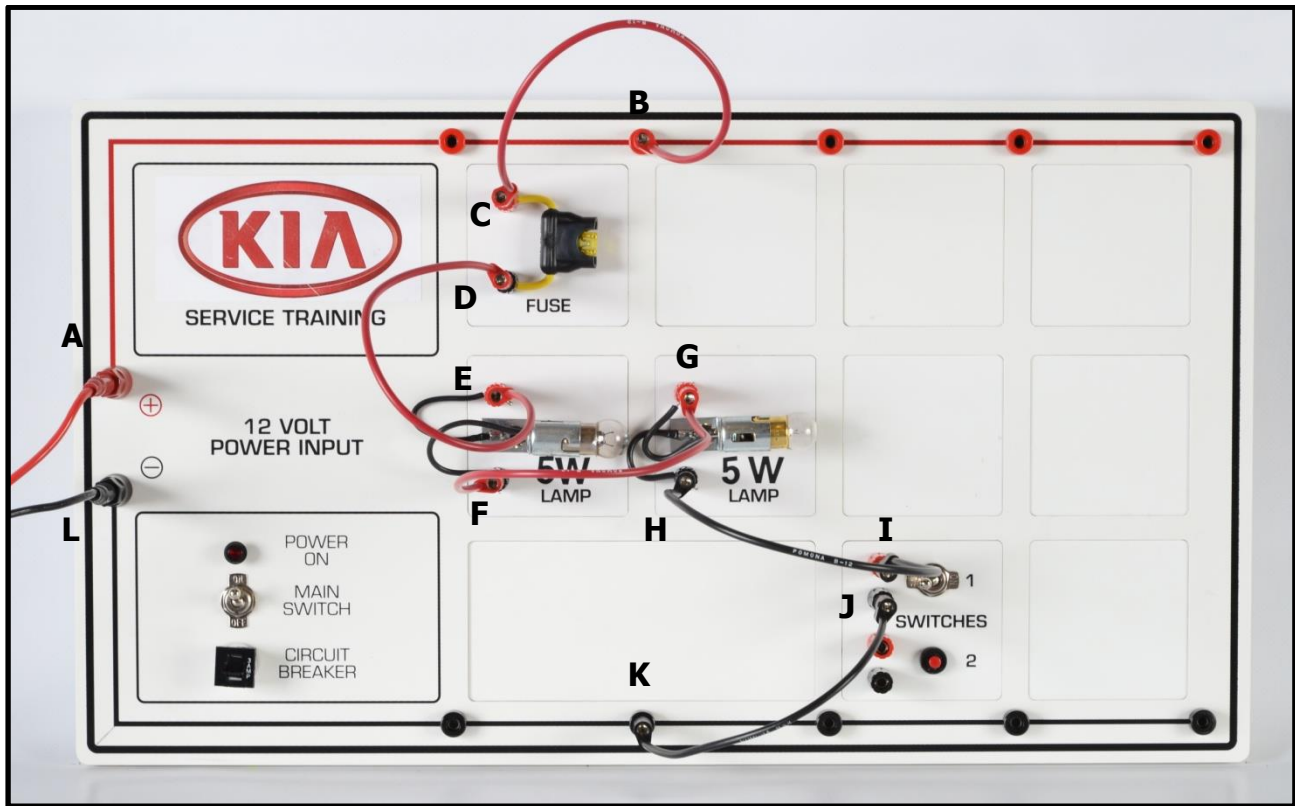


APPLIED VOLTAGE EXERCISE 2

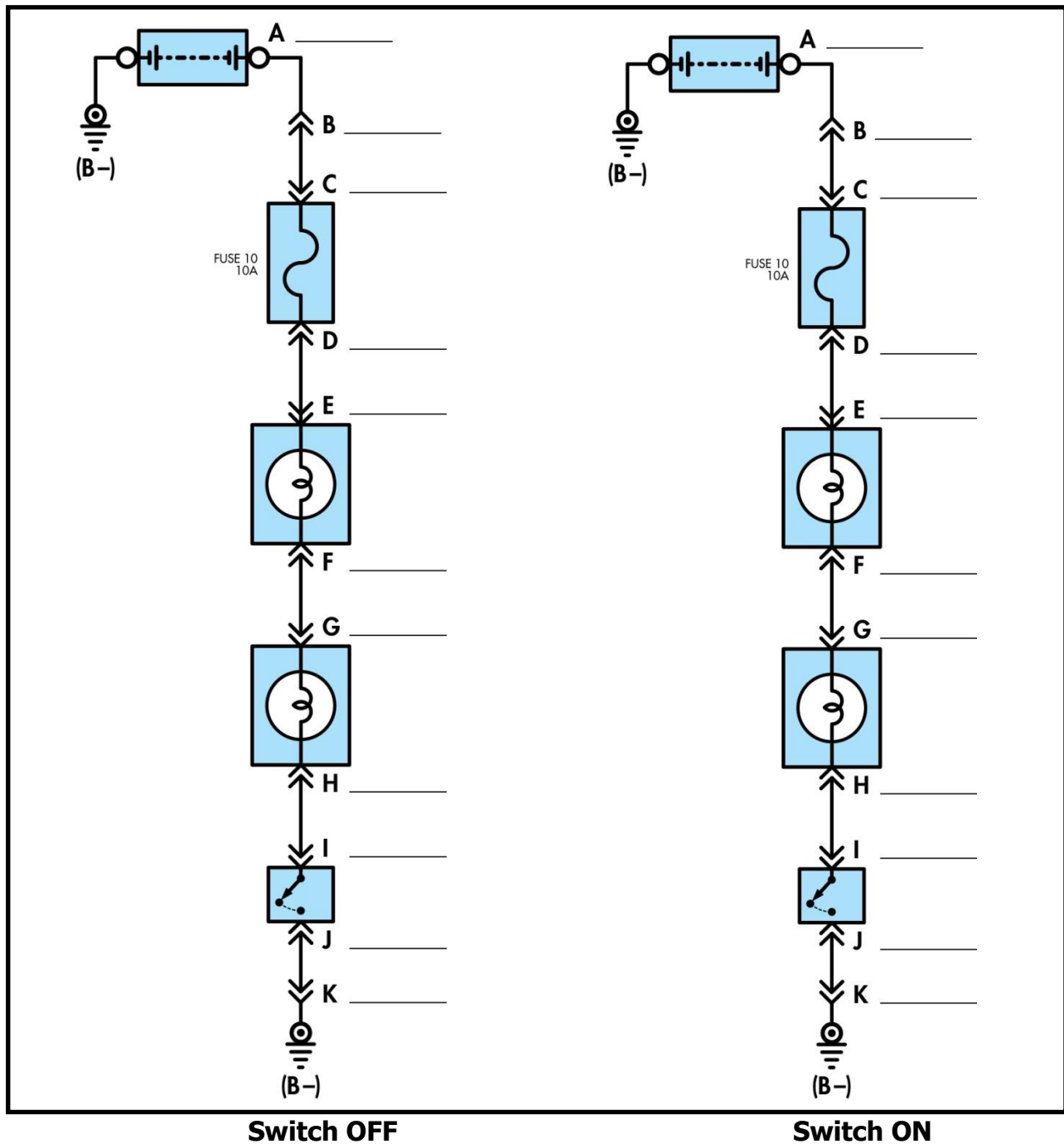
Predicting Applied Voltage

On the circuit schematics above, predict the voltage readings you expect to measure at each location (switch OFF and ON)

Each bulb has the same resistance value.



Building the Circuit On the breadboard, build a series circuit with two lamps as shown above.

**Measuring Applied Voltage**

- ☐ Perform the applied voltage test.
- ☐ Record the Applied Voltage (switch OFF and ON) readings on the wiring diagram above



Wait for further instruction.

[illegible]

RESISTANCE MEASUREMENT

Resistance measurements are made with the circuit OFF and the component disconnected from the circuit.

You typically measure the resistance of a component and compare your reading with the specifications in Service Information.

Resistance tests can be used to determine if a component is:

- Open (high resistance)
- Normal (within specification)
- Shorted (low resistance)



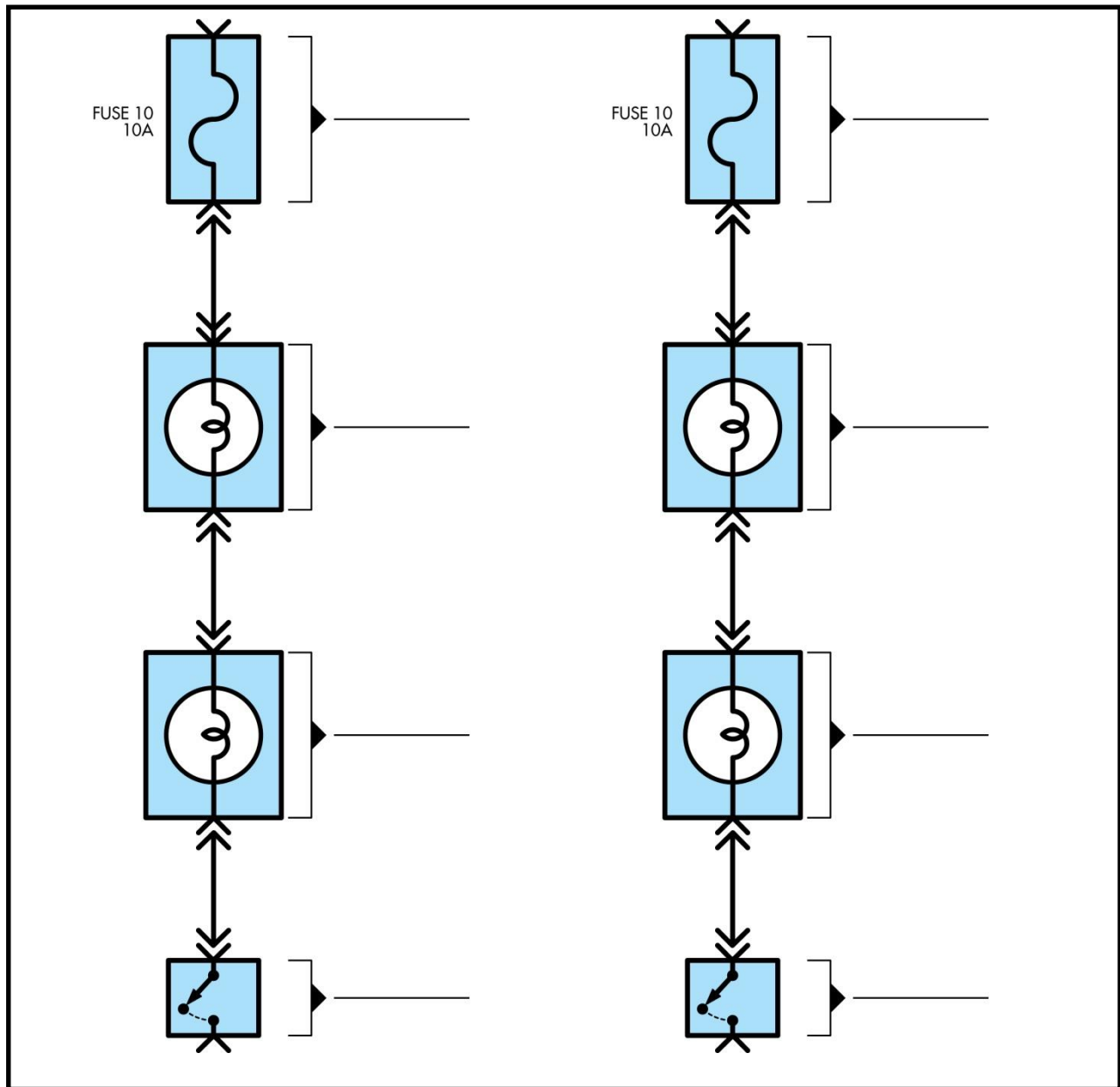
NOTE: A resistance test can be used to isolate shorts in a circuit.

If a fuse blows, there is low resistance in the circuit causing amperage to increase (Ohm's law) beyond the capacity of the fuse.

Using the meter to perform resistance / continuity tests, set the meter so that it reads continuity to ground, and then begin disconnecting connectors. At the point that the meter changes to "O.L" (or Over Limit) you may have isolated the short in the circuit.



VIDEO: Resistance



Switch OFF

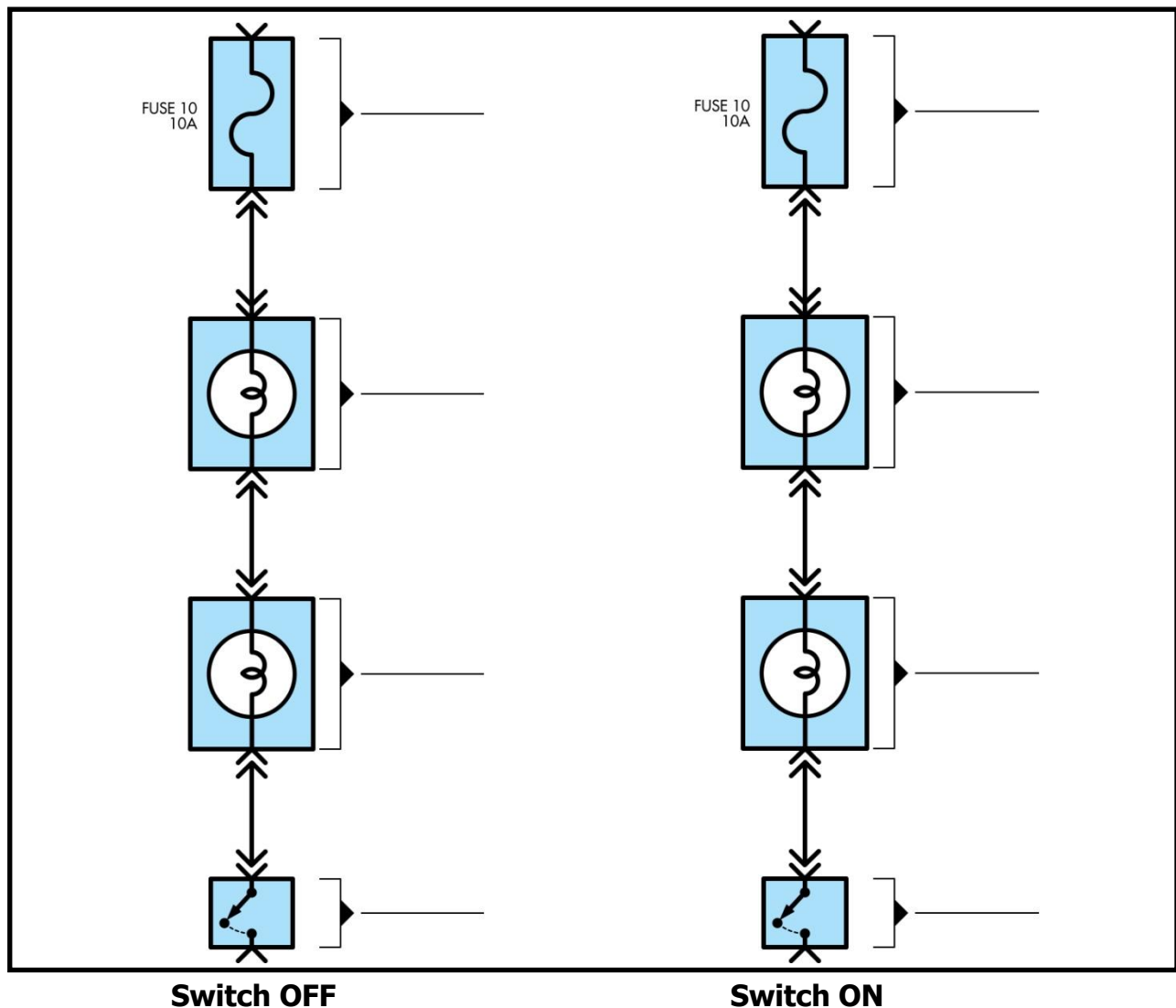
Switch ON

RESISTANCE MEASUREMENT EXERCISE 1

Predicting Resistance

Using the schematics above, predict if each component has (switch ON and OFF):

- NR = No Resistance (Near zero resistance – very low resistance)
- R = Specific resistance (designed into circuit)
- O.L = Over Limit (Open)



Measuring Resistance

Disconnect the power supply from the breadboard.

On the circuit you previously built (2 bulb series), using a DVOM, measure the resistance in each segment of the circuit, or circuit component, with the switch in both the OFF and ON positions. Record your results on the schematic.

- ☐ Measure the resistance of the complete circuit by measuring at point A and point L with the power supply cables disconnected from the workstation and record your reading.

_____ Ω

- ☐ Using the measurement you already made, add the resistance of bulb 1 and bulb 2

Ω Bulb 1 + Ω Bulb 2 = _____ Ω

- ☐ Does the resistance of both bulbs added together equal total circuit resistance?

YES / NO (circle one)

Why or why not? _____



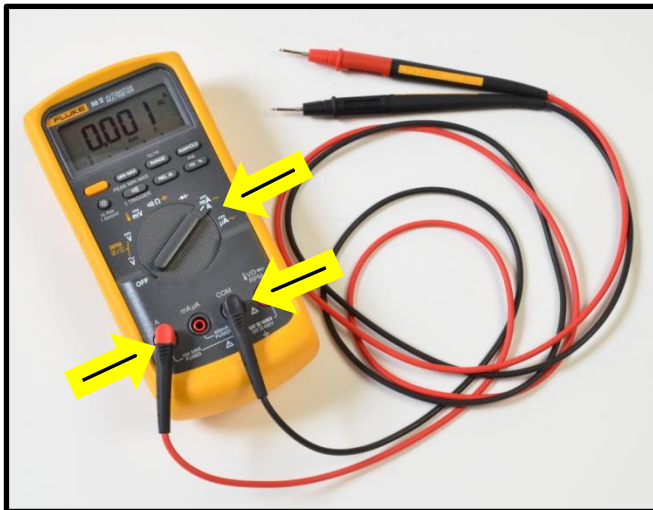
Wait for further instruction.

CURRENT MEASUREMENT

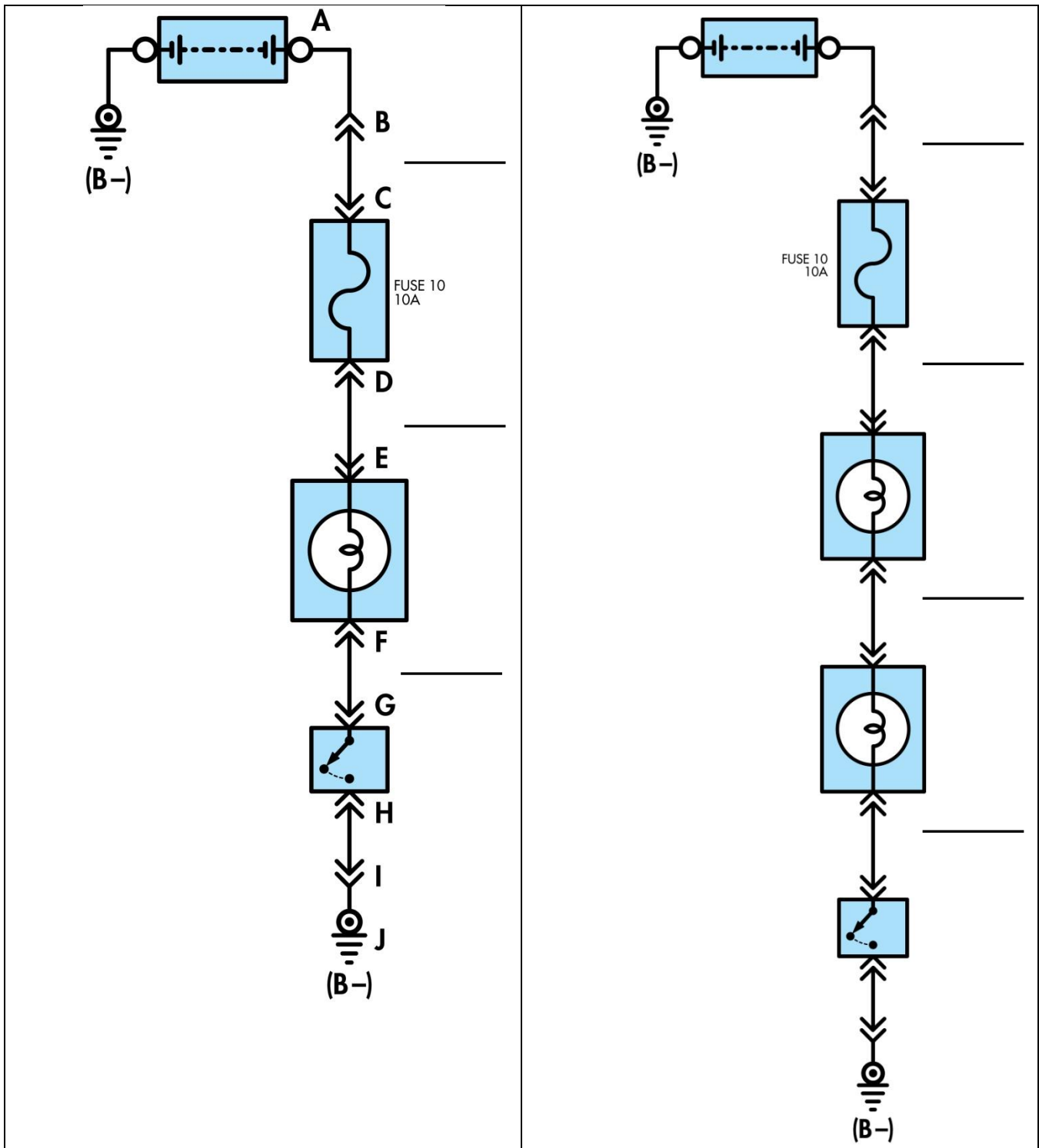
Current is measured by selecting amperage using the rotary switch and moving the red lead to the amperage jack on the DVOM.

The meter is connected in series in the circuit. Think of it as a fused jumper that measures current.

Most DVOMs can safely measure up to 10 amps of current. Many automotive circuits operate at higher current levels than 10 amps (note the fuse rating for the circuit you are working on). If the amperage of the circuit exceeds your DVOM's measurement capabilities, do not attempt to take the reading. Use a meter with an inductive clamp. Most DVOM's have a fuse for the amperage section of the unit to protect you and the DVOM when current exceeds DVOM measuring capabilities. Failure to follow proper procedures can damage vehicle components, the DVOM, and cause injury.



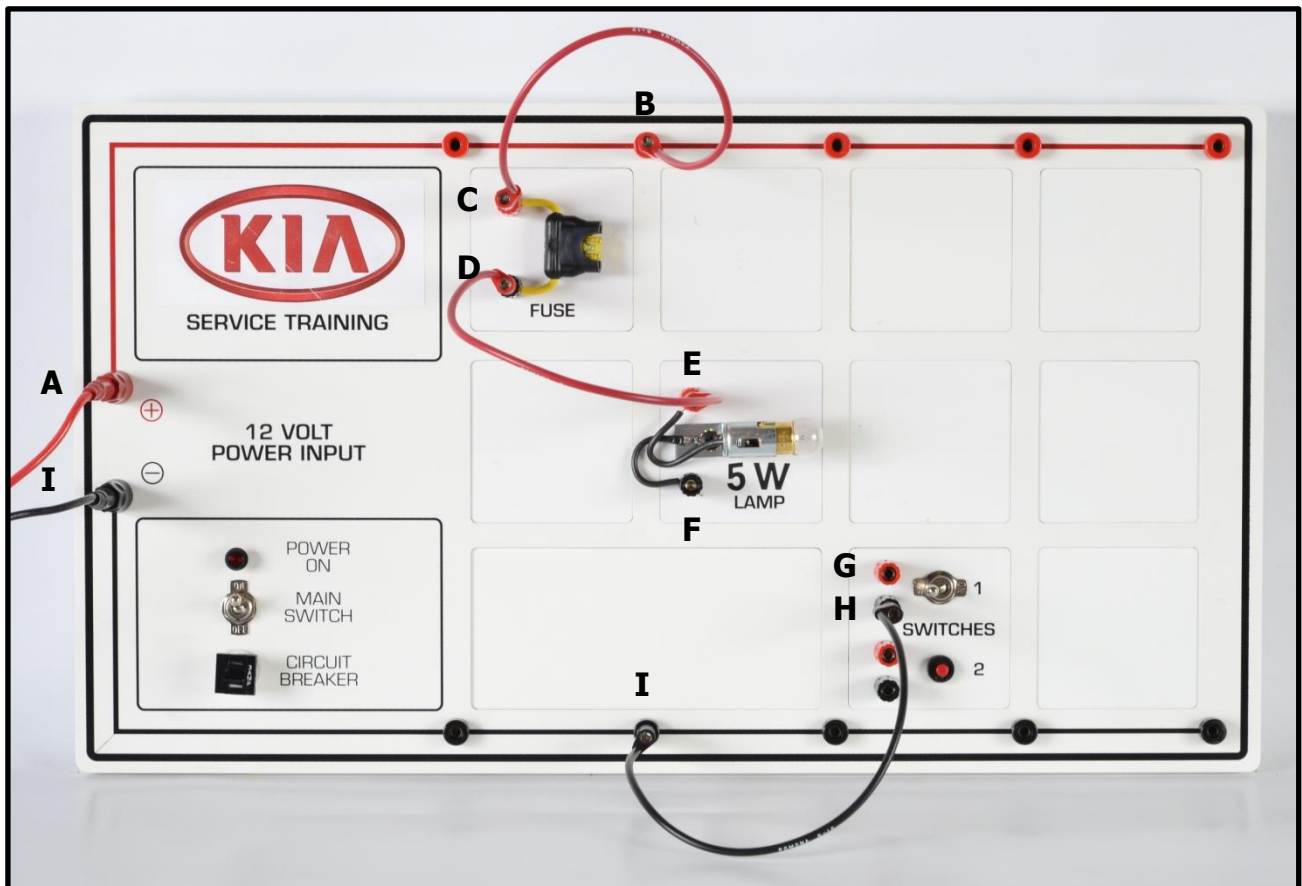
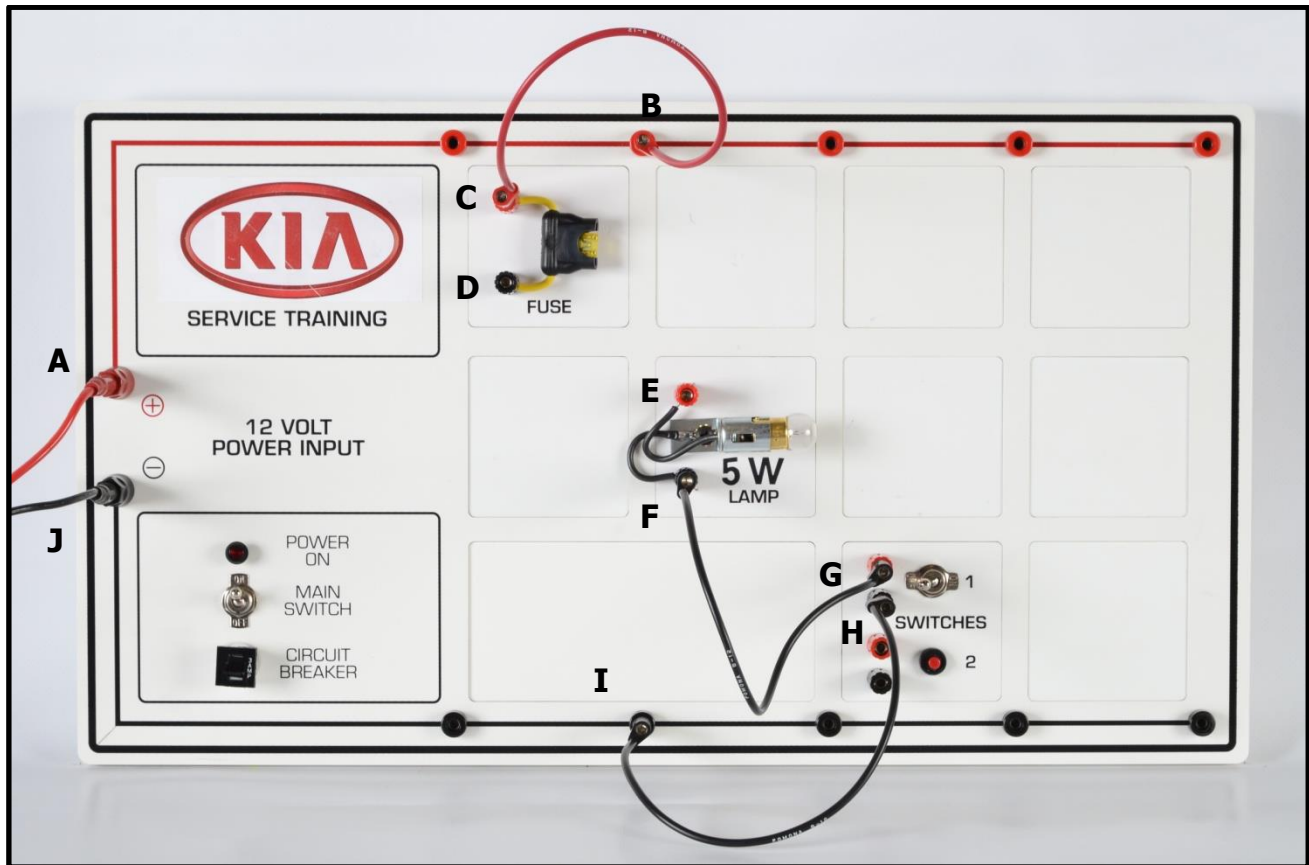
NOTE: Measuring voltage with the test leads in the amperage position may blow the internal fuse in the meter.



Predicting Current (Amperage)

Predict the amperage for each circuit at the locations listed above.

Assume each circuit's total amperage is 1.0 amp.



**CURRENT MEASUREMENT
EXERCISE 1**

Measure amperage as follows:

- ☐ Select DC amps on your DVOM
- ☐ Move the red lead on the DVOM to the amperage jack (if there is more than one, always start with the highest available)
- ☐ Create an **open** circuit on the breadboard by **removing the wire between point D & E**
- ☐ Connect the red lead of the DVOM to point D
- ☐ Connect the black lead of the DVOM to point E
- ☐ Turn the power supply ON (the bulb should light)
If the bulb does not light, have your instructor check the circuit and meter connections.
- ☐ With the bulb ON, record the amperage reading below:
_____A _____mA
- ☐ Turn the circuit OFF
- ☐ **Reconnect the wire between D & E**
- ☐ **Remove the wire between point F & G.** Connect the DVOM red lead to test point F (use an alligator lead if necessary)
- ☐ Connect the black lead of the DVOM to the wire going to point G
- ☐ Turn the power supply ON. What is the amperage reading? _____A _____mA

Is this reading the same as the previous amperage reading you made? YES / NO (circle one)

How many paths for current are there in a series circuit? _____

Is the current the same anywhere you measure in a series circuit? YES / NO (circle one)

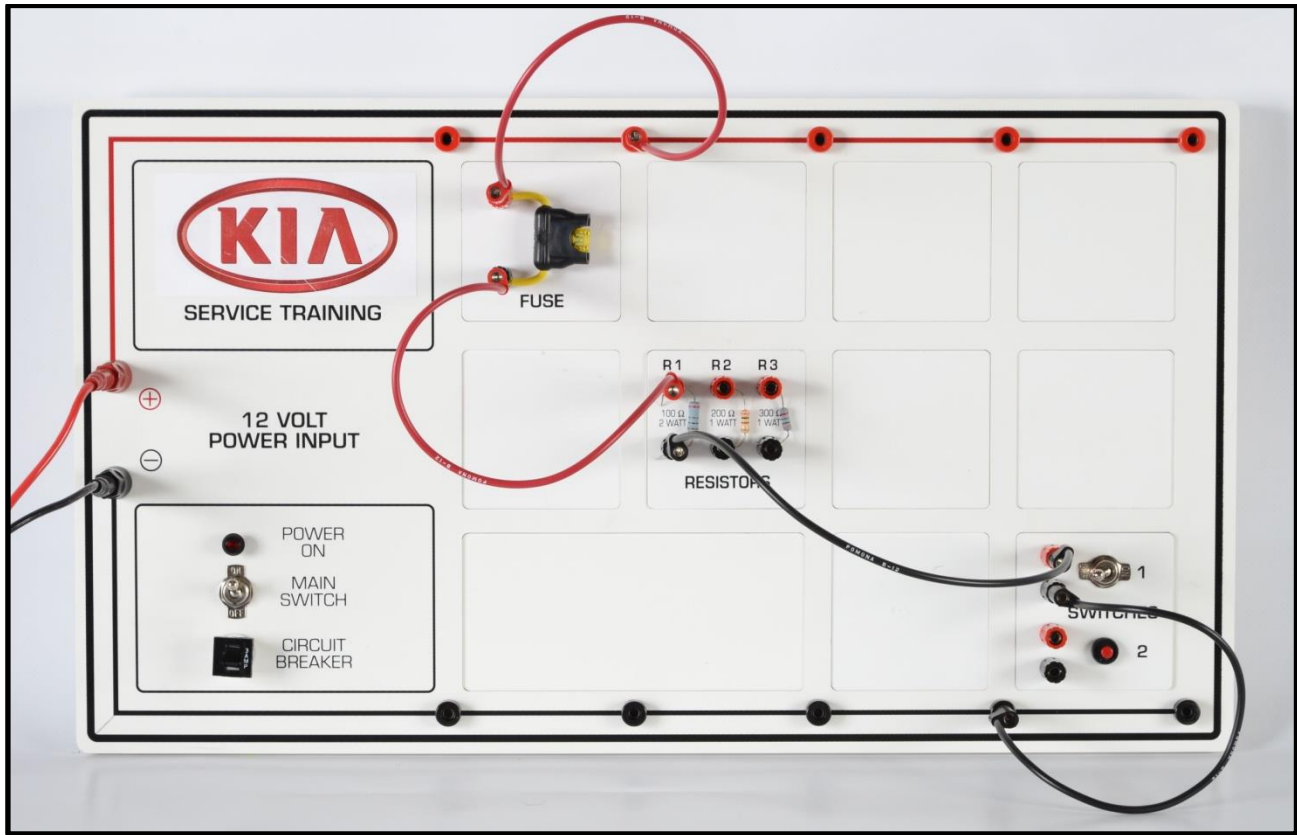


CAUTION: After completing your amperage measurement, place the red lead back in the Volt/Ohms jack on your DVOM. This is a good habit to practice. If you leave the lead in the amperage jack on the DVOM, and you then go to measure voltage, you will have a short circuit through the meter. This can damage the DVOM, blow the fuse in the DVOM, and cause vehicle damage or injury if you cause a spark while measuring source voltage at the battery.



Wait for further instruction.

[illegible]



VERIFYING OHM'S LAW



In this activity, you will observe the relationship between current and resistance.

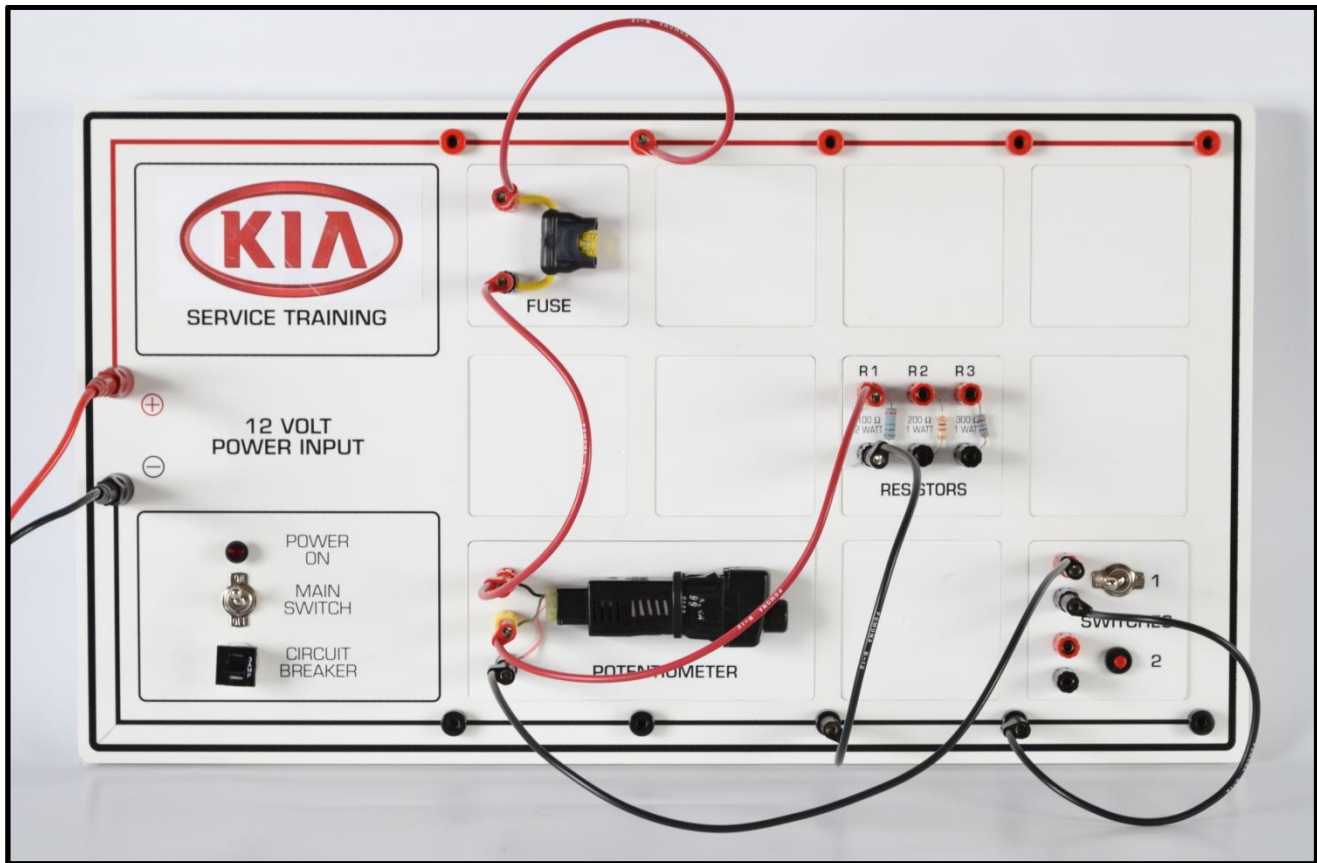
- ☐ Build the circuit shown above connecting R1 with lead wires
- ☐ Measure and record the amperage below
- ☐ Repeat individual amperage measurements with R2 & R3

R1 (100Ω) _____ A _____ mA

R2 (200Ω) _____ A _____ mA

R3 (300Ω) _____ A _____ mA

Explain what you discovered about the relationship between voltage, current, and resistance in a circuit.



VERIFYING OHM'S LAW



In this activity, you will observe the relationship between voltage and current.

- ☐ Build the circuit shown above connecting R1 with lead wires
- ☐ Use the potentiometer to set the voltage at the top of R1 to the value of V1 listed below
- ☐ Measure and record the amperage
- ☐ Repeat individual amperage measurement with V2 & V3

V1 (4Vdc) _____ A _____ mA

V2 (8Vdc) _____ A _____ mA

V3 (12Vdc) _____ A _____ mA

Explain what you discovered about the relationship between voltage, current, and resistance in a circuit.

SERIES CIRCUIT SUMMARY

In this section:

- You built series circuits using the breadboard.
- You predicted, measured, and verified the following values:
 - Voltage
 - Resistance
 - Amperage

Series Circuit Rules

From your series circuit practice, you have verified the following series circuit rules:

- The sum of the voltage drops equals source voltage
- There is one current path therefore current is the same at every point in the circuit
- Total circuit resistance equals the sum of each resistance in the circuit

PROGRESS CHECK



1. The three elements needed for an electrical circuit are source, _____ and ground.
 - A. circuit protection
 - B. load
 - C. switch
 - D. wiring harness
2. In an electrical circuit, what causes a fuse to blow?
 - A. Low resistance
 - B. High resistance
 - C. High current flow
 - D. Both A and C
3. Which of the following is the unit of measure for resistance?
 - A. Volts
 - B. Amperage
 - C. Ohms
 - D. Watts
4. Voltage _____ as it pushes through resistance.
 - A. slightly increases
 - B. does not change
 - C. drops
 - D. doubles
5. When measuring amperage in a circuit, how should the ammeter be connected?
 - A. In series
 - B. In parallel
 - C. Across the battery
 - D. From ground to the point being tested

PROGRESS CHECK



6. 125mV = _____ V

- A. .125
- B. 1.25
- C. 12.5
- D. 125

7. 25.5K Ω = _____ Ω

- A. 2.55
- B. 255
- C. 2,550
- D. 25,500

8. In a series circuit, how many paths does current have to flow?

- A. As many paths as there are loads in the circuit
- B. One path for current flow
- C. Three separate sources and one ground
- D. Current does not flow in a series circuit

Automotive Electrical Diagnosis

Electrical Circuits

TARGET AUDIENCE The target audience for this module consists of Kia service technicians who have basic understanding of automotive electricity.

MODULE GOAL Upon completion of this module, you will be able to list expected measurements and measure voltage, resistance, and amperage in different types of electrical circuits.

MODULE OBJECTIVES Following the instructions in this module, and given a Digital Volt-Ohm Meter (DVOM) and electrical “breadboard”, you will be able to:

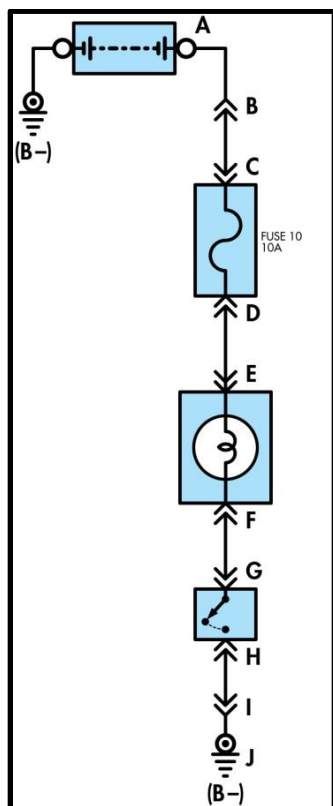
- Build parallel electrical circuits and accurately measure the following with 100% accuracy:
 - Applied Voltage
 - Voltage Drop
 - Amperage
 - Resistance
- Diagnose abnormal voltage readings in a series and parallel circuits and list the root cause with 100% accuracy.

MODULE INSTRUCTIONS Carefully read the material, take notes based on the classroom discussion, and study each illustration.

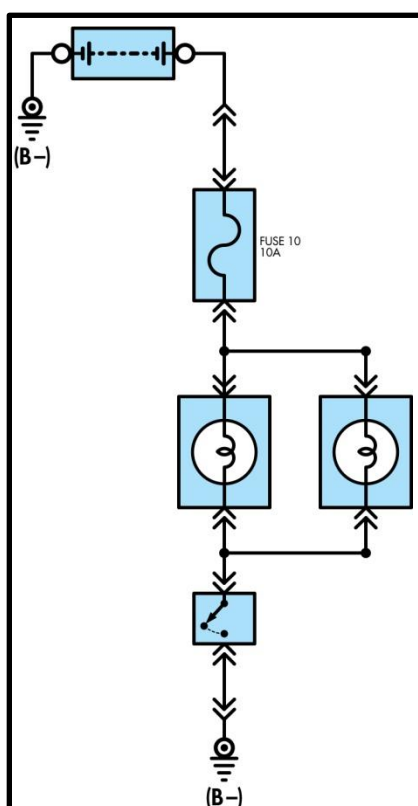
REQUIRED MATERIALS In order to complete this module, you will need the following items:

- DVOM
- Electrical “breadboard”
- Colored highlighters (pink/red, yellow, green)

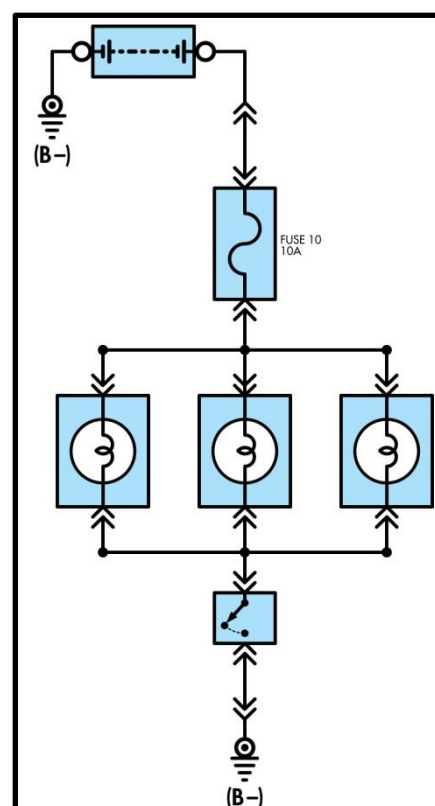
TIME TO COMPLETE Approximately 1.75 hours



Circuit A



Circuit B



Circuit C

PARALLEL CIRCUITS

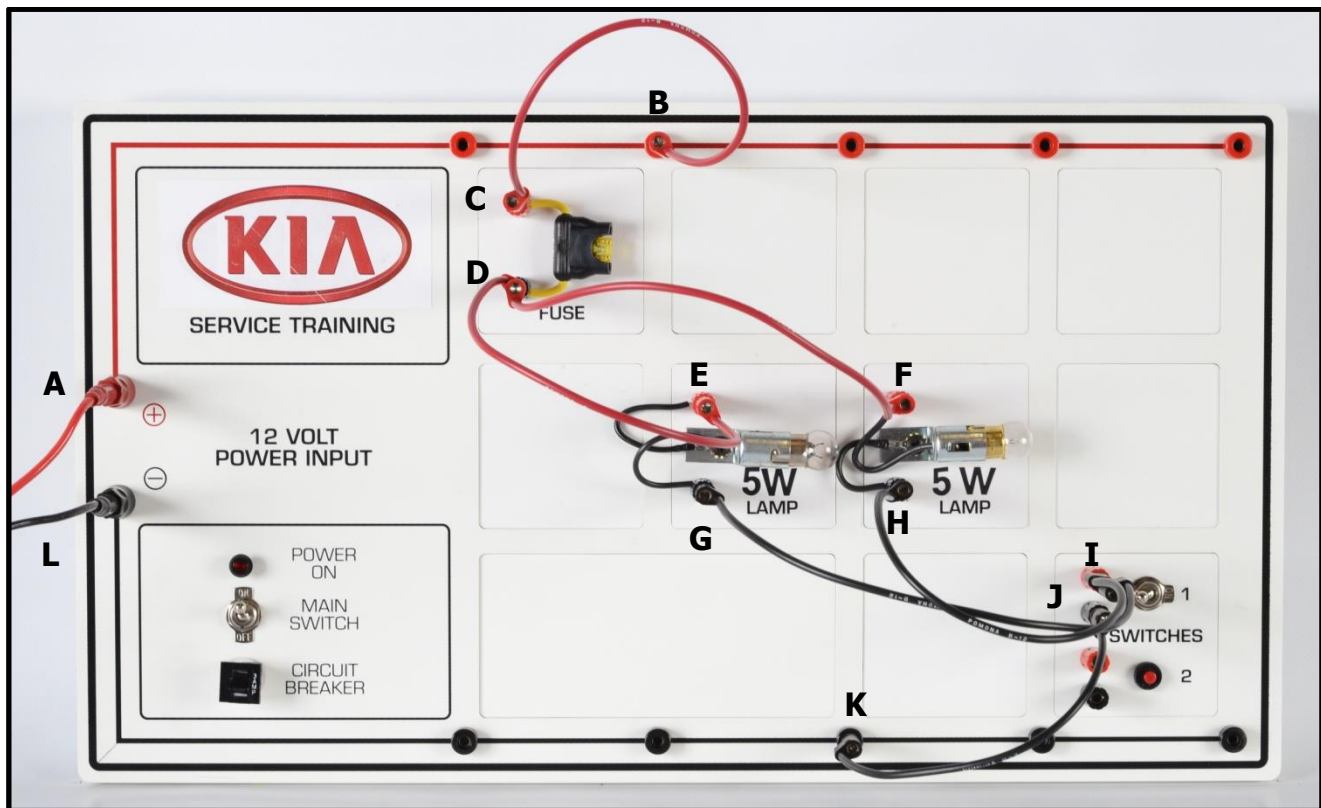
A parallel circuit is created when each load has an individual source of power and ground, or path for current. The paths for current are often called branches (or legs).

Parallel Circuit Rules

1. **Voltage** - Each branch drops source voltage (some loss expected due to connections).
2. **Resistance** - The total circuit resistance is always less than the lowest individual resistance.
3. **Current** - The total circuit current is equal to the sum of each of the individual branch currents.

Use the schematics above to determine and record the number of current paths and loads in each circuit.

Circuit	A	B	C
Current Paths			
Loads			



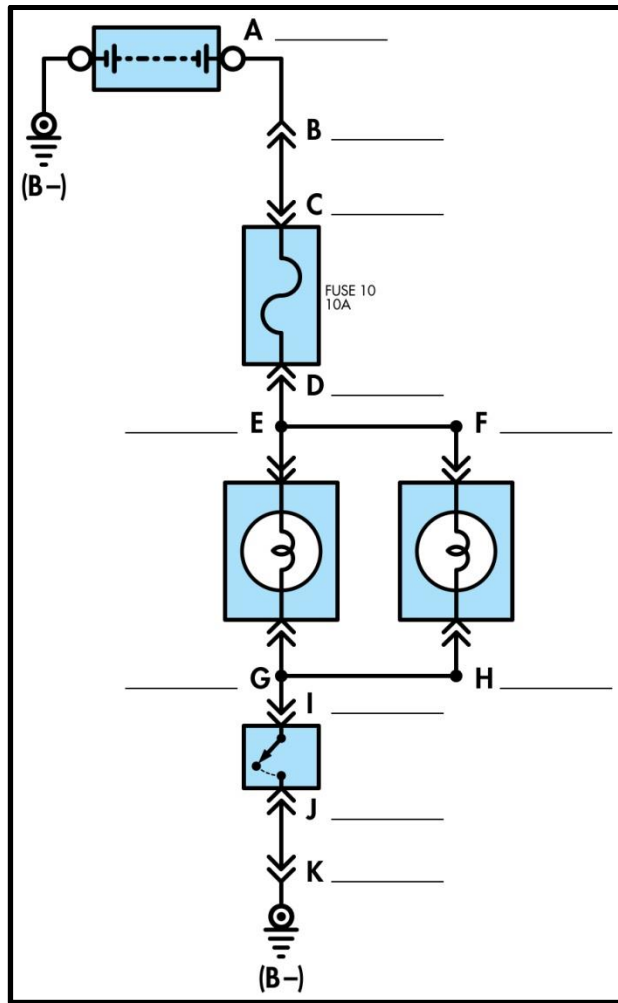
PARALLEL CIRCUIT MEASUREMENT EXERCISE 1

Voltage

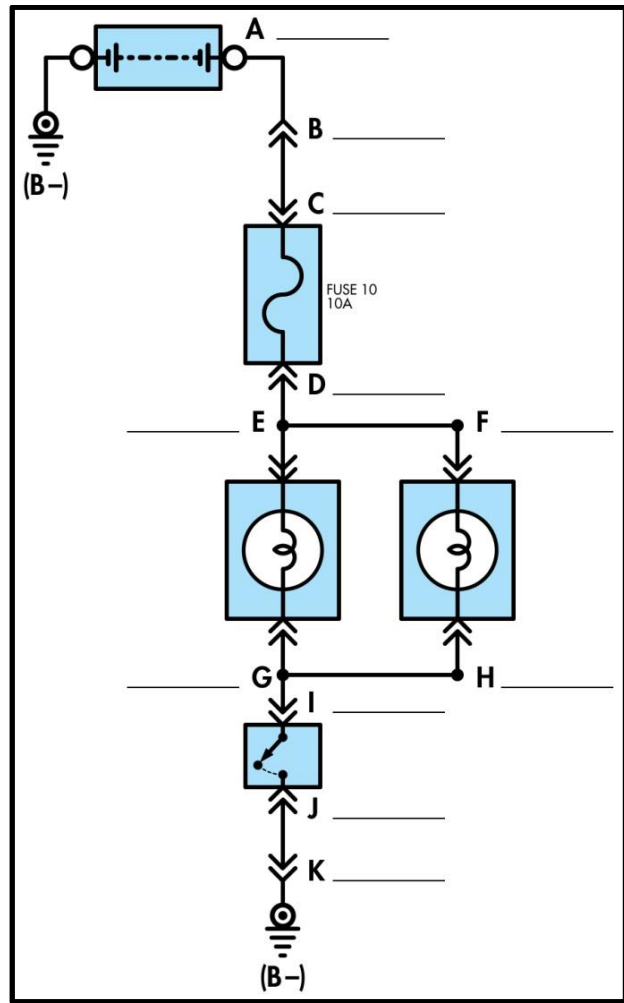


Begin your parallel circuit work by assembling the parallel circuit on your breadboard as shown above:

- Turn the circuit ON. Does each bulb light? YES / NO (circle one)
- If NO, ask your instructor for assistance.
- Remove the wire going to one bulb.
- Does the other bulb remain ON? YES / NO (circle one)



Switch OFF



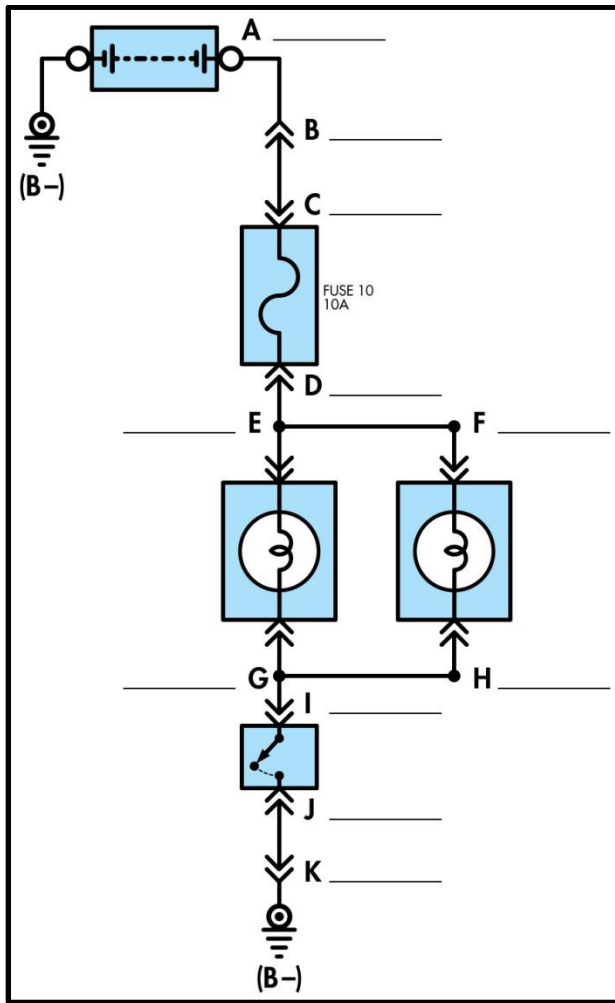
Switch ON

**APPLIED VOLTAGE
(PREDICT)**

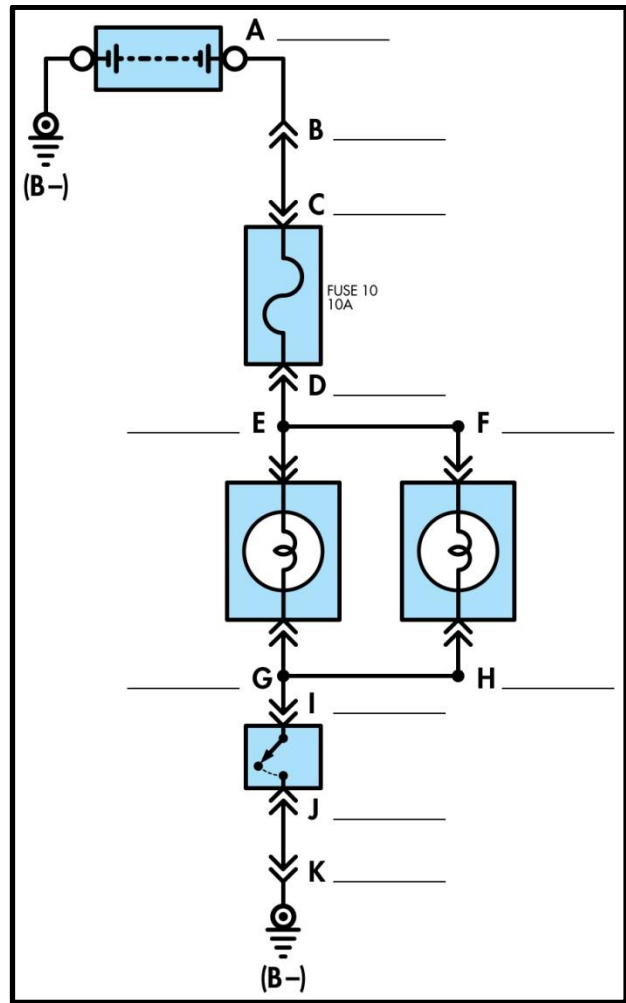
All loads in a parallel circuit receive source voltage. The voltage is not divided like in a series circuit.

On the schematics above:

- ☐ Predict and record the voltage readings you expect to measure at each location (switch OFF and ON)



Switch OFF

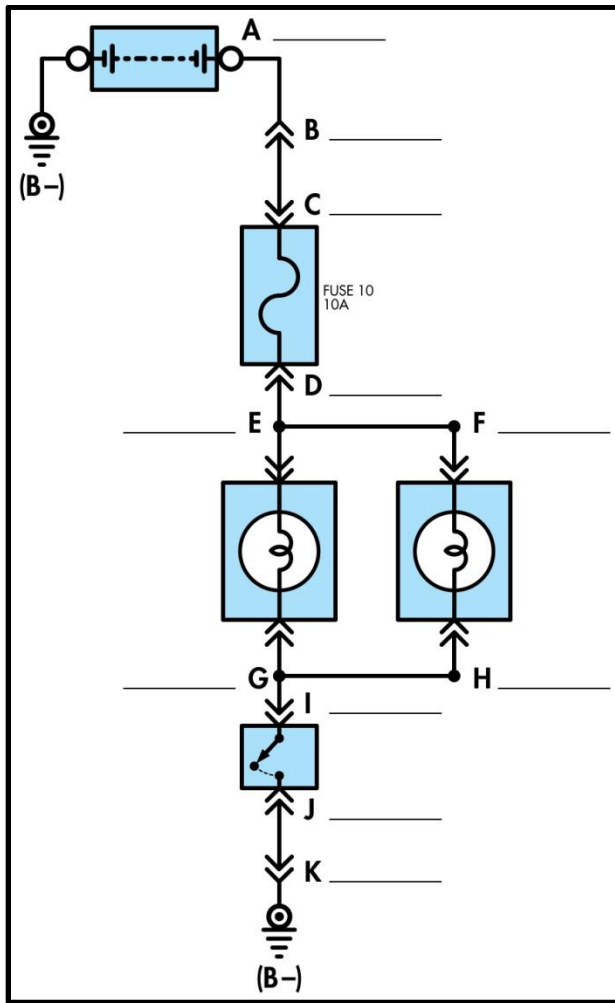


Switch ON

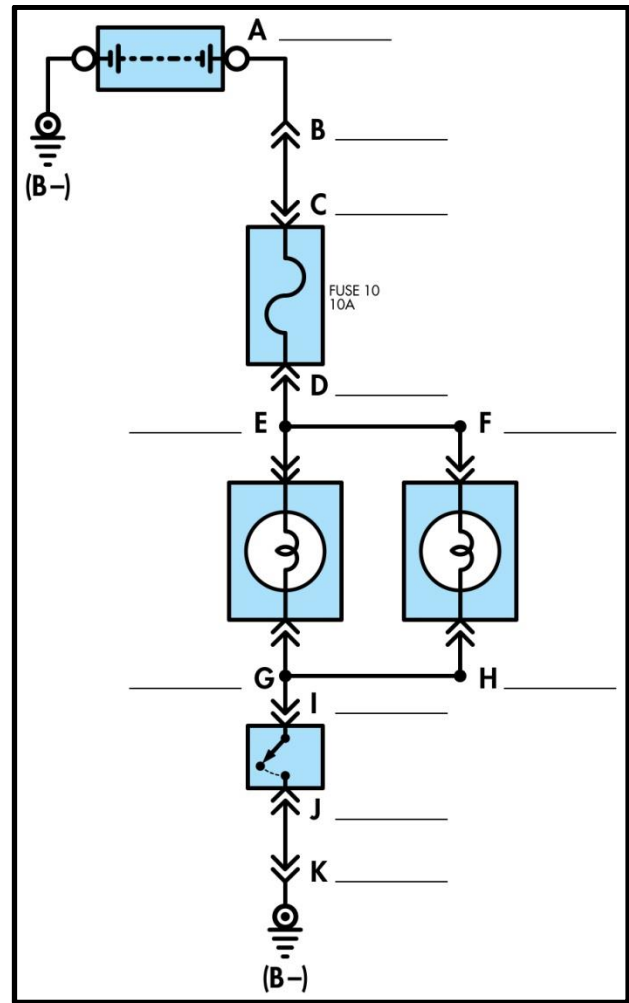
APPLIED VOLTAGE (MEASURE)

Measure the applied voltage as follows:

- ☐ Set the DVOM to measure Direct Current (DC) Voltage
- ☐ Place the DVOM ground (black) lead to the ground connection (K) at the bottom of the breadboard
- ☐ Place the DVOM positive (red) lead at each test point
- ☐ Record the Applied Voltage (switch OFF and ON) readings on the circuit schematics above



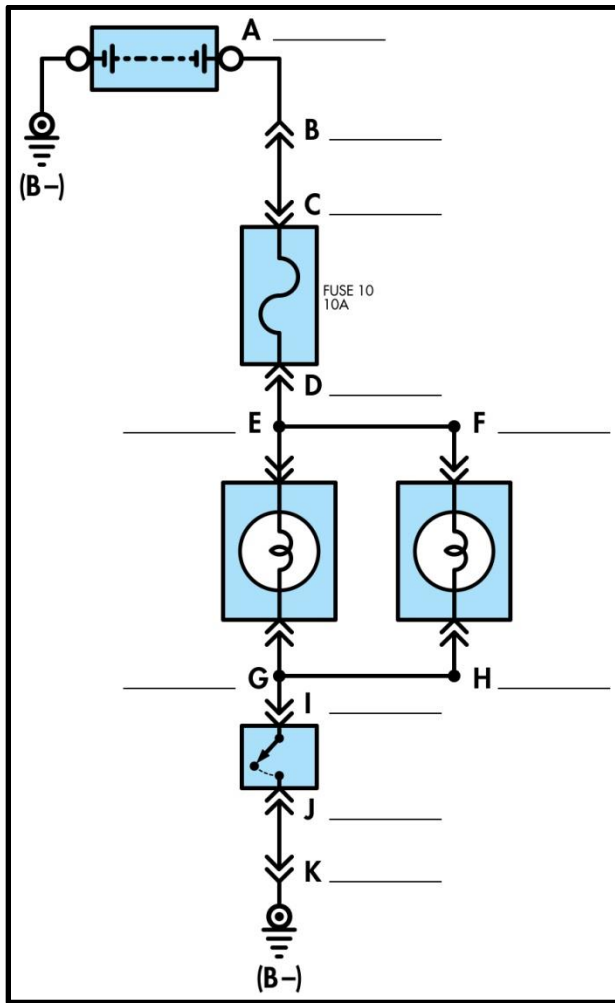
Switch OFF



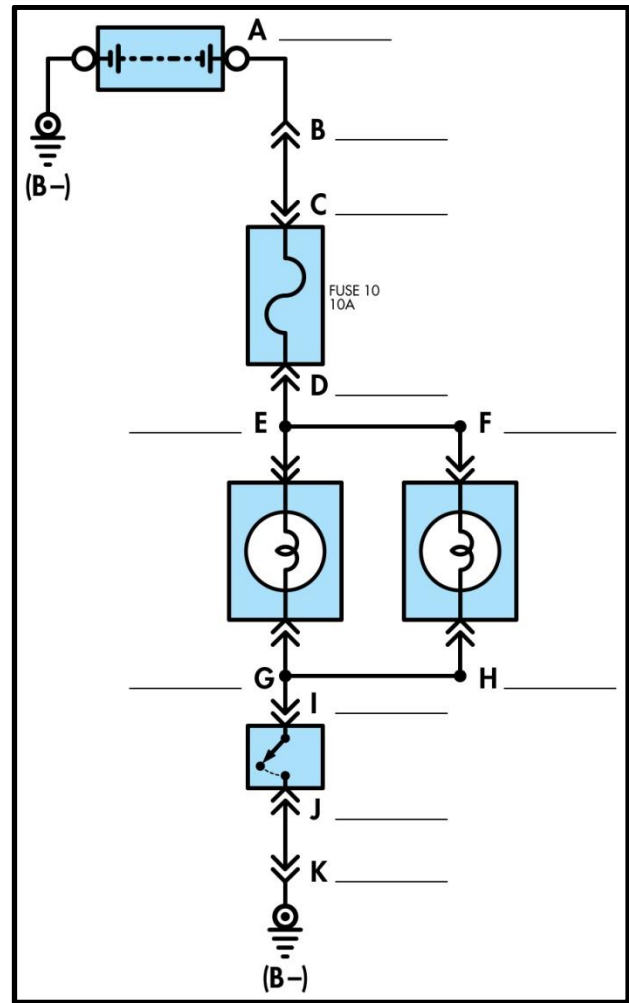
Switch ON

**APPLIED VOLTAGE
(PREDICT)**

- ☐ Replace one bulb in the circuit with a **(10 Watt bulb)**
- ☐ Predict the available voltage at the points shown in the circuit schematics



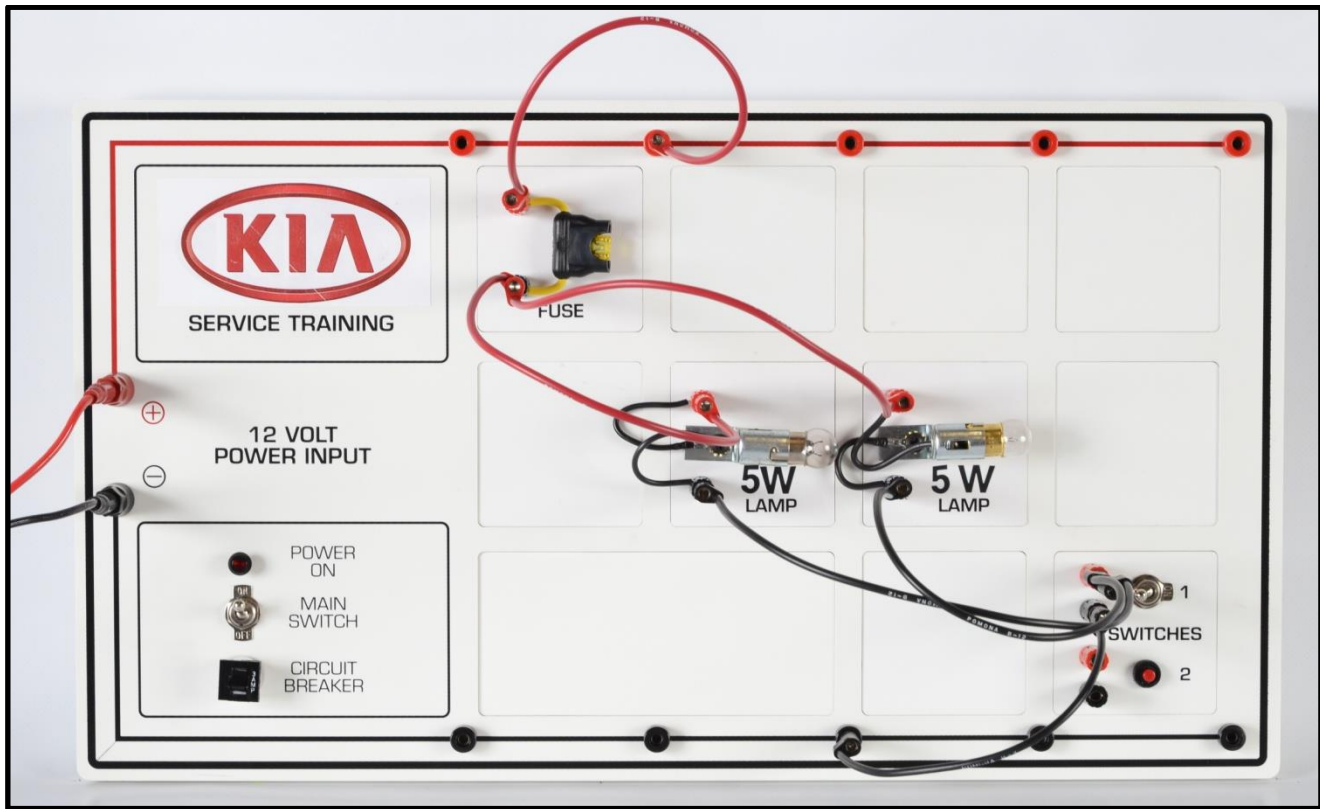
Switch OFF



Switch ON

**APPLIED VOLTAGE
(MEASURE)**

- ☐ Measure the available voltage with a **(10 Watt bulb)**
- ☐ Record your readings on the circuit schematics above



In a parallel circuit, each load has its own path for current. Total circuit amperage will equal the sum of the current in each branch of the parallel circuit.

PARALLEL CIRCUIT MEASUREMENT EXERCISE 2

Amperage



Return the circuit back to normal with two five watt bulbs

Measure parallel circuit amperage at the points indicated as follows:

- ☐ Set your DVOM to read DC Amps.
- ☐ Measure the current flow for the complete circuit and record your reading below.

Total current flow: _____

- ☐ Measure the current flow at the fuse for each leg of the circuit, record your readings below.

Current flow through bulb one: _____

Current flow through bulb two: _____

- ☐ Add the current flow through bulb one and bulb two together

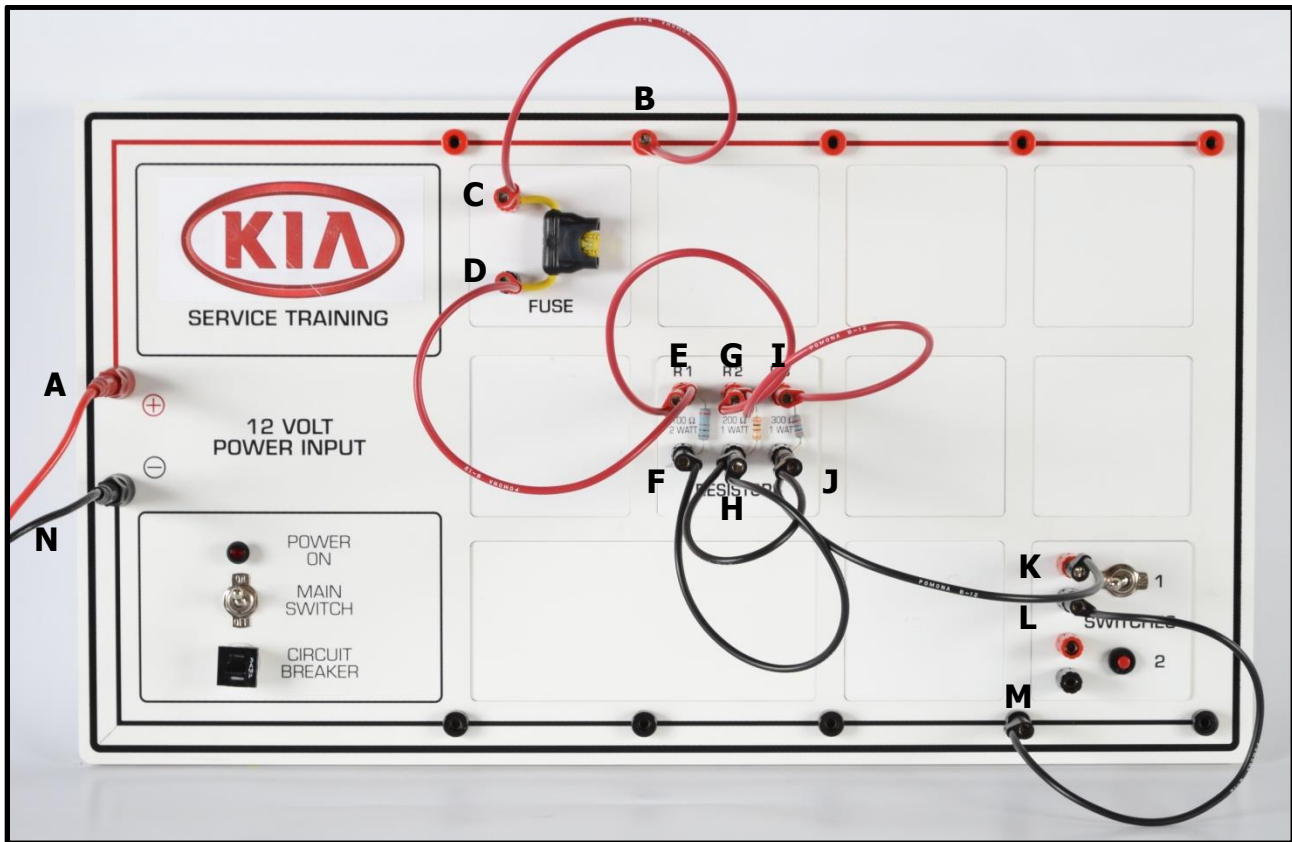
Does it equal the total current flow you measured?
YES / NO (circle one)

Parallel Circuit Rule

The total circuit current is equal to the sum of the individual branch currents.



Wait for further instruction.



PARALLEL CIRCUIT MEASUREMENT EXERCISE 3

Resistance



- ☐ Connect the three (3) resistors in parallel as shown on the breadboard above.

- ☐ **Disconnect power at points A & N.**

- ☐ Measure total circuit resistance using your DVOM at points E and J: _____ Ohms

In a parallel circuit, each branch has its own path for current. In the circuit above, the smallest resistance is ≈ 100 Ohms (Ω). The other two branches are ≈ 200 and ≈ 300 Ohms respectively

- ☐ Is the total circuit resistance more or less than the resistance of any branch?

More / Less (circle one)



NOTE: When a branch is added in a parallel circuit the total:

- Resistance decreases
- Current increases



Parallel Circuit Rule

Total circuit resistance is always smaller than the smallest individual resistance.



Wait for further instruction.

**PARALLEL CIRCUIT
SUMMARY**

In this section:

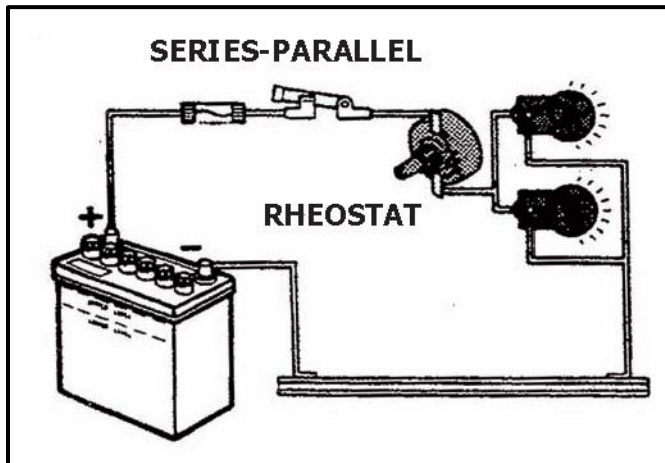
You created a parallel circuit on the electrical workstation.

You predicted the values, and then you measured and verified those voltages.

Parallel Circuit Rules

From your parallel circuit practice, you have verified the following parallel circuit rules:

- **Voltage** - Each branch drops the approximate source voltage (some loss expected due to connections)
- **Resistance** - The total circuit resistance is always less than the lowest individual resistance
- **Current** - The total circuit current is equal to the sum of the current in each circuit branch



- Combines characteristics of series and parallel circuits
- Some components are in series and others are in parallel
- Loads are normally in parallel

SERIES-PARALLEL CIRCUITS Normally in a series-parallel circuit, the power source, control and protection devices are in series and the loads are in parallel.

If an open occurs in the series portion, the whole circuit will lose current flow.

If an open occurs in a parallel branch, the current flow will remain in the series portion and the remaining parallel branches.

The instrument panel lighting circuit is one example of a series-parallel circuit.

Series-Parallel Circuit Rules Electrical behavior in all series-parallel circuits is governed by the rules shown in the illustration.

These rules apply to all series-parallel circuits regardless of the type and number of load devices.

- Current in the series section equals sum of parallel branch circuits
- Total circuit resistance equals parallel resistance plus series resistance
- Voltage applied to the parallel portion equals applied source voltage minus voltage drop across series portion

VOLTAGE DROP

The Voltage Drop test is a valuable test that you can perform during electrical diagnosis. Voltage drop measures the voltage used by resistance in sections of the circuit.

This includes the:

- Power side of the circuit
- Load in the circuit
- Ground side of the circuit

Voltage Drop, tests the circuit while it is operating. The Voltage Drop Test allows us to find any unwanted resistance that may have been missed by other tests.

**VIDEO:** Voltage Drop**Example**

As we stated earlier in this module, the circuit uses (or loses) voltage as current flows through resistance in the circuit:

- We start with approximately 12.6V
- A minimal amount of voltage is used by the resistance of connections in the power side of the circuit as the current flows to the load
- The majority of the voltage is used by the resistance to operate the load in the circuit
- The remaining voltage is used by the resistance of the connections in the ground side of the circuit as the current flows back to the battery
- Any unwanted resistance in the circuit will cause a greater than normal voltage drop



NOTE: Voltage Drop is the voltage differential in a portion of a circuit. This voltage differential is caused by resistance in the circuit.



MEASURING VOLTAGE DROP

Voltage Drop test across the Power Side of the circuit would require that the Red Test Lead starts as close to the power supply (positive terminal) and the Black Test Lead on the positive side of the load device.

For example, to check for resistance between the positive battery terminal and clamp, the positive meter lead would go on the positive battery post, and the negative meter lead would go on the terminal or clamp.



NOTE: The power and groundside of a circuit should have a voltage drop near 0.0 volts. However, each connection, fuse, switch, and splice may drop a small amount of voltage. The integrity of any connections will affect the amount of voltage dropped in a circuit.

A Voltage Drop test across the load would require the Red Test Lead to be on the positive side of the load and the Black Test Lead on the negative side of the load. The reading should be close to source voltage.

A Voltage Drop test across the groundside of the circuit requires the Red Test Lead to go to the negative side of the load device and the Black Test Lead as close to battery ground as possible. The reading should be as close to 0.0V as possible. A voltage drop more than 0.5V may indicate a bad ground or resistance in the circuit between the load device and the battery ground.



NOTE: It may not always be possible to reach to the source from the load device unless long lead adaptors are used.



NOTE: A Voltage Drop test can only be performed on a circuit, which has a complete path for current to flow from source to ground. Voltage cannot drop without current flow. If current is not present the circuit is open, such as when the switch is OFF, the load is open or a wire is cut.

Typical Circuit Voltage Drops

Everything has resistance. The wires, connectors, and switch contacts that make up a circuit all have some resistance. In an undamaged, properly designed circuit, normal resistance is small enough that it does not keep the load from working. If the circuit is functioning properly, all voltage drops to and from the load will be acceptable.

A general rule of thumb for voltage drop includes the following:

- Starting and charging circuits (high current >20A)
0.5 volts
- Lighting and solenoid circuits (low current 1-20A)
0.1 volts
- Computer sensor circuits (very low current < 1A)
0.050 volts

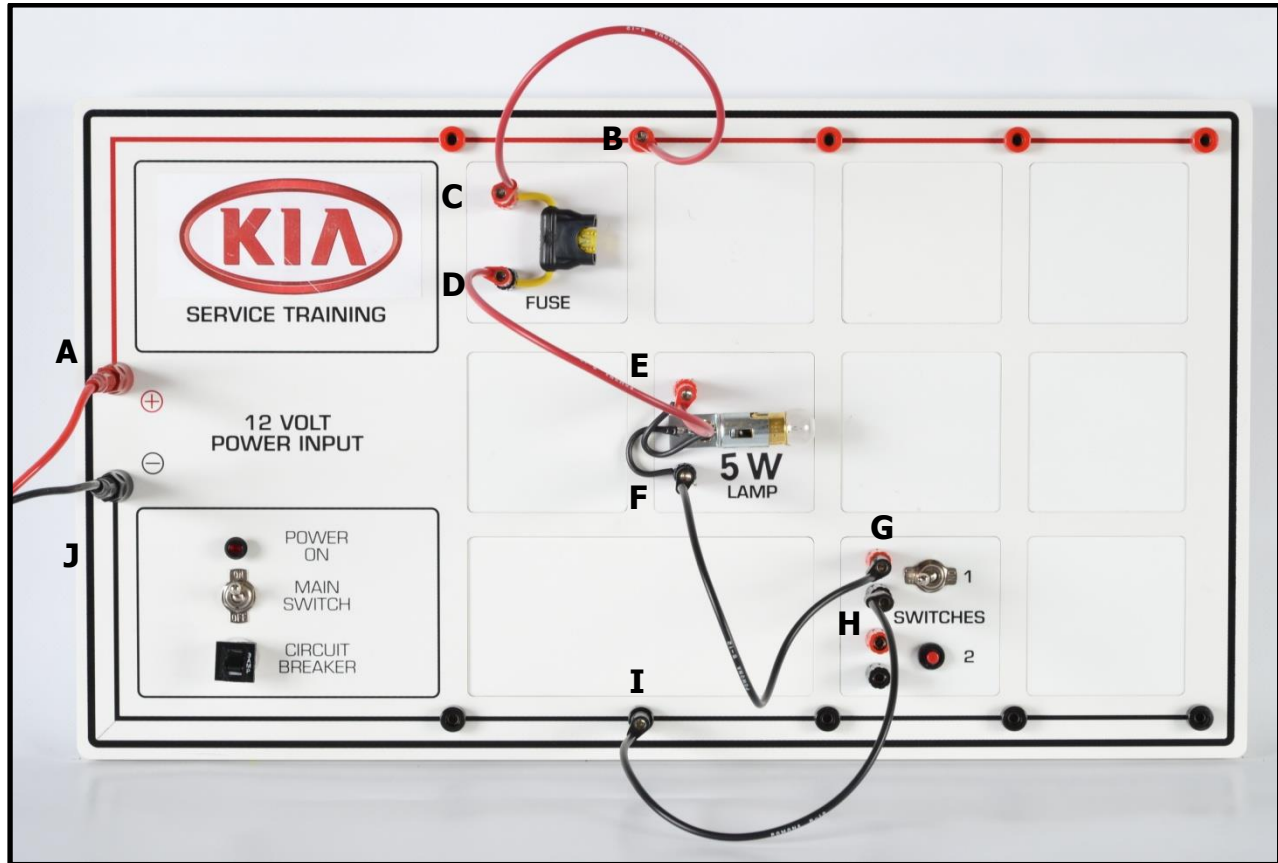
The larger the wire, the larger the voltage drop limit.

Anything higher indicates a need for inspection and possible repair.

Repair usually includes cleaning corrosion from connections, repairing faulty terminal crimps, or tightening connectors and ground terminals.

On a rare occasion, it is also possible that the wire size

is too small for the load.



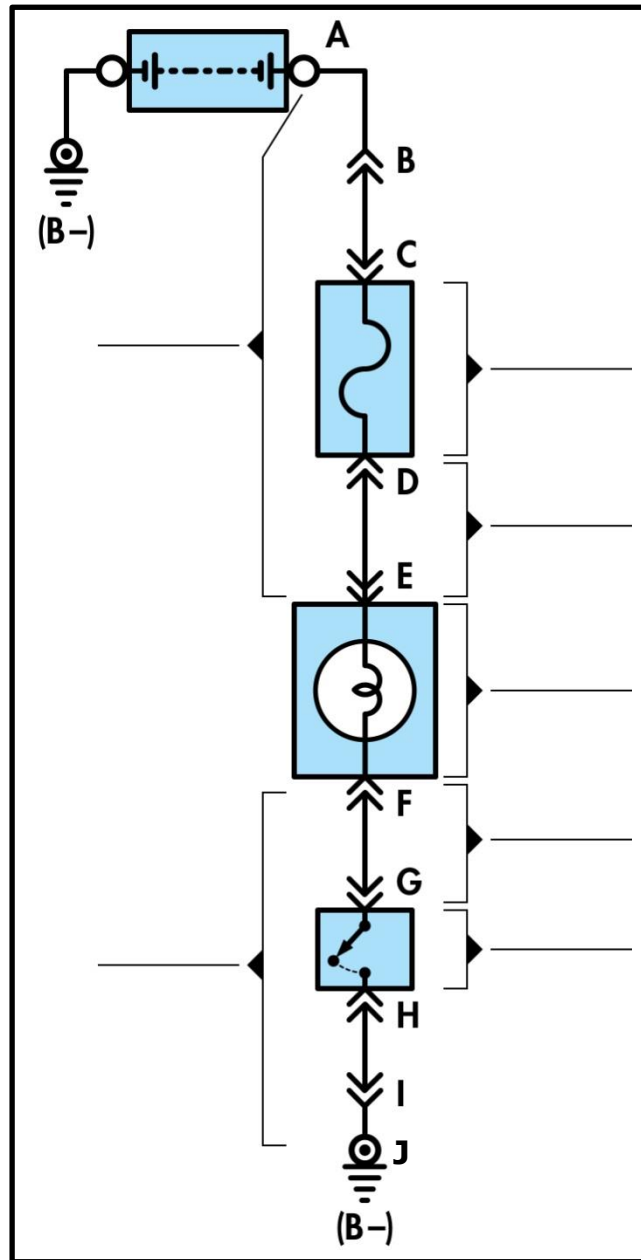
VOLTAGE DROP MEASUREMENT EXERCISE 2 Measuring Voltage Drop



Assemble your breadboard to match the circuit above. Prepare your DVOM to take Voltage Drop tests and complete the information in the schematic on the next page.

Measure the source voltage (circuit ON): _____

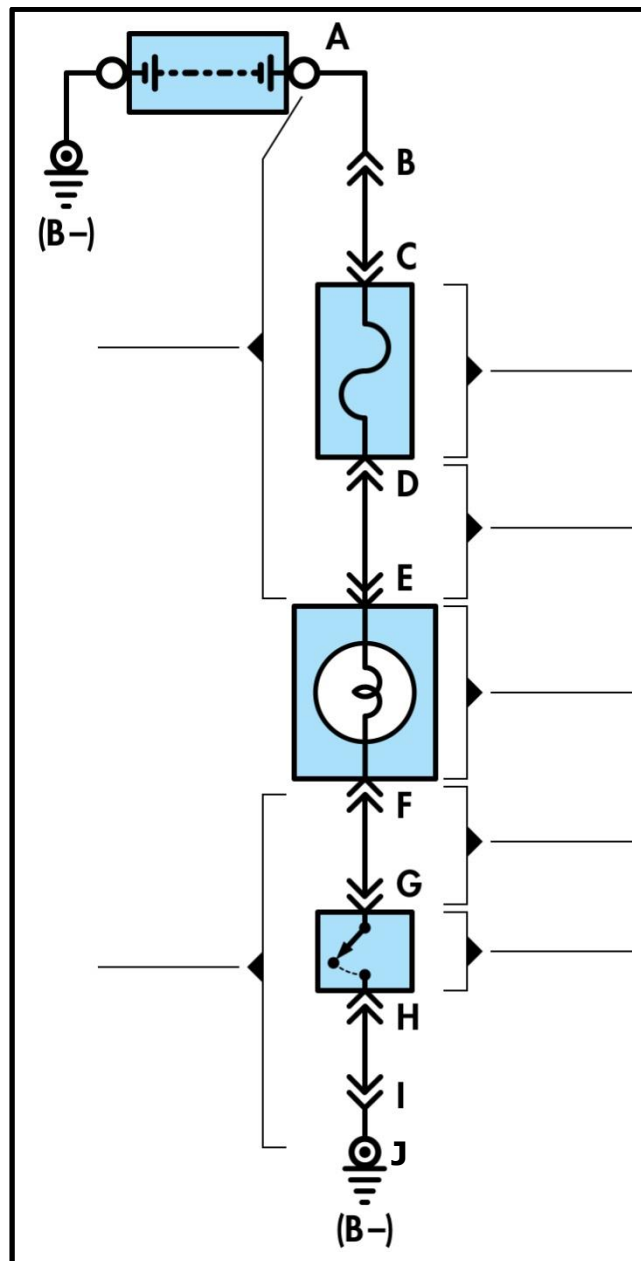
Measure the voltage drops in the circuit and use the schematic on the following page to record your results.



**VOLTAGE DROP
MEASUREMENT
EXERCISE 1
Predicting Voltage Drop**

Use the schematic above to predict the voltage drop in the circuit. Remember, the circuit must be **ON** for any voltage to drop so you have to imagine the switch is closed.

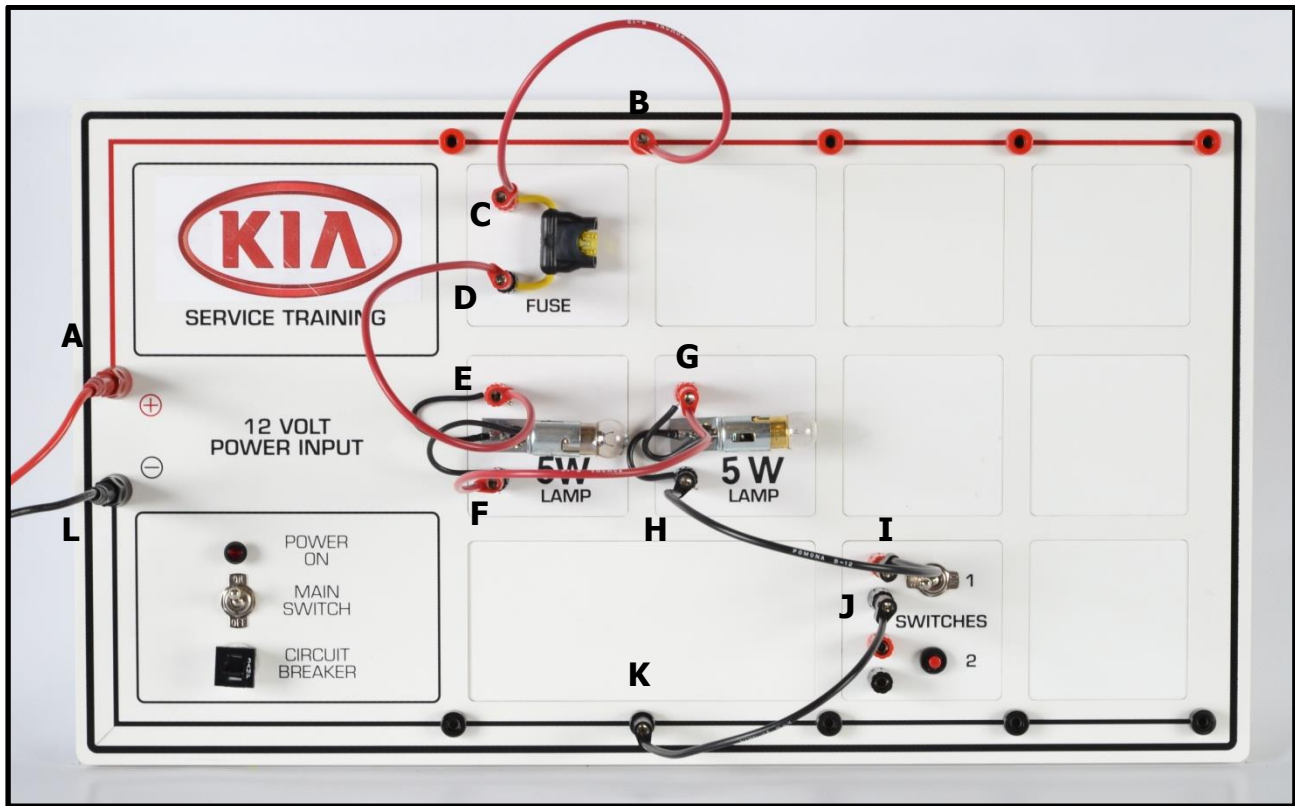
What symptom would you expect to see if there was a 5V voltage drop between points G and H in the circuit?



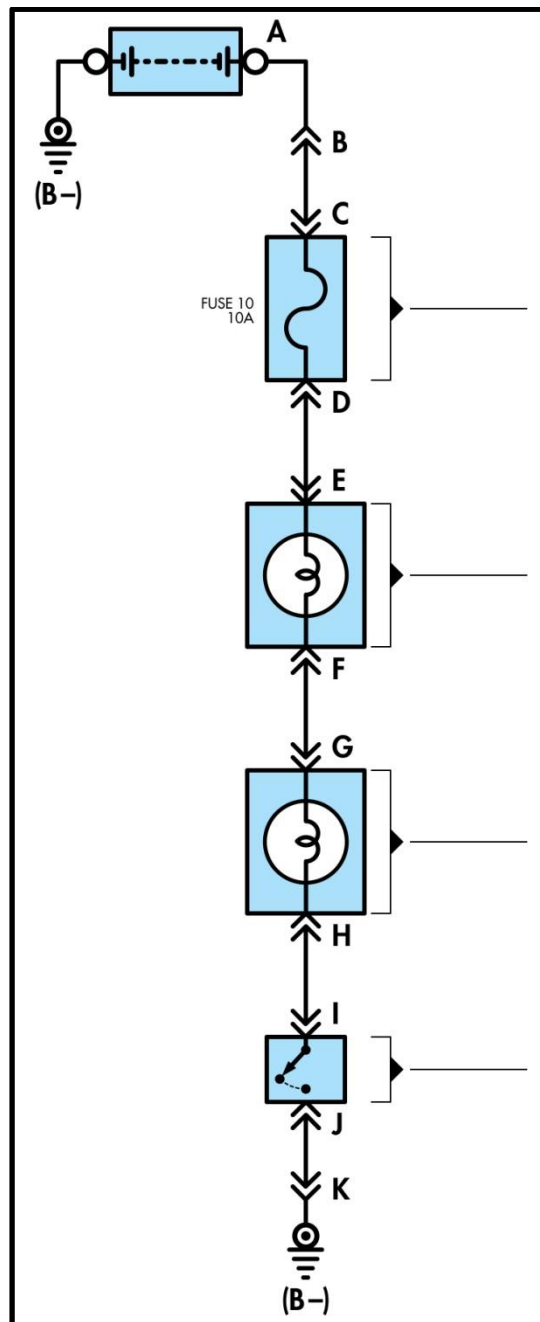
1. Does the total voltage drop equal the source voltage? YES / NO (circle one)
2. What is the voltage drop on the source side of the circuit? _____
3. What is the voltage drop on the groundside of the circuit? _____



Wait for further instruction.

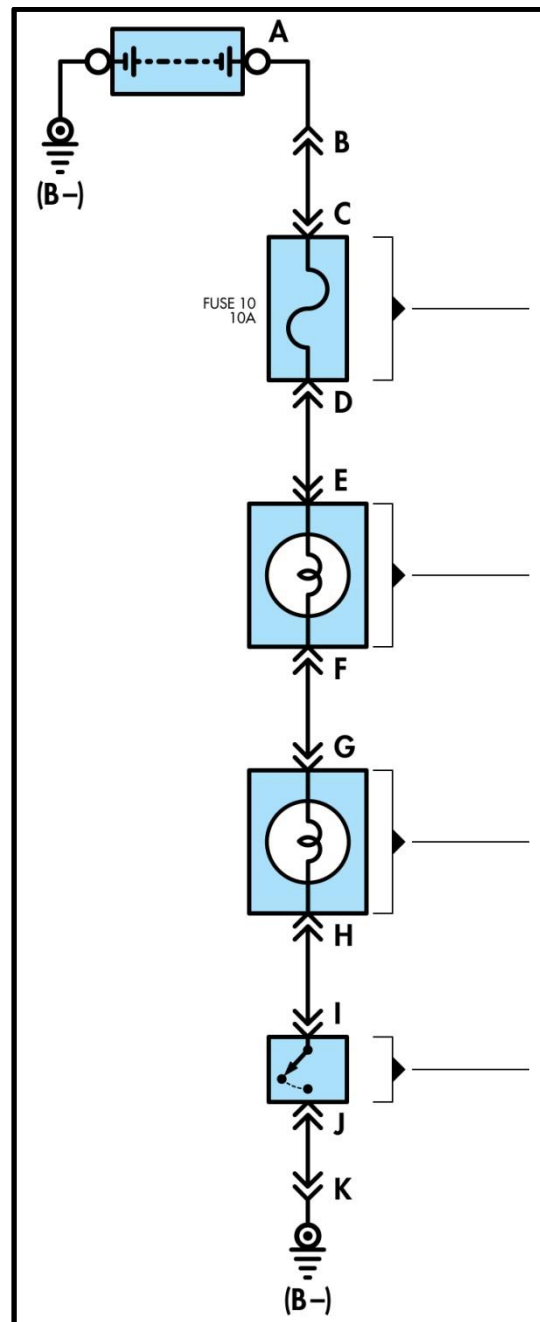
**Building the Circuit**

On the breadboard, build a series circuit with two lamps as shown above.



**VOLTAGE DROP
MEASUREMENT
EXERCISE 3
Multiple Loads (Series)**

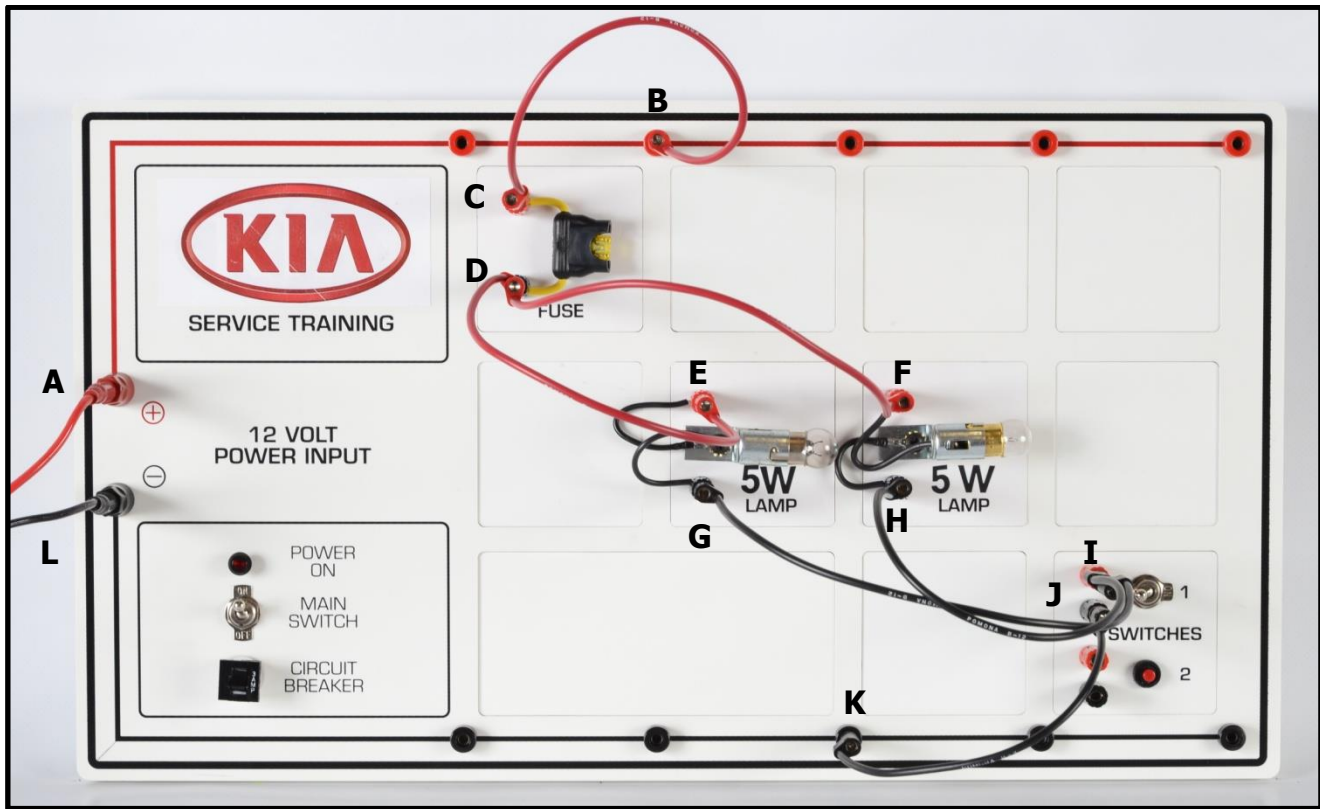
Assuming that the bulbs are the same wattage, predict the Voltage Drop in the schematic above.



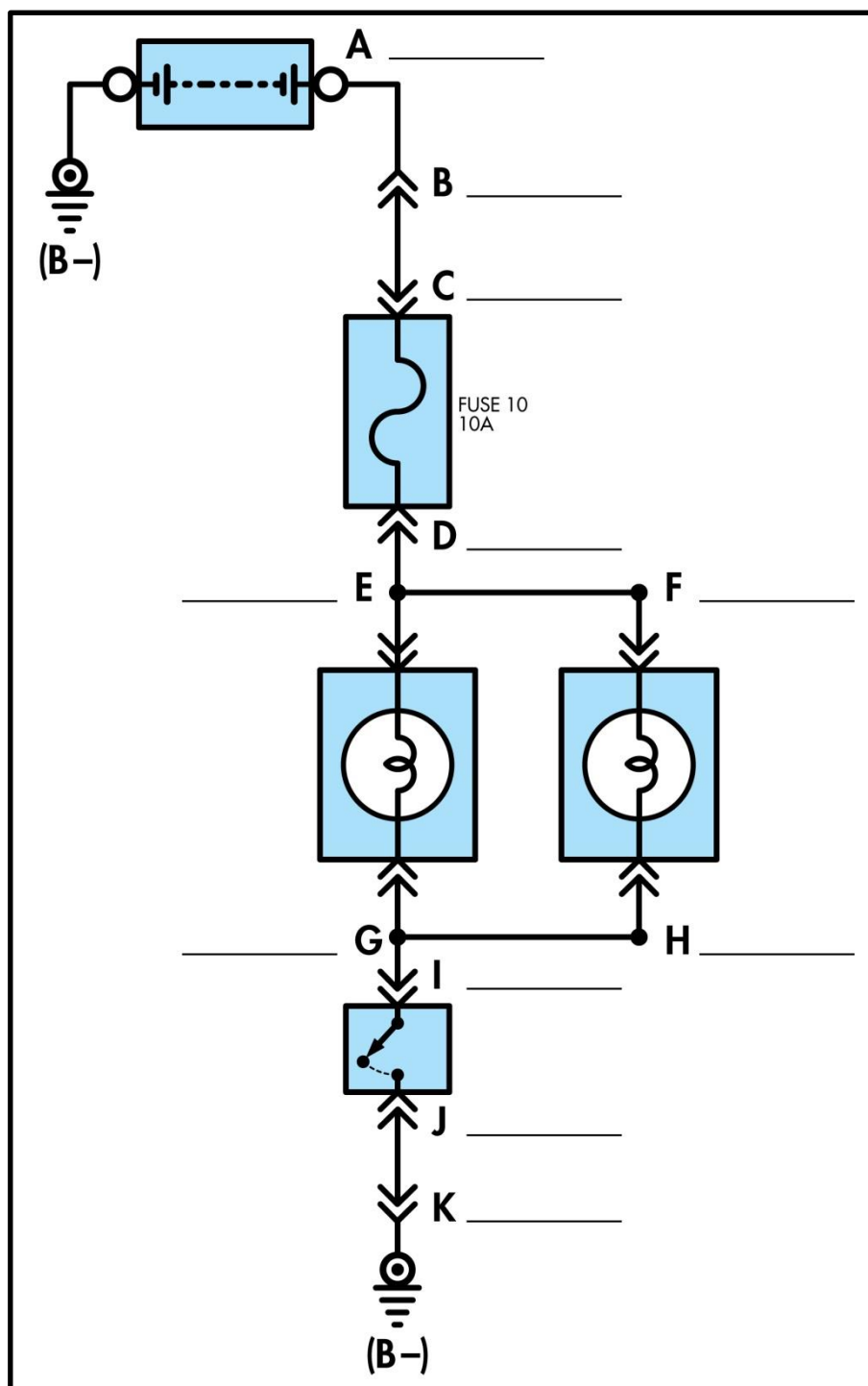
- On the breadboard, measure the voltage drops as indicated in the schematic and record your results.



Wait for further instruction.

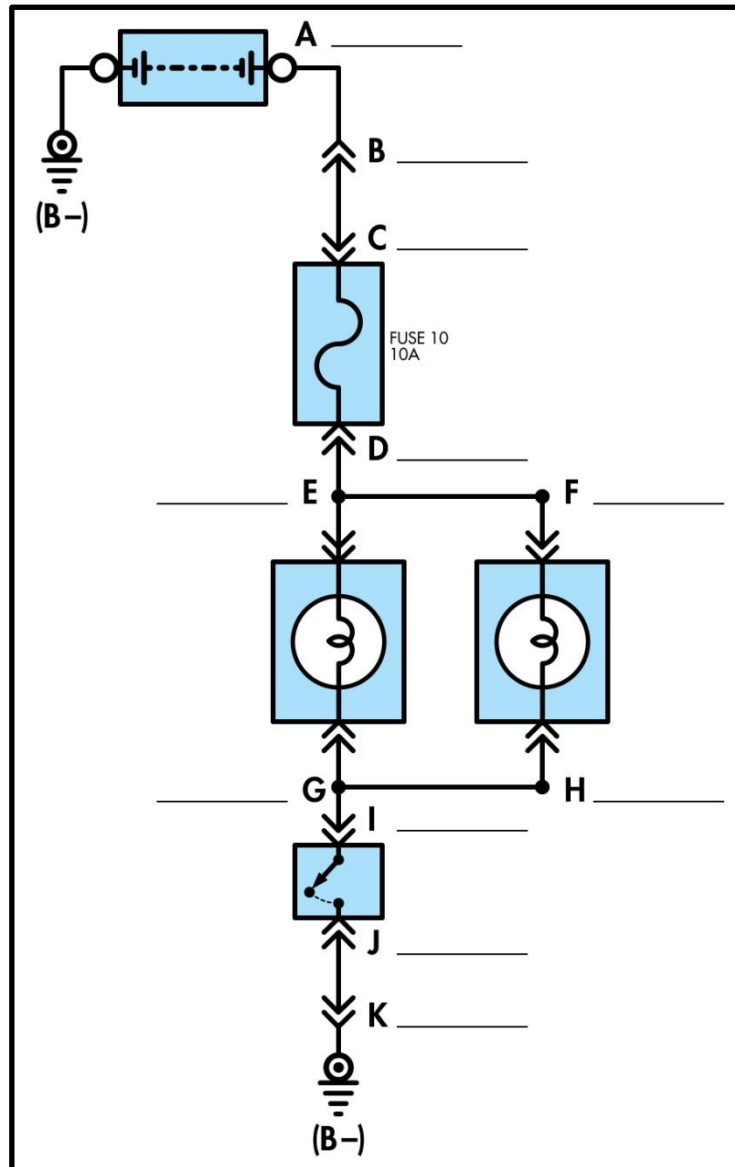


- Build the circuit above on the breadboard.



VOLTAGE DROP MEASUREMENT
Exercise 4

- ☐ Predict the Voltage Drop across each load in the parallel circuit above.



- ☐ Measure the Voltage Drop across each load and record your readings in the provided spaces.

Does each voltage drop equal source voltage?

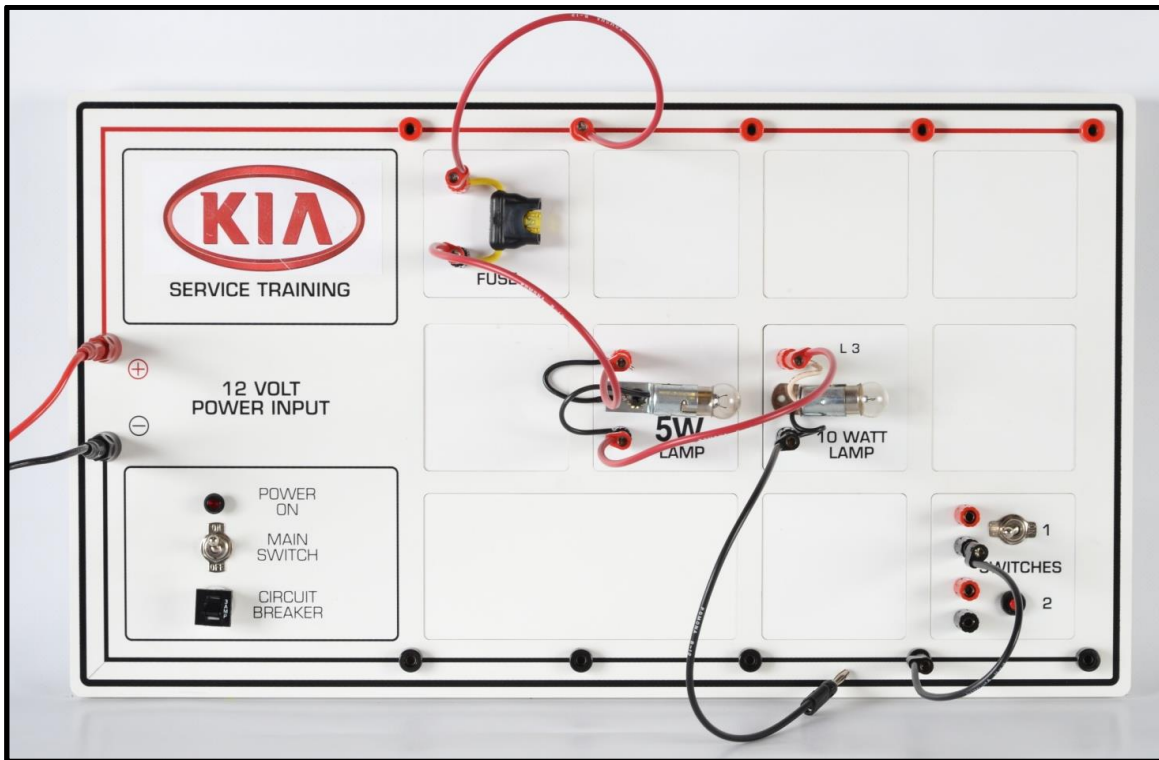
YES / NO (circle one)

Parallel Circuit Rule

Each load drops approximately source voltage.



Wait for further instruction.



OPEN CIRCUITS

An open circuit does not have a complete path for current. Since there is no current, there is no voltage drop across the loads, and the components cannot operate. Troubleshooting an open circuit is made using the available voltage test.

- ☐ Turn the circuit ON (even though it is not operating) and place your negative meter lead on a known good ground.
- ☐ Place the positive lead at convenient test points in the circuit.
- ☐ You will see source voltage up to the point of the open circuit.
- ☐ After the open circuit point, you will see 0 volts.

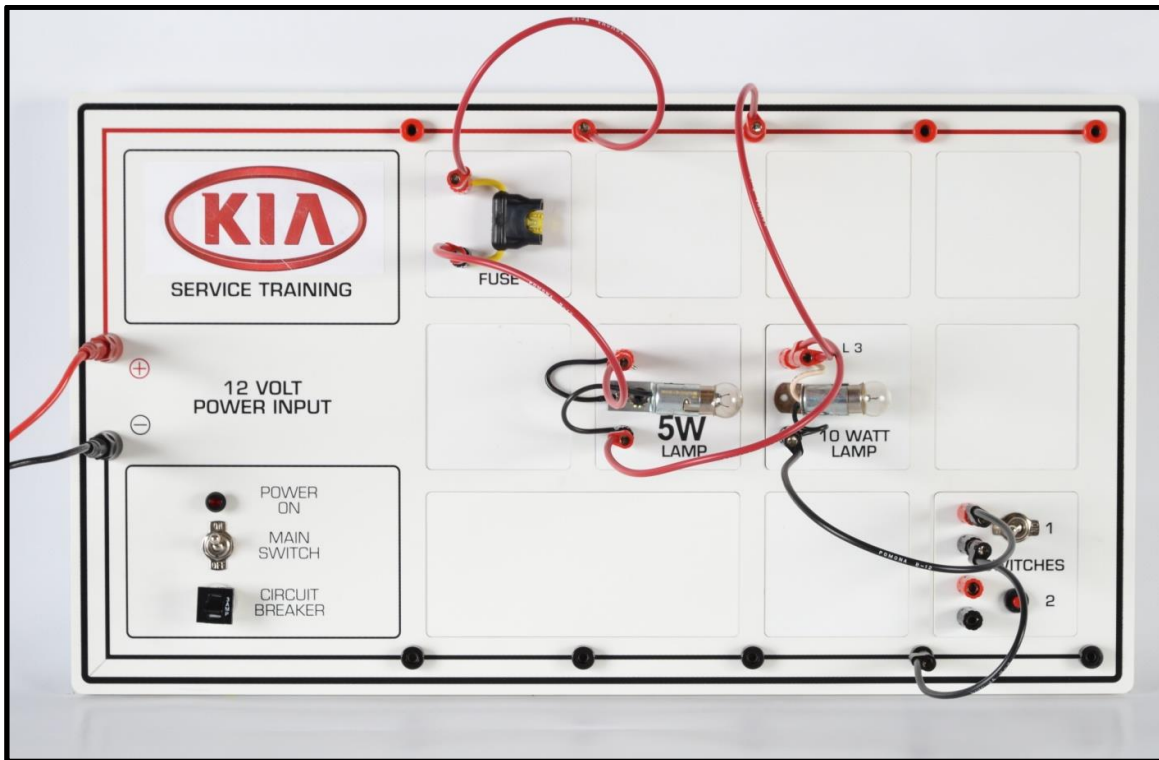
VOLTAGE POTENTIAL

Often referred to as Open Circuit Voltage. This is voltage before doing any work. Measuring a battery that is not connected is measuring Open Circuit Voltage. If you measure identical voltage at multiple points in a non-working circuit, you are seeing voltage potential before the open.



FEEDBACK: Can you do a Voltage Drop test on an open circuit? Why or why not?





SHORT TO POWER

In some cases, there can be a short to power. This can be caused by wire, connector, or component damage. The symptom is an unrelated load operating when the selected circuit is turned ON.

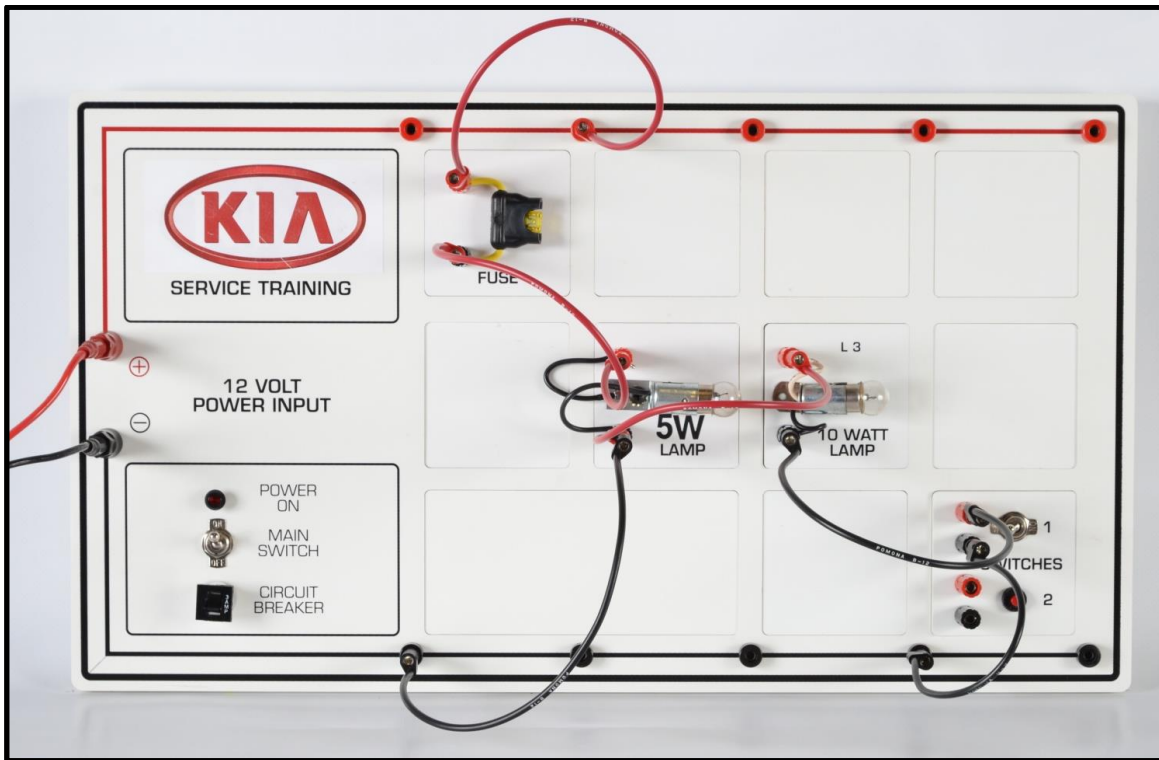
Components such as bulbs, motors, solenoids, and relays have varying resistance determined by their design. The service information provides the component resistance value in the related diagnostic section.

A short to power is when a wire in the circuit touches a positive wire. If the short to power is in the control side of the circuit, a load device such as a bulb may illuminate unexpectedly.

If the short causes the load to be bypassed, the lower resistance in the circuit will increase the amperage and often blow a fuse or other circuit protection device.



NOTE: There are many useful tools (short detectors) available from the aftermarket that can help you diagnose shorts to power and shorts to ground.



SHORT TO GROUND

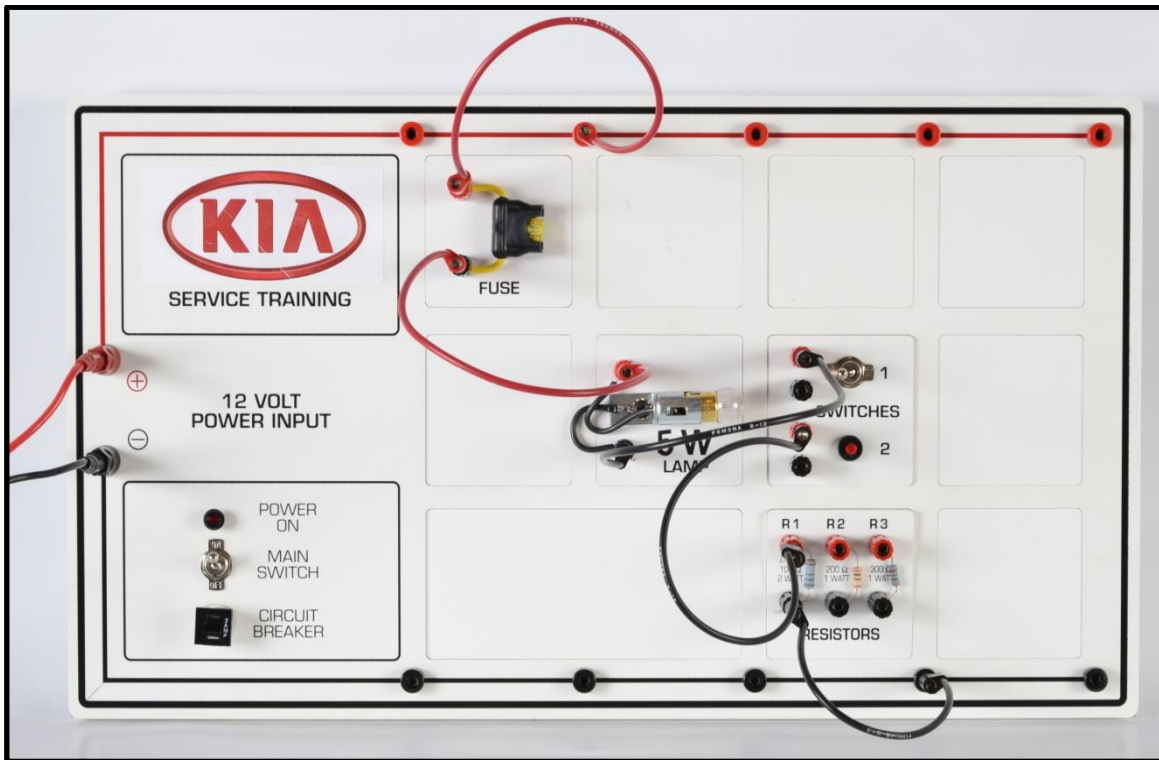
The most common short circuit occurs when a damaged wire touches ground. If this damaged wire is before the load, the circuit resistance decreases to almost 0 and the resulting increase in current results in a blown fuse or an open circuit breaker.



WARNING: Never attempt to repair a circuit by removing a blown fuse, fusible link, or circuit breaker, and installing a jumper lead. Doing so could result in vehicle damage and personal injury.

How do you diagnose a short circuit? Here is a method that allows you to isolate the location of the short:

- Remove the fuse
- Use a 12V test light connected to the fuse holder of the affected circuit (the test light should illuminate)
- Isolate sections of the affected circuit by disconnecting harness connectors
- When the test light turns OFF, you have isolated the section of the circuit with the short



EXCESSIVE RESISTANCE

In a circuit, an increase in resistance results in an increase / decrease in amperage.

Many circuit problems result from an excessive resistance on the groundside of the circuit because:

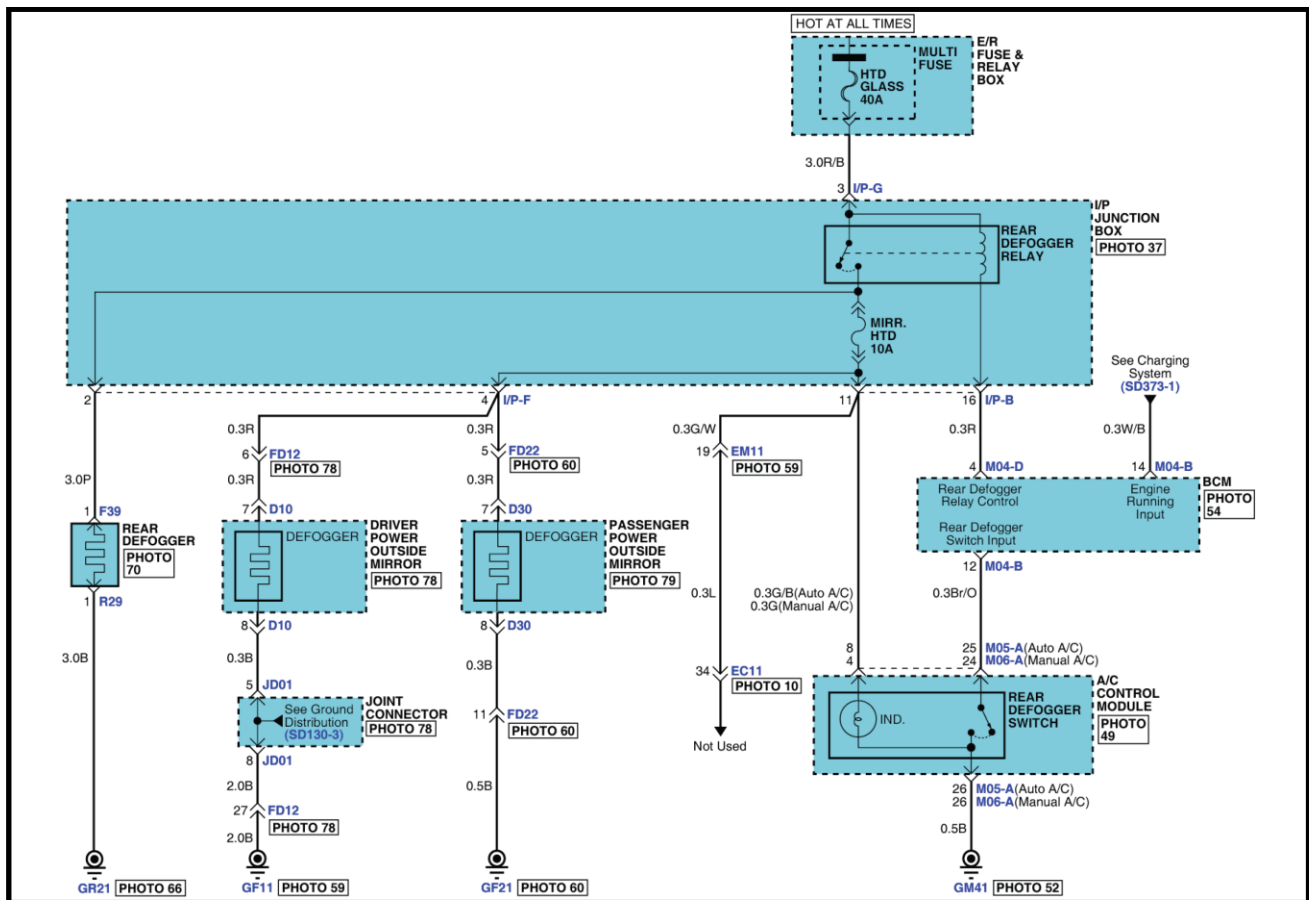
- The vehicle chassis, frame and/or body serve as the ground point for most circuits
- Many of the ground points are exposed to the elements which can lead to corrosion
- Many grounds are made using a bolt which may work loose over time

- ☐ Build breadboard circuit shown.

How does the high resistance on the groundside of the circuit affect the operation of the circuit?

- ☐ Power off the circuit and move the resistance to the power side of the circuit to simulate corrosion in a connector.

How is the circuit affected differently than the resistance on the groundside of the circuit?



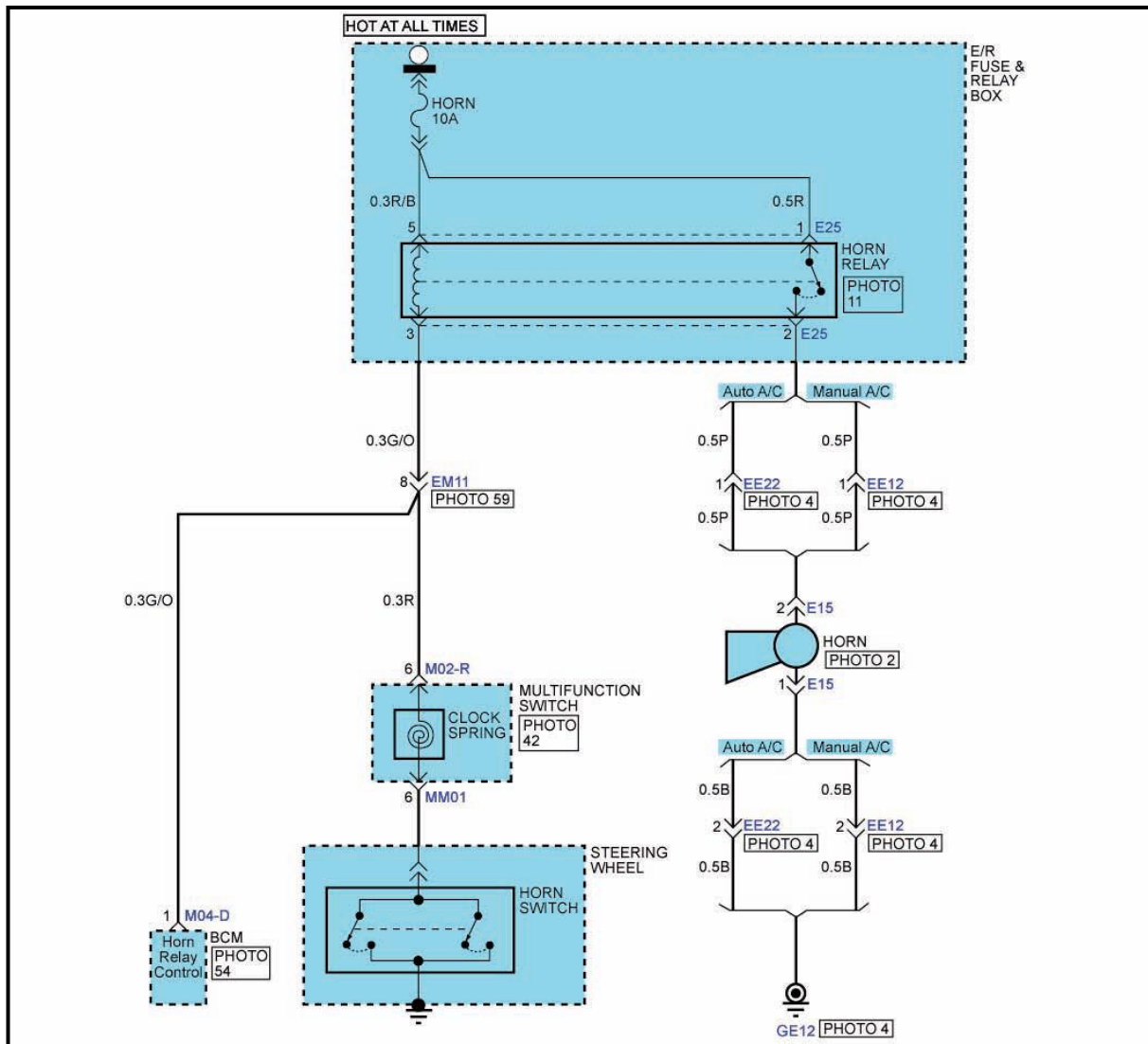
POWER FLOW DIAGRAMS

It is very important to diagnose electrical concerns with a plan. When beginning on an electrical concern, the first step is to print out the wiring diagram (also known as Electrical Troubleshooting Manuals - ETMs).

Once you have the wiring diagrams printed, the next step is to identify the three sections on the wiring diagram:

- Power side of the circuit
- Load device
- Ground side of the circuit

Have 3 colored pens available for the following exercises.

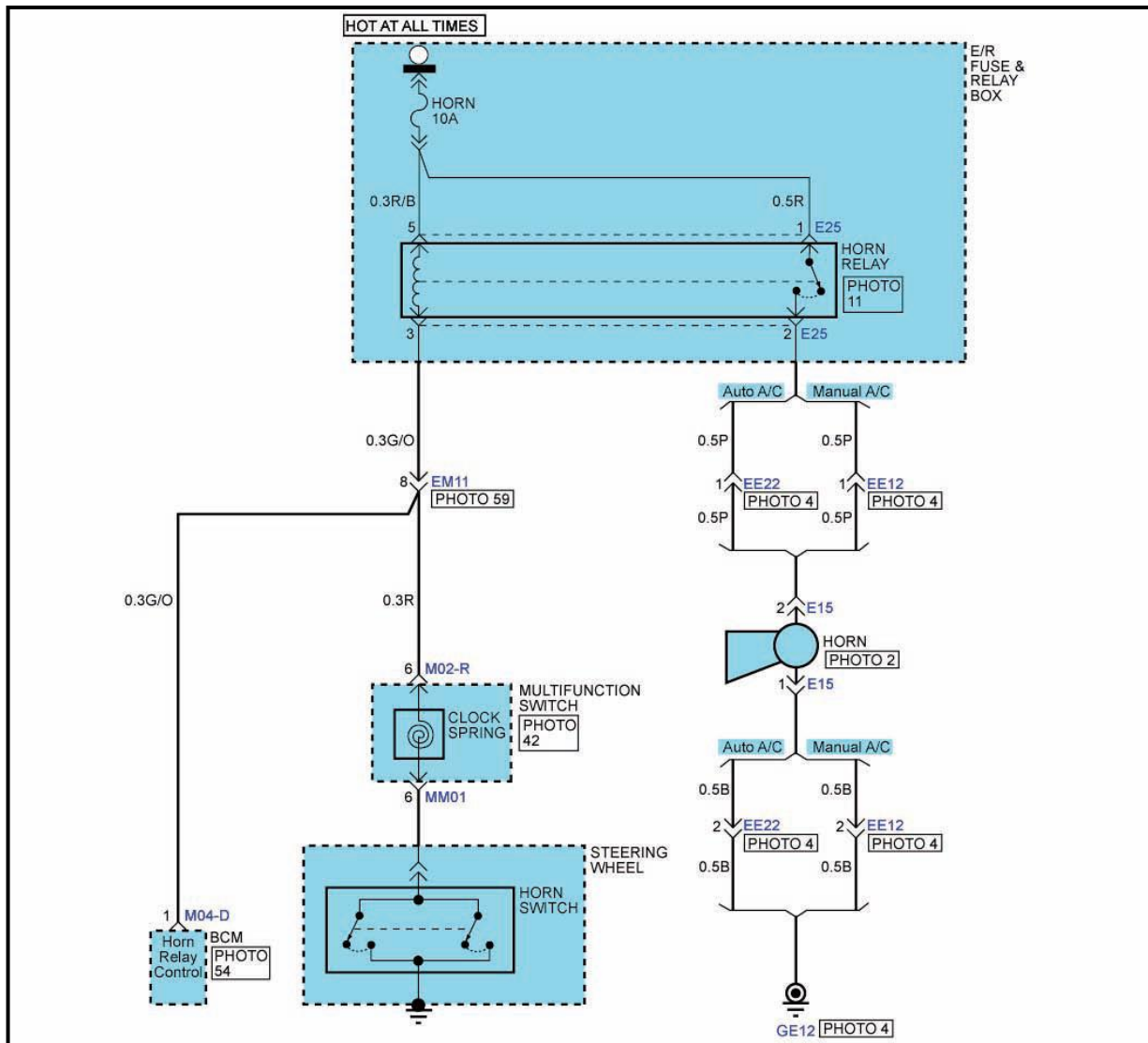


With the Horn Circuit switch **OPEN/OFF**:

- ☐ Draw the power side of the circuit in Pink / Red
- ☐ Draw the load in Yellow
- ☐ Draw the ground side of the circuit in Green

What is the voltage at connector EM11, Pin 8?

What is the voltage at E15, Pin 2? _____



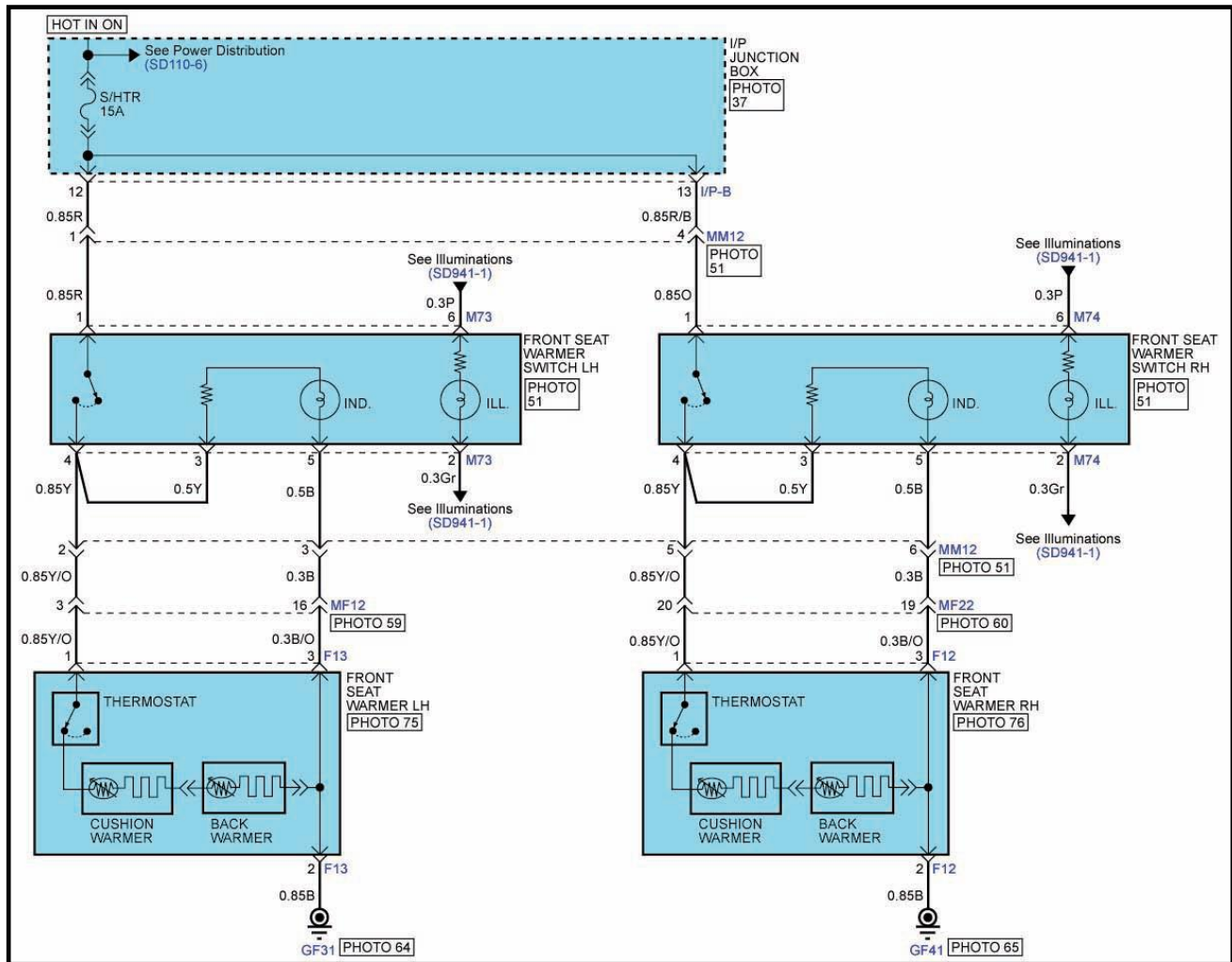
With the Horn Circuit switch **CLOSED/ON**:

- ☐ Draw the power side of the circuit in Pink / Red
- ☐ Draw the load in Yellow
- ☐ Draw the ground side of the circuit in Green

What is the voltage at connector EM11, Pin 8?

What is the voltage at E15, Pin 2? _____

Discuss with your Instructor when completed.

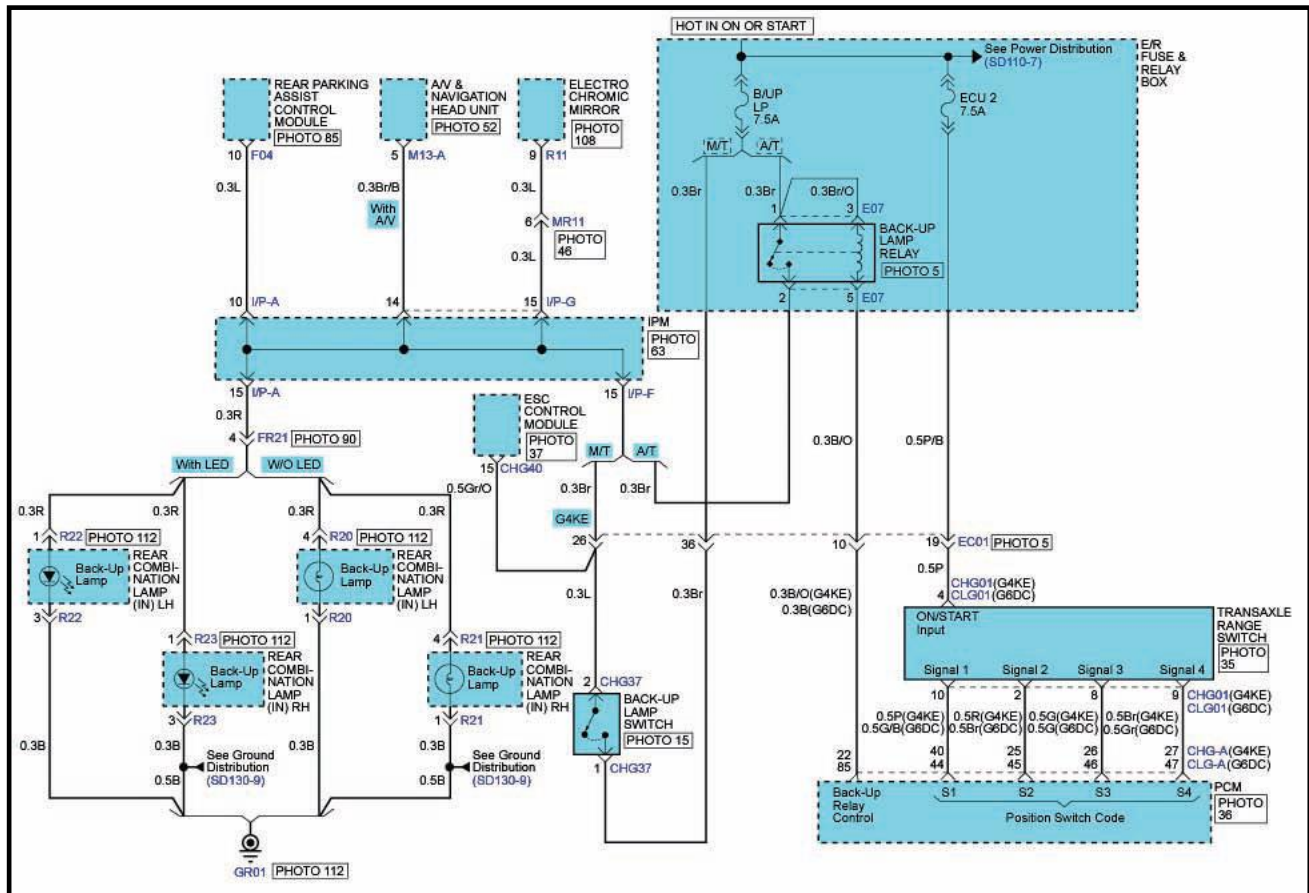


With the Seat Warmer switch **LH CLOSED/ON**
(Key is in the ON Position):

- ☐ Draw the power side of the circuit in Pink / Red
- ☐ Draw the load in Yellow
- ☐ Draw the ground side of the circuit in Green

What is the voltage at connector F13, Pin 1?

What is the voltage at F13, Pin 3? _____

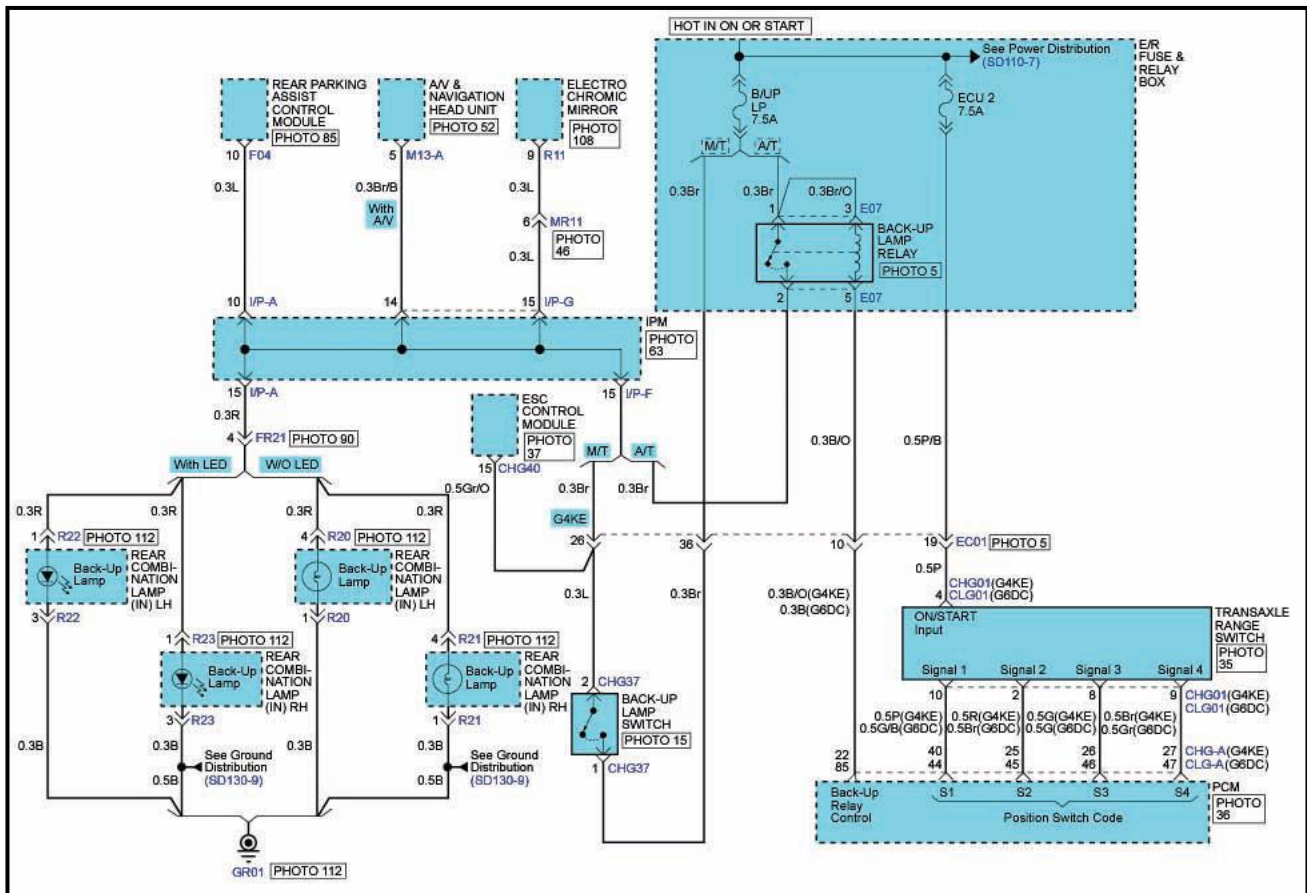


With the Back Up Lamps switch **OPEN/OFF**
(Key is in the ON Position):

- ☐ Draw the power side of the circuit in Pink / Red
- ☐ Draw the load in Yellow
- ☐ Draw the ground side of the circuit in Green

What is the voltage at connector R23, Pin 3?

What is the voltage at FR21, Pin 4? _____



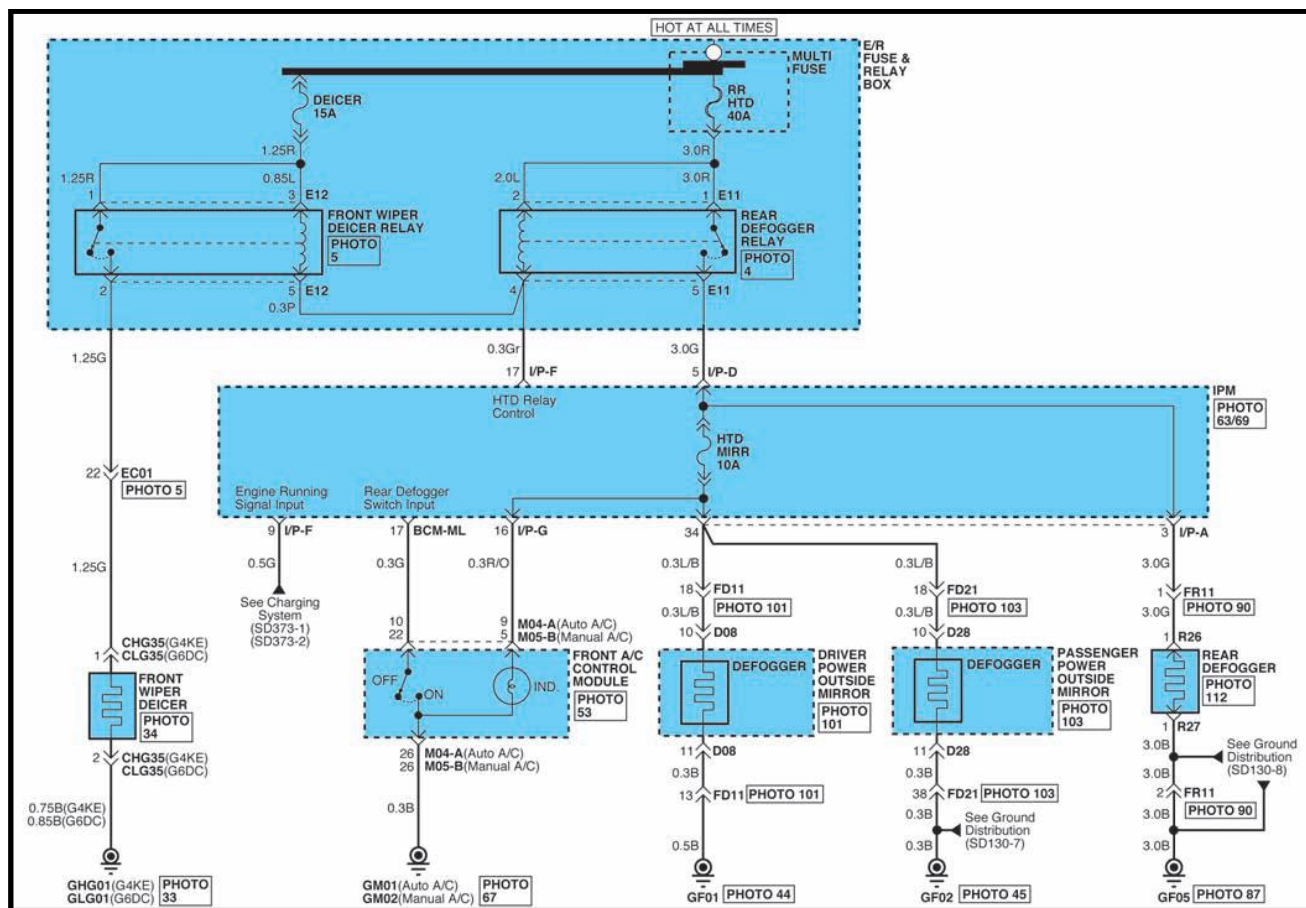
With the Back Up Lamps switch **CLOSED/ON**
(Key is in the ON Position):

- ☐ Draw the power side of the circuit in Pink / Red
- ☐ Draw the load in Yellow
- ☐ Draw the ground side of the circuit in Green

What is the voltage at connector R23, Pin 3?

What is the voltage at FR21, Pin 4? _____

Discuss with your Instructor when completed.



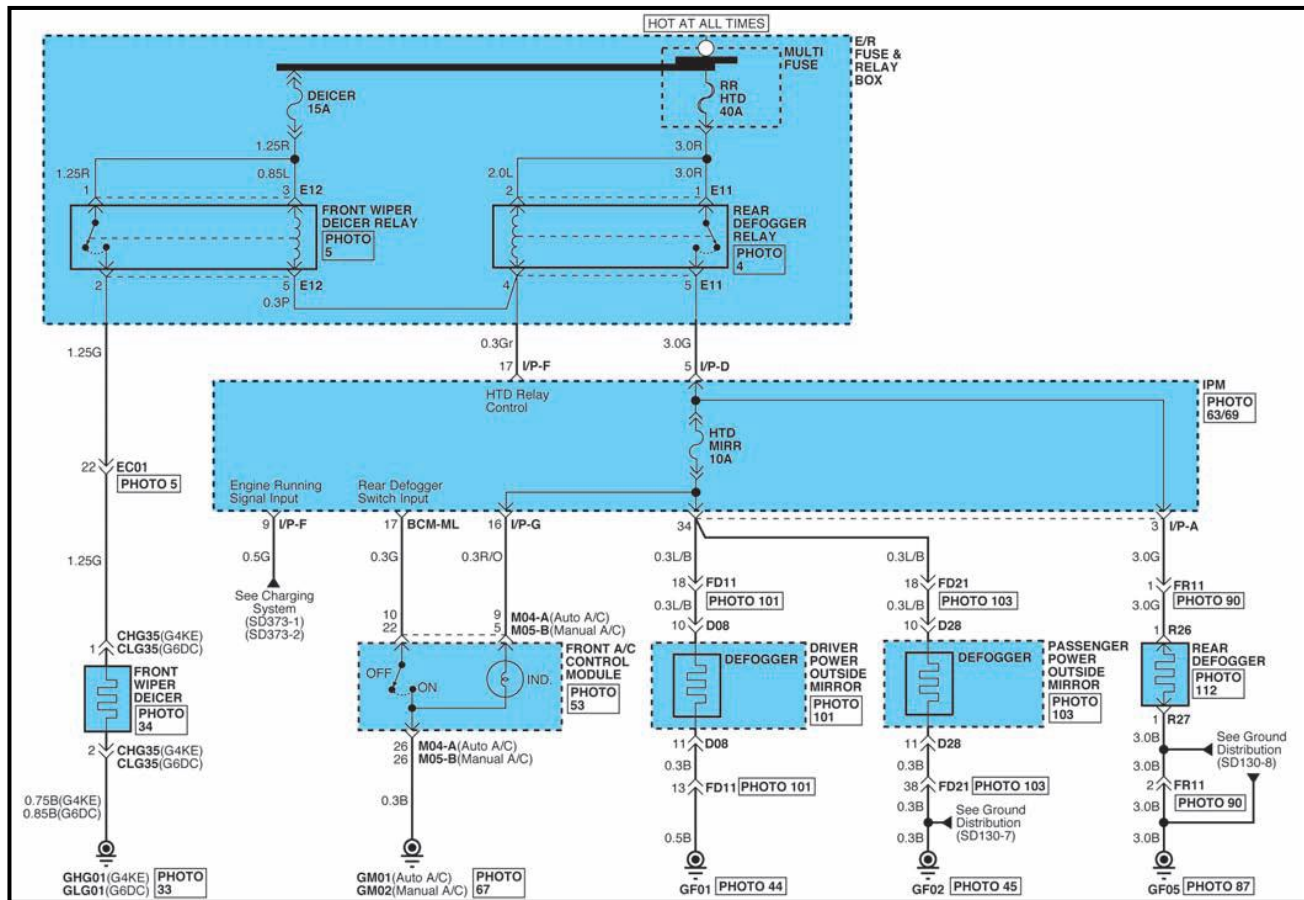
With the Rear Defogger Switch Input **OPEN/OFF***
(Key is in the ON Position):

- ☐ Draw the power side of the circuit in Pink / Red
- ☐ Draw the load in Yellow
- ☐ Draw the ground side of the circuit in Green

What is the voltage at connector D08, Pin 10? _____

What is the voltage at I/P-F, Pin 17? _____

** The IPM activates the HTD Relay Control based on the Rear Defogger Switch Input.*



With the Rear Defogger Switch Input **CLOSED/ON**
(**Key is in the ON Position**):

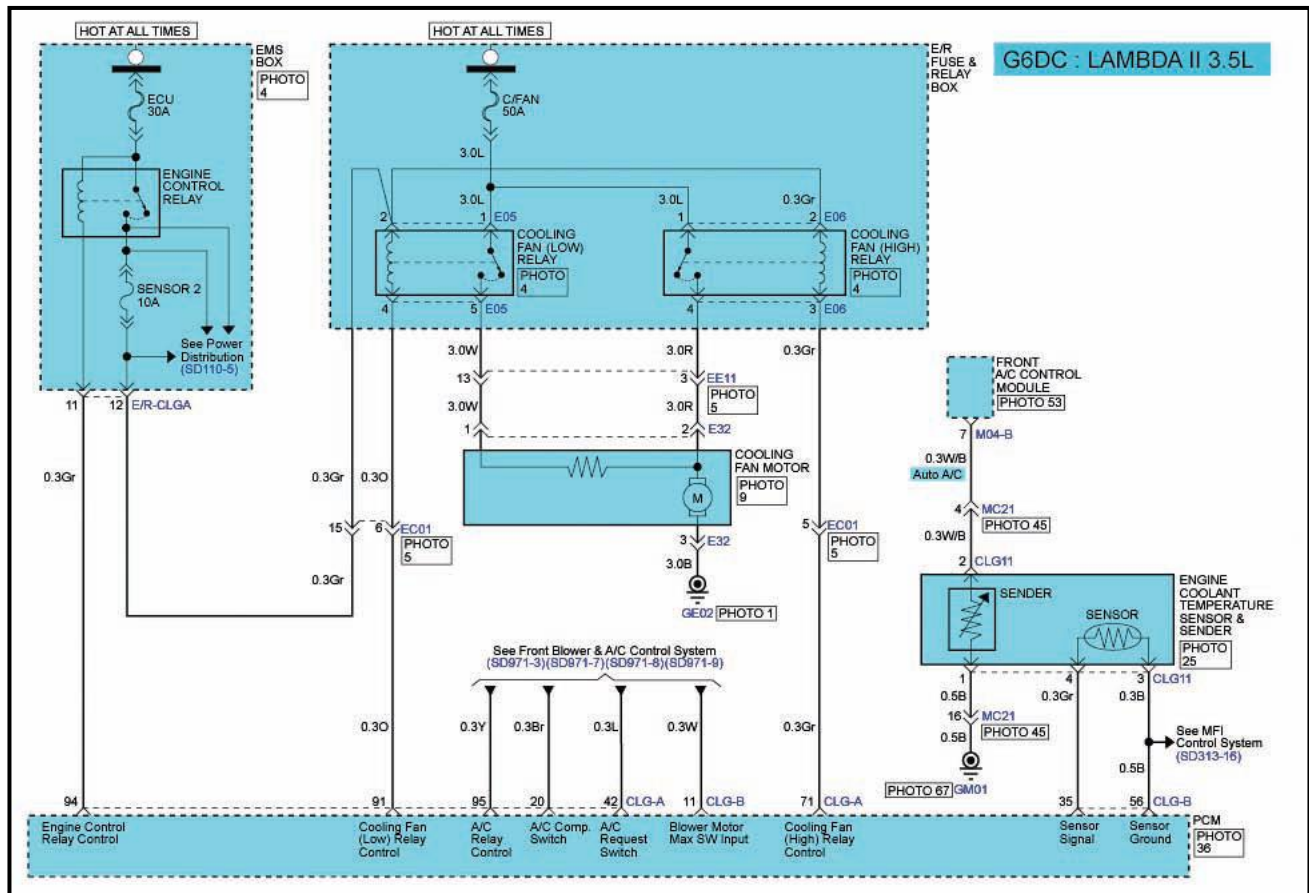
- ☐ Draw the power side of the circuit in Pink / Red
- ☐ Draw the load in Yellow
- ☐ Draw the ground side of the circuit in Green

What is the voltage at connector D08, Pin 10?

What is the voltage at I/P-F, Pin 17? _____

Discuss with your Instructor when completed.

** The IPM activates the HTD Relay Control based on the Rear Defogger Switch Input.*

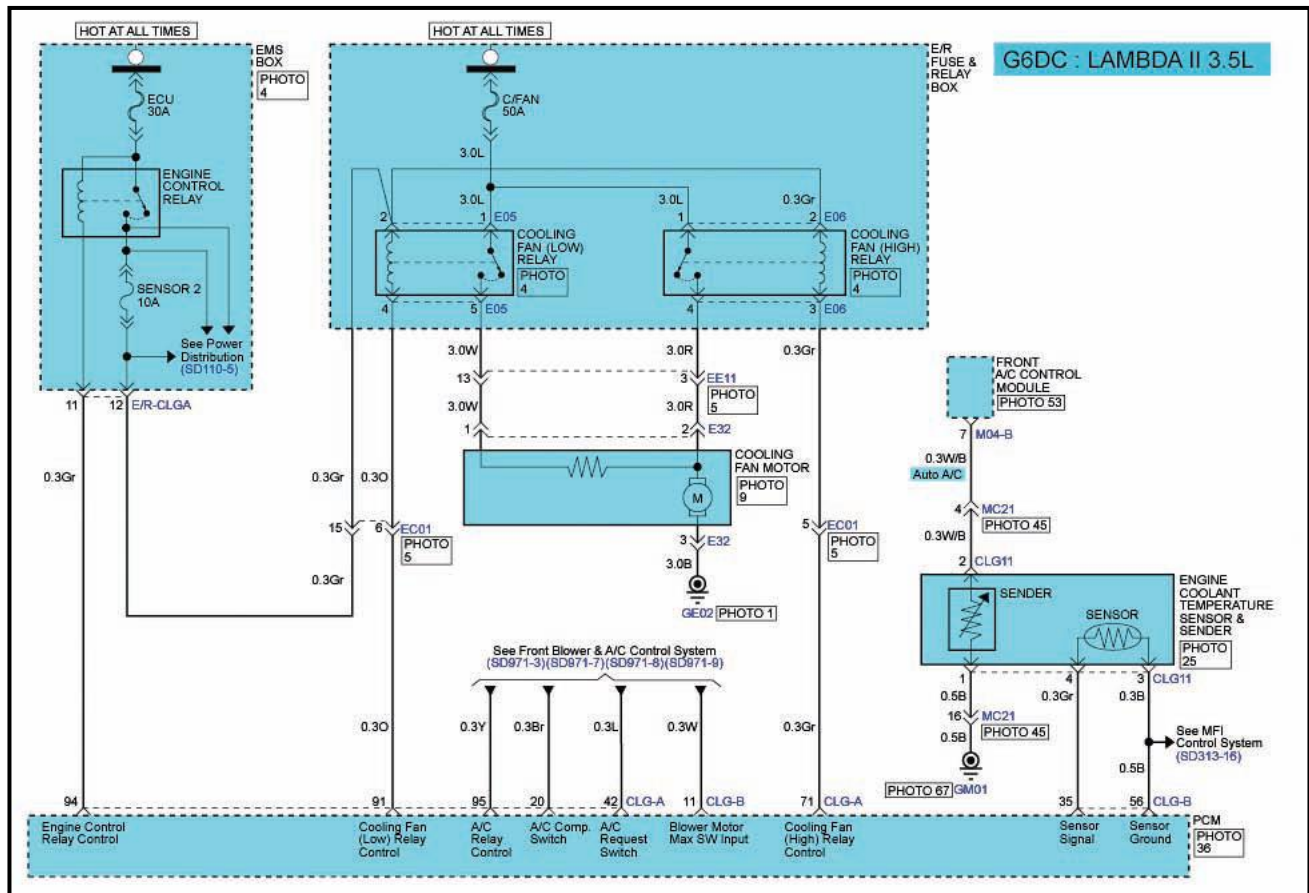


With the Engine Control Relay and Cooling Fan (High) Relay (Non-Energized) **OPEN/OFF**:

- ☐ Draw the power side of the circuit in Pink / Red
- ☐ Draw the load in Yellow
- ☐ Draw the ground side of the circuit in Green

What is the voltage at connector E32, Pin 2?

What is the voltage at CLG-A, Pin 94?



With the Engine Control Relay and Cooling Fan (High) Relay (Energized) **CLOSED/ON**:

- ☐ Draw the power side of the circuit in Pink / Red
- ☐ Draw the load in Yellow
- ☐ Draw the ground side of the circuit in Green

What is the voltage at connector E32, Pin 2?

What is the voltage at CLG-A, Pin 94?

Discuss with your Instructor when completed.

CIRCUIT DIAGNOSIS



Electrical issues usually are caused by the following:

- An open circuit
- A short to power
- A short to ground
- Excessive resistance

When diagnosing a circuit, your plan should include Split Circuit Theory and Common Point Analysis.

Split Circuit Theory - is finding the easiest access point to perform a test and thereby cutting the circuit in half. The load is a good place to start, if accessible, as it separates the power supply portion of the circuit from the ground portion.

Common Point Analysis - is understanding the circuit and selecting a location that affects one or all branches within a circuit. For example, if both tail lamps were out, then Common Point Analysis would check the circuit before it separates into two circuits. If only one of the tail lamps were out, then check after the circuit separates.

**KIA'S 5-STEP
TROUBLESHOOTING
PROCESS**

- 1. Confirm the Customer's Concern (Confirm)**
- 2. Analyze & Diagnose the Customer's Concern (Diagnose)**
- 3. Find the Root Cause (Troubleshoot)**
- 4. Repair the Root Cause (Repair)**
- 5. Verify the Repair (Verify)**

1. Confirm

It is essential you have an accurate description of the customer's concern and that you are able to **Confirm the Customer's Concern**. This includes what the vehicle is doing, and the conditions it is doing it under (i.e., hot, cold, accelerating, braking). Sometimes the customer's concern is a normal characteristic of the vehicle. If so, it is important to have the service advisor explain normal vehicle operation to the customer. If the customer is describing an abnormal concern, confirm it first-hand before continuing. Sometimes the customer may describe the concern in a way that leads you down the wrong diagnostic path. You must understand and experience exactly what the concern is if you are to repair it.

In many cases, you may have to **road test** the vehicle to duplicate the symptoms they experience. During your road test, look for related symptoms and abnormalities. Are there any indicator or warning lights lit? Are there any abnormal noises? Does the vehicle handle in an unusual manner? Try to experience the symptoms first hand. Note the symptoms, but do not begin disassembly or testing until you have verified the concern and narrowed down the problem.

Also, note what is working. This will help you to performing a common point analysis.

- 2. Analyze** Once you have verified a malfunction does exist, you can now **Analyze & Diagnose the Customer's Concern**. First, determine whether it is related to some other obvious problem such as a previous repair, collision damage or from adding on an aftermarket accessory (e.g. car alarm or audio system).

The malfunction may be in either the electrical component or the circuit itself. At this time, you should perform some preliminary checks. Based on the symptoms and your understanding of how the system operates, identify one or more possible causes. Use the troubleshooting information in the Service Manuals and ETMs as a guide.

There are other valuable resources like Technical Service Bulletins (TSBs), Tech Times, and Pit Stop newsletter materials. Be familiar with the latest information and always check for TSBs before going any further.

Locate and identify the appropriate wire diagram for the circuit(s) in question and break the circuit into three parts:

1. Power (source)
2. Load (accessory or electrical component)
3. Ground (return path)

Identify the possible causes for a particular malfunction, and then consult the troubleshooting section in the Service Manual. From this information, create a logical procedure to diagnose the symptom.

Write out a list of possible causes and of tests you can perform that will confirm or eliminate each item on your list as the root cause. Now organize the list starting with the most likely to the least likely. Here are some key questions that help when analyzing a wiring diagram:

- Which part of the circuit is most likely the source of the malfunction, power, load, or ground?
- Are there other systems that are affected by the malfunction?
- Which part of the circuit is common to both?
- Is there a system that is functioning properly which shares part of the circuit?
- If so, what part of the circuit can be ruled out as the cause?
- What are some easy points to begin troubleshooting (i.e. easy to access connections, sensors, actuator, fuse block, etc.)

3. Troubleshoot

Now you will need to **Find the Root Cause**. Work through your list starting with the most likely cause first. Isolate electrical circuits, if possible, in your tests. Look for the easy points to access the circuit and work to the difficult access points as required. Following these troubleshooting steps will save you valuable time and help assure an accurate determination of the root cause of the customer's concern.

If you were unable to find the root cause, stop, go back to step 2; analyze & diagnose the customer's concern. The last thing you want to do is just "throw parts at it". Have the concerns looked at by a fresh set of eyes, the shop foreman, a fellow technician, or call the Tech Line.



NOTE: If you spend 30 minutes or more, or if it's a second warranty repair for the same concern, contact Techline.

4. Repair After identifying the root cause of the malfunction, **Repair the Root Cause.** Use the recommended repair procedures described in the service manual and/or ETM. Never take shortcuts that may cause personal injury or damage to the vehicle. Be careful during the repair not to create other concerns by damaging connectors or components that may have lost some of their elasticity over time.

5. Verify Most importantly, **Verify the Repair.** Be certain that it has corrected the customer's concern by performing the system checks as documented by the repair order. Operate any related systems and verify no new malfunctions appear.

During your test drive, operate the vehicle under the same conditions that caused the concern to occur and verify the repair is complete. If the concern reoccurs, go back to step 2 and walk through the process again paying close attention to what may have been overlooked. Occasionally, causes are missed and concerns slip through unresolved. Skipping this step will lead to dissatisfied customers, be certain the repair(s) corrected the concern before returning the vehicle to the customer.

Do not release the vehicle to the customer if you are not sure it is fixed. If you need more time, get the service advisor involved.

Make it your personal goal to fix-it-right the first time, every time!

SUMMARY Vehicles contain many electrical circuits. The circuit design may be series, parallel or series-parallel. It is important that you are familiar with and have the ability to recognize the circuit type. Having knowledge of circuit types helps determine the circuit operation as well as your approach in diagnosing a circuit problem.

PROGRESS CHECK



1. What happens to the total current in a parallel circuit when one of the branches is removed?
 - a. Current stays the same
 - b. The current will always go up
 - c. The current will always go down based on decreased resistance
 - d. Current changes to equal the remaining resistance
2. The voltage drop in each branch of a parallel circuit will equal:
 - a. source voltage
 - b. $\frac{1}{2}$ of source voltage
 - c. twice the source voltage
 - d. the resistance in that branch of the circuit
3. In a parallel circuit, how many paths is/are there for current flow?
 - a. One
 - b. There is an individual path for each load
 - c. There is an individual path for each control device
 - d. One path for the source and multiple paths for ground
4. What indicates a problem when performing a voltage drop test?
 - a. low resistance
 - b. excess voltage
 - c. low current flow
 - d. unwanted resistance

PROGRESS CHECK



5. Total resistance in a parallel circuit is:
 - a. equal to the sum of the individual resistance in each branch
 - b. always less than the lowest individual resistance
 - c. equal in each branch of the circuit
 - d. $\frac{1}{2}$ the total circuit resistance in each branch
6. When tracing power flow diagrams, your first goal is to locate:
 - a. source
 - b. load
 - c. ground
 - d. All of the above
7. What is a circuit that doesn't have a complete path to ground called?
 - a. Shorted
 - b. Open
 - c. Clipped
 - d. Incomplete
8. Corroded terminals in a wiring harness can cause which of the following?
 - a. Short circuit
 - b. Excessive resistance
 - c. Excessive current flow
 - d. Blown fuse

Electrical Components Theory Module

TARGET AUDIENCE The target audience for this module consists of Kia service technicians who have a basic understanding of automotive technology.

MODULE GOAL Upon completion of this module, you will be able to troubleshoot the following electrical system components: fuse, relay, solenoid, lamp, horn, audio speaker, and electrical motor.

MODULE OBJECTIVES

- Given an electrical workstation, you will build a horn and blower motor circuit, including the use of a switch, relay, and resistor block, noting the function of each component with 100% accuracy.
- Given an electrical workstation and a Digital Volt-Ohm Meter (DVOM), you will identify normal voltage readings in a horn and blower motor circuit with 100% accuracy.
- Given an electrical workstation and a Digital Volt-Ohm Meter (DVOM), you will troubleshoot an open or excessive resistance fault in a horn and blower motor circuit with 100% accuracy.

MODULE INSTRUCTIONS Carefully read through the material, take notes based on the classroom discussion and study each illustration. Throughout the module there will be Progress Check questions for you to answer. You may use the modules to answer the questions.

REQUIRED MATERIALS In order to complete this module, you will need the following items:

- Digital Volt-Ohm Meter (DVOM)
- Electrical Workstation

TIME TO COMPLETE Approximately 1.25 hours

ACRONYMS	ATC – Closed Blade Fuse
	ATO – Open Blade Fuse
	ECU – Electronic Control Unit
	EVAP – Evaporative Emission Control System
	HVAC – Heating Ventilation & Air Conditioning System
	LED – Light Emitting Diode

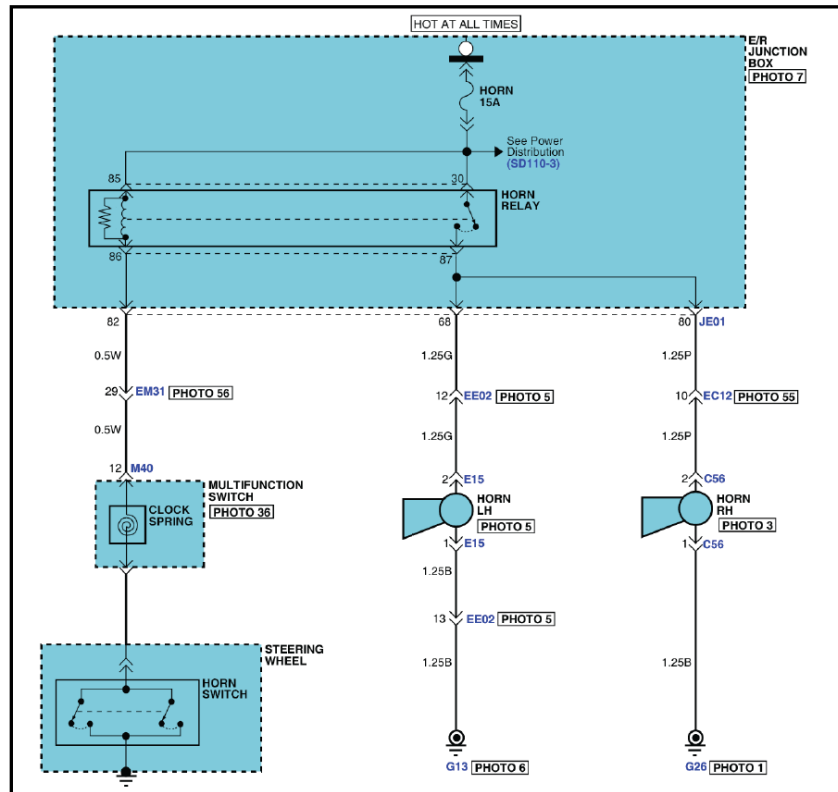
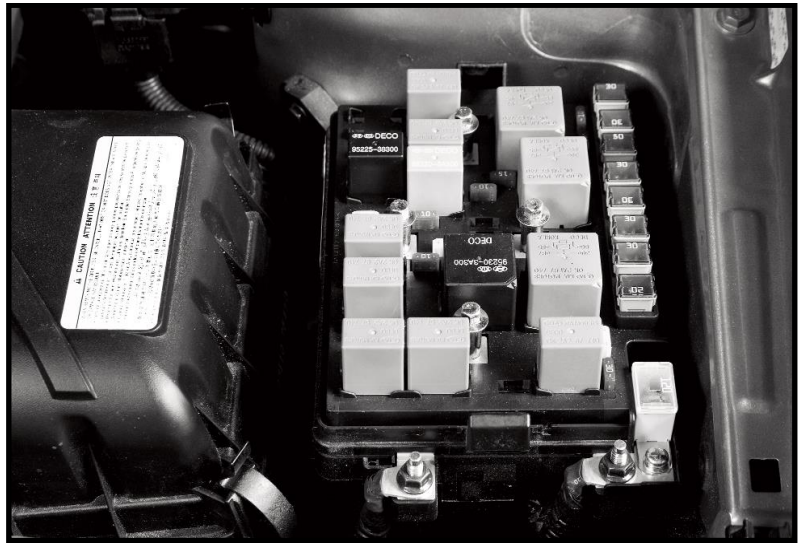
INTRODUCTION

In an earlier module, we looked at the three types of circuits. Now it is time to take a closer look at some of the common components found in many vehicle circuits including fuses, diodes, relays, solenoid, lamps, horns, and gauges.

A solid understanding of the components that make up the various circuits will enable you to quickly determine the root cause of a customer concern. This module will describe the operation, function, and diagnosis of the most common circuit components and how they are used in a circuit.

Electrical components typically fall into the following categories:

- Circuit Protection Devices
- Conductors
- Control Devices
- Load Devices



CIRCUIT PROTECTION DEVICES

The function of a **circuit protection** device is to create an _____ in the circuit, which is the result of _____ caused by _____ in the circuit.

FUSES

Function & Operation

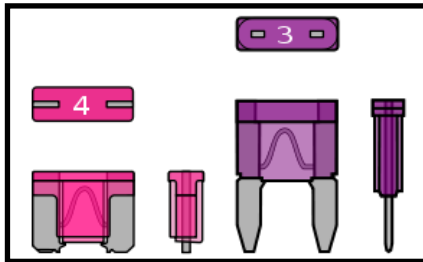
Fuses are thin wires that melt very easily when overheated by excessive current. They are built-in weak spots in a circuit designed to protect circuit components and wiring from damage.

Fuse ratings are based on the current flow needed to operate the circuit while protecting it from excessive current flow. For example, a circuit that would normally draw 7 amps might have a 10 amp fuse to protect it.

TYPES

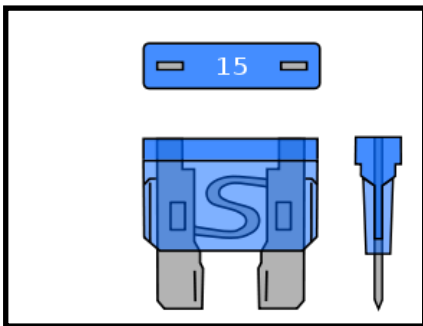
Fuses come in a variety of shapes and sizes including:

- Low profile mini fuses
- Mini fuses
- ATO/ATC fuses
- Low Profile Slow Blow fuse
- Multi fuse
- Main fuse



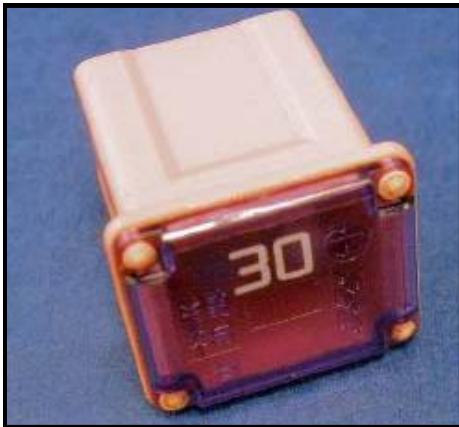
“Low profile Mini” and **“Mini”** fuses are smaller size version of the ATC fuse.

The fuse material is molded into plastic shells that are also color-coded to indicate the current rating. The current rating is molded on the top of the fuse. Never replace a fuse with one of a higher rating; circuit damage may occur.



ATC fuses are sealed to prevent accidental ignition of any fuel vapors that may accumulate in the engine compartment, from the arc created when the fuse blows. Only ATC fuses should be used under the hood and in areas where fuel vapors may accumulate.

ATO fuses are open on the bottom and should only be used in the interior of the vehicle.



Low Profile JCASE "Slow Blow" fuses are used for circuits that may have a current spike that exceeds the fuse rating. This typically included circuits with solenoids and motors.



Multi-fuses are more than one fuse in a single housing and protect multiple circuits individually. The fuses in a multi-fuse assembly cannot be replaced individually.



Main fuses are located at the battery positive (+) terminal. Examples of these fuses include:

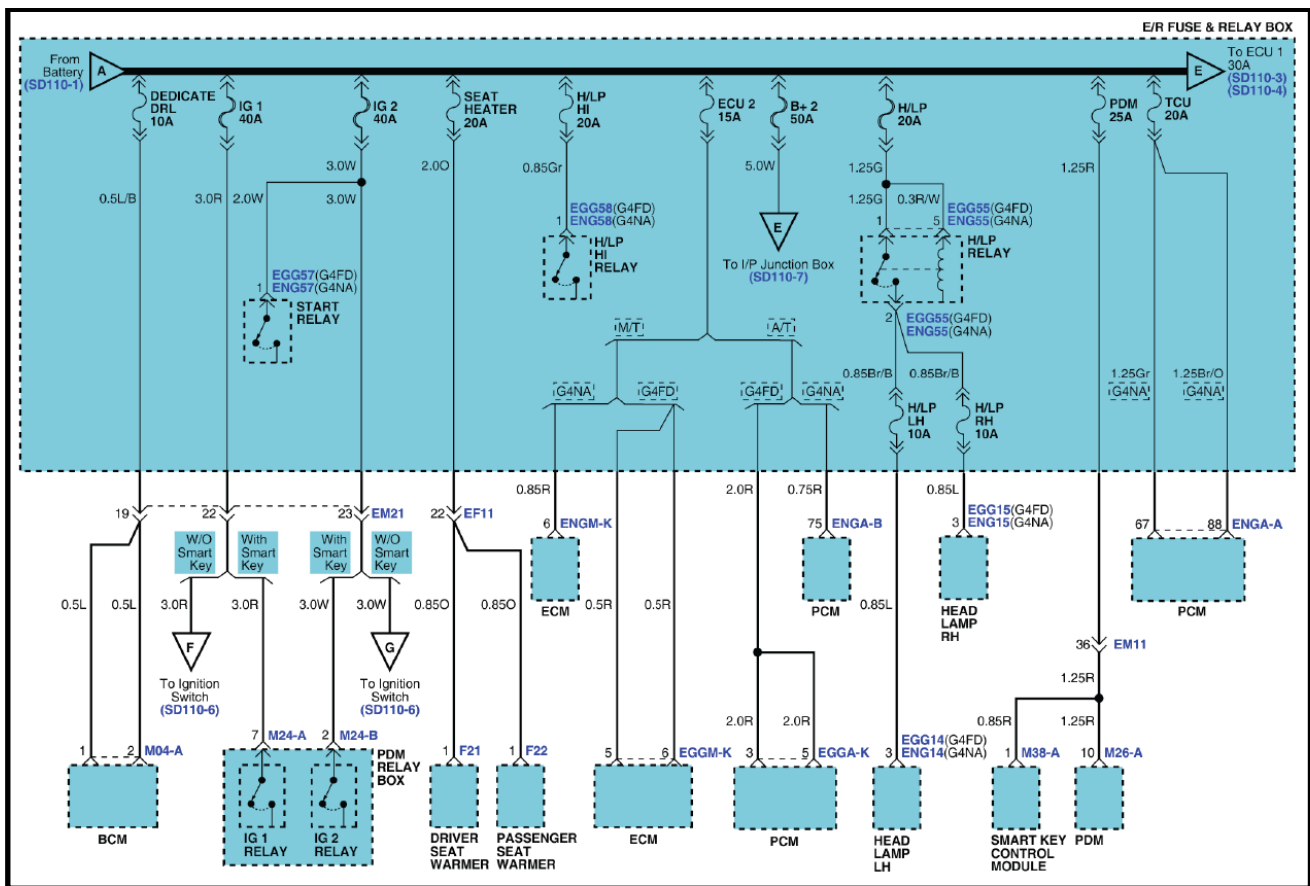
- Alternator
- Cooling Fan

A **fusible link** is simply a short piece of wire approximately four (4) AWG sizes smaller than the wire that is being protected.

For example, an AWG 16 fusible link might be used to protect AWG 12 wiring. Electrical fusible links are common in high-current automotive applications. The wire in an electrical fusible link is encased in high-temperature fire-resistant insulation to reduce hazards when the wire melts.

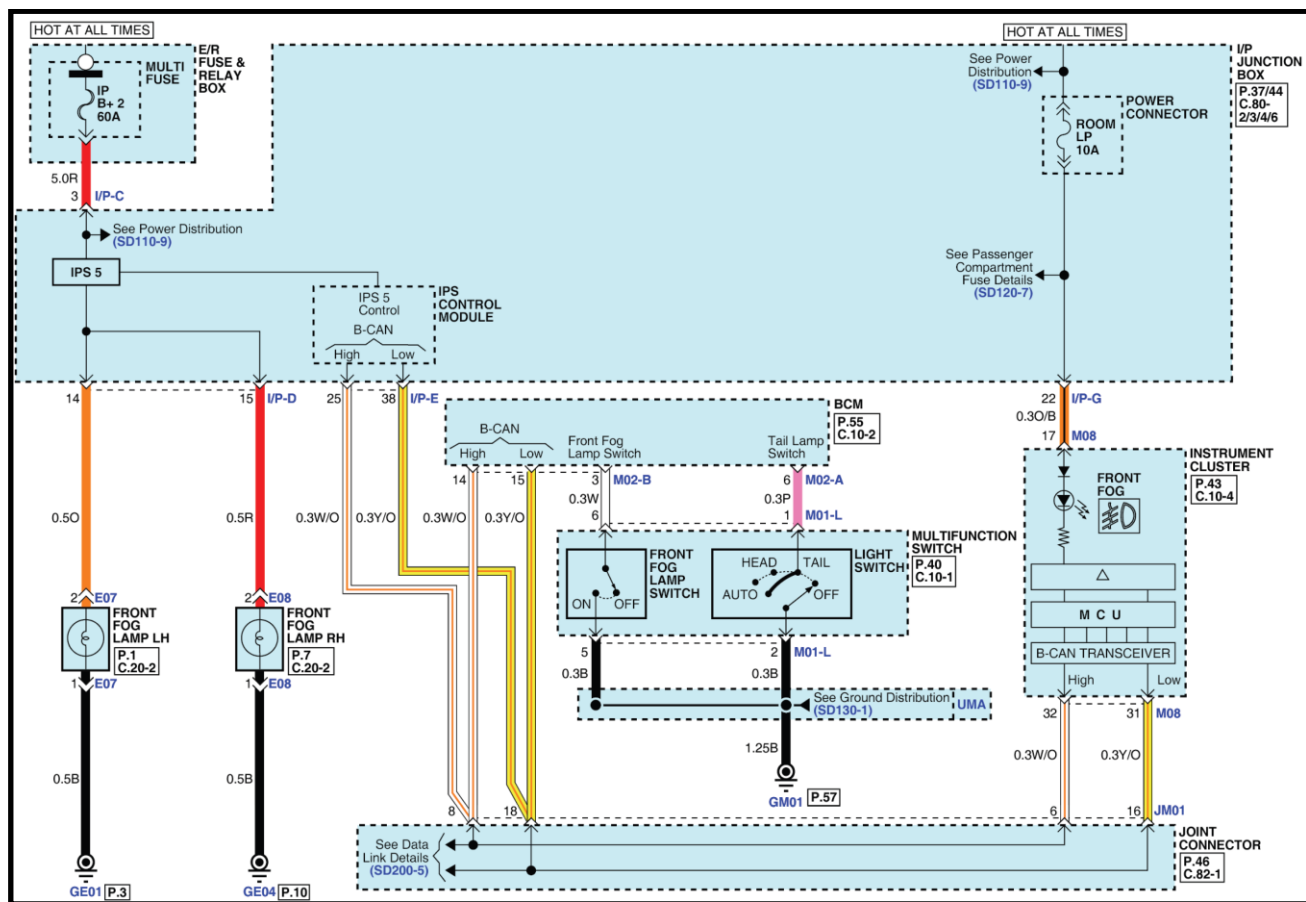
Fuses/Fusible links can fail due to the following:

- _____
- _____
- _____
- _____
- _____



The Power Distribution Schematic Diagrams, located in the ETM, are a good place to determine which fuse/fusible link protects which circuit.





INTELLIGENT POWER SWITCH

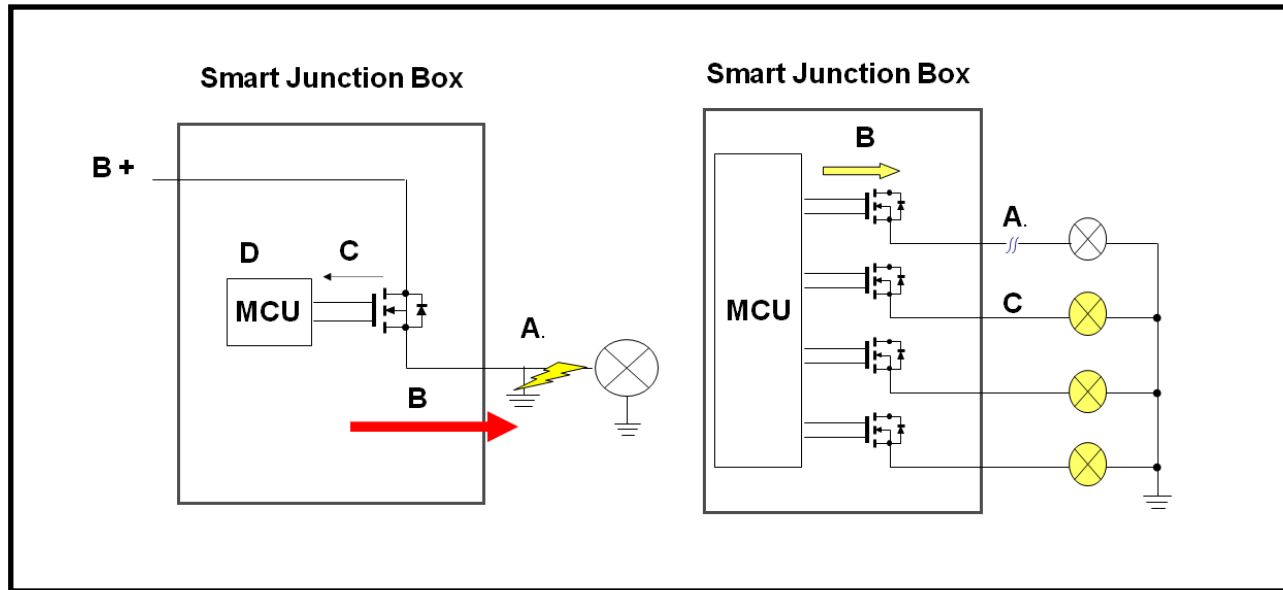
The Intelligent Power Switch (IPS) provides circuit protection by monitoring current flow. The switch is integrated on printed circuit boards within various modules on Kia vehicles. A circuit that is protected by an IPS will feature "IPS" inside the module.

If the current flow exceeds preset limits, the IPS will shut down the current flow to the affected circuit. If there is no problem when the circuit is repaired, and the ignition cycled OFF/ON, the IPS resets itself so the load devices operate properly again.

Depending on the vehicle and the circuit, the IPS can set a DTC if the circuit is open, shorted to ground or both while the fault is present.



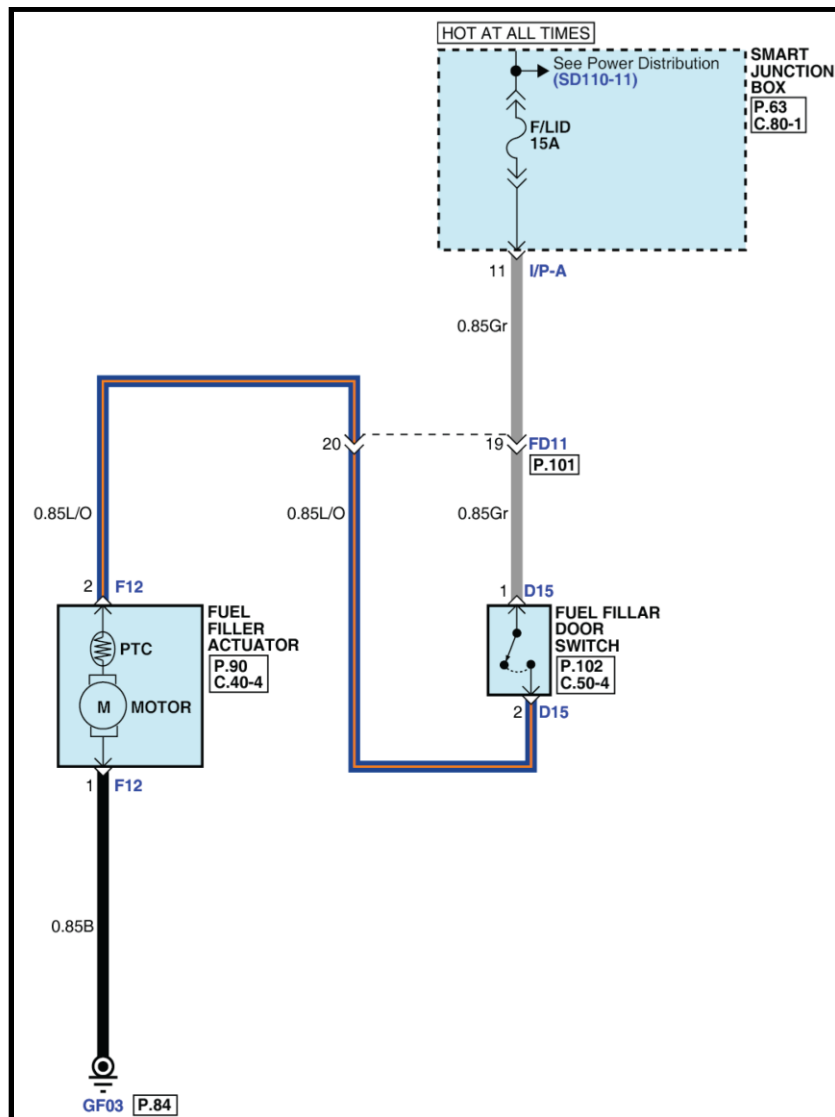
FEEDBACK: What would happen to the IPS if a customer installed an aftermarket bulb in the vehicle? How could a different wattage bulb affect the IPS operation?

**Programmable Current Limit****Open Current Limit****TYPES OF IPS****Programmable Current Limit (PCL)**

- A. Short to ground
- B. Excessive current flow
- C. Monitors current information
- D. Prevents control while excessive current is detected and record lamp status

Open Current Limit (OCL)

- A. Open circuit when the turn signal lamps are activated.
- B. Current flow not detected.
- C. Blinking rate on the remaining lamp changes (example: turn signal lamps blink faster).



POSITIVE TEMPERATURE COEFFICIENT RESISTOR Function & Operation

PTC resistor (or thermal resistor) is a special type of circuit protection that increases resistance as its temperature increases.

When the polymer material in the resistor is heated from excessive current flow, it expands, causing the circuit to "open". The PTC remains "open" as long as voltage is applied to the circuit. It resets when voltage is removed and the polymer cools.

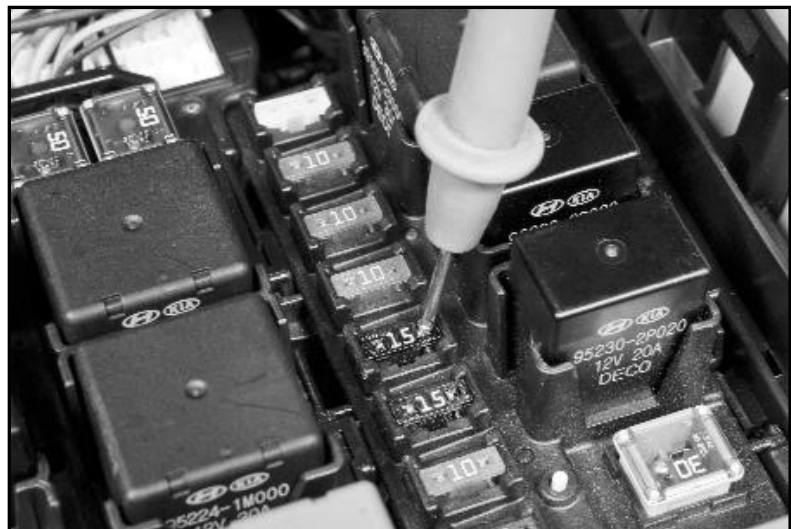
PTCs are soldered in place and cannot be serviced separately. The component that contains the PTC must be replaced if a PTC fails.

TROUBLESHOOTING

If an electrical system is inoperative, use the power distribution schematic diagram to determine the circuit protection devices in the circuit.

If one system is inoperative, check the protection for that device. If multiple circuits are inoperative, use the diagram to determine the protection device that protects all of the circuits.

Fuses, fusible links and PTC resistors can be checked by:

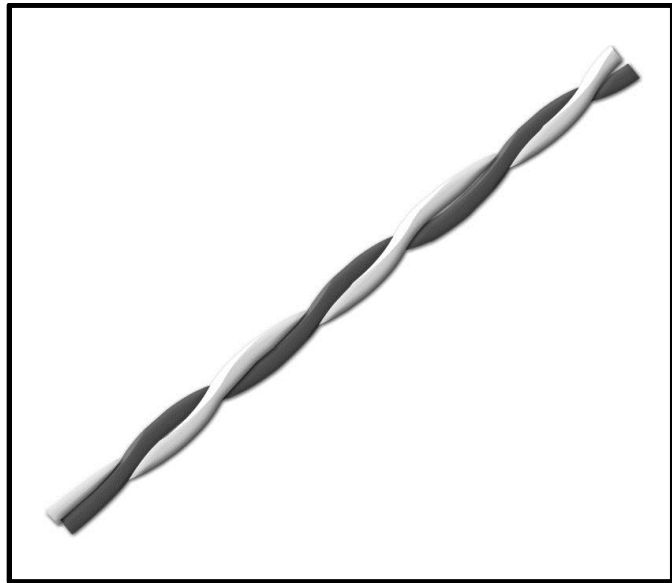


Use the DVOM and a visual inspection to verify the condition. Do not assume a circuit protection device is good because it looks good.



NOTE: If a blown fuse is found, read the fuse. Inspect the fuse to determine if the fuse wire is just broken due to the load operating near its maximum capacity, or did a major short cause the fuse wire to vaporize.

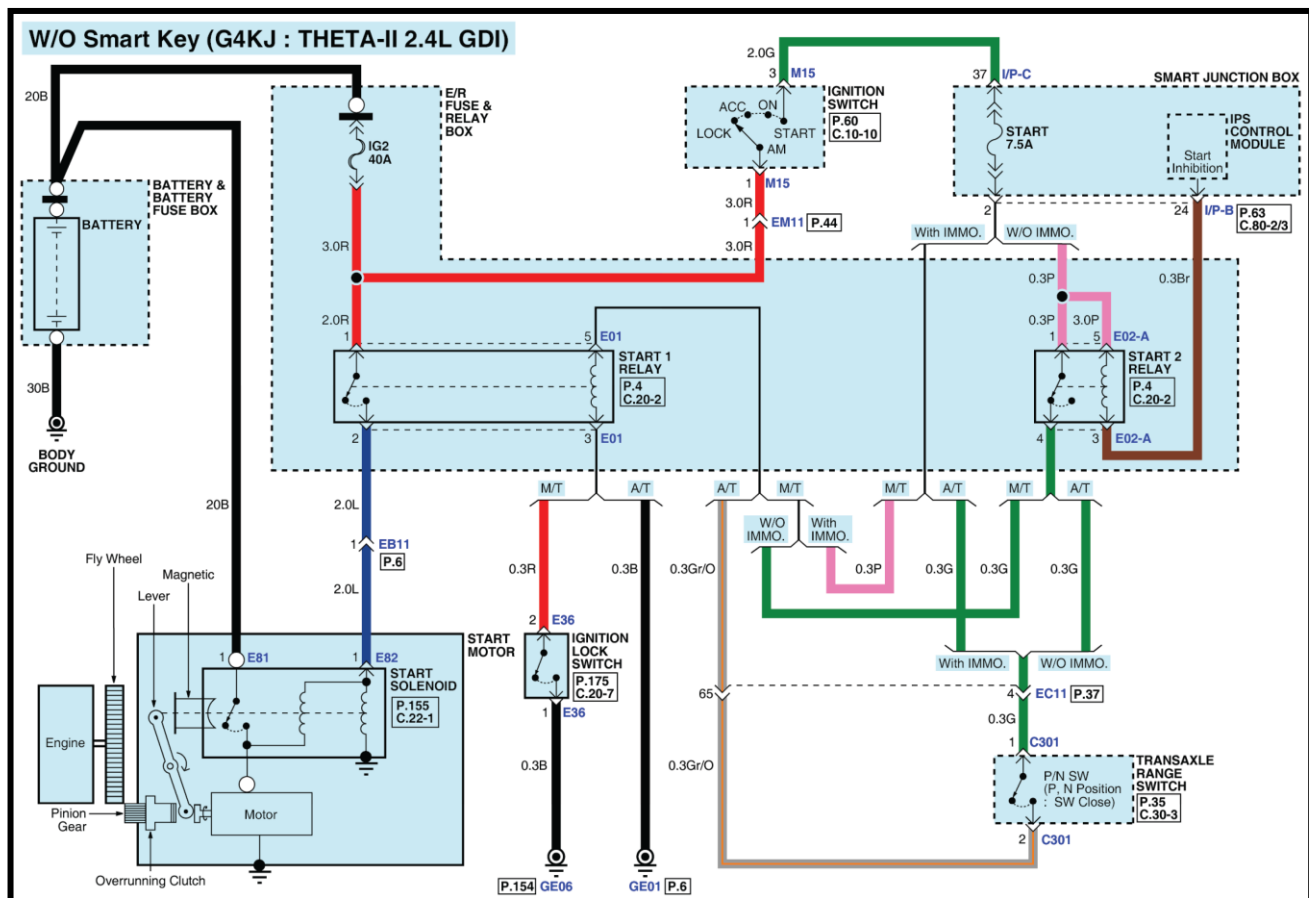
NOTES:



CONDUCTORS Conductors are made of copper wire that allow current to flow freely.

The insulation protects the wire from shorts or grounding.

Twisted pairs of wires are often used in circuits to reduce electro-magnetically induced voltages.



AWG-to-INCHES-to-METRIC CONVERSION TABLE

AWG Size	Metric Size mm ²
24	0.2
20	0.5
18	0.8
16	1
14	2
12	3
10	5
8	8
6	13
4	19
2	32
0	52

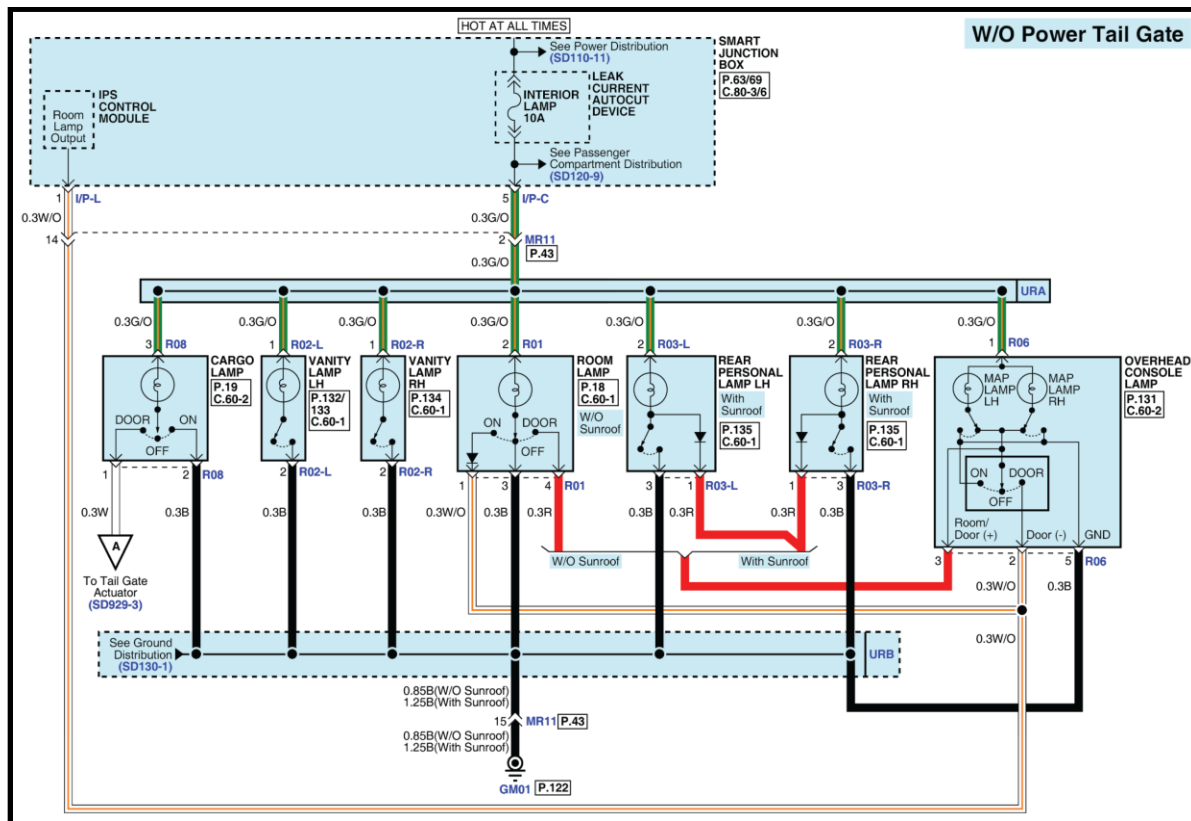
Automotive wiring typically has multiple strands to increase the wire's ability to be flexible and conduct electricity.

Two types of wire size charts are used in automotive applications:

- American wire gauge (AWG)
 - The larger the number, the smaller the wire size
- Millimeter Squared (mm²)
 - The larger the number, the larger the wire size

For comparison purposes:

- Standard automotive wiring is approx. 20 AWG or 0.5 mm²
- Average battery cable is approx. 4 AWG or 19 mm²



CONTROL DEVICES

Control devices typically turn circuits ON or OFF. They can also control a system function. For example, changing the A/C mode or blower fan speed.

Control devices typically:

- Open the circuit to turn it OFF
- Close the circuit to turn it ON

Control devices may also limit the current in a circuit or only allow current to flow in one direction.

Examples of control devices:

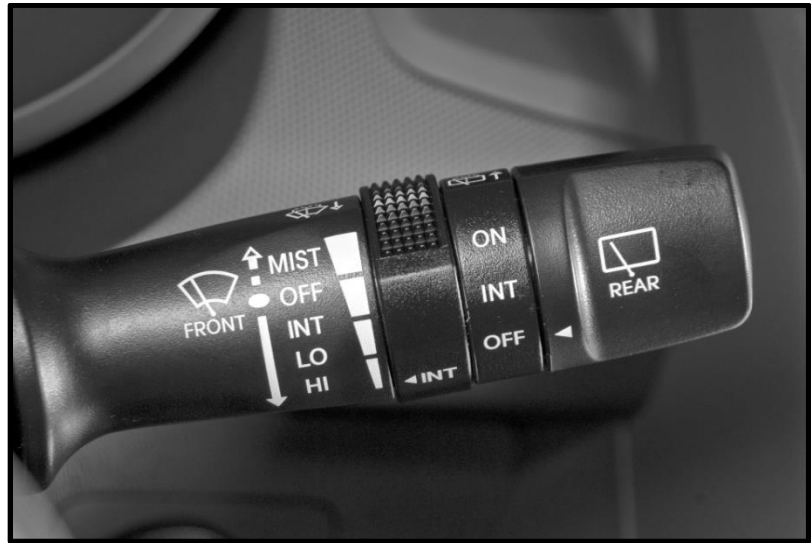
- **Switch** – Brake light
- **ECU** – Transistor to control an output device (solenoid)
- **Relay** – Fog light
- **Intelligent Power Switch (IPS)**

Control devices can also be used as an _____

or as an _____ device in computer controlled circuits.

SWITCHES

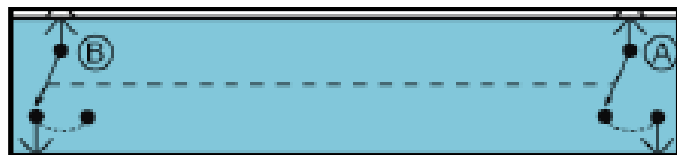
Function & Operation



A switch is one of the most common control devices. Switches vary from the simple, a door switch, to the complex, a multi-function switch. The simplest switch has two positions, OPEN - no continuity and CLOSED - continuity. Switches can be opened and closed manually, mechanically, electrically or by pressure. Switches may be normally open or normally closed.

A switch is placed in a series position in the circuit. When the switch is open, no current flows and the circuit is OFF. When the switch is closed, there is a complete circuit for current and the circuit is ON.

A switch may be used as an input to an Electronic Control Unit (ECU) which in turn analyzes other inputs and then decides when to turn the circuit ON or OFF.



Normally Closed (NC)

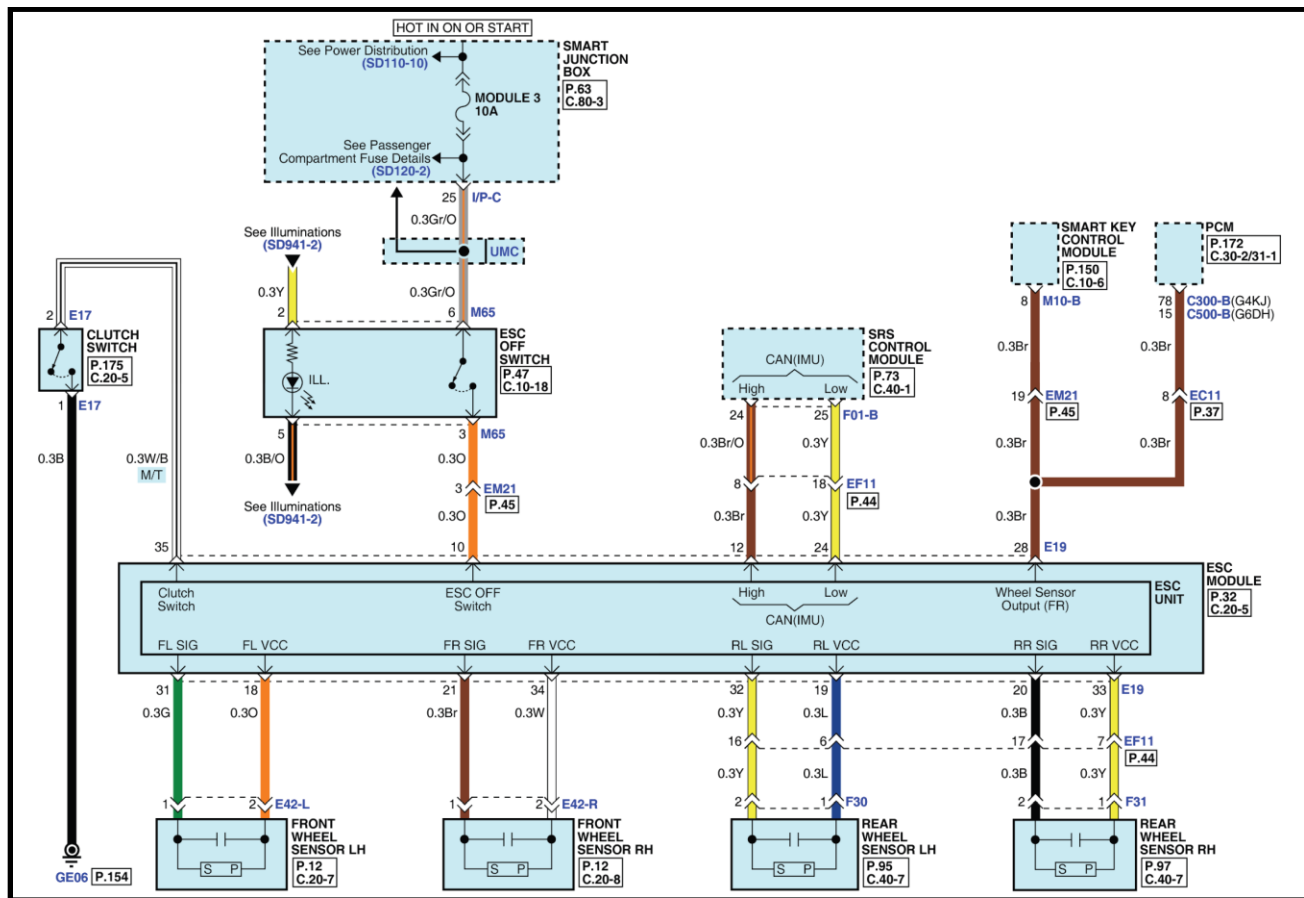
Normally Open (NO)

Switches can be either Normally Open (NO) or Normally Closed (NC) type. They are shown in circuit diagrams in their "normal, at-rest" position.



You can test the switch for continuity by:

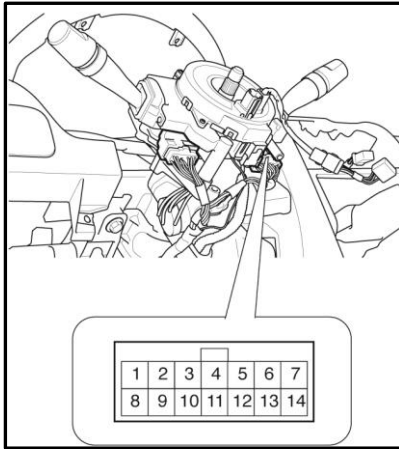
- Continuity should only exist when the switch is in the ON position. Unless the switch pin-out chart indicates resistance, the tests should have little or no resistance.



You can also test the switch using a voltage drop test.

The advantage of the voltage drop test is that you are testing the switch under normal operating conditions.

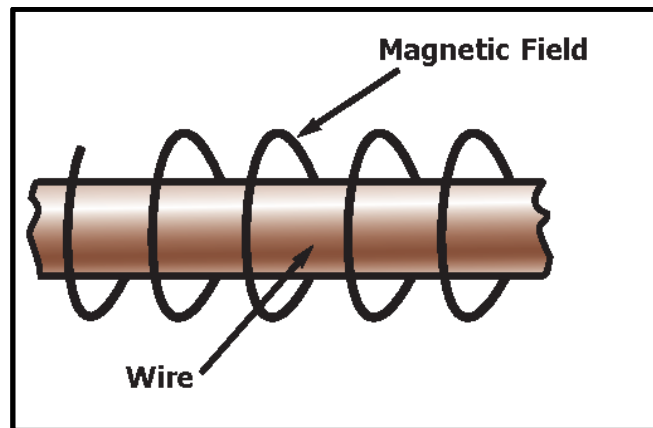
- Using the appropriate ETM schematic and the switch in the ON position measure the Voltage Drop across the switch terminals.
- GOOD SWITCH: Voltage Drop of less than 0.2V when the switch is closed.
- BAD SWITCH: Voltage Drop of more than 0.2V when the switch is closed indicates excessive resistance in the switch or the connector.
- Switch Open/OFF: Voltage Potential measured across the switch contacts confirms the switch is open.



Terminal Position	1	2	3	4	5	6	13	14
MIST				○—○				
OFF		○—○						
INT		○—○			○—○		○—○	○—○
LOW		○—○			○—○			
HI	○—○				○—○			

Using the pin-out table above, record the switch position when continuity is measured between the following terminals?

1. Terminals 2 and 3 _____
2. Terminals 4 and 5 _____
3. Terminals 1 and 5 _____
4. Terminals 13 and 14 _____

**ELECTROMAGNETISM**
Inductors

Many automotive components use electromagnetism to create a magnetic field to do work or control operation of fluid flow.

How is this magnetic field created?

_____ is the flow of electrons

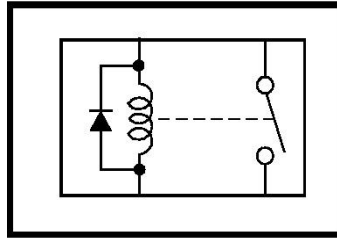
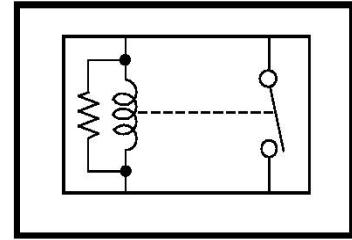
The flow of electrons creates a _____

Coiling the wire concentrates the _____
increasing its strength.

One example of this is the electro-magnet. Wrapping an insulated wire around an iron core and then connecting the wire to a power source. The coil of wire becomes an inductor when connected to a power source.

Examples of other inductive devices include:

- Coils
- Transformers (Step Up / Step Down)
- Solenoids
- Relays
- Fuel injectors
- Motors
- Stepper motors

**Suppression Diode****Suppression Resistor****Spike Suppression**

When an inductor is turned off, the magnetic field that has built up collapses very rapidly. This creates a voltage spike like an ignition coil. Many inductive devices have a clamping diode or resistor in the circuit to block this voltage spike from traveling back to an electronic control unit (ECU). These diodes and resistors are critical to proper circuit operation.

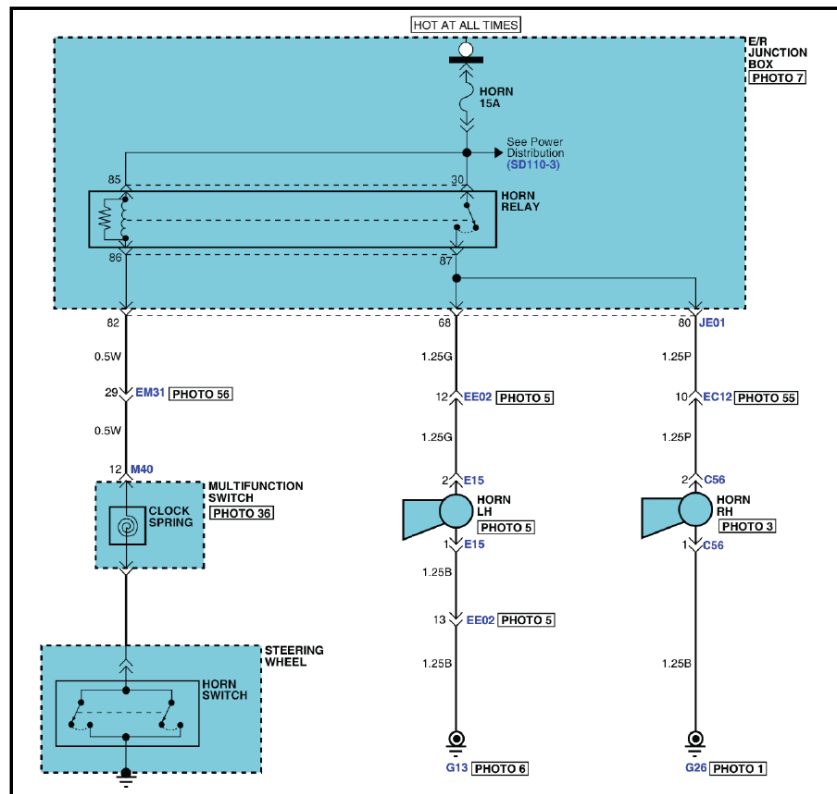
We will review diodes in more detail later in this module.



NOTE: Sensitive ECUs must have spike suppression or ECU damage may occur. Always check for aftermarket or misplaced relays on circuits with sensitive electrical components.



CAUTION: Relays should only be replaced with the same type of relay. Never substitute a relay that has a clamping diode with a relay not equipped with a clamping diode or damage to the ECU may occur.



RELAYS Function & Operation

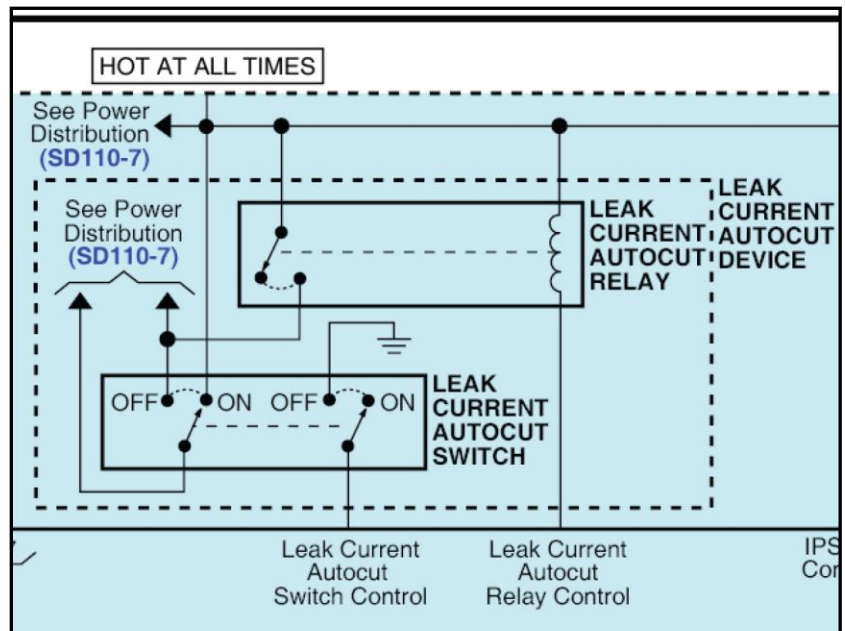
A relay is a remote control switch that can control a _____ amount of current using a _____ amount of current. A relay contains an electro-magnetic coil that closes a set of _____ when the coil is energized.

A relay is especially useful when controlling a high current device. An ECU, using a small amount of current, can energize the coil side of the relay with about 500 mA or 0.5A of current can control circuits that carry 20 to 40 times as much current.

Using a low current signal to control a high current circuit not only allows an ECU to control the circuit, it also reduces the need for heavy-duty control switches and heavy gauge circuit wire to and from the switch.

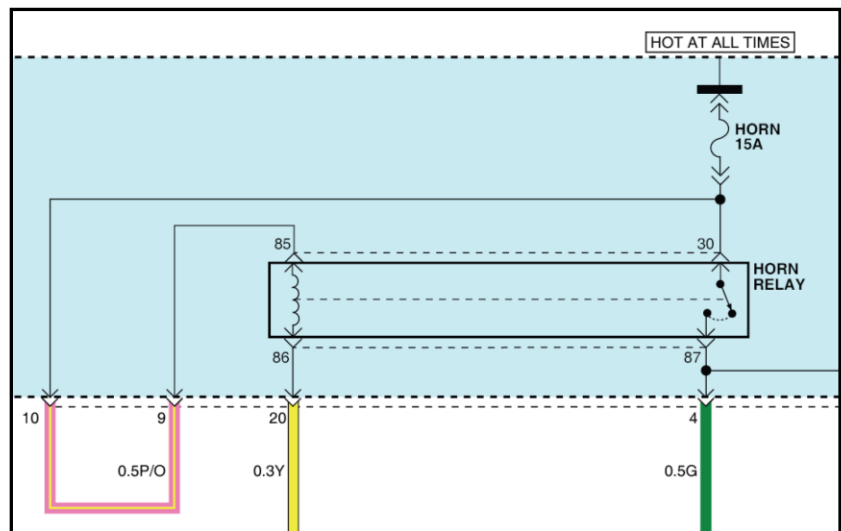


NOTE: Like switches, relays are identified by their "at-rest" position, normally open (NO) or normally closed (NC)



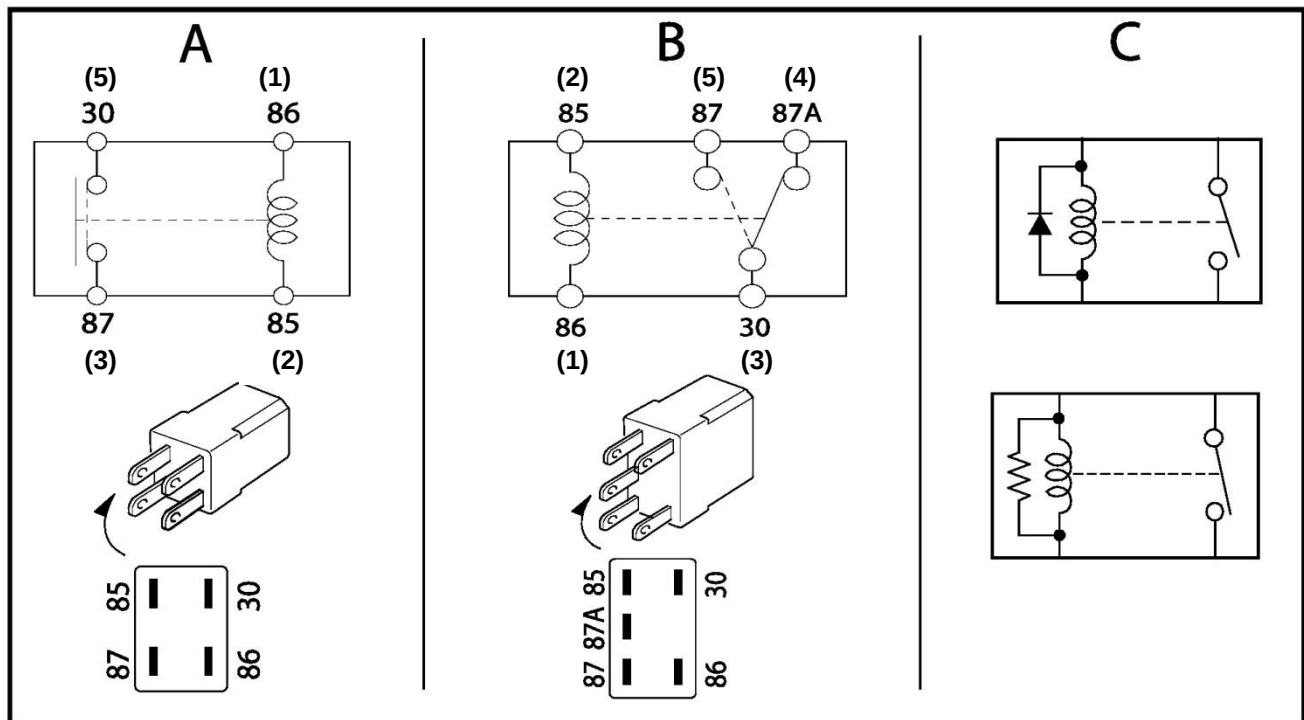
INTERNAL RELAY Internal relays:

- Not serviceable
- No terminal numbers on ETM
- May have Actuation Test using GDS



EXTERNAL RELAY External relays:

- Serviceable
- Terminal numbers on ETM
- Typically has Actuation Test using GDS



Example A Normally, pins 30 and 85 have source voltage applied to them with the IGN key ON. Pin 86 is controlled by a switch or an ECU and provides a path to ground. This energizes the coil and closes the contact points. Now the current can flow through the contact point side of the relay, pins 30 and 87 to operate the load device.

Example B Shows a 5 pin relay. This is sometimes referred to as a crossover relay. The current is directed to pin 87A (NC) when the relay is OFF and to pin 87 (NO) when it is ON. These relays are sometimes used on 2-speed cooling fans.

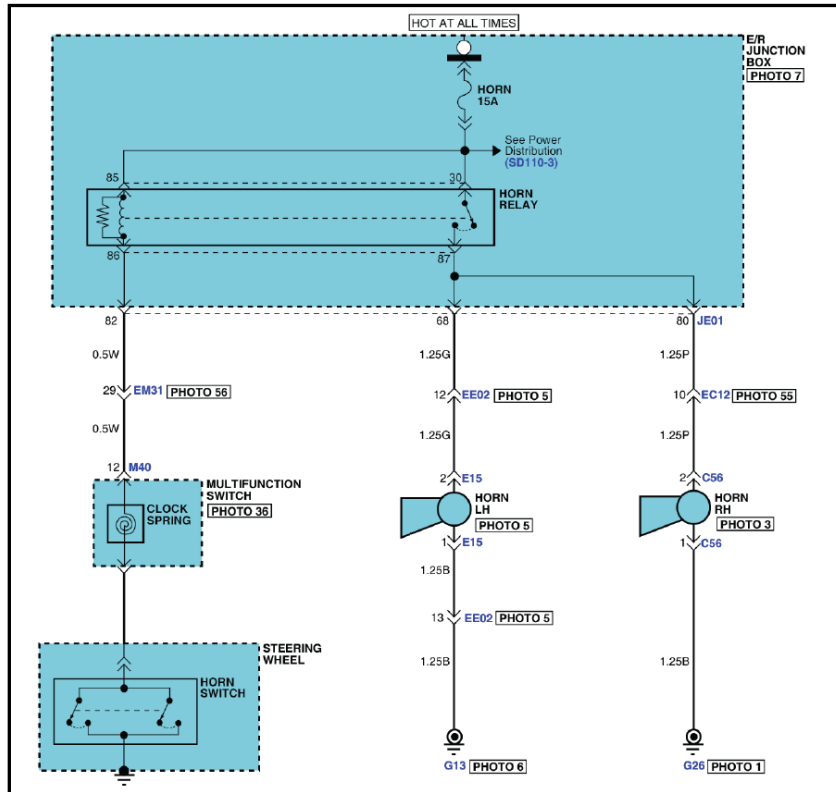
Example C Notice that the top relay has a diode in parallel with the coil, while the bottom relay has a resistor in parallel with the coil. The diode and resistor are used to suppress the voltage spike that is induced when power is removed from the coil side of the relay. Do not interchange these two types of relays. If polarity is reversed across the relay with the diode, the diode will be damaged and the relay will no longer suppress the voltage spike.



NOTE: Relay pin numbers and how the relay is wired into the circuit may vary. Always refer to the appropriate service information when troubleshooting.



Troubleshooting



Relays are two separate circuits, control (coil), and switch.

Whenever possible test the relay in the car, energized and current flowing through the circuits.

Most relays have four terminals.

The coil side of the relay terminals are ____ and ____.

The switch side of the relay terminals are ____ and ____.

Step 1 - While turning the ignition switch ON or activating the relay, a click should be heard coming from the relay. A click would indicate that the control side of the relay is functioning properly and to focus your attention on the switch side of the relay (terminal 30 and 87).



NOTE: If the relay does not click, then go to Step 2. If the relay clicks, then jump to Step 3.

Step 2 - No click indicates either a lack of power on terminal 85, no ground control on terminal 86, the relay coil is open, or the switch is stuck in the relay. To test:

- Measure the resistance of the relay coil. It should measure between 50 and 100 ohms. If not, replace the relay.
- Place a DVOM on terminal 85 with the ignition ON and note the voltage (battery voltage should be present here).
- No voltage would indicate an open on the power supply of the relay control circuit.
- If voltage is present then check the voltage on the ground side terminal 86 with the relay energized.
- If voltage is present, the ground control circuit to the relay is open.
- If there is no voltage at terminal 86, and no audible click from the relay when activated, then the coil side of the relay is open and the relay must be replaced.
- If the relay tests OK electrically, the switch may be stuck. NO relay contacts, test for continuity at terminals 30 and 87 with no power applied to the relay. Continuity indicates that the switch is stuck.

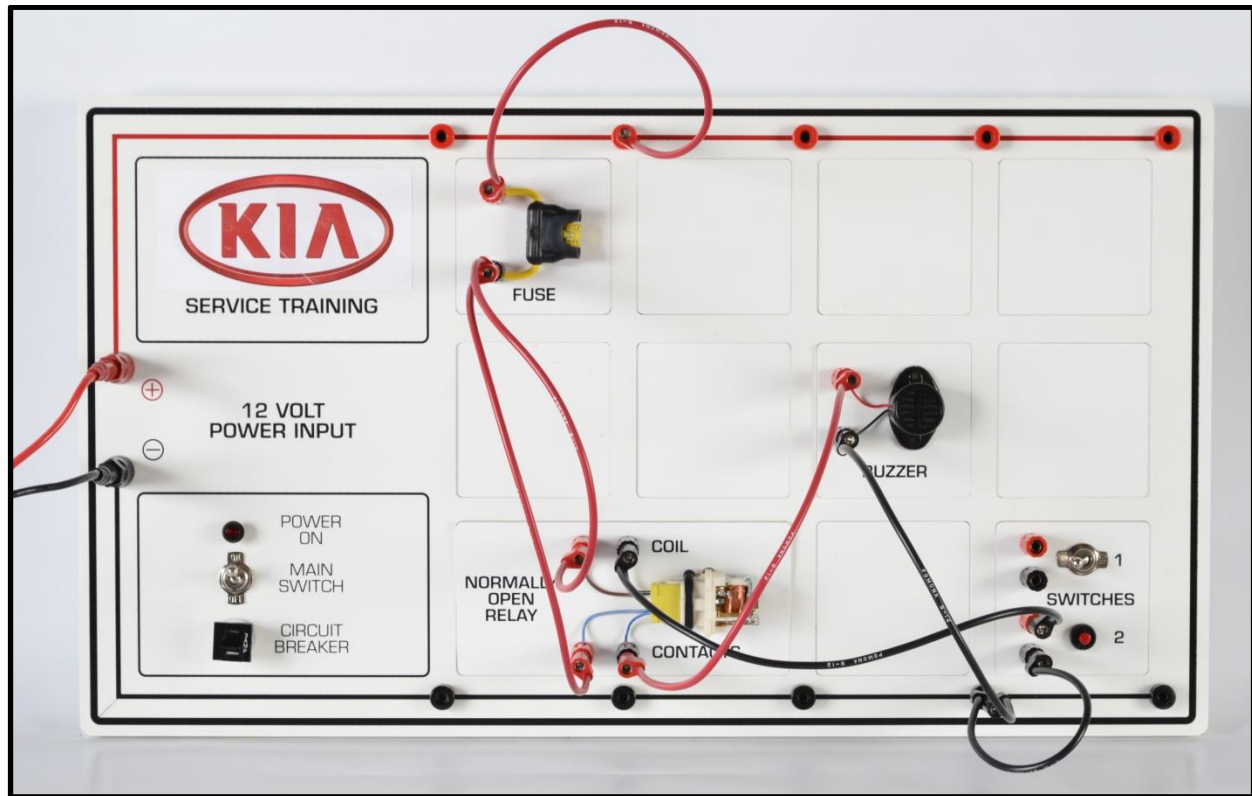
Step 3 – If the relay clicks when activated, proceed to check the voltage on terminal 30 when the relay is activated.

- Battery voltage at this point shows that the circuit from the battery feed to the relay is good.
- No voltage or low voltage indicates an open or high resistance on the main battery feed to the relay terminal 30 and the circuit will need to be repaired.
- If proper voltage is observed at terminal 30, then check voltage on terminal 87 with the relay energized.
- No voltage would indicate an open in the relay and the relay should be replaced. Battery voltage would indicate that the circuit from the relay terminal 87 to the load (that the relay controls) is open and the circuit needs to be repaired.



VIDEO: Relay





Horn Circuit

Build the horn circuit as shown. Connect the power source to the breadboard and turn ON the main switch.



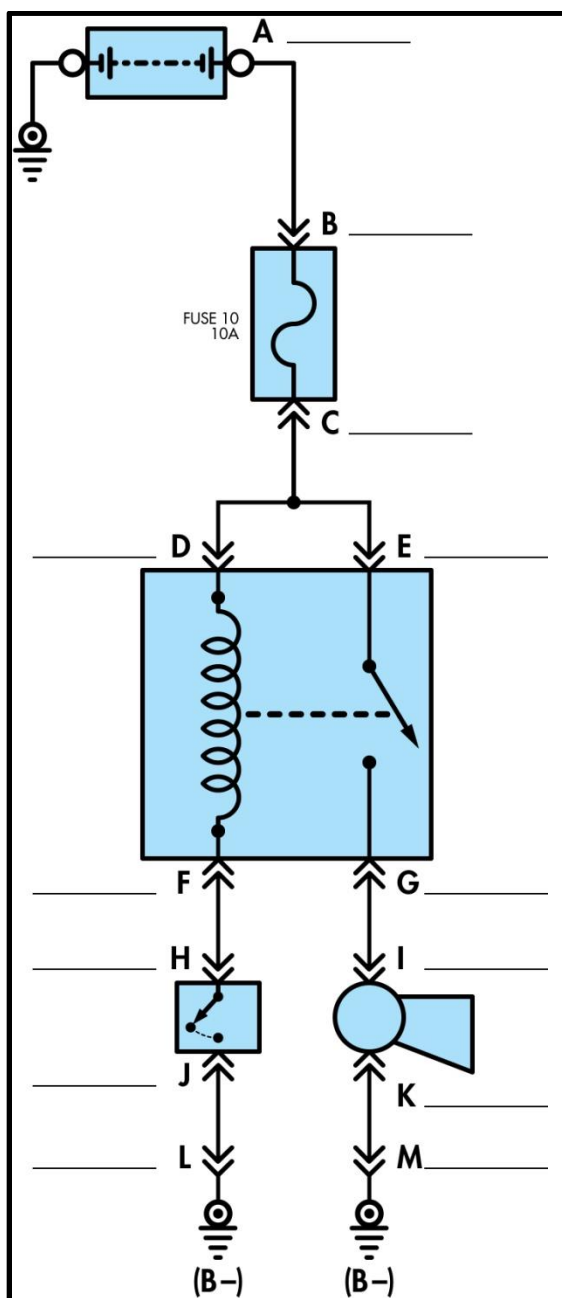
1. What are the load devices in this circuit?

2. How many switches are in this circuit? Explain.

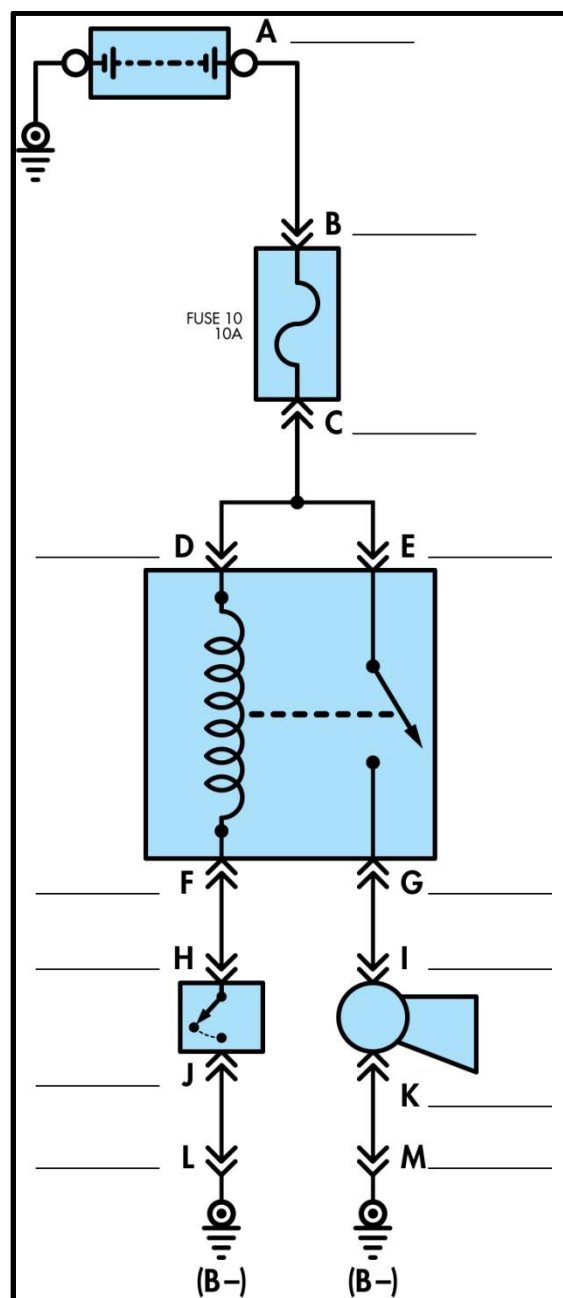
3. Is the switch in series with the horn?

YES / NO Explain: _____

4. Is this one or two circuits? Explain.



Predict



Measure

Troubleshooting Relays

Using the horn circuit you built on the previous page:

- ☐ Predict the voltage readings when the circuit is **CLOSED (ON)** using the diagram above.
- ☐ Turn the circuit ON and use the DVOM to measure the voltage readings and current flow. Note your readings on the diagram.

DIODES

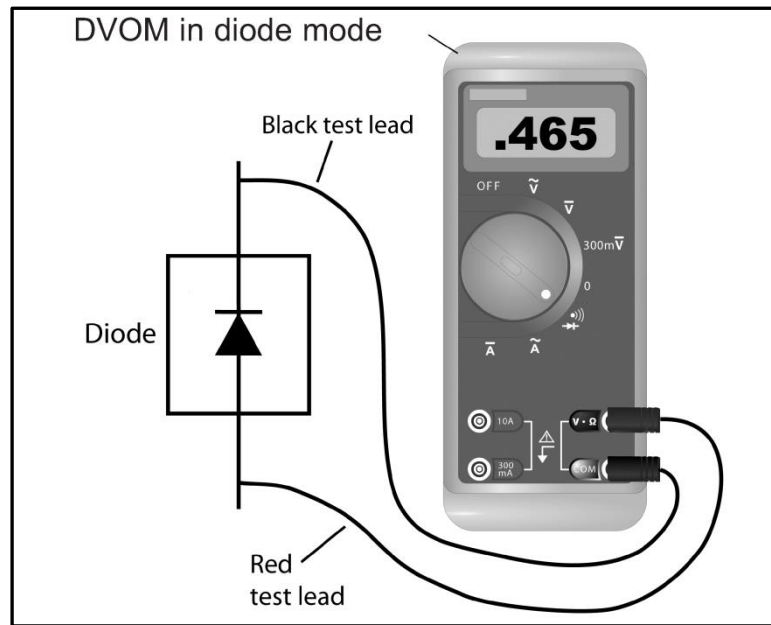
A **diode** is a semiconductor device that allows current to flow _____.

Function & Operation

It can be compared to a _____ which permits a fluid to pass in one direction and stops it in the other.

Diodes are commonly used to control voltage spikes and in the generator to rectify or change the AC voltage into DC voltage. A diode is also used in the generator to prevent the battery from discharging back through the windings in the generator.

Other types of diodes include Zener diodes, which allow current to flow in both directions above a certain voltage, and Light Emitting Diodes (LEDs).

**Troubleshooting**

Since current should flow through a diode in only one direction, use the diode test feature of a DVOM to verify its operation.

- Power down the circuit and isolate the diode from the rest of the circuit.
- Select diode test and connect the leads.
- With most DVOM's when the leads are connected in one direction, the meter will beep and display the forward bias voltage required to "turn on" the diode i.e., the voltage required which will allow current to flow in one direction.
- When the leads are reversed, the meter will display O.L.
- If the diode measures no continuity or O.L in both directions, it must be replaced.

The bias voltage for most diodes is around 0.5V.



NOTE: A diode is said to be "forward biased" when current is allowed to flow and "reverse biased" when current does not flow.



CAUTION: Diodes are directional. Installing a diode backward will cause the circuit to malfunction, may damage the circuit, and may damage the diode.

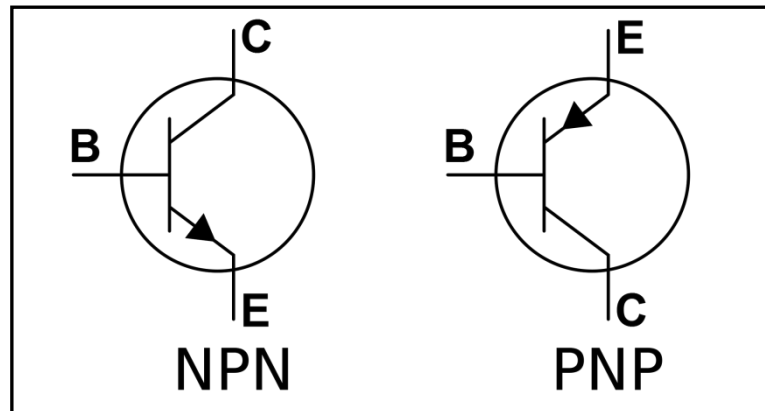
**DIODE INSPECTION**

- ☐ Locate the diode in your workstation unit.
- ☐ Set your DVOM to the diode test function.
- ☐ Place the red lead on one end of the diode, and the black lead on the other end. Record your reading here:

- ☐ Reverse the DVOM leads and record your reading here:



NOTE: A good diode will show approximately 0.5V when forward biased and Over Limit when reverse biased.

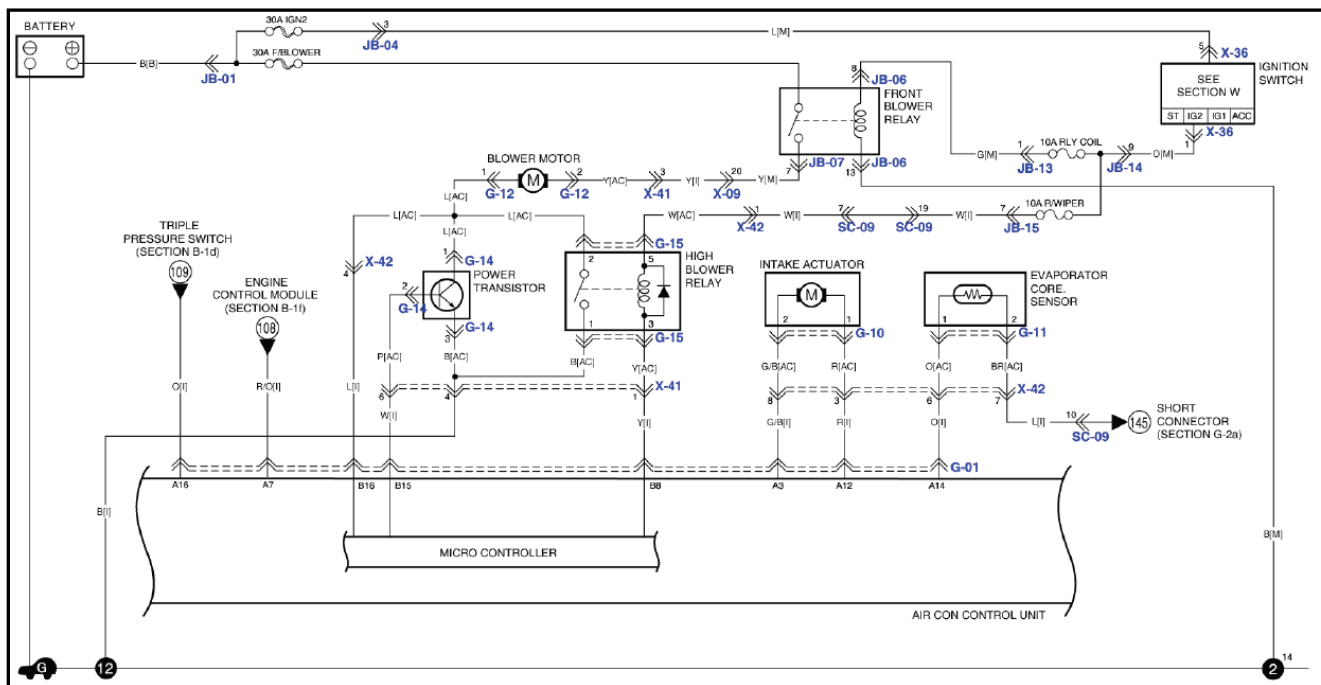


TRANSISTORS

Transistors are an electronic switch.

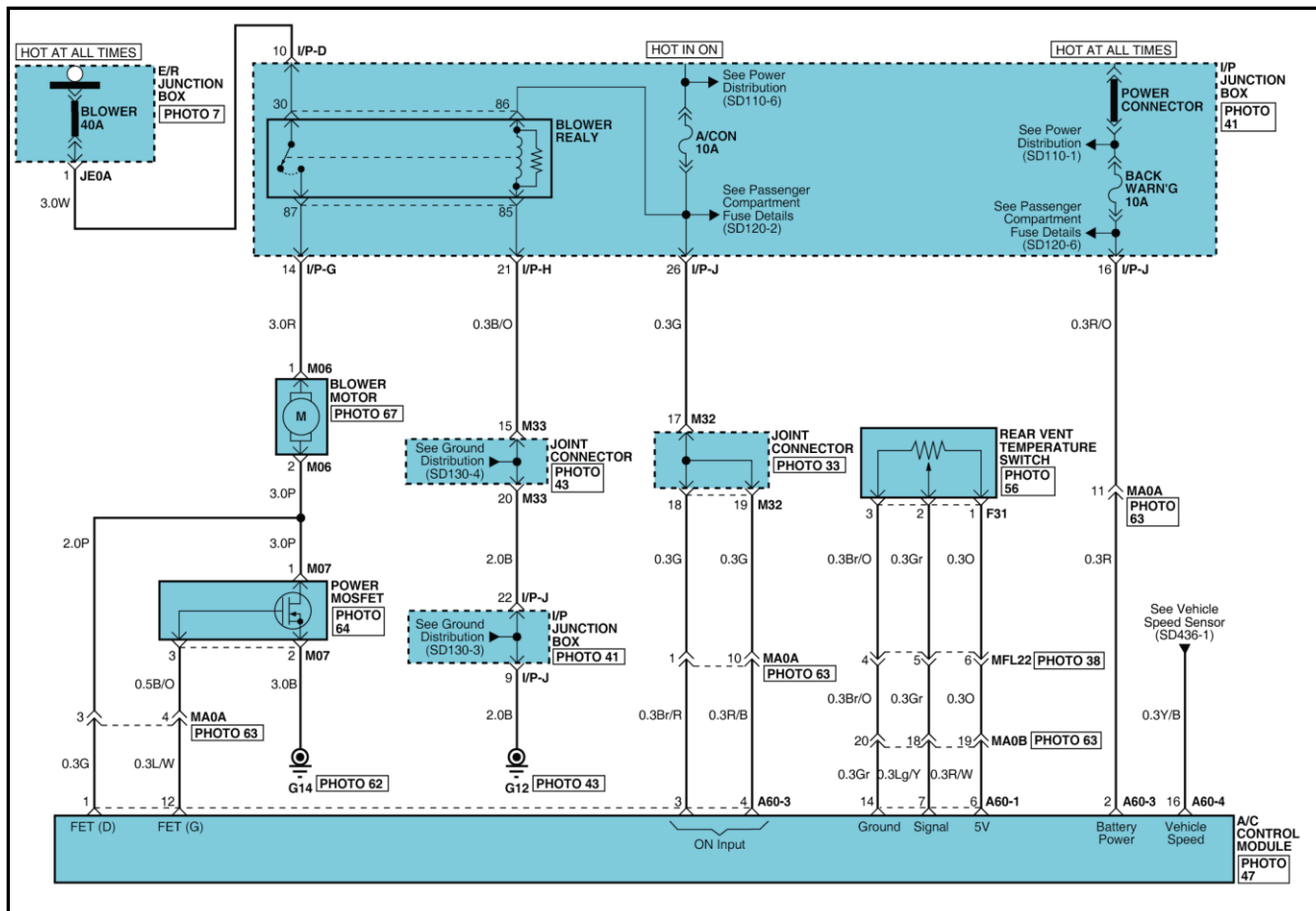
- NPN used for ground side control
- PNP used for power side control

In a transistor, a small amount of current is used to control the base (electronic switch) which allows a much larger current to flow between the collector and emitter terminals. Similar to a relay controlling a large current.



Early transistors could not lower their resistance to zero. This resulted in a voltage drop of 1.0 to 1.5 volts dc. They required a relay to bypass the transistor to reduce the voltage drop to zero. In this example, it allows the blower to run at high speed.





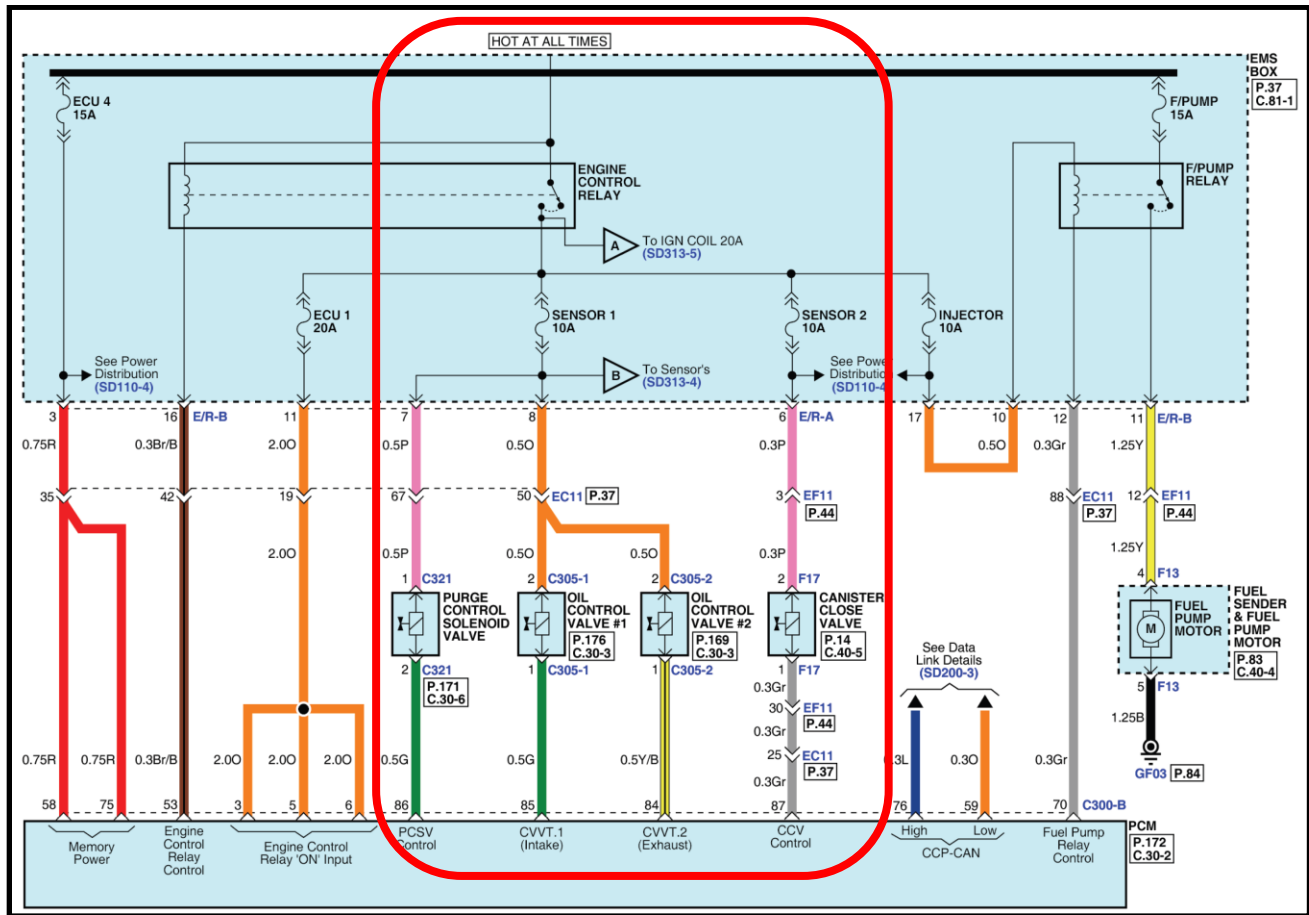
MOSFET A MOSFET provides better control than the standard transistor. It is able to lower its resistance to near zero and does not require a bypass relay.

A MOSFET is typically found in Air Conditioning systems in place of the blower transistor.



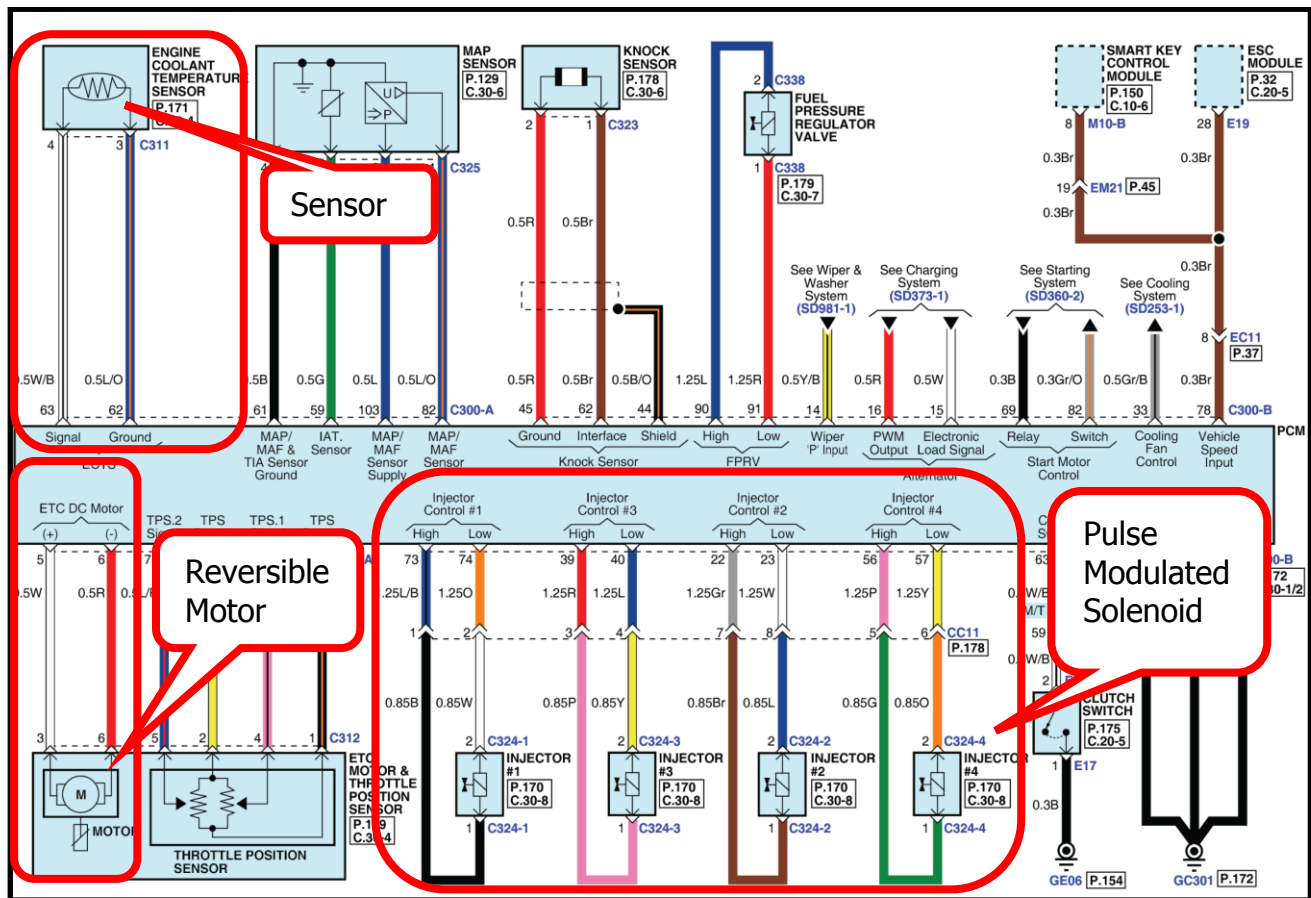
While you cannot often diagnose the ECU itself, its important that you identify the role it plays in the curcuit so that you can properly diagnose the circuits the ECU controls.

When you identify the load and it is connected directly to ground, and the ECU is before the load, the ECU is controlling the power side of the circuit.



Groundside Controlled

When you identify the load and it is connected directly to power, and the ECU is after the load, the ECU is controlling the groundside of the circuit.



Power and Ground or Sensor

If you identify a component with resistance in the circuit, and both wires go to the ECU, either the circuit is a sensor circuit, or the ECU is controlling the power and ground as a reversible motor or pulse modulated solenoid.

[illegible]



LOAD DEVICES The load is the device that performs _____, and should have the most _____, and the largest _____ in the circuit.

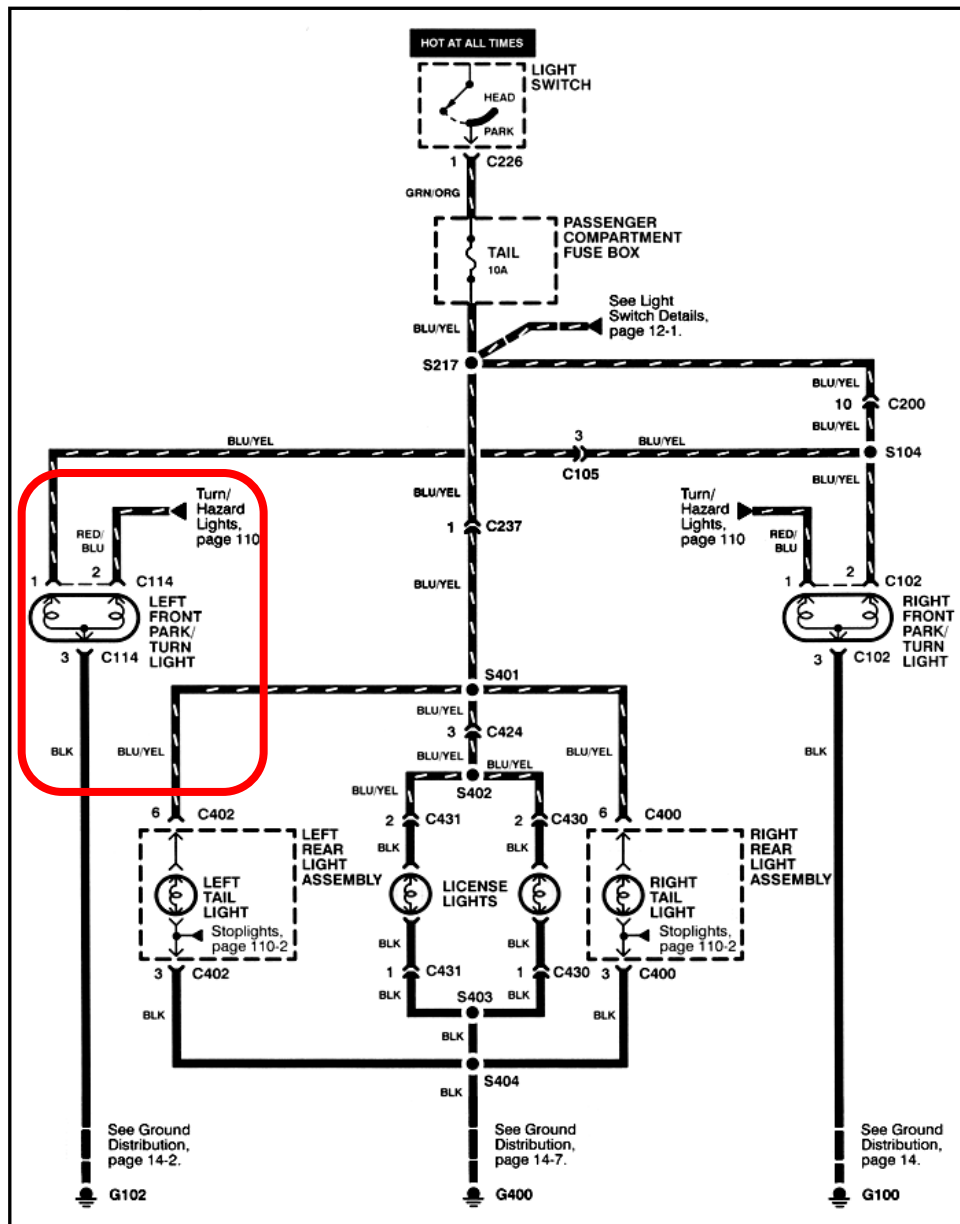
The load converts electrical energy into light, heat or motion that can be used for a specific purpose.

Examples of various load devices include:

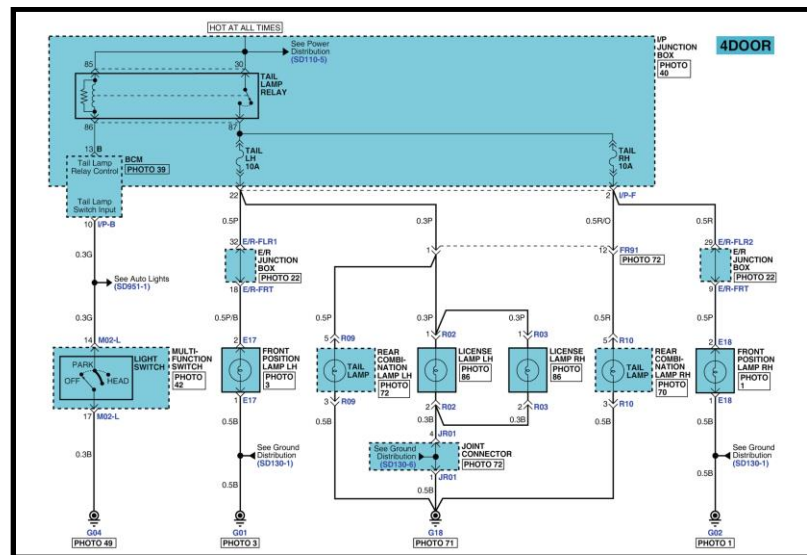
- Lamps
- Solenoids
- Motors
- Resistors



Light bulbs have a resistive wire encased in glass with their circuit ends either connected to the metal base of the bulb, or exposed in the glass tube.



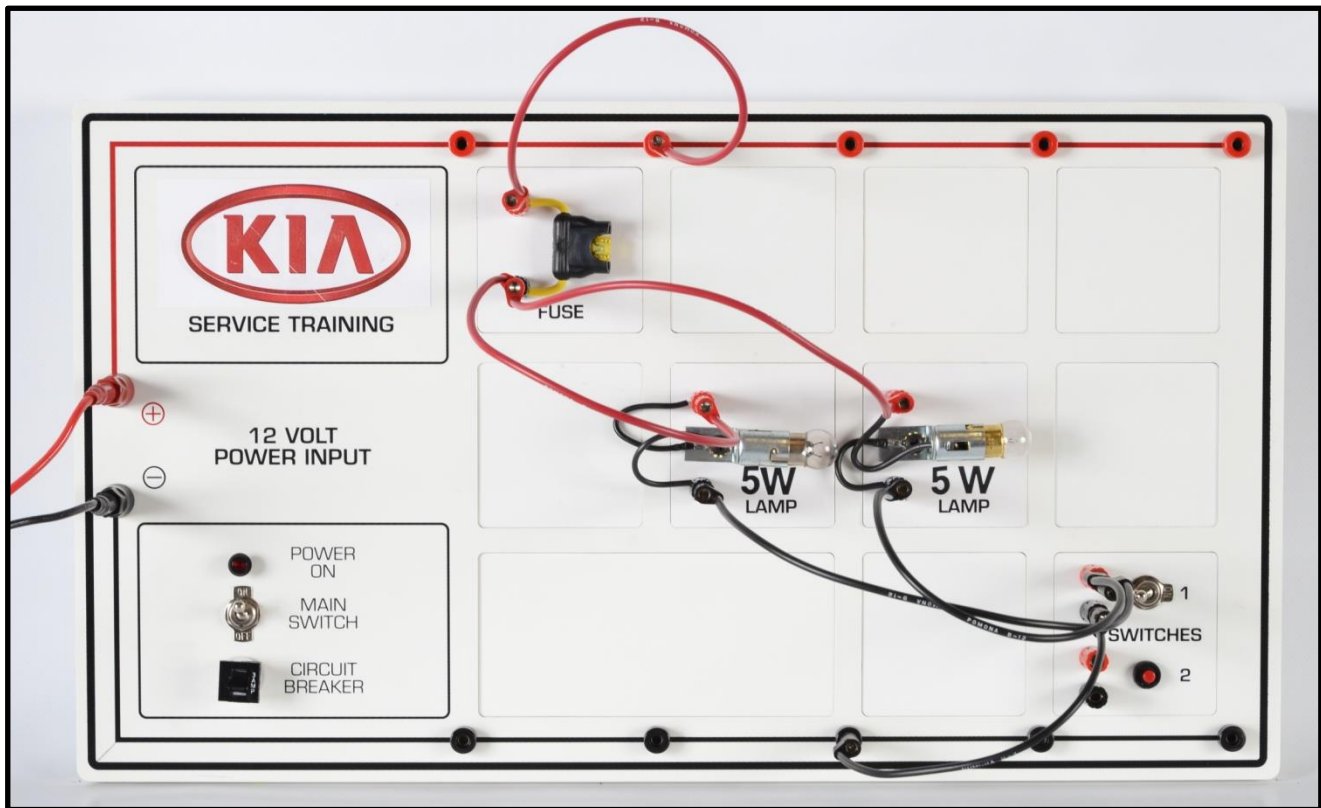
Combination lamps (stop lamps/rear tail lamps) have dual filaments that are essentially two circuits.



Troubleshooting Lamps

Troubleshooting a lamp circuit should be based on the symptom:

- If the lamp doesn't light, check:
 - The fuse
 - The bulb
 - Source voltage
 - Ground
 - The switch
- If the fuse is blown, there is a short to ground that is bypassing the load, or a short at the load. Test the bulb for resistance as they can short internally.
- If the lamp lights but is dim, there is unwanted resistance in part of the circuit. Use a voltage drop test to determine if the resistance is in the source or ground portions of the circuit, and isolate to the fault.
- In most cases, except with dash lamps that are on a dimmer circuit, the bulb should drop close to system voltage. If the voltage drop is less, then measure the voltage drop on the power and ground side of the circuit. The power and ground side should drop less than 0.3V each.



Assemble your breadboard as shown in the example above. Perform a Voltage Drop test across each load device (lamps). What are your readings?

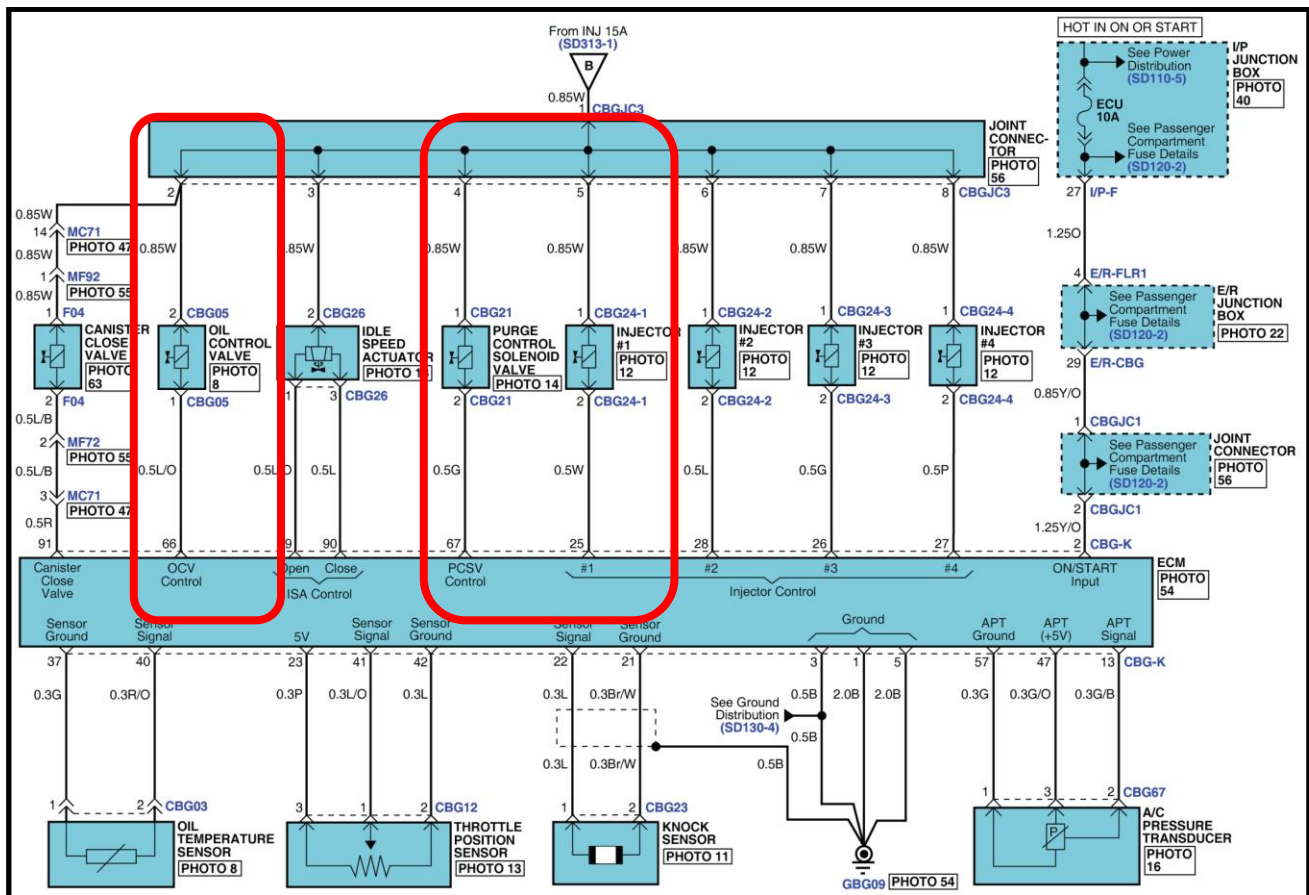
Bulb 1: _____

Bulb 2: _____

Add a resistor in between one of the lamps and the ground side switch. Perform a Voltage Drop test across each load device (lamps). What are your readings?

Bulb 1: _____

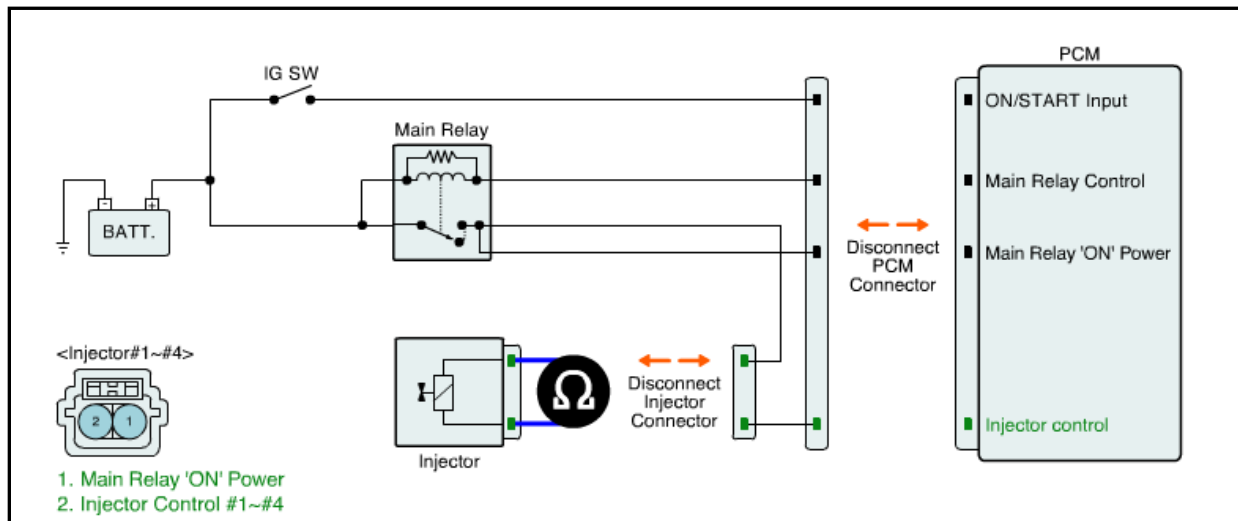
Bulb 2: _____



SOLENOID Function & Operation

A solenoid uses an _____ to move a mechanical plunger. Current flows through a coil of wire to form the electro-magnet. The plunger positioned inside the coil of wire moves when current passes through the windings.

Examples include the purge control solenoid, oil control valve, and fuel injectors.



Troubleshooting

Solenoids are electro-mechanical devices and both the electrical part and mechanical parts need to be tested.

Electrical

The coil winding can be checked with an DVOM for continuity, opens and shorts. The appropriate service information will contain resistance specifications. Some components, like fuel injectors, have a hot and cold specification (i.e., cold 12-14 ohms; hot 16-19 ohms).

Mechanical

Verify the plunger moves when energized. Apply 12V and a ground while verifying plunger movement. Movement can be verified:

- Listening for a clicking sound
- With a vacuum solenoid this can be done by supplying vacuum to one of the ports. See service information for proper operation and testing.



CAUTION: Do not leave a solenoid connected to a constant 12V source. Doing so can damage the component since most are designed for duty cycle or short periods of ON time operation.

MOTORS
Function & Operation

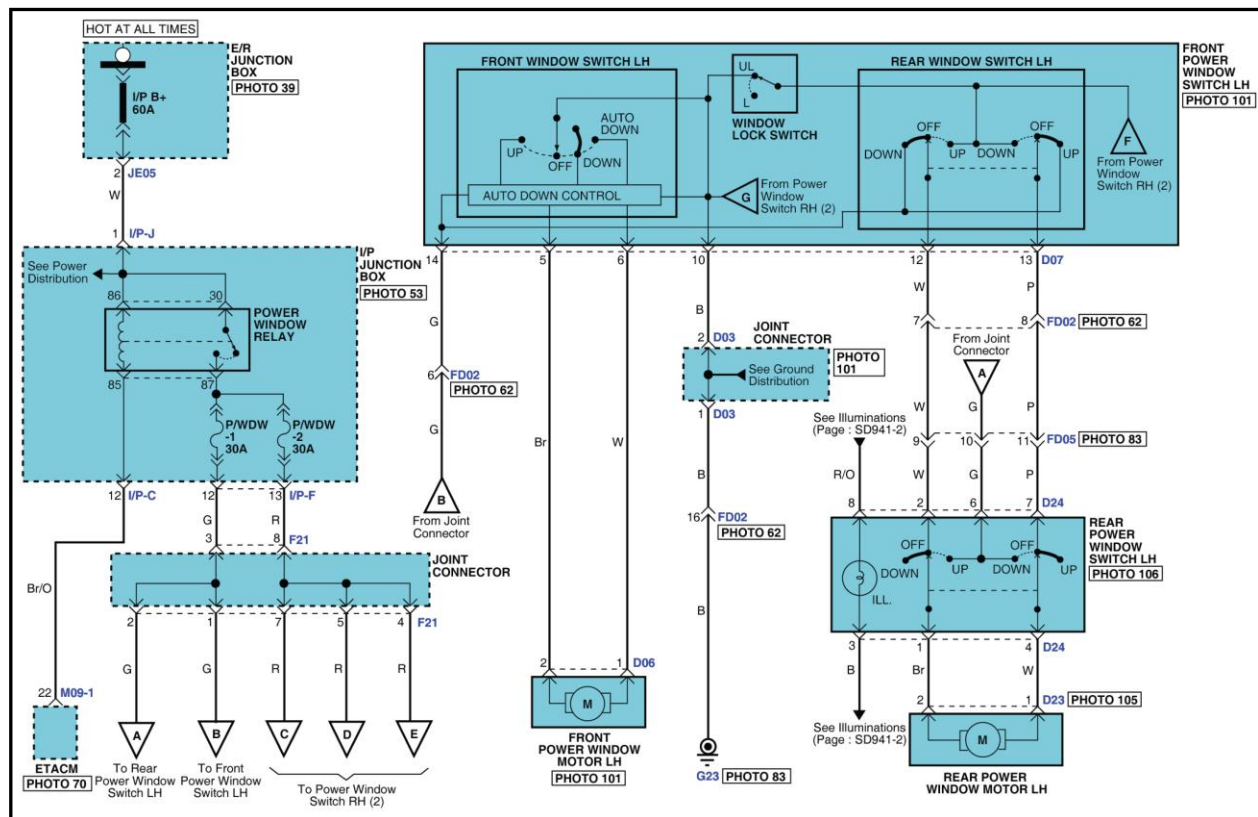
Motors are electro-magnetic devices that convert electrical energy into rotational motion.

Several types of motors are used in Kia vehicles:

- DC brush type
- DC brushless
- AC brushless

When current is applied to the armature, a magnetic field is created and the armature is free to spin within the motor housing. Attached to the housing are either permanent magnets or electro-magnets, called field magnets. As the armature poles react with the field poles, like poles repel, unlike poles attract, causing the armature to spin.

Some motors such as those used in power windows and sunroofs must rotate in two directions. Changing the direction a motor rotates is accomplished by using a switch that changes the polarity (current direction) through the motor.



Troubleshooting

Start troubleshooting with a thorough visual inspection.

- Is there excessive end play or movement in the armature?
- Is there corrosion on the terminals?
- Do the bearings feel smooth?

Follow the service information procedures when testing motors.

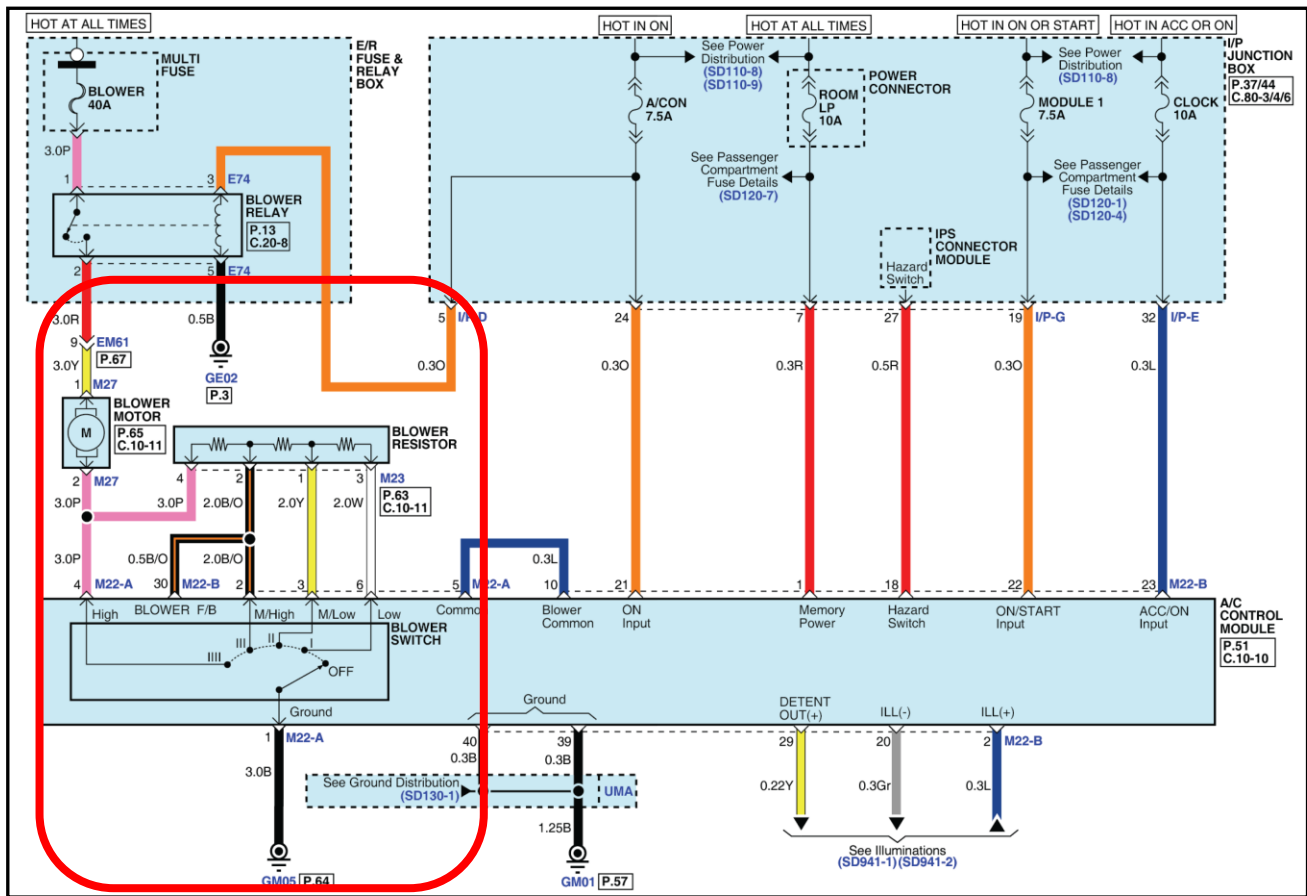
Motors can be bench tested by supplying B(+) and ground to the correct terminals. Larger motors have higher current draw. Be certain you have a large enough power source and the correct gauge jumper wires for the motor being tested.



WARNING: Do not test a fuel pump outside of the fuel tank. When the pump is outside the tank it may ignite the fuel vapors surrounding it.



NOTE: Any force working against the electric motor will cause the amperage draw to increase. This can be caused by a tight motor, bearings, or accessories that require excessive force to rotate.



RESISTORS

Function & Operation

A **resistor** is placed in a circuit to limit, or control, current flow.

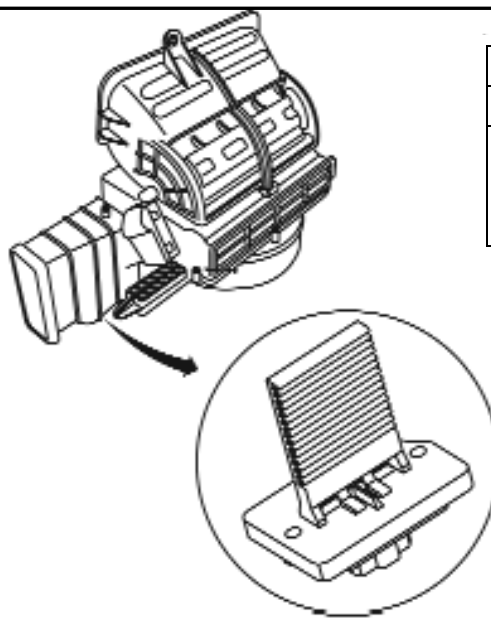
A resistor _____ the voltage by converting some of the electrical energy into heat.

Resistors can be **fixed** or **variable** and are available in a variety of different ratings.

Because resistors oppose current flow, voltage

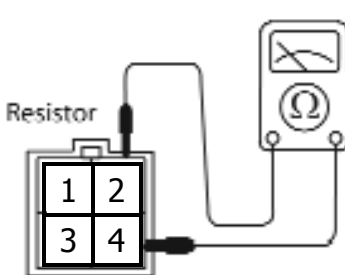
_____ as current flows through the resistor leaving _____ voltage to operate the load.

The blower motor resistor uses this resistance to operate the blower motor at different speeds. At each setting, the voltage that operates the motor is controlled by a different resistor(s).



Terminal	2	1	4	3	Ohms (Ω)
Speed	MH	ML	HI	LO	
Measure between terminals			O	O	
		O	O		
	O		O		

Blower Resistor



Troubleshooting

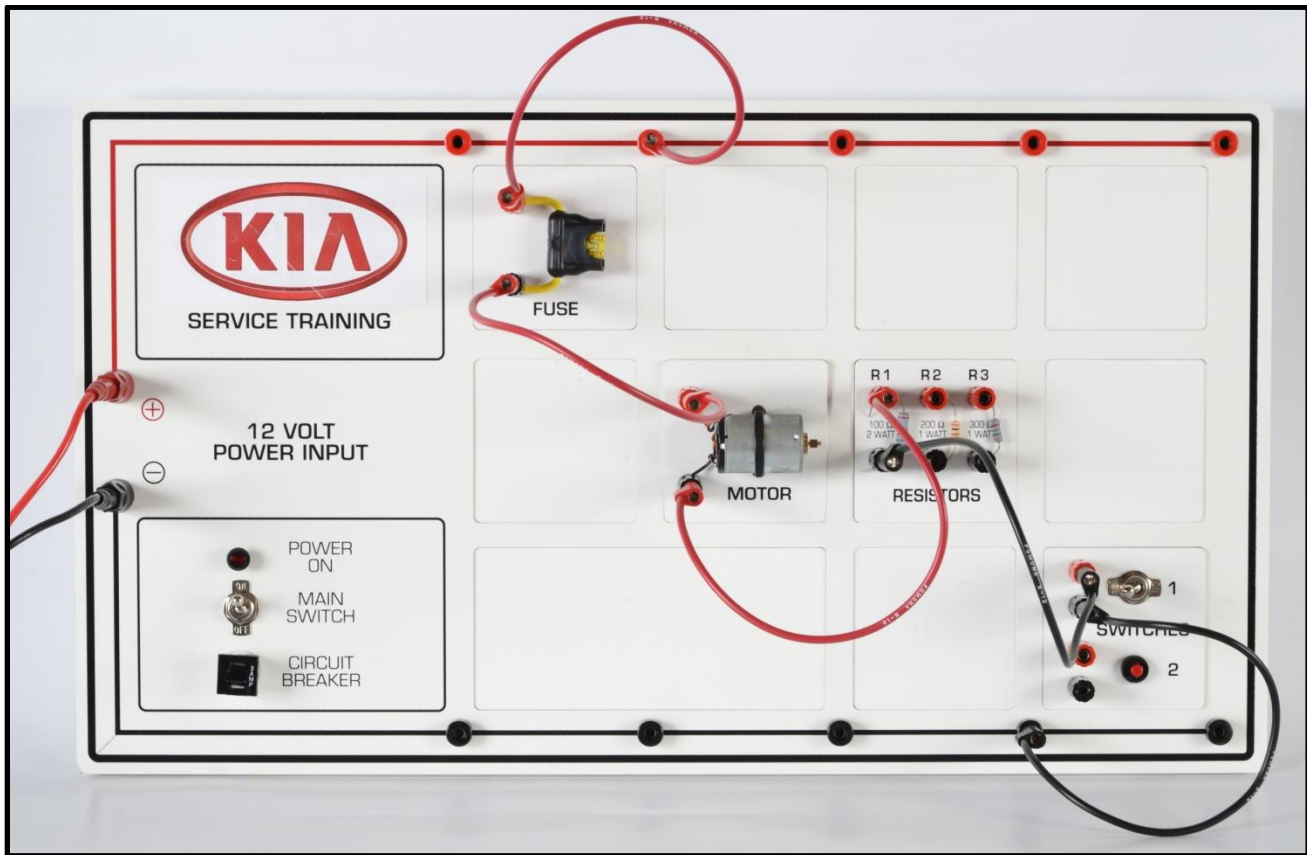
As with most circuits, the troubleshooting approach to the blower motor circuit would depend on the symptom.

- If the blower motor operates at some speeds and not others, the problem is most likely the blower motor resistor. If the resistor checks out good, the switch may be bad.
- If the blower motor does not operate at all, check for source voltage and continuity to ground, including the fuse. Note: Ground circuit resistance will change based on the position of the blower motor switch.
- If the blower motor turns slowly at all speeds, there is either unwanted resistance in the source or ground circuit, or the motor is malfunctioning.

Use the service manual specifications and procedures when diagnosing a blower motor malfunction.



Using the blower motor resistor provided by your instructor, measure the resistance between the terminals and write the resistance in the table above.



Blower Motor Circuit

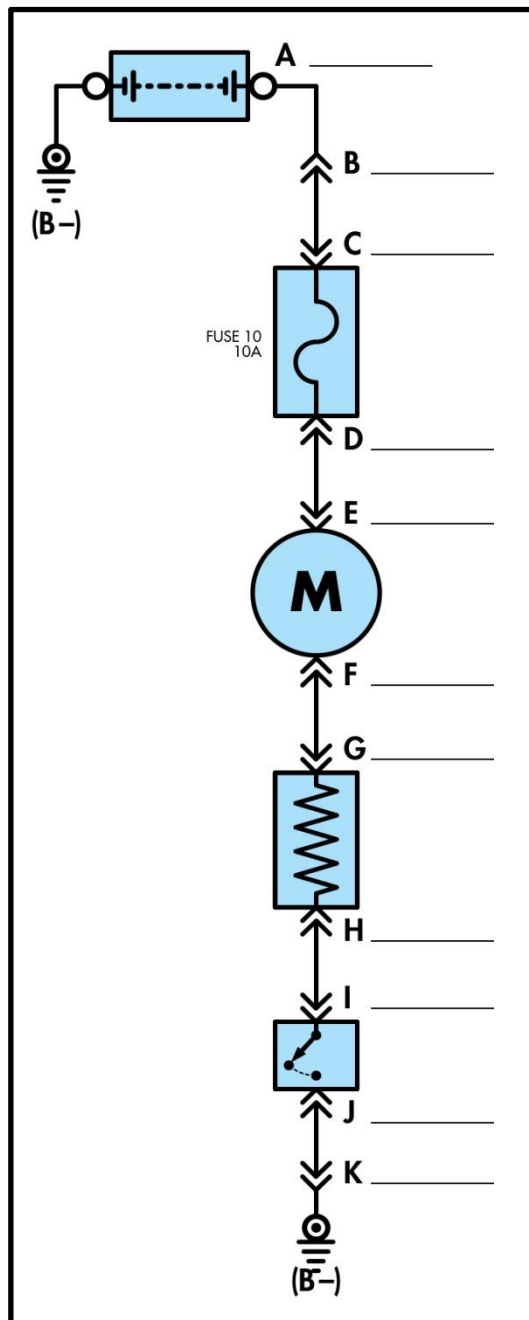
Build the motor circuit as shown. Connect A & L to the power source and turn ON the main switch.



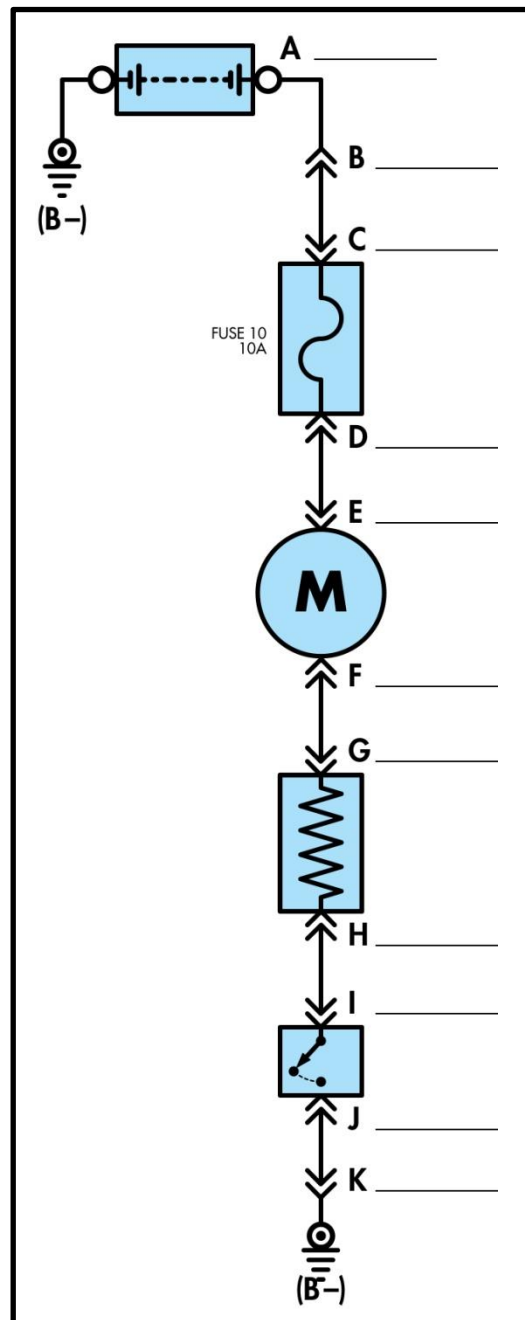
What is the load device in this circuit?

If the resistors were placed on the groundside of the motor, would they still control motor speed?

- ☐ Turn on your DVOM and move the leads to the Amperage setting. Connect the DVOM and record the Amperage reading. _____
 - ☐ Next, apply gentle pressure to the tip of the motor and record the Amperage reading. _____
 - ☐ Finally, remove the resistor block and record the Amperage reading _____
- * Reinsert the resistor block as in the Power Point. Set the DVOM to test Voltage.



Predict



Measure

Troubleshooting Using the blower motor circuit you built on the previous page:

1. Predict the voltage readings when the circuit is CLOSED (ON) using the above diagram.
2. Turn the circuit ON and use the DVOM to measure the voltage readings and current flow. Note your readings on the diagram.



GAUGES Function & Operation

Instrument cluster gauges display critical engine operating parameters to the driver; engine speed, vehicle speed, engine temperature, and fuel level. Based on the gauge readings, the driver can make informed decisions.

There are two parts to a gauge circuit:

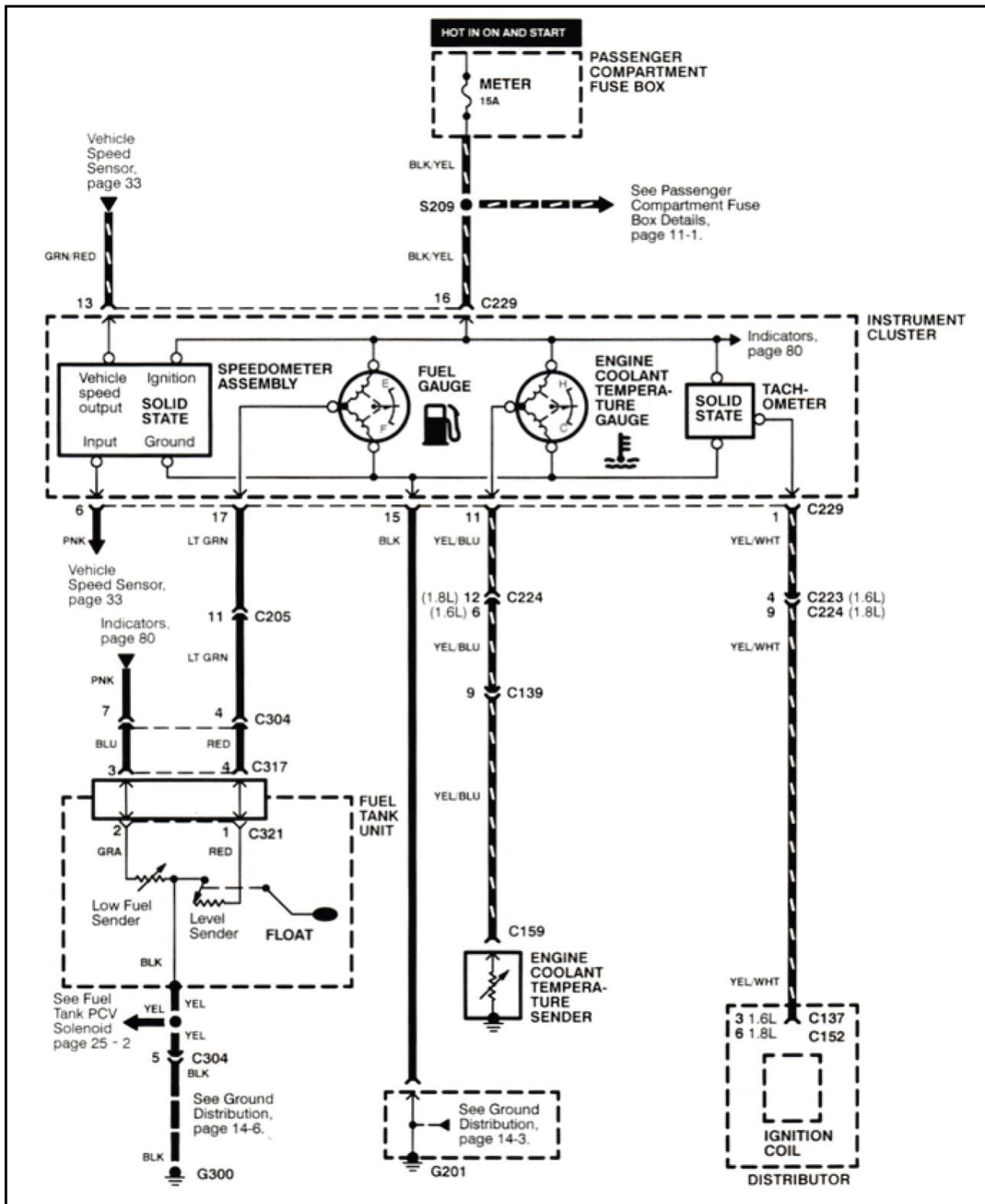
Gauges:

- Magnetic balance coil
- Microprocessor controlled gauge
- Microprocessor controlled display.

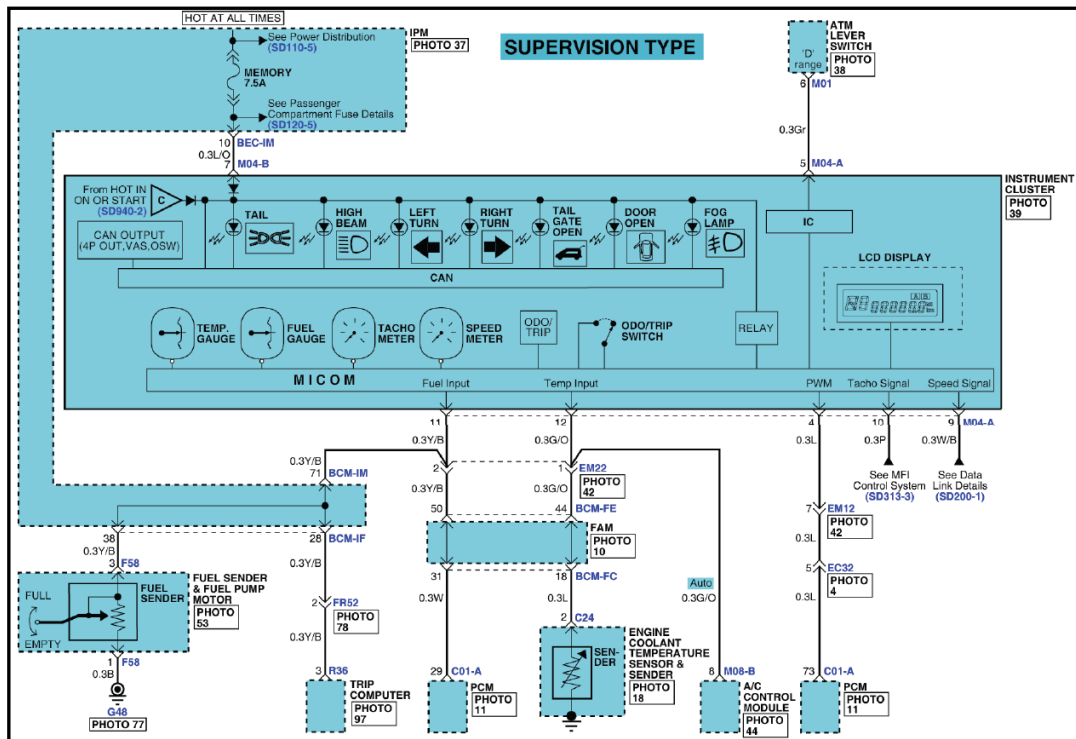
Sending Unit:

- Variable resistor

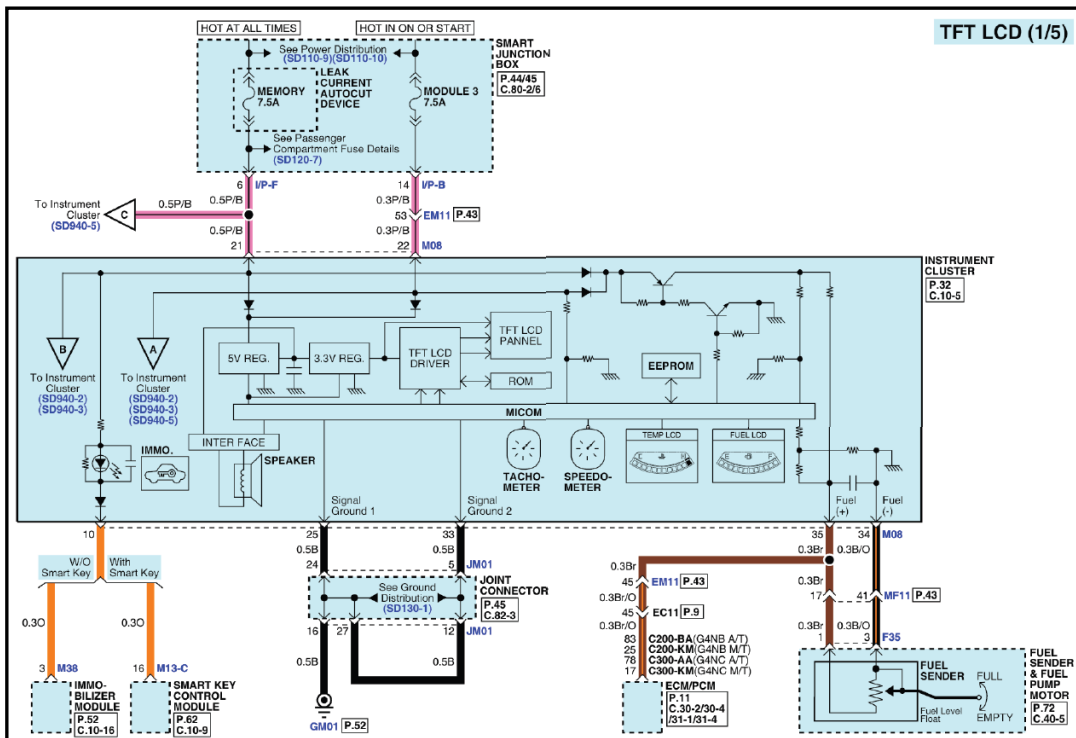
The tachometer uses a signal from the ECM while the speedometer uses a signal from the vehicle speed sensor.



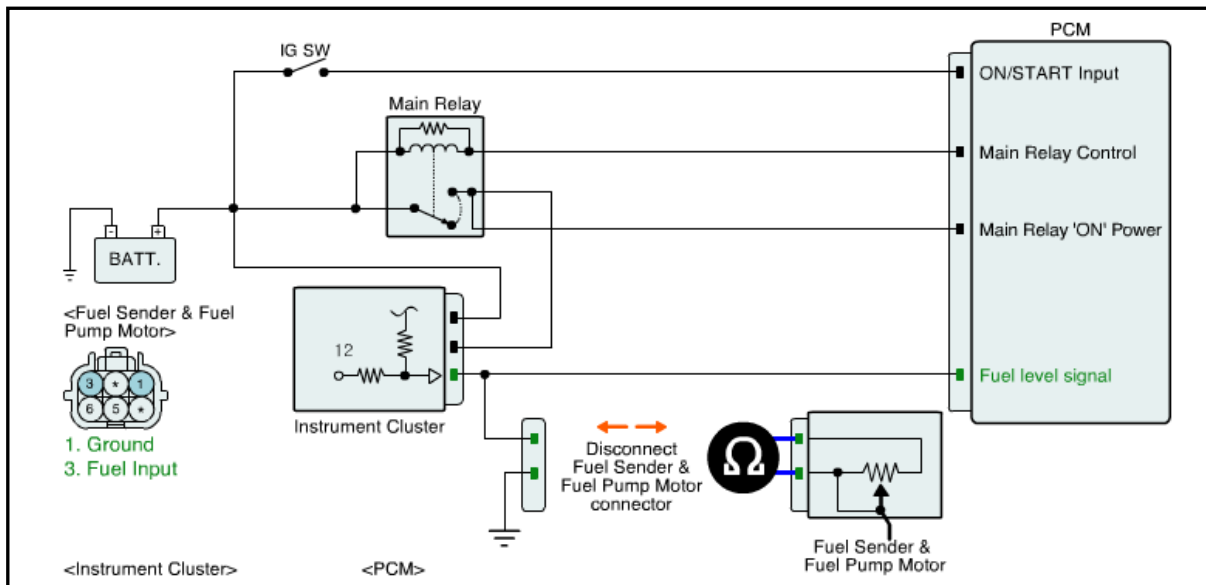
Magnetic Balance Type with sending units



Microprocessor controlled with Engine Temperature Sending Unit



Microprocessor controlled Engine Temperature via CAN



Troubleshooting

A thorough visual inspection is the best way to start your troubleshooting. Are there any damaged wires, terminals, or components? Next, isolate the circuit at a logical test point. Locate the appropriate circuit diagram and determine if the sending unit is on the power or ground side of the circuit.

In the illustration above the fuel sending unit is in the ground side. Follow the service information instructions to test the gauge portion.

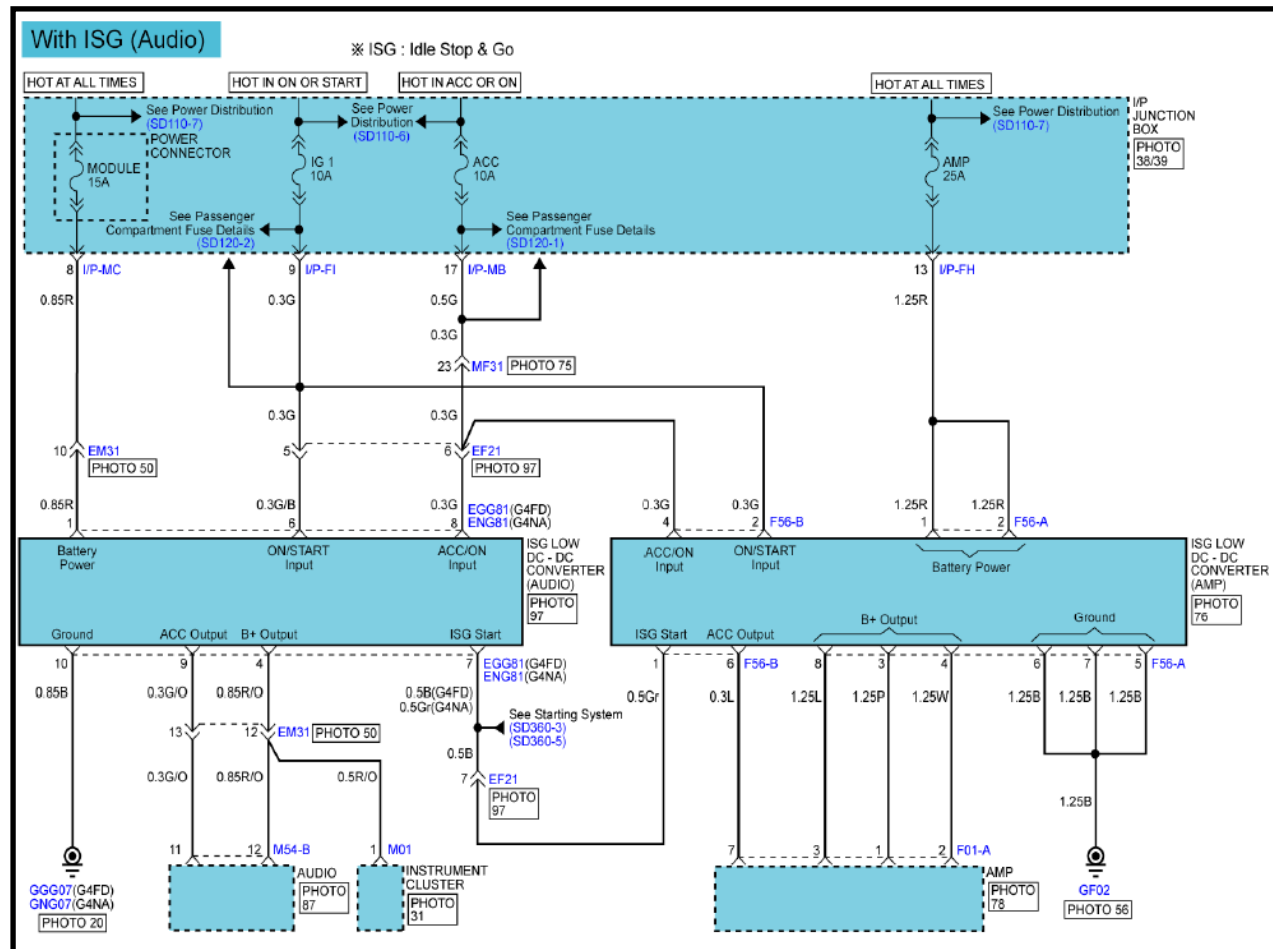
To troubleshoot a sending unit, measure the resistance of the unit as shown in the appropriate service information. If the resistance is out of specification, the root cause may be the sending unit.



NOTE: To quickly check the wiring and processor in a gauge circuit:

- Disconnect the sensor connector from the sending unit. The gauge should go to one end due to the high resistance in the circuit.
- Jump the sensor circuit at the sensor connector using a fused jumper wire. The gauge should go to the other end due to the low resistance in the circuit.

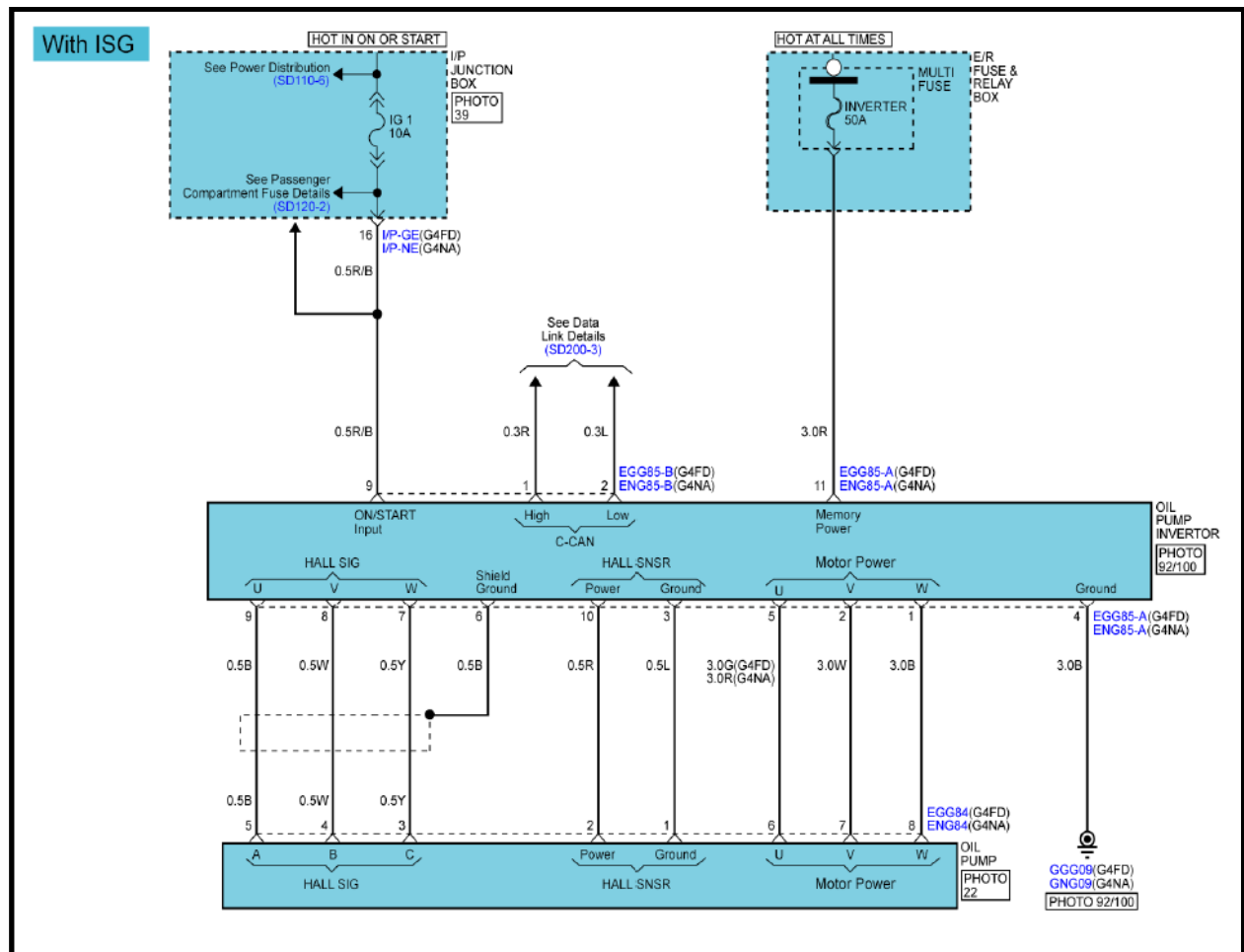
It may take up to 30 seconds for the fuel gauge to change position due to the circuit damping effect that keeps the gauge from moving as the fuel sloshes in the tank.



DC/DC CONVERTOR Function & Operation

The DC/DC converter is used on vehicles with an Idle Stop & Go (ISG) system to compensate for voltage dips caused by more frequent engine starting. The DC/DC converter supplies the by pass relay with a constant voltage during starting, keeping accessories and other electrical systems operating. Based on battery voltage, the CPU decides whether power is supplied by the bypass relay (normal battery voltage) or by the DC/DC converter.

If the audio system turns OFF during ISG "Auto Start" or "Idle Start", the DC/DC converter should be replaced.



ELECTRICAL OIL PUMP Function & Operation

The automatic transaxle (A/T) on Idle Stop & Go (ISG) vehicles includes an Electric Oil Pump (EOP) in addition to the Mechanical Oil Pump (MOP). The EOP provides the oil pressure needed to keep the A/T hydraulic system pressurized for normal operation when the vehicle comes to a stop and the engine shuts off and during low speed driving.

The system uses an oil pump inverter to control EOP operation based on vehicle speed.



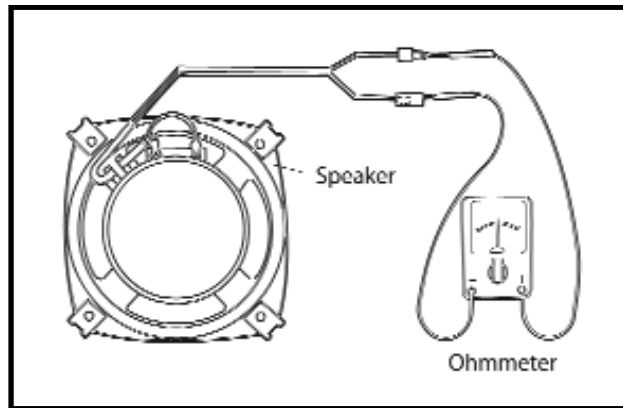
HORNS, BUZZERS AND AUDIO SPEAKERS

Function & Operation

Buzzers, horns and audio speakers use electricity to reproduce sounds. Buzzers and horns contain a plate or flexible diaphragm that vibrates as the electro-magnetic circuit is closed and opened.

An audio speaker contains a voice coil and cone assembly. The cone is attached to a moveable core inside a wire winding. When current from an audio amplifier is sent through the windings, a electro-magnetic field, aided by large permanent magnets, causes the cone to vibrate and create various pressure waves (sound).

Troubleshooting



Since these components use electro-magnetism to operate, use a DVOM to check the continuity of the coil windings. An open or grounded circuit will reduce the magnetic field strength and prevent proper operation.

SUMMARY Basic electrical components can be grouped as circuit protection, circuit control and load devices. Understanding the function and operation of components like resistors, relays, switches, motors, and gauges is critical to properly diagnose and troubleshoot electrical malfunctions.

PROGRESS CHECK

1. The Intelligent Power Switch is an example of a _____ device.

- e. load
- f. input sensor
- g. circuit protection
- h. ground

2. Technician A says that if a “click” can be heard when a relay is energized, the relay is good.

Technician B says that you should check resistance across the switch side when the relay is energized.

Who is correct?

- e. A only
- f. B only
- g. Both A and B
- h. Neither A nor B

3. If a circuit consists of a load device, power source and an ECU, what is the ECU doing?

- e. Reading the load device as a sensor
- f. Providing circuit protection
- g. Controlling the power side of the circuit
- h. Controlling the ground side of the circuit

4. What needs to be done when testing a solenoid?

- e. Test for proper mechanical operation
- f. Test for proper electrical operation
- g. Test the current flow in the circuit
- h. Both A and B

PROGRESS CHECK



5. What is the most likely cause of a blower motor that operates at some speeds and not at others?
- e. High resistance in the ground circuit
 - f. A blown fuse
 - g. A bad blower motor resistor
 - h. Low resistance in the blower motor switch

Automotive Electrical Diagnosis

Theory Module

TARGET AUDIENCE The target audience for this module consists of Kia service technicians who have a basic understanding of automotive technology.

MODULE GOAL Upon completion of this module, you will be able to diagnose basic to moderately complex wiring harness malfunctions.

MODULE OBJECTIVES Upon completing this module and Module 7 (Electrical Diagnosis Guided Practice), you will be able to:

- Identify the elements of a wiring harness while diagnosing a vehicle with an electrical problem, with 100% accuracy
- Identify wire position in a connector based on wiring schematics and harness diagrams while diagnosing a vehicle with an electrical problem, with 100% accuracy
- Properly test electrical readings on wires in a wiring harness and connector while diagnosing a vehicle with an electrical problem, without damaging the connector
- Identify proper wiring harness repair procedures bases on vehicle diagnosis, with 100% accuracy

MODULE INSTRUCTIONS Carefully read the material, take notes based on the classroom discussion, and study each illustration. At the end of the module, there will be Progress Check questions for you to answer. You may use the modules to answer the questions.

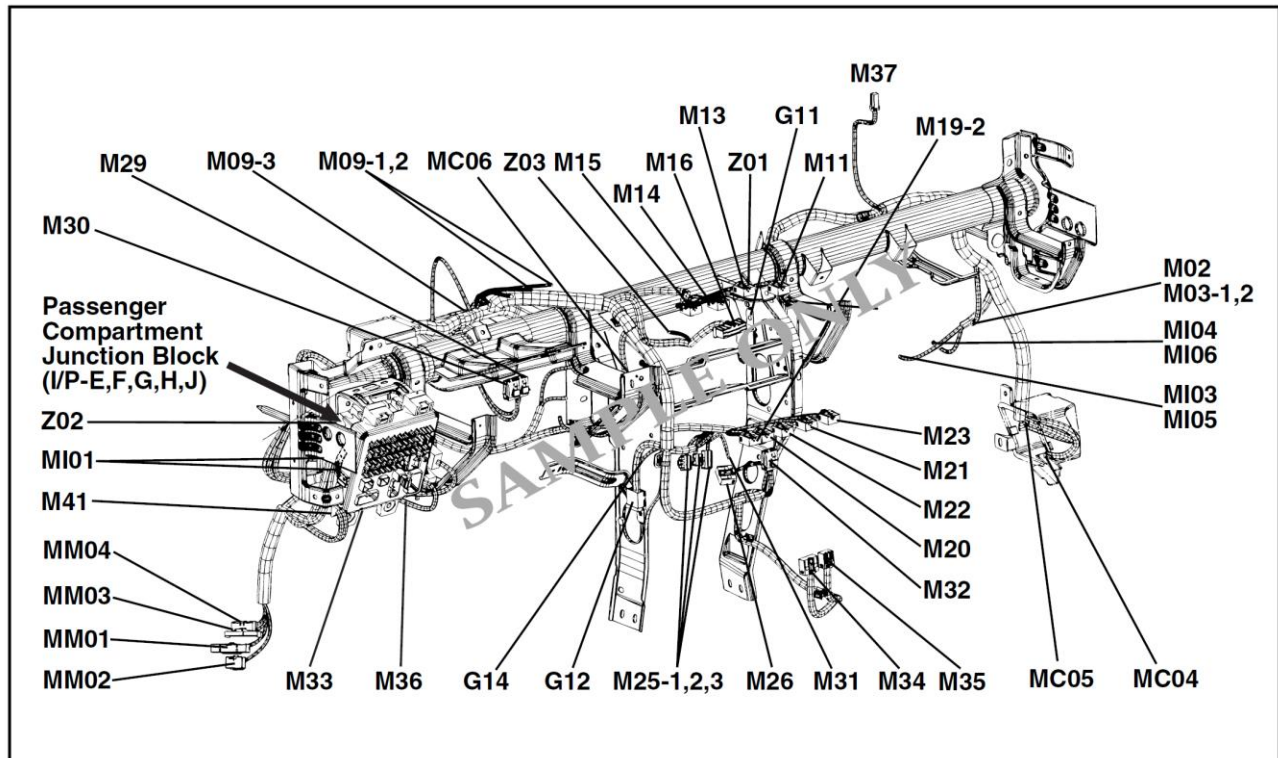
REQUIRED MATERIALS In order to complete this module, you will need the following items:

- Kia Electrical Repair Kit II
- Terminal Removal Tools

TIME TO COMPLETE Approximately 1.5 hours

ACRONYMS

CPA	– Connector Position Assurance
SRS	– Supplemental Restraint System
TPA	– Terminal Position Assurance



INTRODUCTION

PURPOSE

The wiring harness is an important part of any electrical circuit. The harness provides a current path to, from and between electrical components as well as to the battery and chassis ground connections.

Wiring harnesses are bundles of separate wires wrapped together with tape and/or protected by plastic tubing. A vehicle has several wire harnesses. Individual sections can be conveniently diagnosed, repaired, or replaced. Harness sections attach to each other using connectors, which keep the individual wires separate.

Harness Classification

Electrical wiring connectors are classified according to the wiring parts in the Harness Layouts.

Symbol	Harness name	Location
C	Control, Injector EXT., Ignition Coil, Oil Control Valve Harness	Engine Compartment
D	Door Harness	Door
E	Front, Battery, Front End Module Harness	Engine Compartment
F	Floor, Floor Console EXT. Harness	Floor
M	Main, Main Console EXT., Steering Remote Control Harness	Passenger Compartment, Center Console
R	Roof, Sunroof Motor EXT., Tail Gate, Rear Bumper Harness	Roof, Tail Gate, Rear Bumper
S	Seat Harness	Seat

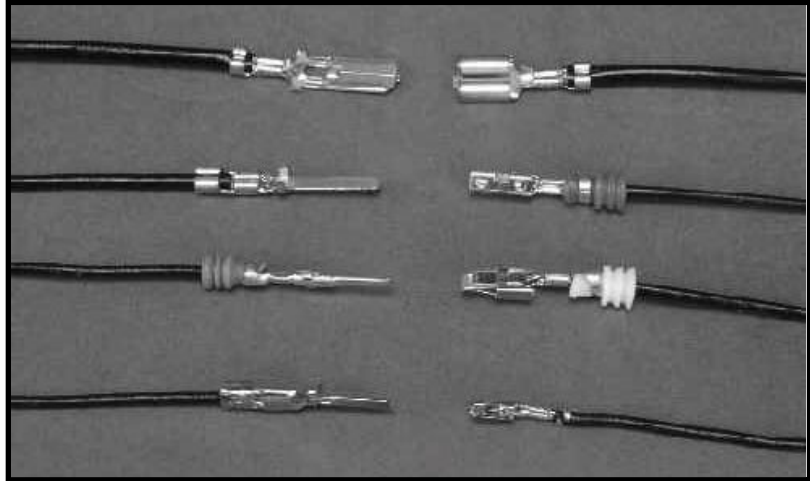
APPLICATION

The table above illustrates the number of harnesses in a typical vehicle. The number of wiring harnesses varies depending on the vehicle and how the vehicle is equipped. Refer to the applicable service information for specific vehicle information.

COMPONENT OVERVIEW

Every wire in a vehicle terminates or ends with a terminal or a ground point. Connectors are used to house the terminals, and sometimes a group of ground wires, which is referred to as a ground block.

TERMINALS



A **terminal** is a metal device mechanically attached (crimped, soldered) to the end of the wire. "Male" and "female"– shaped terminals in each connector half join to continue the current path.

A loose fitting terminal can cause intermittent concerns.

A pin tension gauge can help determine if a terminal is too loose.

STABILANT 22A

Stabilant 22A improves conduction of electrical terminals in harness connectors. Follow the manufacturer's instructions when using this product. When a connection is suspect, Stabilant 22A can be used to provide a good connection. Use it sparingly; a small amount is very effective in providing a positive connection.

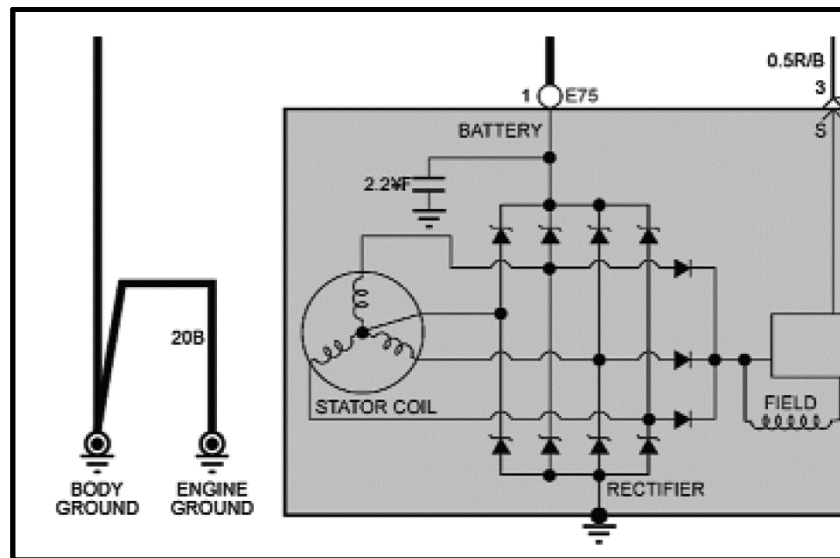


WARNING: Do not use Stavilant 22A on SRS or O2 sensor terminals or connectors.

Stabilant 22A can be sourced from the following:

CARQUEST	- ask for the Standard Motor Products line of products in the Remanufacturing Catalog - the part number is SL 5
NAPA	- ask for the ECHLIN line of products - the part number is CE 1
PEP BOYS, PARTS PLUS, AND O'REILLYS	- ask for the Borg Warner line of products - the part number is CL 85
NIEHOFF	the part number is NL 5
KRAGEN	the part number is LUB 5

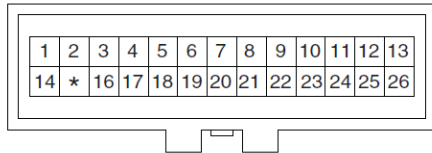




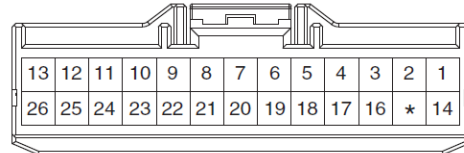
GROUNDS A ground point consists of one or more wires usually connected to an eyelet terminal bolted to the body or frame assembly. In some cases, multiple wires come together into a grounding connector and joined to a single common eyelet. Grounds may also be numbered. A harness ground symbol is different from a unit ground.

FF01 Floor To Floor Console EXT. #1 Harness Connection

- 26 Male / White (KET_090III_26M_W)
- Floor Harness

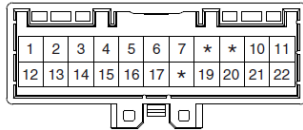


- 26 Female / N/A (N/A)
- Floor Console EXT. #1 Harness

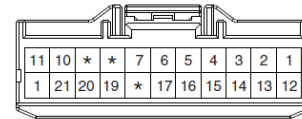


FF02 Floor Console EXT. #1 To Floor Console EXT. #2 Harness Connection

- 22 Male / N/A (N/A)
- Floor Console EXT. #1 Harness

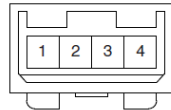


- 22 Female / N/A (N/A)
- Floor Console EXT. #2 Harness

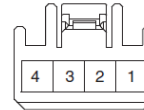


FF03 Floor To Floor Console EXT. #3 Harness Connection

- 4 Male / White (AMP_090III_04M_W)
- Floor Harness

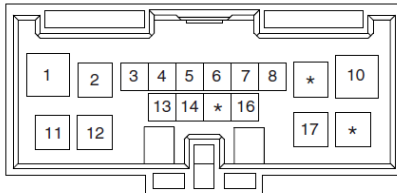


- 4 Female / White (AMP_090III_04F_W)
- Floor Console EXT. #3 Harness

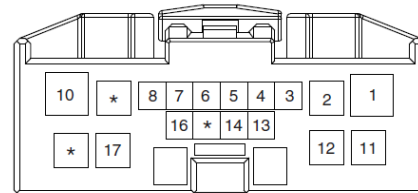


FR11 Floor To Tail Gate Harness Connection

- 18 Male / White (AMP_025060110_18M_W)
- Floor Harness



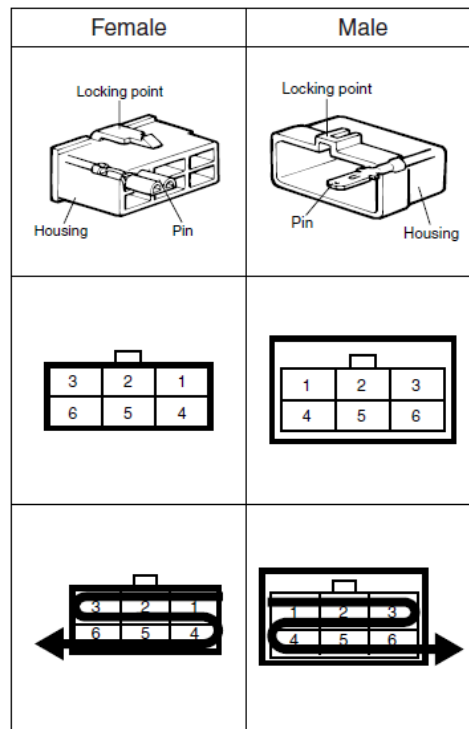
- 18 Female / White (AMP_025060110_18F_W)
- Tail Gate Harness



CONNECTORS

A **connector** is a plastic block that houses and protects a group of wire terminals. Two halves make up a connector. The terminal and connector halves allow a point to separate for testing and replacement.

Harness connectors provide an excellent access point for circuit diagnosis, but be careful. Connectors can be damaged if handled or tested improperly. Separate connector halves by pressing and releasing the lock lever. Do not pull the wires to separate the connector. Hold each connector housing while separating them.

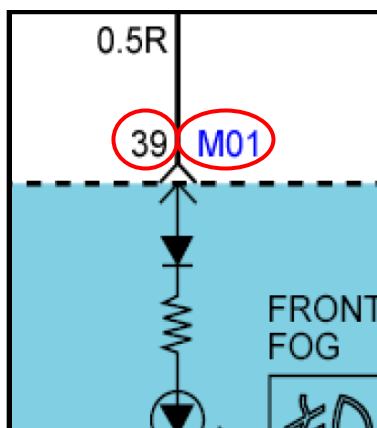


TERMINAL NUMBERING

When looking at the harness side of the connector:

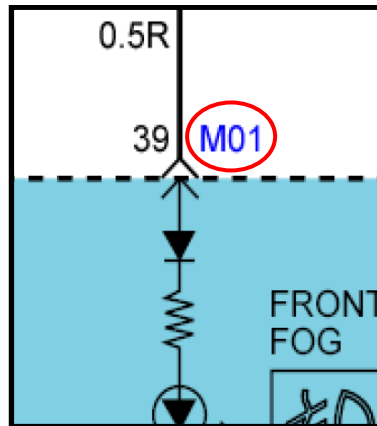
- The male terminals are numbered left to right starting in the upper left hand corner
- Female terminals are numbered right to left starting on the upper right corner
- When facing each other, the terminals are a mirror image and each terminal number will match

All terminal passages in the connector are identified even though some may be empty. Always refer to the appropriate service information for correct numbering.



The terminal number (39), and connector number (M01) are given in the wiring schematic.

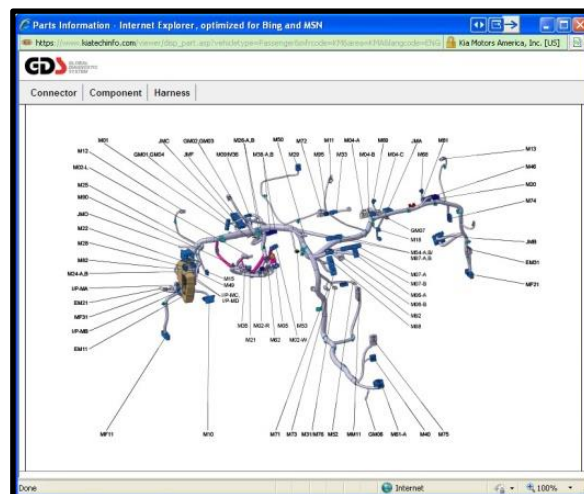
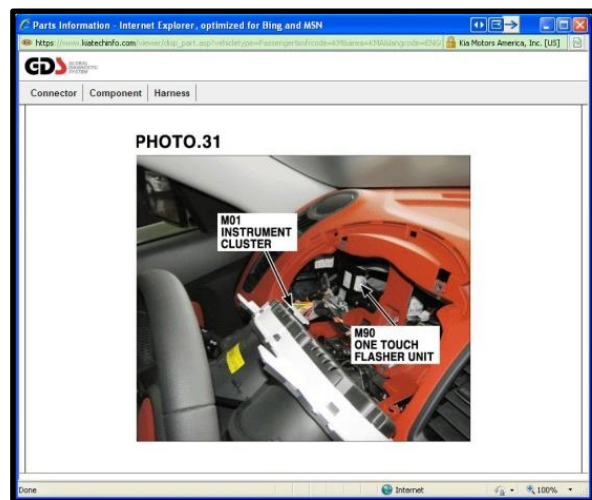
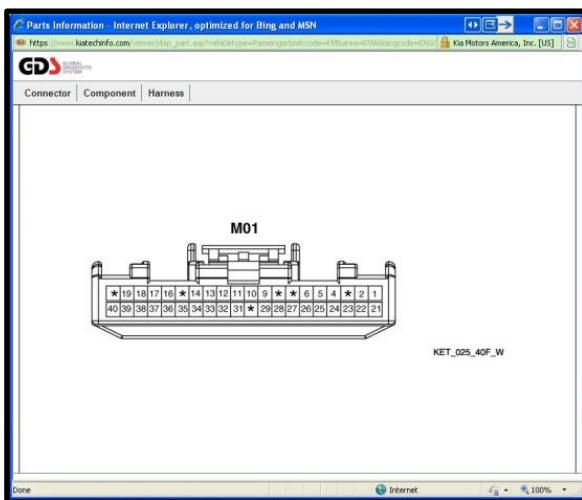




LOCATING HARNESSES, CONNECTORS AND COMPONENTS

In KGIS, ETM, Schematic Diagrams you can view:

- An image of the connector
- A picture of the connector location
- The wiring harness for the connector





⑥ Wire Color Abbreviations

The following abbreviations are used to identify wire colors in the circuit schematics.

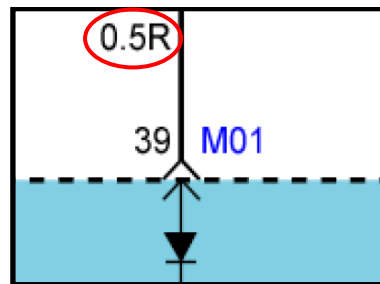
Symbol	Color of wire	Symbol	Color of wire
B	Black	O	Orange
Br	Brown	P	Pink
G	Green	R	Red
Gr	Gray	W	White
L	Blue	Y	Yellow
Lg	Light Green	LI	Light Blue

* (Y)/(B): Black stripe with yellow ground (2 colors)

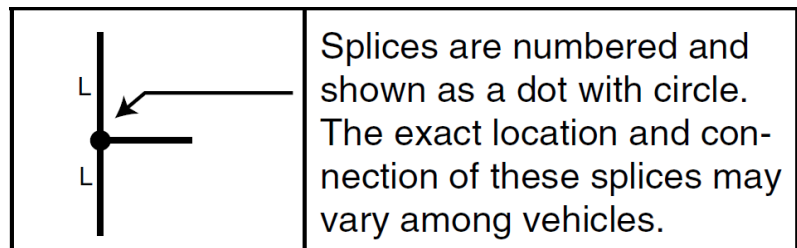
↑ ↑
the color of background the color of stripe

Wiring Colors

Color coded wires provide a convenient (and obvious) method to identify circuit wires. Two-color wires are indicated by a two-letter symbol. The first letter describes the base wire color; the second letter is the stripe color.



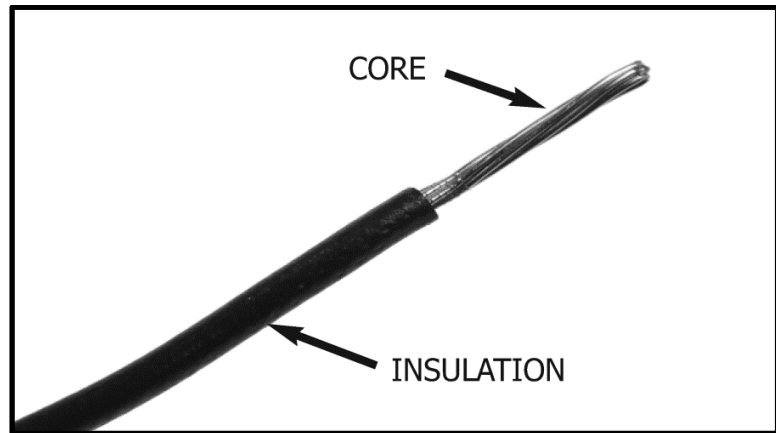
The wire size (0.5mm) and color (Red) is also shown in the wiring schematic.



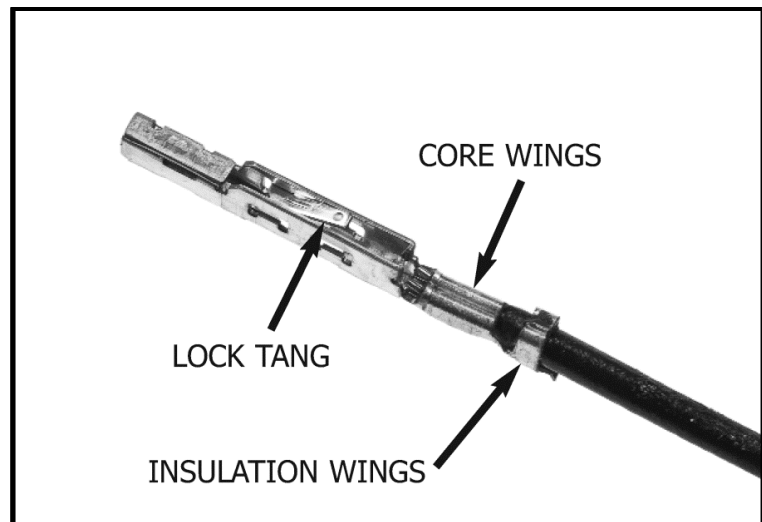
Splices are numbered and shown as a dot with circle. The exact location and connection of these splices may vary among vehicles.

SPLICES

A splice is where two or more conductors meet at a common connection point in a circuit. Do not confuse the splice symbols in a wiring diagram for two conductors crossing over each other (no electrical connection).

WIRE

Comprised of a core of conductive metal, usually copper strands, and covered with electrical resistant insulation.

TERMINAL

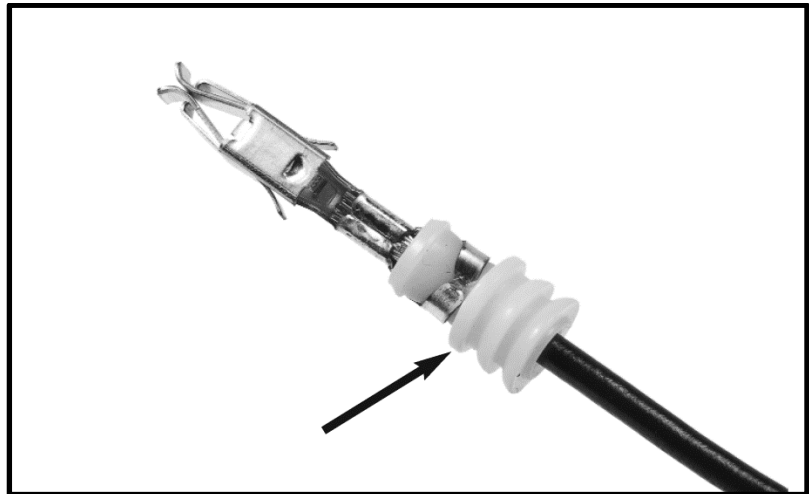
The terminal is metal piece attached to the end of the wire that is plugged into a connector. Connection to a mating terminal or device will complete an electrical connection (Terminal is often used incorrectly to describe a connector).

The lock tangs are part of the terminal that locks it into a connector. Some terminal designs feature multiple lock tangs.

The core wings are part of the terminal that is crimped over the wire core and provides the electrical path between the terminal and the wire core.

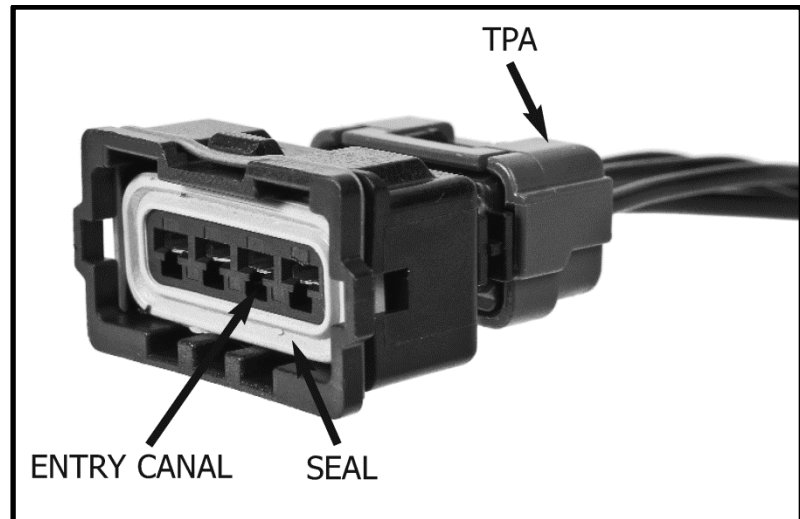
The insulation wings are part of the terminal that is crimped over the insulation (and seal if present).

Terminal Seal



The terminal seal is designed to help prevent contamination from the environment after the terminal is plugged into the connector.

CONNECTOR



A connector is the plastic component which contains one or more terminals and provides a mechanical connection to another connector or component such as a switch, relay, sensor, etc.

Entry Canal

Canal(s) in the connector is where a removal tool is inserted to depress the terminal lock tang(s) in order to remove the terminal. (Some connector designs have plastic lock features used with terminals that do not have tangs. The plastic lock is raised to remove the terminal.)

Terminal design (series) determines removal tool to be used.



Connector Seal

The connector seal is used to protect the terminals from debris and moisture.

Terminal Position Assurance (TPA)

Designed to help prevent contamination from the environment after the connector is mated, and prevent the terminal(s) from backing out of the connector.



CAUTION: When connecting and disconnecting connectors, always make sure that the connector seals and TPA are in place to help protect the terminals.

Connector Position Assurance (CPA)



A device that is inserted after the connector is mated in order to prevent disconnection.

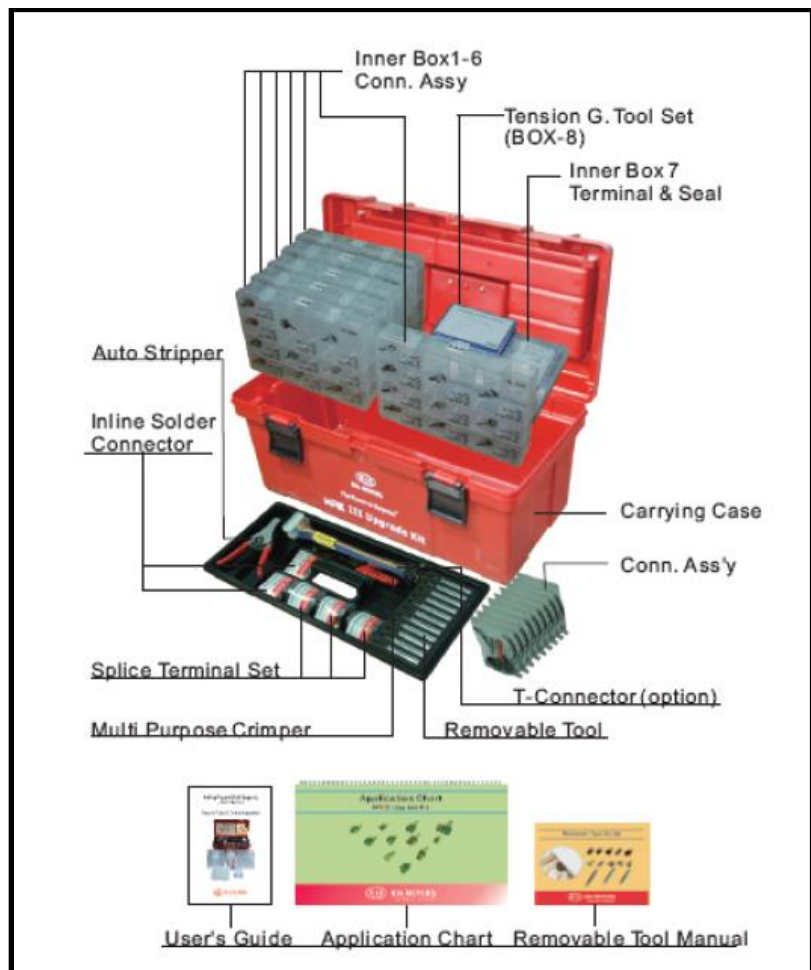
Pigtail



A replacement part consisting of wire(s) that are bare at one end with a connector or device attached to the other.



Wiring Repair Kit II



Wiring Repair Kit III Upgrade

Wire Harness Repair

The Wiring Repair Kit II and Wiring Repair Kit III Upgrade are designed so that you can repair certain wiring harnesses instead of replacing the whole harness.



**T-CONNECTOR**

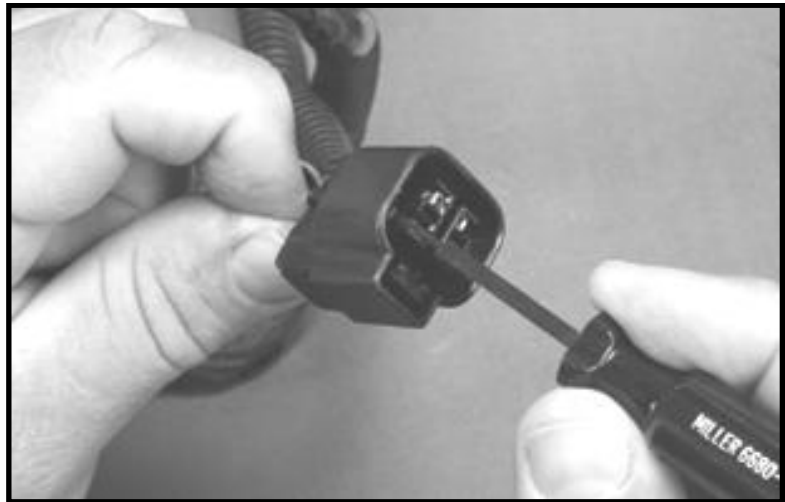
Wiring Repair Kit II contains T-Connectors, which are a set of molded wires that allow you to test live circuits. Each T-Connector has male and female terminals that are inserted into the terminals of a connector. Once connected, the circuit can be powered up and tested. Each kit is equipped with 3 T-Connectors. T-connectors can be purchased separately through Kia parts department.



CAUTION: Remove the negative battery cable (B-) from the battery before performing wire repairs.



WARNING: Never repair SRS connectors and terminals. Replace the complete harness if repairs are required.



Harness Repair Overview

When replacing a damaged wire, verify the wire size is the same or larger than the existing wire. If possible, use the same color wire.

1. When replacing terminals, identify the connector by name, location of locking clip, and the direction to unlock the plug from vehicle charts.
2. When replacing terminals, check the condition of the other terminals.
3. Do not use excessive force on the terminal and do not pry on the terminal with a pick.
4. If the new length of wire that is replacing the old wire is not approximately the same length, problems may occur.
 - a. Too short of wire may cause additional strain on the splice, terminal, or the connector.
 - b. Too long of wire may cause the wire to get pinched or damaged, causing a short circuit.
5. If heat shrink tubing is to be used, install tubing before putting connector on end of wire. Make sure heat shrink tubing is kept away from any area to be soldered as tubing is very sensitive to heat and will shrink prematurely before it can be properly placed.
6. When reinserting terminals into the plug, make sure terminals are locked in the connector.
7. Not all terminals will give a noticeable click.
8. When putting both ends of plugs back together, make sure that both ends are locked in place.

SUMMARY A wire harness is a group of circuit wires that are bundled together. Wire harnesses are joined to other harnesses for ease of assembly, service and diagnosis. They are joined by plastic connectors containing metal terminals inside that allow a continuous circuit path when the connectors are joined.

PROGRESS CHECK

1. Place a check next to each item that is part of a wiring harness:
 - ☐ Terminal
 - ☐ Actuator
 - ☐ Connector
 - ☐ Ground eyelet
 - ☐ ECU
 - ☐ Splice
 - ☐ Stranded wire
2. Connectors join which of the harness elements?
 - A. Grommets
 - B. Terminals
 - C. Fuses
 - D. Grounds
3. The wiring schematics found in the ETM include:
 - A. The wire color
 - B. The connector number
 - C. The pin number of the wire in the connector
 - D. A picture of the location of the connector
 - E. All of the above
4. How are male terminals numbered?
 - A. Right to left starting in the lower right hand corner
 - B. Exactly the opposite of the female terminals
 - C. Left to right starting in the upper left hand corner
 - D. It changes by connector type
5. What color is a wire labeled (Br) on the wiring schematic?
 - A. Brown
 - B. Blue with a red stripe
 - C. Black with a red stripe
 - D. None of the above

Automotive Electrical Diagnosis Guided Practice Module

TARGET AUDIENCE	The target audience for this module consists of Kia service technicians who diagnose vehicles with automotive electrical concerns.
MODULE GOAL	Completing this module prepares you to diagnose and troubleshoot automotive electrical concerns using the appropriate diagnostic tools.
MODULE OBJECTIVES	Upon completion of this module, you will be able to perform the task in each guided practice.
MODULE INSTRUCTIONS	Carefully read through the material and take notes based on the classroom discussion. Throughout the module there will be activities for you to participate in.
REQUIRED MATERIALS	In order to complete this module, you will need the following items: • Pen or Pencil • Safety Glasses • Digital Volt-Ohm Meter • Vehicles as assigned by your instructor
TIME TO COMPLETE	Approximately 2.5 hours

GUIDED PRACTICE 1

The instructor will assign the vehicle and tell you what the customer's concern is.



Vehicle: _____

Symptom: _____

☐ Go to the vehicle assigned by your instructor.

☐

☐ What were the results when you verified the concern?

☐

☐ List any codes using the GDS:

☐ Using KGIS, print the schematic and highlight the power flow for the suspected circuit.

☐ On the wiring diagram, identify the test points you would check to diagnose the concern, and indicate what results you would expect when performing the test.



☐ Show the Instructor your wiring diagram with power flow illustrated and your test points.

☐ After the instructor has reviewed your test points, perform the tests on the vehicle. DO NOT perform any additional tests without your instructor's approval.

☐ Based on your test results, identify on your wiring diagram the suspected area of concern.

- ☐ Answer the following questions:

Based on your test results, what is the cause of the problem?

What additional tests (if any) do you need to perform to further diagnose the problem?

What would you do to repair the problem?

How would you verify that your repair fixed the problem?



- ☐ Review the answers to the above questions with your instructor. Reset your test area and keep your wiring diagrams for scoring later in the classroom.

Instructor's Initials _____

GUIDED PRACTICE 2

The instructor will assign the vehicle and tell you what the customer's concern is.



Vehicle: _____

Symptom: _____

☐ Go to the vehicle assigned by your instructor.

☐ What were the results when you verified the concern?

☐ List any codes using the GDS:

☐ Using KGIS, print the schematic and highlight the power flow for the suspected circuit.

☐ On the wiring diagram, identify the test points you would check to diagnose the concern, and indicate what results you would expect when performing the test.



☐ Show the Instructor your wiring diagram with power flow illustrated and your test points.

☐ After the instructor has reviewed your test points, perform the tests on the vehicle. DO NOT perform any additional tests without your instructor's approval.

☐ Based on your test results, identify on your wiring diagram the suspected area of concern.

- ☐ Answer the following questions:

Based on your test results, what is the cause of the problem?

What additional tests (if any) do you need to perform to further diagnose the problem?

What would you do to repair the problem?

How would you verify that your repair fixed the problem?



- ☐ Review the answers to the above questions with your instructor. Reset your test area and keep your wiring diagrams for scoring later in the classroom.

Instructor's Initials _____

GUIDED PRACTICE 3

The instructor will assign the vehicle and tell you what the customer's concern is.



Vehicle: _____

Symptom: _____

☐ Go to the vehicle assigned by your instructor.

☐ What were the results when you verified the concern?

☐ List any codes using the GDS:

☐ Using KGIS, print the schematic and highlight the power flow for the suspected circuit.

☐ On the wiring diagram, identify the test points you would check to diagnose the concern, and indicate what results you would expect when performing the test.



☐ Show the Instructor your wiring diagram with power flow illustrated and your test points.

☐ After the instructor has reviewed your test points, perform the tests on the vehicle. DO NOT perform any additional tests without your instructor's approval.

☐ Based on your test results, identify on your wiring diagram the suspected area of concern.

- ☐ Answer the following questions:

Based on your test results, what is the cause of the problem?

What additional tests (if any) do you need to perform to further diagnose the problem?

What would you do to repair the problem?

How would you verify that your repair fixed the problem?



- ☐ Review the answers to the above questions with your instructor. Reset your test area and keep your wiring diagrams for scoring later in the classroom.

Instructor's Initials _____

GUIDED PRACTICE 4

The instructor will assign the vehicle and tell you what the customer's concern is.



Vehicle: _____

Symptom: _____

☐ Go to the vehicle assigned by your instructor.

☐ What were the results when you verified the concern?

☐ List any codes using the GDS:

☐ Using KGIS, print the schematic and highlight the power flow for the suspected circuit.

☐ On the wiring diagram, identify the test points you would check to diagnose the concern, and indicate what results you would expect when performing the test.



☐ Show the Instructor your wiring diagram with power flow illustrated and your test points.

☐ After the instructor has reviewed your test points, perform the tests on the vehicle. DO NOT perform any additional tests without your instructor's approval.

☐ Based on your test results, identify on your wiring diagram the suspected area of concern.

- ☐ Answer the following questions:

Based on your test results, what is the cause of the problem?

What additional tests (if any) do you need to perform to further diagnose the problem?

What would you do to repair the problem?

How would you verify that your repair fixed the problem?



- ☐ Review the answers to the above questions with your instructor. Reset your test area and keep your wiring diagrams for scoring later in the classroom.

Instructor's Initials _____

GUIDED PRACTICE 5

The instructor will assign the vehicle and tell you what the customer's concern is.



Vehicle: _____

Symptom: _____

☐ Go to the vehicle assigned by your instructor.

☐ What were the results when you verified the concern?

☐ List any codes using the GDS:

☐ Using KGIS, print the schematic and highlight the power flow for the suspected circuit.

☐ On the wiring diagram, identify the test points you would check to diagnose the concern, and indicate what results you would expect when performing the test.



☐ Show the Instructor your wiring diagram with power flow illustrated and your test points.

☐ After the instructor has reviewed your test points, perform the tests on the vehicle. DO NOT perform any additional tests without your instructor's approval.

☐ Based on your test results, identify on your wiring diagram the suspected area of concern.

- ☐ Answer the following questions:

Based on your test results, what is the cause of the problem?

What additional tests (if any) do you need to perform to further diagnose the problem?

What would you do to repair the problem?

How would you verify that your repair fixed the problem?



- ☐ Review the answers to the above questions with your instructor. Reset your test area and keep your wiring diagrams for scoring later in the classroom.

Instructor's Initials _____

GUIDED PRACTICE 6

The instructor will assign the vehicle and tell you what the customer's concern is.



Vehicle: _____

Symptom: _____

☐ Go to the vehicle assigned by your instructor.

☐ What were the results when you verified the concern?

☐ List any codes using the GDS:

☐ Using KGIS, print the schematic and highlight the power flow for the suspected circuit.

☐ On the wiring diagram, identify the test points you would check to diagnose the concern, and indicate what results you would expect when performing the test.



☐ Show the Instructor your wiring diagram with power flow illustrated and your test points.

☐ After the instructor has reviewed your test points, perform the tests on the vehicle. DO NOT perform any additional tests without your instructor's approval.

☐ Based on your test results, identify on your wiring diagram the suspected area of concern.

- ☐ Answer the following questions:

Based on your test results, what is the cause of the problem?

What additional tests (if any) do you need to perform to further diagnose the problem?

What would you do to repair the problem?

How would you verify that your repair fixed the problem?



- ☐ Review the answers to the above questions with your instructor. Reset your test area and keep your wiring diagrams for scoring later in the classroom.

Instructor's Initials _____

[illegible]

Automotive Electrical Diagnosis

Theory Module

TARGET AUDIENCE The target audience for this module consists of Kia service technicians who have a basic understanding of automotive electrical.

MODULE GOAL Upon completion of this module, you will be able to diagnose basic battery/starting/charging system malfunctions.

MODULE OBJECTIVES

- Given this module and a multiple-choice test, you will identify the operation of the battery/starting/charging system components with 100% accuracy
- Given this module and a multiple-choice test, you will identify the function(s) and variation(s) of the battery/starting/charging system components with 100% accuracy
- Given this module and a multiple-choice test, you will identify the diagnostic procedures related to battery/starting/charging system components with 100% accuracy

MODULE INSTRUCTIONS Carefully read the material, take notes based on the classroom discussion, and study each illustration. Throughout the module, there will be Progress Check questions for you to answer. You may use the modules to answer the questions.

REQUIRED MATERIALS None

TIME TO COMPLETE Approximately 2.0 hours

ACRONYMS

AC	– Alternating Current
BCM	– Body Control Module
CCA	– Cold Cranking Ampere
EPMS	– Electrical Power Management System
IBSM	– Intelligent Battery Sensor Module
PCM	– Powertrain Control Module
SOC	– State of Charge

INTRODUCTION

The battery, starting, and charging systems are the foundation of the vehicle's electrical system.

- The battery is the source of electrical energy
- The starting system rotates the crankshaft when starting the engine
- The charging system charges the battery and provides a power source once the engine is running.

In this module, we will look at the function, operation, troubleshooting, and diagnosis of these systems and key components. Knowing how these systems operate will help you diagnose system malfunctions quickly and more efficiently.

PURPOSE

The purpose of the battery is to supply electrical energy to:

- Operate electrical accessories when the engine is not running
- Crank the engine
- Provide electrical energy to run the fuel, ignition, and accessory systems while the engine is running.

APPLICATION

The physical size and location as well as the amount of energy produced/used by the systems varies from vehicle to vehicle based on the size of the engine and the amount of electrical accessories on the vehicle.



BATTERY FUNCTION

The battery performs the following major functions:

1. When the engine is OFF, the battery provides energy to operate the lighting and accessory systems.
2. When starting the engine, the battery provides the electrical energy to operate the starter motor and the ignition system.
3. When the engine is running the battery acts as a voltage stabilizer to even out voltage spikes within the electrical system.

Battery Types

Kia uses two types of sealed batteries:

- Closed Maintenance Free (CMF) lead-acid type, also called a wet-cell battery
- Advanced Glass Mat (AGM)

LEAD ACID

The CMF lead-acid battery

- Uses lead peroxide and sponge lead plates housed separately in a plastic container
- Each group of plates is called "cells"
- The cells are submerged in an electrolyte solution consisting of water and sulfuric acid
- The chemical reaction between the electrolyte and the plates in each cell produces voltage
- Each fully charged cell generates 2.11 volts
- Six cells connected in series produce a total of 12.66 volts

**ADVANCED GLASS MAT
AGM**

Advanced Glass Mat

- Used in Idle, Stop & Go (ISG) vehicles, and vehicles with large electrical loads such as power sliding doors, and power lift gates
- Electrolyte is absorbed into glass mats held in place by capillary action
- Keeping the plates "moist" with electrolyte, reduces gassing
- Delivers higher power and efficiency than other battery types

**Cold-Cranking Ampere
(CCA)**

CCA rating - Number of amperes the battery can deliver at 0°F for 30 seconds while maintaining a voltage of 7.2 or greater.

Reserve Capacity (RC)

RC rating - The length of time in minutes that a fully charged battery can deliver 25 Amps while maintaining a minimum of 10.5 volts at 80°F.

Amp-Hour (AH)

AH rating - The amount of current in amperes times 20 hours that a battery can deliver while maintaining a voltage of at least 10.5 volts at 80°F. Example 54 Amp Hours:

- $2.7 \text{ Amps} \times 20 \text{ hours} = 54 \text{ Amp Hours}$
- $54 \text{ Amp Hours} / 20 \text{ hours} = 2.7 \text{ Amps}$

TROUBLESHOOTING & DIAGNOSIS

There are a number of tests to run when troubleshooting the battery to determine its state of charge and its ability to supply current. The state of charge test checks the battery's overall condition while the current test measures the battery's ability to deliver adequate cranking power.

Visual Inspection

The first step in diagnosing a battery related concern is a visual inspection. The chart below describes the parts of a good visual inspection.

Inspection Item	Solution
Cracks in case Distorted case Loose or broken terminals	Replace battery
Cracked or broken cables or connections	Replace as needed
Corrosion on terminals	Clean with wire brush
Dirt or acid on top of case	Clean with a mixture of water and baking soda
Rubber protector on positive terminal	Check for proper coverage
Loose battery hold down Loose cable connections	Tighten as needed



Open Circuit Voltage Test

The open circuit voltage (OCV) test checks the available voltage supplied by the battery.

Using a DVOM:

- Place the positive red lead on the positive battery terminal (B+)
- Place the negative black lead on the negative battery terminal (B-)
- If the battery has just been charged, turn the headlamps ON for one minute to remove any surface charge, turn the headlamps OFF, then wait 5 - 10 minutes and retest
- A fully charged battery should produce about 12.66 volts.



CAUTION: Always wear safety glasses when working around batteries. Hydrogen gas may be expelled during charging or testing and is highly explosive! Keep all sparks away from the battery and never smoke near a battery.

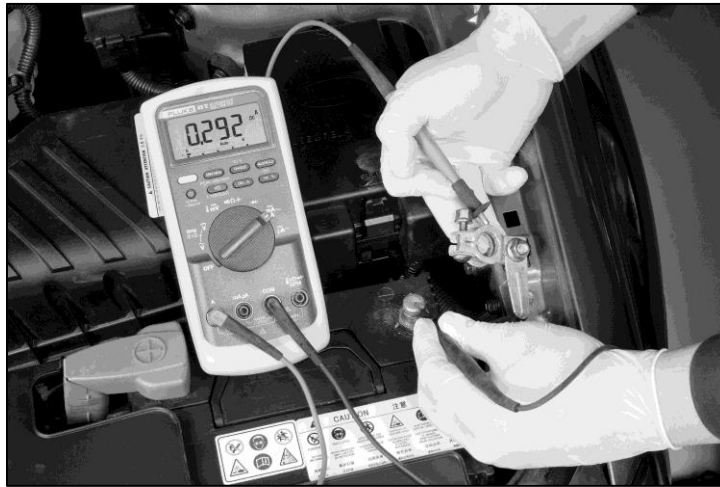
OPEN CIRCUIT VOLTAGE	STATE OF CHARGE
12.80	REMOVE SURFACE CHARGE AND RETEST
12.70	
12.66	100%
12.58	90%
12.51	80%
12.45	70%
12.35	60%
12.28	50%
12.20	40%
12.11	30%
12.04	20%
11.97	10%
11.89	0%

Results The OCV values should be about 12.66 volts. If the reading is below 12.45 volts possible reasons include:

- Vehicle sitting for extended period of time
- Insufficient battery charge prior to testing
- Corroded battery terminals
- Potential low charging voltage condition
- Parasitic draw (Ignition-Off Draw) condition exists
- Defective battery



VIDEO: Open Circuit Voltage Test

**Parasitic Draw**

All vehicles have a constant small amount of parasitic current draw due to computer memories and other electronic components and circuits.

A parasitic draw is excessive when the current draw is too high and discharges the battery over a short period of time.

This is usually caused by a circuit or component that remains powered up when it should not be.

Parasitic Current Inspection and Specifications can be found in:

- Engine Electrical System
- Charging System
- Battery
- Repair Procedure

With the key out of the ignition and all loads off, there should be less than a 50mA draw after 20 minutes.

If a specification is not available, the follow equation can be used to calculate an approximate maximum parasitic draw in mA.

$$\text{Reserve Capacity} \times .25 = \text{Maximum Parasitic Draw}$$



CAUTION: Use the A (amps) range on the DVOM to test parasitic draw then convert the reading to mA (milliamps).

You may blow the fuse in the DVOM if you use the milliamp input and you get a higher current flow than expected. This may be due to the amount of time it takes for some computers to go to "sleep."



To test for excessive parasitic draw:

- Turn all accessories and the ignition OFF, then remove the key and key fob from the vehicle
- Close all doors. This will allow all the on-board computers to go into "sleep" mode
- Disconnect the hood switch to allow the BCM or PCM to go to sleep
- Disconnect the negative (B-) battery cable from the battery post
- Connect a jumper wire between the negative terminal (B-) and the negative clamp
- Connect a DVOM set to measure AMPS in series between the negative cable and the negative post
- Wait up to 20 minutes to allow the computers to enter sleep mode and all circuits to power OFF
- Remove the jumper wire and read the amperage draw
- Compare measured value with service information specifications



NOTE: To prevent accidentally disconnecting the battery while connecting the ammeter:

- Attach a "Jump Box" **not a battery charger** to the positive (B+) and body ground
- Connect the ammeter between the battery negative terminal and the negative cable
- After the DVOM is connected, remove the Jump Box

If the parasitic draw is excessive, a load device is on or a computer is not going to sleep.

Check the obvious first, interior lamps, glove box lamp, and trunk lamp.

If the root cause is not found, use the Power Distribution wiring diagram in the ETM to determine which fuse (load protection devices) provide power to which circuits.

Start with the fuse that controls the most circuits and disconnect them one at a time, to narrow down which circuit is causing the excessive parasitic draw.

Use the wiring diagrams in the ETM to continue to narrow down the circuit causing the draw.



VIDEO: Parasitic Draw



System Ground Voltage Drop

Using the voltage drop method for checking system grounds is useful for finding a component or circuit with excessive resistance.

Unwanted resistance in the ground circuit changes voltage and amperage values.

Multiple sensors and actuators often rely on common grounds, which may affect their operation.

System ground voltage drop readings are normally less than 0.10 volts.

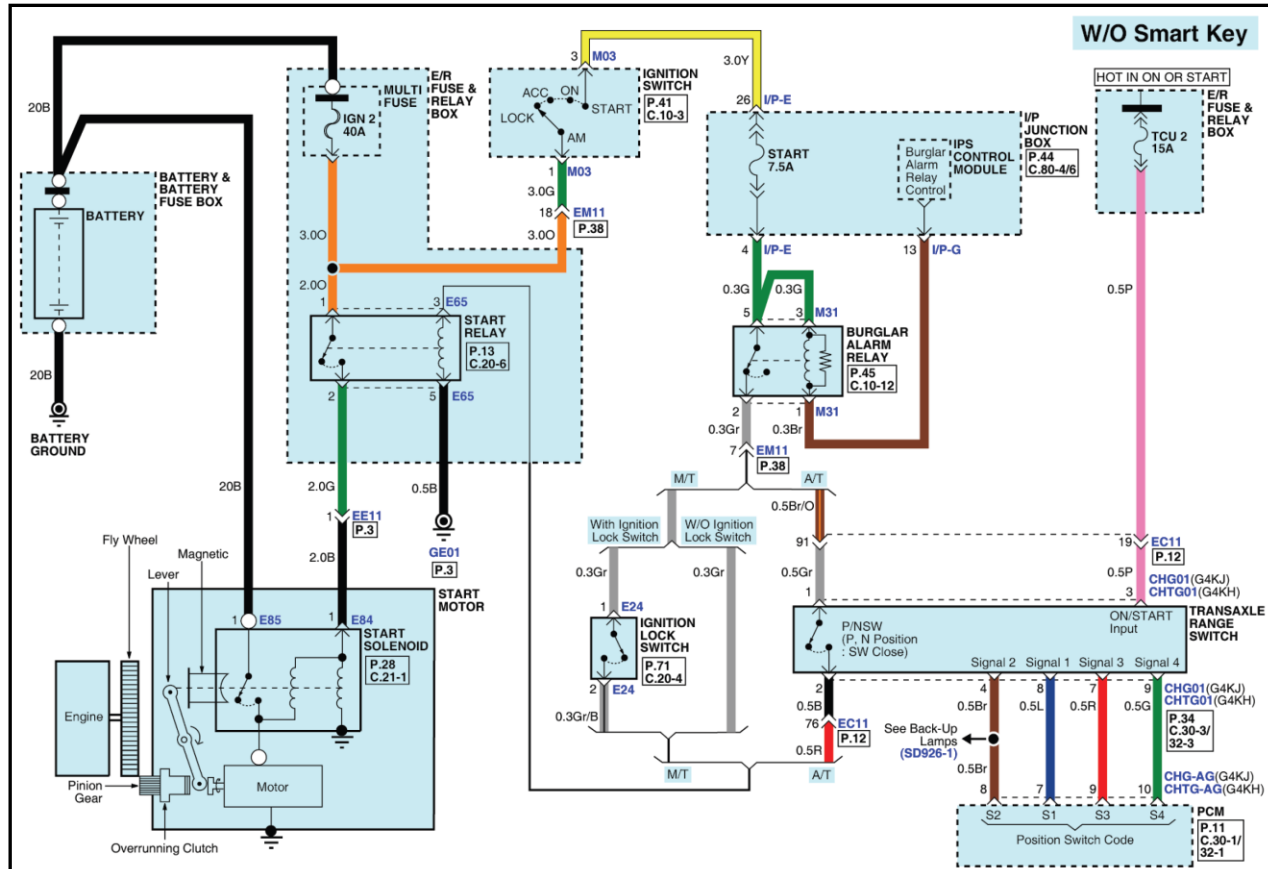
To perform the system ground voltage test:

- Place the DVOM negative black test lead on the battery negative post (B-)
- Place the positive DVOM red test lead on an area near the ground connection (typically body sheet metal or the engine block)
- Next, turn the Key On Engine Running and turn on the following:
 - High beam headlights
 - Windshield wipers
 - Radio
 - HVAC blower on high
- The DVOM should read 0.10 volts or less.

A voltage drop more than 0.10 volts indicates possible corrosion in the ground connections or in the ground cable. Clean all connections and re-test.



STARTING SYSTEM

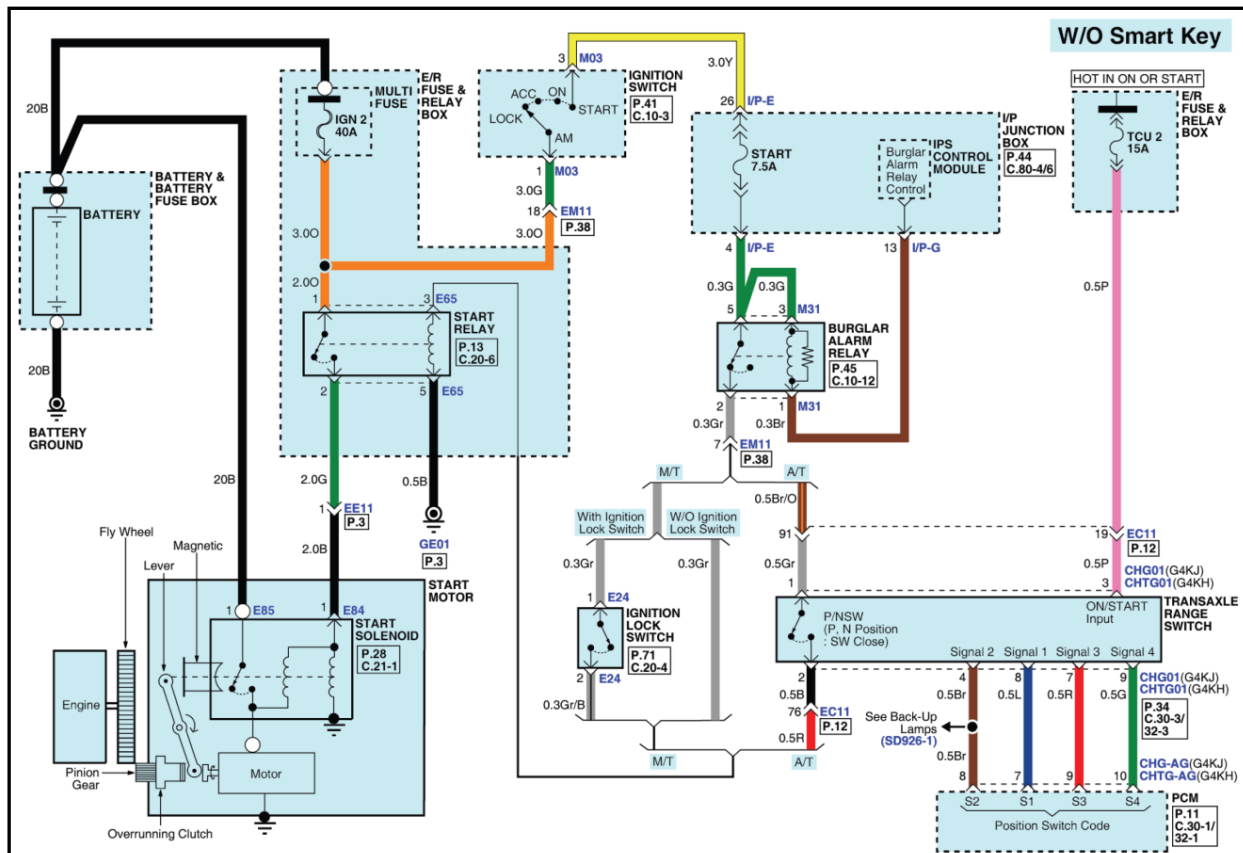


Overview

The **starting system** consists of the battery, starter motor, and electrical control components.

- The ignition switch remotely energizes the starter motor through the start relay.
- Vehicles with an automatic transaxle/transmission, the A/T range switch allows the starter to run only in the Neutral (N) or Park (P) positions.
- Vehicles with a manual transaxle/transmission have a switch on the clutch pedal that allows the starter to energize only when the clutch pedal is depressed.
- When the starter is engaged, pull-in and hold-in windings in the solenoid pulls in a plunger which causes a pinion gear to mesh with the ring gear on the flywheel.
- When the plunger is pulled in it closes a set of contacts which supplies power to the motor to crank the engine

STARTER RELAY



Function The **starter relay** is used to control the current to the starter solenoid. As with any relay, the starter relay is used to control a large amount of current with a small amount of current.

Operation When the driver operates the ignition switch to the START position:

- Current flows from the switch through either the range switch or the clutch pedal position switch when the contacts are closed
- Current then flows to the control side of the relay to ground.
- The small amount of current on the control side energizes an electro-magnet that closes contact on the switch side of the relay.
- These contacts carry a large amount of current from the battery to the starter solenoid through the battery cables.
- This current energizes the starter solenoid and supplies current to the starter motor to crank the engine.

**Battery Cranking Voltage**

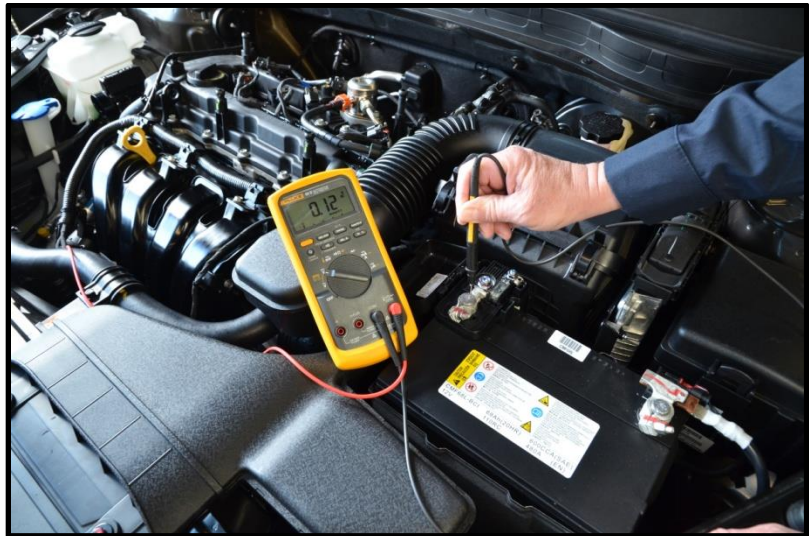
The Battery Cranking Voltage Test checks the condition of the battery while cranking the engine.

To perform the battery cranking voltage test:

- Connect the DVOM
 - Red test lead to the positive battery terminal post (B+)
 - Black test lead to the negative battery terminal post (B-)
- Remove the fuel pump relay and run the engine until it stalls
- Crank the engine for 5 seconds and record the lowest readings from the DVOM.

Good	10.0 – 11.5 vdc
Marginal	9.6 – 9.9 vdc
Test Battery *	< 9.6 vdc

* Test battery using the GR8 Multitasking Battery and Electrical Diagnostic Station



Engine Positive Voltage Drop

The Engine Positive Voltage Drop test checks for unwanted resistance between the battery and battery cable terminal on the starter solenoid.

To complete the test:

- Connect the DVOM
 - Red test lead on the positive battery post (B+)
 - Black test lead on the positive post of the starter solenoid
- Remove the fuel pump relay and run the engine until it stalls
- Crank the engine and record the DVOM readings.

Good	0.10 – 0.30 vdc
Marginal	0.30 – 0.50 vdc
Requires Further Testing *	> 0.50 vdc

* Run voltage drop tests on each part of the circuit to isolate the source of unwanted resistance.



NOTE: It is possible to see an initial starter “voltage spike” over 0.50 volts, it should drop while the starter motor is cranking.



Engine Ground Voltage Drop (Cranking)

The Engine Ground Voltage Drop test tests for unwanted resistance in the groundside of the starter circuit.

To perform the test:

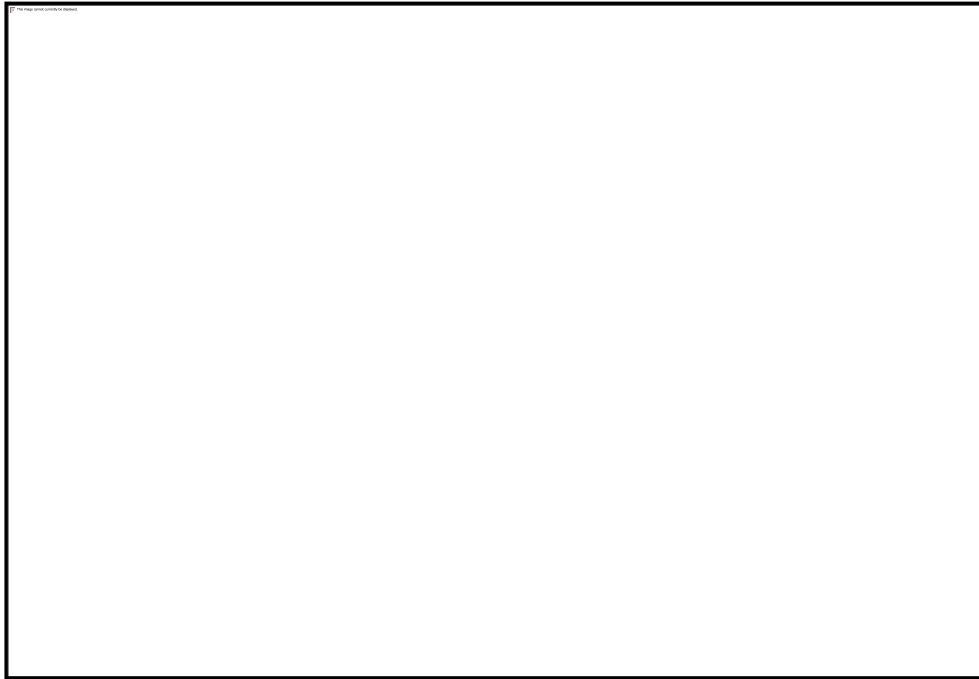
- Connect the DVOM
 - Black test lead on the negative battery post (B-).
 - Red DVOM test lead onto the engine block or head (**do not use the fuel rail or bolts attached to plastic cam cover**).
- Remove the fuel pump relay and run the engine until it stalls
- Crank the engine and record the DVOM readings

Good	0.10 – 0.30 vdc
Marginal	0.30 – 0.50 vdc
Requires Further Testing *	> 0.50 vdc

* Run voltage drop tests on each part of the circuit to isolate the source of unwanted resistance.



NOTE: It is possible to see an initial starter “voltage spike” over 0.50 volts but it should drop while the starter motor is cranking.



CHARGING SYSTEM

The **Charging System** consists of the battery and a belt driven generator that provides the vehicle's electrical energy needs when the engine is running.

When the engine is running, the drive belt rotates the generator rotor to produce electricity.

The generator converts mechanical energy into electrical energy by rotating electro-magnets to produce electricity.

A voltage regulator limits the output of the generator.

A charging indicator (warning) lamp displays the status of the charging system.

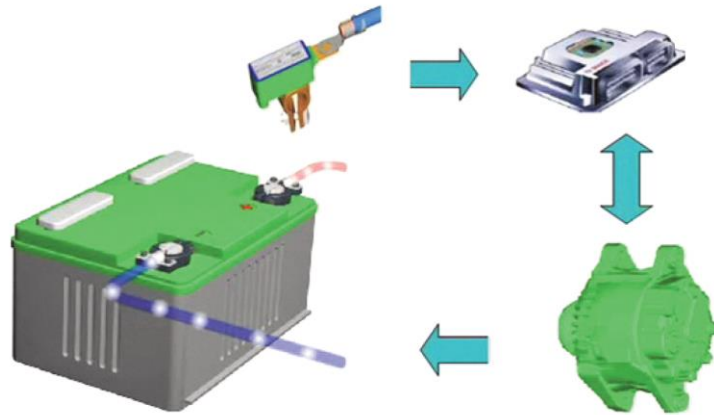
The voltage and current produced by the generator keeps the battery charged and provides the electrical energy to operate the vehicle electrical systems.

The amount of current and voltage produced is limited by the voltage regulator.

The regulator may be located in the generator or in the PCM.

When the generator is not producing enough electricity (belt broken or other malfunction), the warning lamp goes ON to alert the driver.

ELECTRICAL POWER MANAGEMENT SYSTEM



The Electrical Power Management System (EPMS) changes the way the generator is managed.

It takes full use of the battery's capacity, engine power, and braking performance while maximizing battery life and increasing fuel economy.

The battery uses the existing technology of a lead-acid battery, and a generator that is specific to EPMS and is controlled by the ECM/PCM.

The EPMS fully charges the battery and stops charging during heavy acceleration for improved performance.

During braking, the EPMS increases the charging rate from the generator.

This increases the load on the crankshaft, which helps with engine braking as well as increasing the charge rate to the battery.



INTELLIGENT BATTERY SENSOR MODULE

The EPMS uses an Intelligent Battery Sensor Module (IBSM) located on the negative battery terminal.

It monitors the following to determine the battery's State Of Charge (SOC):

- Battery temperature
- Available Voltage
- Current Flow

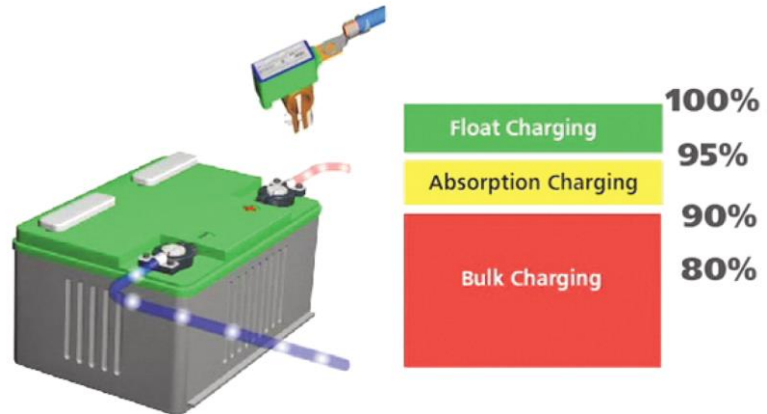
Since the battery's State Of Charge is a measure of how full the battery is charged, the EPMS controls the charge rate to best optimize battery efficiency and engine performance.

A vehicle without an EPMS typically maintains only an 80% charge on the battery.

With EPMS, a 95%-100% charge can be achieved.



FEEDBACK: What possible effects could battery corrosion have on the Intelligent Battery Sensor Module?



HOW EPMS WORKS

The generator's output voltage is controlled by the EPMS, which minimizes battery gassing during the charging stages.

Each State Of Charge zone requires a different charging strategy. By controlling the battery current, the EPMS limits gassing and provides a deeper charge compared to a conventional system.

Reducing battery gassing minimizes the need for battery maintenance and maximizes battery life.

The EPMS takes action on the following thresholds:

- **95% ~ 100%** - is considered fully charged. Charging voltage and current are minimized in a condition known as Float Charging. This action reduces fuel consumption and battery gassing.
- **90% ~ 95%** - is the Absorption Charging stage. During this stage, the charging voltage is moderated to provide a deep charge while minimizing gassing.
- **0% ~ 90%** - is the Bulk Charging stage. Charging takes priority over fuel economy and the EPMS provides high charging current to the battery. The IBSM monitors battery temperature and charging current. When the battery warms up in temperature, the voltage remains high but the current is limited to protect the battery plates from too much current.

EPMS Disable Conditions

Under most circumstances, the EPMS controls the battery charge rate.

There are conditions when the EPMS does not control the charge rate.

These include the following:

- **Headlights On** – The EPMS does not cycle the charge rate of the system because drivers might notice a fluctuation in the headlight intensity as the charging voltage is reduced.
- **HVAC Blower On Max** – Float charging is disabled as varying the charge rate would affect blower fan speed operation.
- **Excessively High/Low Battery Temperature** – Excessive temperatures affect the operation of the battery. Low temperatures reduce the battery's capacity (Amp Hours). Hot temperatures can create an excessive load on the system that may create a hazardous condition.
- **EPMS Error** – When a calculation error is determined within the EPMS, the system will not cycle the charge and will default to conventional charging.



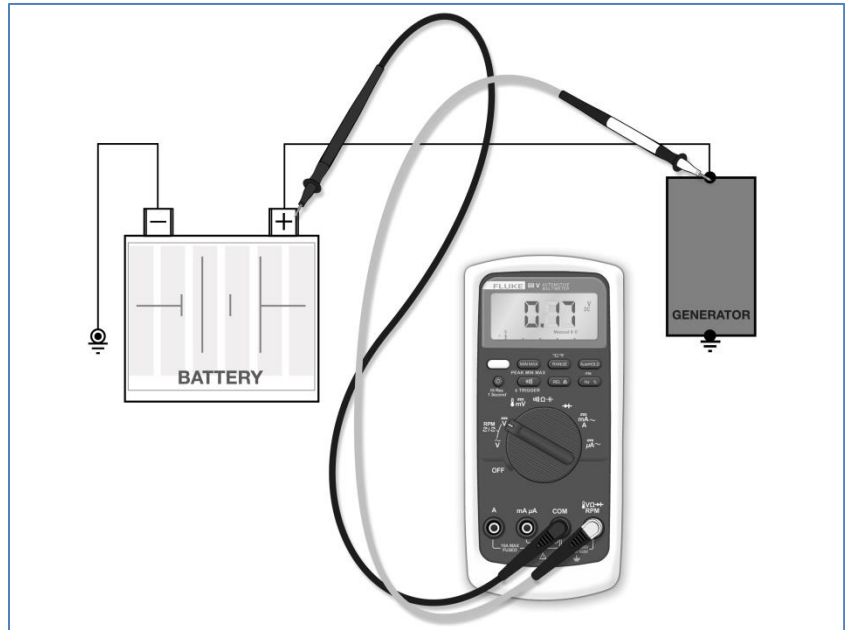
CHARGING VOLTAGE TEST

This quick test allows you to determine if the charging system is charging the battery.

- Perform an open circuit voltage test. An ideal battery voltage should be 12.66 volts
- Start the vehicle and raise the engine rpm to about 2000 rpm
- The result should be 1.0 - 2.0 volts above the Open Circuit Voltage test results
- If the reading is less than 1.0 volt above the recorded Open Circuit Voltage test, then further charging system testing is needed



NOTE: Charging voltage should only be measured at the posts of the battery, not on the clamps or the B+ Terminal at the back of the generator.



Positive Side Voltage Drop

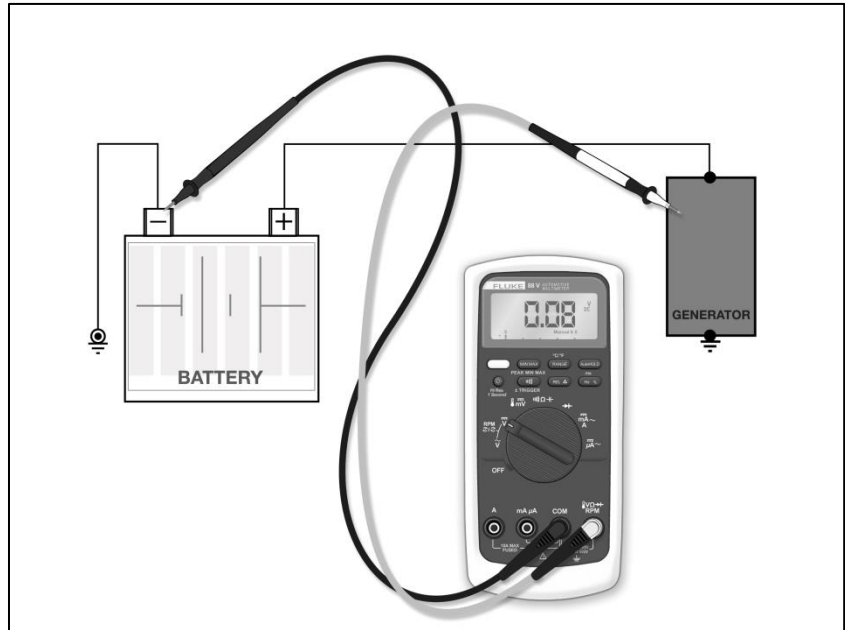
The positive side voltage drop test checks for unwanted resistance from the positive battery post to the B+ terminal on the generator.

To perform the test:

- Connect DVOM
 - Black lead on the positive battery post (B+)
 - Red lead on the B+ Terminal on the back of the generator
- With the engine running, raise the engine to 2000 RPM.
- A voltage drop of greater than 0.25 volts indicates that there is resistance on the B+ side of the charging system
- Inspect and clean connections and inspect the wire itself for excessive resistance



NOTE: If the voltage drop is still greater than 0.25 volts after the connections have been cleaned and the wire inspected, then the battery may be drawing excessive charging current. This could be caused by an internal battery failure that may pass a GR8 battery test. The excessive current causes heat in the generator, which may ultimately create generator failure. Check the vehicle's prior history to see if generators have been recently replaced on the vehicle.

**Voltage Drop Ground Side**

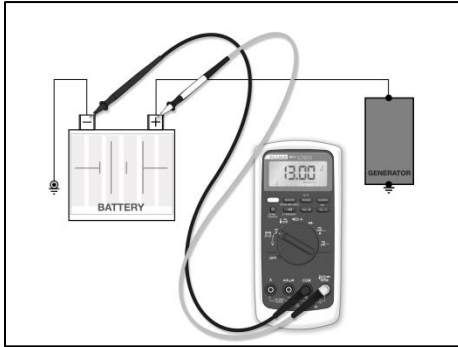
Another quick check that can be performed after checking the voltage drop on the positive side of the charging system is to check the groundside of the charging system.

To perform this test:

- Connect the DVOM
 - Black lead from the DVOM to the negative battery terminal
 - Red lead to the case of the generator
- Raise the engine to 2000 rpm
- Test results should be less than 0.15 volts
- Readings greater than 0.15 volts indicate unwanted resistance in the groundside of the charging system circuit
- Check all electrical wiring and connections, clean the generator mounting-bracket and mounting bolts to the engine block, and retest

CHARGING VOLTAGE TEST FOR UNDERCHARGE

To measure the charging system's capacity there must be a heavy load placed on the system. This test is useful to determine if the generator is capable of the charging demands of the vehicle.

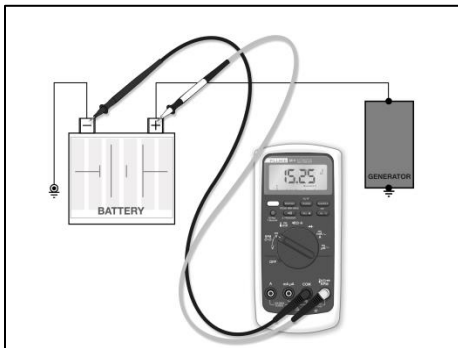


To perform this test:

- Connect DVOM
 - Black lead on the negative post
 - Red lead on the positive post
- With the engine running, raise the RPM to 2000
- Turn on the headlamp high beams, the radio, the HVAC blower motor on high, and the wipers on a High-speed
- The reading should be greater than 13.4 volts
- A reading less than 13.4 volts may indicate that the charging system wiring or generator is faulty

Charging Voltage Test For Overcharge

An overcharge condition may cause considerable damage to the vehicles battery, electrical circuits, and components, including sensitive circuits inside computers.



To check for an overcharge condition:

- Start the engine and keep the engine RPM at 2000
- Turn off all electrical loads
- Place the DVOM leads onto the battery
 - Black lead on the negative post
 - Red lead on the positive post
- If the reading is above 15 volts, then an overcharge condition may exist
- An overcharge condition may indicate that the voltage regulator may be defective

Trouble Symptom	Probable Cause	Remedy
Charging warning indicator does not light with ignition switch "ON" and engine OFF	Fuse blown Light burned out Wiring connection loose Electronic voltage regulator faulty	Check fuses Replace light Tighten loose connections Replace voltage regulator
Charging warning indicator does not go out with engine running (Battery requires frequent recharging)	Drive belt loose or worn Battery cables loose, corroded or worn Fuse blown Fusible link blown Electronic voltage regulator or generator faulty Wiring faulty	Adjust tension or replace drive belt Repair or replace cables Check fuses Replace fusible link Test generator Repair wiring
Overcharge	Wiring connection loose or open circuit Fusible link blown Poor grounding Electronic voltage regulator or generator faulty Worn battery Electronic voltage regulator faulty Voltage sensing wire faulty	Tighten loose connection or repair wiring Replace fusible link Repair Test generator. If faulty repair or replace Replace battery Replace voltage regulator Repair wire

Service Information

Refer to the latest service information for additional charging system diagnosis procedures. Here is an example of a trouble symptom chart:

Recharging Rate Chart (for fully discharged batteries)						
RC Rating	AH Rating	5A	10A	20A	30A	40A
less than 75	less than 48	10 hrs	5 hrs	2.5 hrs	2 hrs	Not Recommended
75-115	48-75	15 hrs.	7.5 hrs	3.25 hrs	2.5 hrs	2 hrs
115-160	75-100	20 hrs.	10 hrs	5 hrs	3 hrs	2.5 hrs
160-245	100-150	30 hrs	15 hrs	7.5 hrs	5 hrs	3.5 hrs

Battery Recharging

To recharge a battery, current must be applied to the battery in the opposite direction of the discharge flow. This restores the imbalance of electrons in the battery.

There are three methods of charging the battery:

- Fast charging
- Slow charging
- Trickle charging

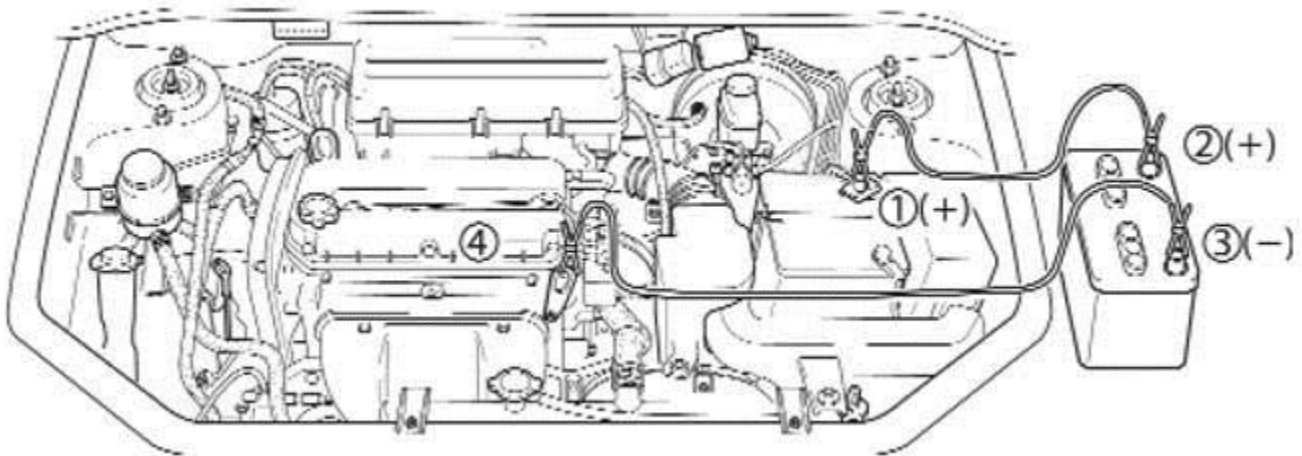
Fast charging applies a high current rate to the battery for a short period of time. Fast charging should not be used on a completely discharged battery to prevent damaging it.

Slow charging recharges a battery by using a reduced current level for a longer period of time. The maximum charging current should be less than 1/10th of the battery capacity.

Trickle charging is a very low rate of charging used to maintain a battery in a fully charged condition.



NOTE: A battery may lose its charge while the vehicle sits on the lot waiting to be sold. To avoid customer embarrassment or wrongfully replacing a battery, be sure to periodically start and run the engine to maintain full battery charge.

**Jump Start**

Occasionally, it may be necessary to jumpstart a vehicle with a discharged battery using a vehicle with a good battery. Follow these procedures to reduce the risk of personal injury and/or damage to the vehicle(s):

1. Position the vehicle so the jumper cables can reach both batteries. Do not allow the vehicles to touch each other.
2. Turn all accessories OFF in both vehicles.
3. Turn the ignition OFF in both vehicles before connecting the jumper cables.
4. Connect one end of the positive jumper cable to the positive terminal of the discharged battery. Connect the other end of the positive jumper cable to the positive terminal of the booster battery.
5. Connect one end of the negative jumper cable to the negative terminal of the booster battery. Connect the other end of the negative jumper cable to the engine ground of the vehicle with the discharged battery.
6. Start the engine of vehicle with the booster battery and allow it to idle.
7. Start the vehicle with the discharged battery.
8. Remove the jumper cables in the reverse order they were connected; negative jumper cables first, then positive cables.



CAUTION: Personal injury or vehicle damage can occur if jumpstarting is done incorrectly.



GR8 MULTITASKING BATTERY AND ELECTRICAL DIAGNOSTIC STATION

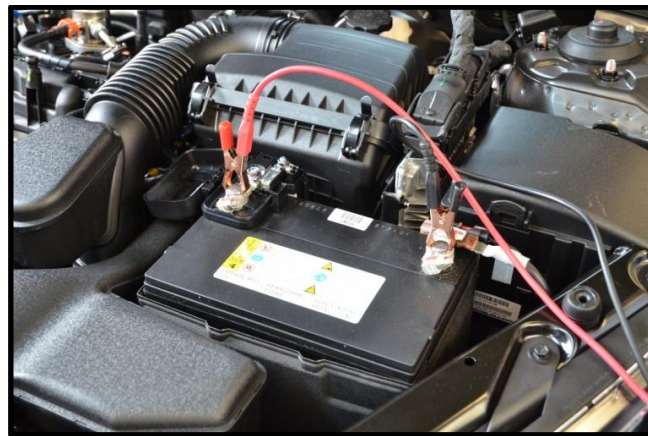
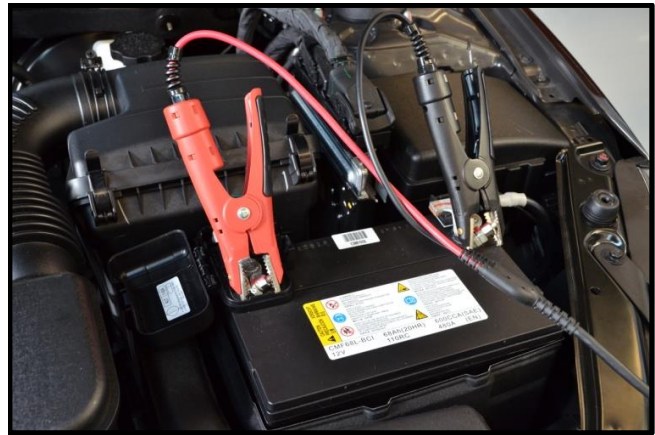
Kia currently uses the GR8-1299 to test batteries, starting, and charging systems. The GR8 uses two methods to test a battery

- Conductance test, determines the battery's ability to store electricity
- Load test, determines the battery's ability to provide a specific open circuit voltage under a resistance type load

In most cases, the GR8 can test a battery as received without recharging it for testing.

Here is a brief overview of the testing procedure.
Before testing:

- Make sure all vehicle accessory loads are OFF and the ignition is OFF
- If the battery is not installed, clean the battery posts with a wire brush



Use the content of the classroom discussion, and the **GR8-1299 Instruction Manual on your computer** to fill in the blanks in the following section of this module.

**GR8 COMPONENTS AND
CONTROLS**

The GR8-1299 has 3 sets of clamps, label the type of clamp shown above.

CONNECTING TO THE BATTERY

Connect the charging clamps to the battery.

Do not connect either clamp to the _____.

If you connect the clamps in the wrong polarity

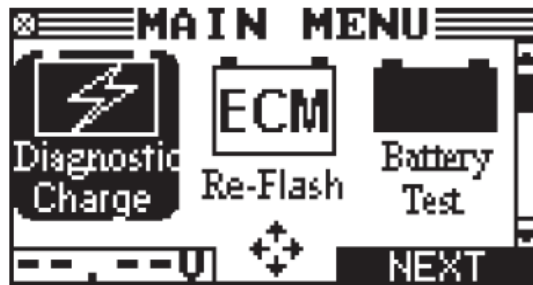
(red to negative or black to positive),

the analyzer _____,

and displays _____.

Make sure _____ of the clamps are gripping the terminals.

Poor connections will prevent testing, and the analyzer displays the message _____.



DIAGNOSTIC CHARGE

The diagnostic charge determines the internal condition of the battery before attempting to charge it.

The charger applies a(n) _____

at a known frequency across the terminals and

measure the _____ to

determine the battery health.

The diagnostic charge is the most thorough test and should always be performed when the battery is questionable.

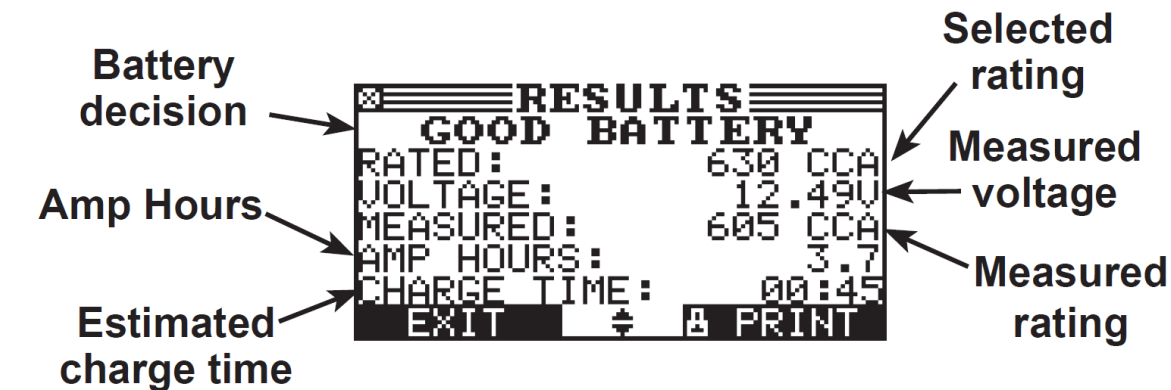
All Diagnostic Charge tests must include:

- _____
- _____
- _____
- _____

CHARGING MODES

Diagnostic Charging has four modes.

- _____ – to determine if the battery is good or needs replacement.
- _____ – to determine if the battery can be recovered if the initial analysis is negative.
- _____ – Once the GR8 determines that the battery is good, needs charging, and is safe to charge, it proceeds to the diagnostic charge mode. During the diagnostic charge, the battery is charged and the GR8 applies an electrical load to the battery and measures the CCA to determine if the battery needs replacing.
- _____ – occurs at the end of the charge cycle to charge a good battery to capacity.



DIAGNOSTIC CHARGING TEST RESULTS

BATTERY DECISION (list the three battery decisions)

- _____
- _____
- _____

RATED

- the battery's rated CCA

VOLTAGE

- battery voltage

MEASURED

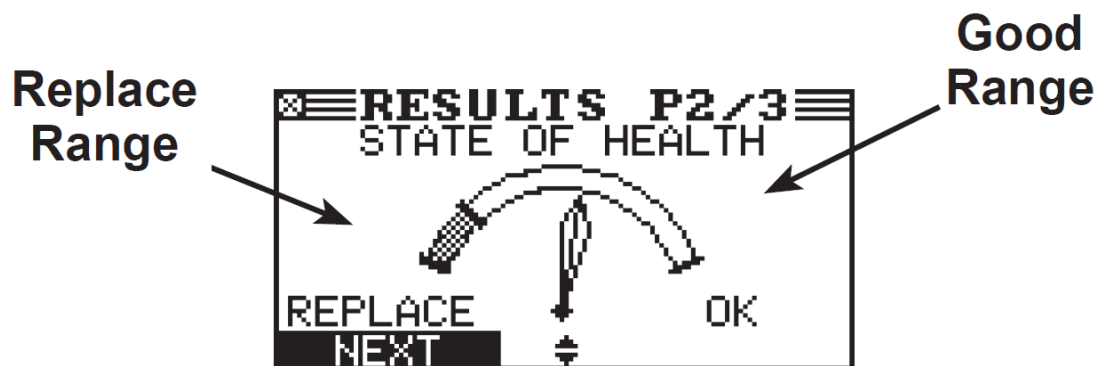
- measured CCA

AMP HOURS

- An indication of the total amount of energy the battery will deliver at a constant rate of discharge before it drops below 10 volts

CHARGE TIME

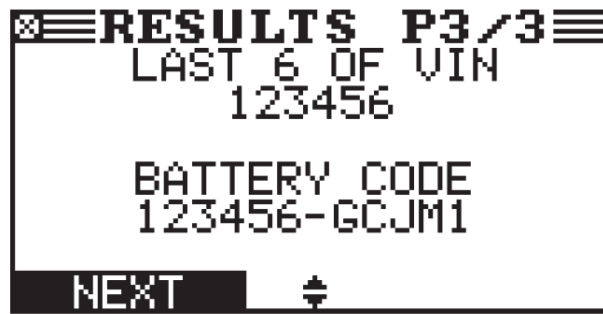
- time needed to completely charge the battery



STATE OF HEALTH

The battery state of health shows general condition of the battery and its ability to deliver the specified performance as compared with a fresh battery.

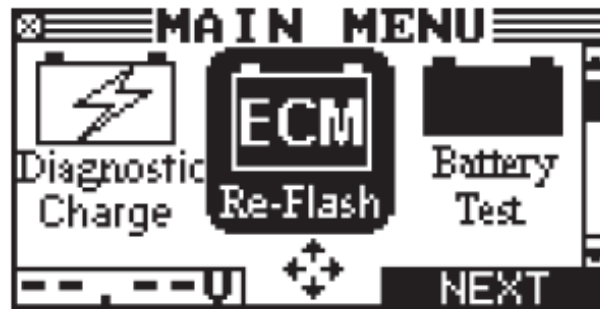




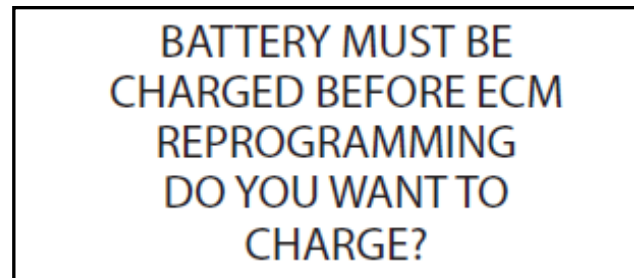
BATTERY CODE The last six digits of the VIN and battery code for warranty purposes.



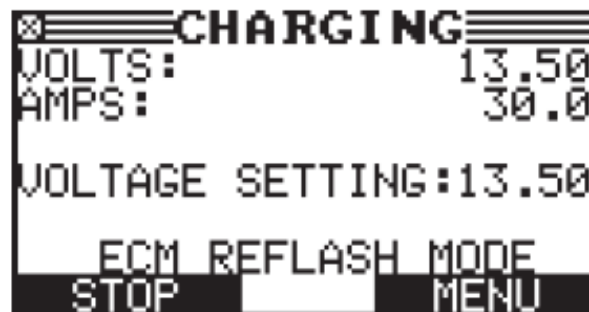
NOTE: A REPLACE BATTERY result may also mean that there is a poor connection between the battery and the vehicle. If you tested the battery using the System Test, disconnect the battery cables and retest the battery using the Battery Test before replacing it.

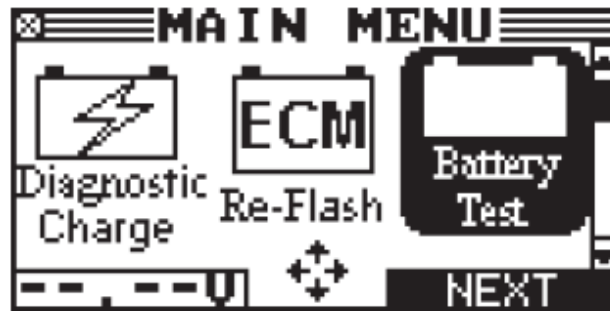


ECM Re-Flash The ECM Re-Flash mode maintains the battery voltage at _____ volts using the _____ clamps allowing for uninterrupted power during programming of computers or retaining vehicle system settings.



The message above will be displayed if the battery needs to be charged before entering the ECM Re-Flash Mode.





Battery Test

The battery test performs the Conductance Test using the test clamps. It is a shorter test than the diagnostic charge and does not load test or charge the battery.

The battery test guides you through the steps of selecting test parameters and interpreting the results.

During the battery test, the GR8 may display Additional Test Requirements which are messages and instructions to complete analysis of the battery's condition.

Additional Test Requirements include the following (provide the title for each requirement):

- _____:
 - Computer or ignition noise is present or a parasitic draw has been detected.
 - The battery will need to be retested.
 - Be sure that all electrical loads and the ignition are OFF.
 - The tester will automatically retest the battery when the noise is no longer detected.
- _____:
 - The battery is weak or has just been charged
 - A fully charged battery should stabilize quickly
 - Weak batteries should be charged and retested
- _____:
 - Runs automatically when further analysis is required

Test results include:

- _____
- _____

Based on the test results, the following battery decision table indicates the recommended action.



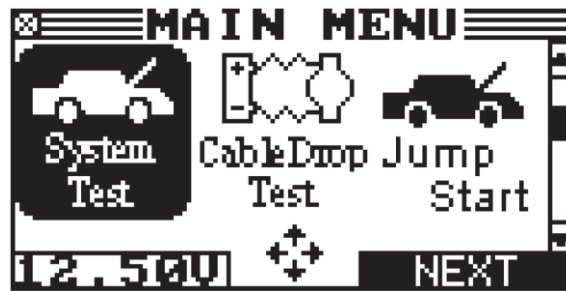
Battery Condition	Recommended Action
GOOD BATTERY	Return the battery to service
GOOD-RECHARGE	Fully recharge the battery and return it to service
CHARGE & RETEST	Fully charge the battery and retest. If CHARGE AND RETEST appears after fully charging the battery, replace the battery.
REPLACE BATTERY	A REPLACE BATTERY result may also mean that there is a poor connection between the battery and the vehicle. If you tested the battery using the System Test, disconnect the battery cables and retest the battery using the Battery Test before replacing it.



NOTE: When you start a new battery test, the last test results in memory are overwritten. Record or print the previous test results if you need to retain them.



CAUTION: Never connect both sets of clamps to the battery at the same time. Only the test clamps (large clamps) would be connected to the battery terminals when performing a battery test.

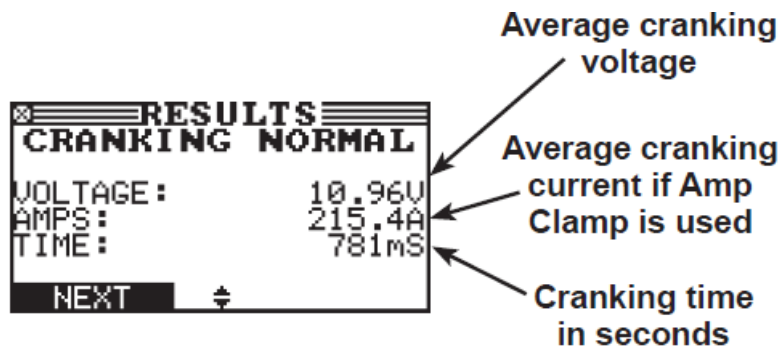


System Test The system test tests the following systems:

- _____
- _____
- _____

It uses the _____ Clamps to perform all the tests.

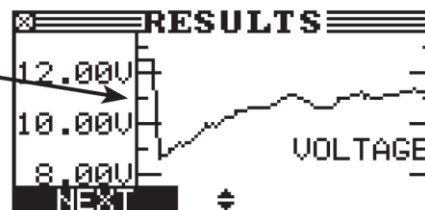
Battery test results are the same as those for the diagnostic charge and/or battery test.



Starter test results include:

- Average cranking voltage
- Average cranking current (if Amp Clamp connected)
- Cranking time during the test in seconds

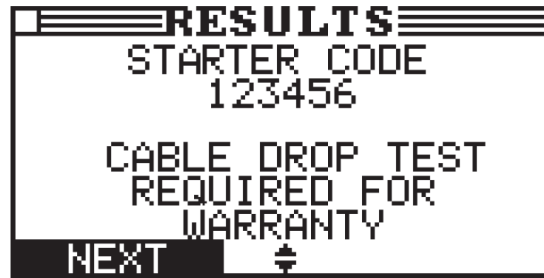
Y axis = System performance: cranking voltage



X axis = Time

- System performance cranking voltage





- Starter code for warranty purposes
- Cable drop test reminder

Based on the test results, the following starter decision table indicates the recommended action.

Starter Test Results	Recommended Action
_____	The starter voltage is normal and the battery is fully charged.
_____	The starter voltage is low and the battery is fully charged.
_____	The starter voltage is low and the battery is discharged. Fully charge the battery and repeat the starter system test.
_____	Replace the battery before testing the starter.
_____	The engine did not start and the test was canceled.
_____	The GR8 did not detect the vehicle's starting profile and skipped the starter test.
_____	Test data was inconclusive using the side post. Retest using side post adapters.
_____	Data was inconclusive using the jump-start post. Retest at the battery terminals.

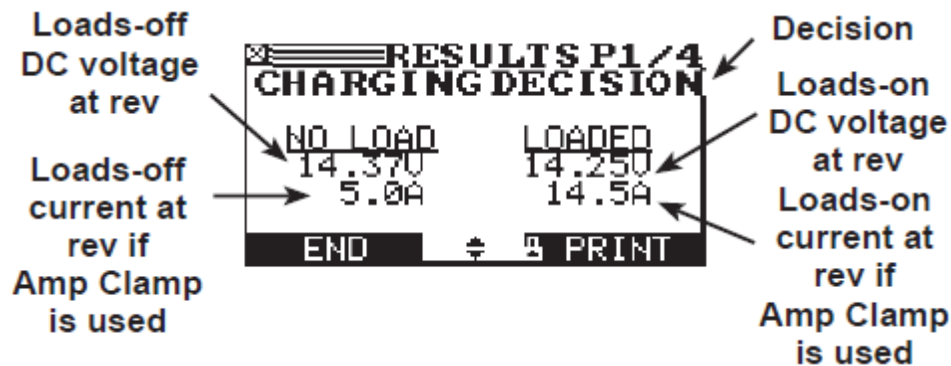


NOTE: In some cases, the GR8 may not detect the vehicle's starting profile and will display the options STARTED and NO START. Select STARTED to continue with an alternator test. Select NO START to end the test.

- CHARGING SYSTEM TEST** To complete the charging system test you will need to:
- Raise rpm
 - Return to idle
 - Turn loads ON and OFF

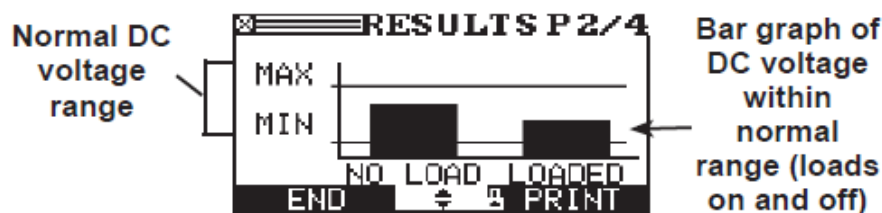
It is easier if you are in the vehicle to follow the test procedures and instructions on the GR8 to complete the test.

Charging system test results include.

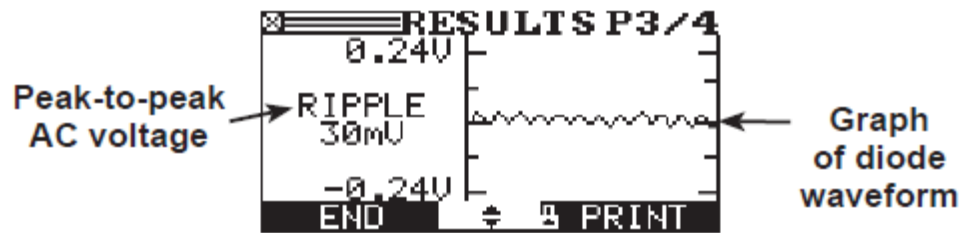


Load on and off current is displayed when amp clamp is used

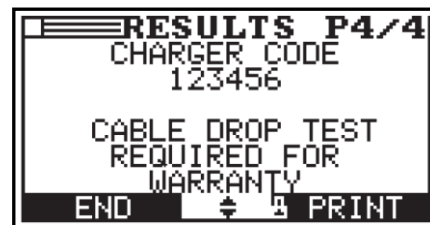
- Charging Decision
- Charging voltage, no load
- Charging voltage, load
- Amps, not load (if amp clamp used)
- Amps, loaded (if amp clamp used)



- Bar graph of normal voltage range and NO LOAD/LOAD test results



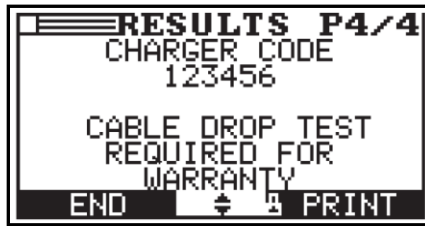
- Diode waveform



- Charger code for warranty
- Cable drop test reminder

Based on the test results, the following charging decision table indicates the recommended action.

Charging System Test Results	Recommended Action
NO PROBLEMS	The system is showing normal output from the generator. No problem detected.
NO VOLTAGE	The generator is not providing charging voltage to the battery. • Check the generator drive belt and belt tension • Check all system electrical connections • Check the voltage regulator (if equipped) • If all checks out good, replace the generator
LOW VOLTAGE	The generator is not providing enough voltage. • Check the generator drive belt and belt tension • Check all system electrical connections
HIGH VOLTAGE	The voltage output exceeds the normal limits. • Ensure that there are no loose connections and that the ground connection is normal • Check the voltage regulator (if equipped)

**CABLE DROP TEST**

If the starter or charging system tests indicates a potential problem, you can perform a cable drop test to determine if the condition is caused by unwanted resistance in the system's circuits.

The cable drop test is a voltage drop test. There is no need to run the engine as the GR8 sends a signal through the circuit being tested and calculates the voltage drop based on the results.

The preset tests are:

- _____
- _____
- _____
- _____



DO NOT USE THE CHARGING CLAMPS!!!

The cable drop test requires both sets of test leads to be connected:

- _____ connected to the circuit being tested
- _____ across the battery terminals

After entering the required data and pressing "NEXT", the GR8 will display TESTING and a stopwatch while it evaluates the circuit.

```

BATTERY GROUND
CIRCUIT DROP: 0.08V
PASS
+ DROP: 0.00V
- DROP: 0.08V
CODE 123452345
END A PRINT

```

```

STARTER CIRCUIT
CIRCUIT DROP: 0.30V
PASS
+ DROP: 0.15V
- DROP: 0.15V
CODE 123452345
END A PRINT

```

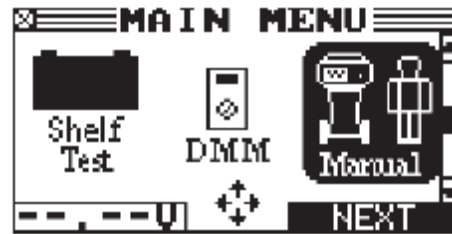
```

ALT CIRCUIT
CIRCUIT DROP: 0.30V
PASS
+ DROP: 0.15V
- DROP: 0.15V
CODE 123452345
END A PRINT

```

Test results include:

- Circuit Tested
- Circuit Drop (voltage drop total)
- PASS/FAIL
- + Drop (voltage drop on the positive side of the circuit)
- – Drop (voltage drop on the negative side of the circuit)
- Code (needed for a warranty claim)



MANUAL CHARGING

The GR8 can manually charge the battery using the Charging Clamps based on your input.

If the battery is safe to be charged, the battery can be charged at selected levels.

You can set limits for _____

If you select CURRENT, what is the available range?

If you select VOLTAGE, what is the available range?

What is the minimum charge duration? _____

What is the maximum charge duration? _____

What do you select if you want to charge longer than the maximum charge duration? _____



When the manual mode ends, the GR8 displays the final charge status based on the selected charge criteria.

An alarm sounds every 30 seconds until the clamps are disconnected, or the "NEXT" key is pressed.

GR8 TESTS The GR8 Multitasking Battery and Electrical Diagnostic can perform the following tests.

Data input required for each tests can be found in the GR8 Data Input table.

GR8 Data Input Table

Data to be Entered	Last 6 of VIN	Battery IN/OUT of vehicle	Vehicle Type	Battery Type	CCA	Amp Clamp Option	Type of Test
GR8 Test							
Diagnostic Test	✓	✓	✓	✓	✓	✓	✓
ECM Re-Flash				✓			
Battery Test	✓	✓	✓	✓	✓		
System Test	✓	✓	✓	✓	✓	✓	
Cable Drop Test					✓ *		✓
Manual Charge				✓	✓ **		

* Rated amperage of the circuit being tested

** Set charging voltage or current limit

NOTES:



Load Testing

A battery load test checks the battery's ability to deliver voltage under a load. Always follow the service manual information when testing a battery. To perform a battery load test:

- Apply a 300A load for 2 seconds to remove any surface charge.
- Connect a DVOM to the battery terminals to monitor battery voltage
- Use a load tester and apply a load equal to 1/2 the Cold Cranking Amps (CCA) rating of the battery for 15 seconds. CCA specs can generally be found on the battery's label
- Record the DVOM reading at the end of the 15 second test with the load still applied
- Disconnect the load

The voltage should be above 9.6V with ambient temperature above 68° F (acceptable voltage is based on ambient temperature, see appropriate service information).



CAUTION: Never apply a load to the battery for more than 15 seconds as it may damage the battery or the load tester.



Starter Draw Test

If you experience a starter motor cranking slowly, perform a starter current draw test.

To complete the test:

- Connect the positive and negative cable leads from the tester unit to the vehicle's battery, and place the inductive clamp onto the positive cable that leads to the starter motor
- Remove the fuel pump relay and crank the engine until the engine stalls
- Set the tester timer on the tester with enough time to safely enter the vehicle. Crank the engine until the tester completes the test. Record your results.

Although Kia does not publish starter draw test results, a reading between 160 - 190 Amps is normal on a cold engine. Results will be affected by engine size (cylinders, compression and displacement), and the temperature at which the test is performed

If the reading is below 160 Amps, look for high resistance in the starter cables and wiring by performing voltage drop tests on the positive and groundside of the starter system.

If the reading is above 190 Amps, the starter motor may be encountering mechanical resistance such as a tight engine or defective starter motor components.



VIDEO: Starter Draw Test



Current Output Test

A series of more traditional load tests may be performed if the GR8 Multitasking Battery and Electrical Diagnostic Station is not available.

To perform a generator load test:

- Connect the tester leads to the battery
- Connect the amp probe to the generator output cable (at least 6 inches from the generator if possible)
- Set the timer on the tester and press the Generator Output Test button

The tester will begin to test the generator and the results will be displayed on the screen.

The reading for a good generator will be within 10% of the manufacturer specification

If the results are greater than 10%, suspected components include the generator, regulator, wires, or connections.

Always refer to the latest service information for exact procedures and specifications.



NOTE: Some Kia vehicles are regulated internally within the ECM and are not regulated externally.

SUMMARY The battery, starting, and charging systems are an important part of the vehicle electrical system. The battery supplies the source voltage to operate the starter motor and all electrical accessories when the engine is OFF.

The starting system relies on the condition of the battery and the starting system circuits in order to crank the engine under different operating conditions and temperatures.

When the engine is running, the charging system must recharge the battery and carry the electrical load of all vehicle circuits.

System diagnosis is made easier using the GR8 Multitasking Battery and Electrical Diagnostic Station which can be used to troubleshoot the condition of the battery, starting and charging systems.

Helpful troubleshooting charts and procedures can be found in the service information.

This information can make your troubleshooting steps more efficient in pinpointing the root cause of an electrical malfunction in these circuits.

PROGRESS CHECK

1. Which of the following is an advantage of testing a battery using the GR8?
 - A. Indirectly checks battery capacity by measuring circuit resistance
 - B. The meter reads battery electrolyte directly to determine state of charge
 - C. Current output is read directly on the display
 - D. It can determine how much life is left in the battery, which is measured by months.
2. What does a battery's Cold Cranking Amps (CCA) refer to?
 - A. The lowest temperature under which the battery can still start the vehicle
 - B. The temperature under which the battery stops providing voltage
 - C. The amps the battery can deliver at 0° F for 30 seconds while maintaining a voltage $\geq 7.2V$
 - D. The amps the battery can deliver at 0° F for 30 seconds while maintaining a voltage $\geq 12.66V$
3. Typically, a parasitic draw test should not be greater than 0.050A.
 - A. True
 - B. False
4. What component completes the circuit between the battery and the starter motor as well as engages the pinion gear to the engine flywheel?
 - A. Overrunning clutch
 - B. Ignition switch
 - C. Ignition switch relay
 - D. Starter solenoid

PROGRESS CHECK



5. The GR8 Multitasking Battery and Electrical Diagnostic Station can be used to:
 - A. test the battery
 - B. test the charging system
 - C. Test the starting system
 - D. All of the above
6. The battery converts chemical energy into electrical energy.
 - A. True
 - B. False
7. What is the minimum open circuit voltage reading on a fully charged battery?
 - A. 12.00
 - B. 12.22
 - C. 12.45
 - D. 12.66

NOTES:

Automotive Electrical Diagnosis

Guided Practice Module

TARGET AUDIENCE The target audience for this diagnostic module consists of Kia service technicians who have a good understanding of automotive electrical and have completed the *Battery, Starting, and Charging Systems Theory Module*.

MODULE GOAL This module will give you the opportunity to demonstrate the skills required to diagnose and troubleshoot battery, starting, and charging system concerns.

MODULE OBJECTIVES Given a Digital Volt-Ohm Meter, the GR8-1299, and a GDS Station, you will perform battery, starting, and charging system tests with 80% or greater accuracy.

Optional Given a Digital Volt-Ohm Meter, and a load-type electrical system tester, you will perform battery, starting, and charging system tests with 80% or greater accuracy.

MODULE INSTRUCTIONS Follow all shop safety rules
Working in teams of two, each technician will perform each test.

- The first time through technician 1 will perform the test and technician 2 will assist and record the results.
- The second time through technician 2 will perform the test and technician 1 will assist and record the results.

REQUIRED MATERIALS In order to complete this module, you will need the following items:

- Digital Volt-Ohm Meter (DVOM)
- GR8 Battery and Electrical Diagnostic Station
- Load Type Electrical System Tester (if available)
- Pen or Pencil
- 1 Soul (AM), 1 Optima (TF), 1 Sedona (VQ), 2 Sorento (XM)

TIME TO COMPLETE Approximately 2.5 hours

GUIDED PRACTICE 1

Assigned Vehicle: _____

Battery State of Charge

- ☐ Locate a DVOM and bring it to your assigned vehicle
- ☐ Confirm that **all loads are OFF** and that the **ignition is in the OFF position**
- ☐ Set the DVOM to **Volts DC** then place the test leads in their appropriate ports
- ☐ Place the DVOM **red test lead on the battery positive terminal** and the DVOM **black test lead on the battery negative terminal**
- ☐ Record your results

- ☐ What is the battery's state of charge?

Are these results acceptable?

YES NO

- ☐ Have the second technician in your team perform the test

Instructor's Initials _____

OPEN CIRCUIT VOLTAGE	STATE OF CHARGE
12.80	REMOVE SURFACE CHARGE AND RETEST
12.70	
12.66	100%
12.58	90%
12.51	80%
12.45	70%
12.35	60%
12.28	50%
12.20	40%
12.11	30%
12.04	20%
11.97	10%
11.89	0%

GUIDED PRACTICE 2 Assigned Vehicle: _____**Parasitic Draw Test**

- ☐
- Use GDS to answer the following questions:

What is the maximum parasitic draw specification?

How long should you wait before recording your test results? _____

- ☐ Locate a DVOM and bring it to your assigned vehicle
- ☐ Confirm that **all loads are OFF** and that the **ignition switch is in the OFF position**
- ☐ Set the DVOM to **DC Amperage** and place the test leads in their appropriate ports to test for AMPS



CAUTION: If a voltage reading is taken with the meter leads in the amp ports, the meter's fuse may blow.

- ☐ Loosen the clamp on the negative battery terminal
- ☐ 1. Place the DVOM **black test lead on the negative battery post**
2. Place the DVOM **red test lead on the negative battery clamp**
3. Lift the clamp off of the negative battery post

- ☐ Record the results _____

What are the results in mA? _____

Are these results within an acceptable range?

YES NO

- ☐ Have the second technician in your team perform the test

Instructor's Initials _____

GUIDED PRACTICE 3

Assigned Vehicle: _____

**System Ground
Voltage Drop**

- ☐ Locate a DVOM and bring it to your assigned vehicle
- ☐ Confirm that all loads are OFF and that the ignition switch is in the OFF position
- ☐ Set the DVOM to **DC Voltage** and set the test leads to measure voltage
- ☐ With the **Key ON** and **Engine Running**, turn on the vehicle's loads including the Electric Defroster, HVAC Blower set to HIGH speed, Windshield Wipers on HIGH, and High Beams ON
- ☐ 1. Place the DVOM **black test lead on the ground terminal of the battery**
2. Place the DVOM **red test lead on the vehicles engine ground**
- ☐ System ground voltage drop readings are normally less than 0.10 volts
- ☐ Record your results _____

What are the results in mV? _____

Are these results within an acceptable range?

YES NO

- ☐ Turn the Key OFF and all loads off

Have the second technician in your team perform the test

Instructor's Initials _____

GUIDED PRACTICE 4

Assigned Vehicle: _____

**Battery Cranking
Voltage Test**

- ☐ Locate a DVOM and bring it to your assigned vehicle
- ☐ Confirm that all loads are OFF and that the ignition is in the OFF position
- ☐ Set the DVOM to **DC Voltage** and place the test leads in their appropriate ports
- ☐ Remove the fuel pump relay
- ☐ Start the car
- ☐ Let the engine stall
- ☐ 1. Connect the DVOM to the battery
- ☐ 2. Place the DVOM **black test lead on the ground terminal of the battery**
- ☐ 3. Place the DVOM **red test lead on the positive terminal of the battery**
- ☐ Have your teammate **crank the engine** for 10 seconds and record the lowest reading

Was the reading within an acceptable range?

YES NO

- ☐ Have the second technician in your team perform the test
- ☐ Do not replace the fuel pump relay

Instructor's Initials _____

Battery Cranking Voltage Test

Good	10.0 – 11.5 vdc
Marginal	9.6 – 9.9 vdc
Test Battery*	< 9.6 vdc

GUIDED PRACTICE 5

Assigned Vehicle: _____

**Engine Ground (Cranking)
Voltage Drop**

- ☐ Locate a DVOM and bring it to your assigned vehicle
- ☐ Confirm that all loads are OFF and that the ignition is in the OFF position
- ☐ Set the DVOM to **DC Voltage** and place the test leads in their appropriate ports

Fuel pump relay should already be removed from the previous test. If not.

- ☐ Remove the fuel pump relay
- ☐ Start the car
- ☐ Let the engine run until it stalls
- ☐ 1. Connect the DVOM to the battery
- ☐ 2. Place the DVOM **black test lead on the ground terminal of the battery**
- ☐ 3. Place the DVOM **red test lead on the metal surface of the engine**

This test is performed with the engine cranking.

- ☐ Crank the engine for 10 seconds and record the lowest reading _____

What are the results in mV? _____

Was the reading within an acceptable range?
(specifications on next page)

YES NO

- ☐ Have the second technician in your team perform the test
- ☐ Do not replace the fuel pump relay

Instructor's Initials _____

Engine Ground Voltage Drop

Good	0.10 – 0.30 vdc
Marginal	0.30 – 0.50 vdc
Requires Further Testing*	> 0.50 vdc

GUIDED PRACTICE 6

Assigned Vehicle: _____

**Engine Positive (Cranking)
Voltage Drop**

- ☐ Place the vehicle on a lift
- ☐ Locate a DVOM and bring it to your assigned vehicle
- ☐ Confirm that all loads are OFF and that the ignition is in the OFF position
- ☐ Set the DVOM to **DC Voltage** and place the test leads in their appropriate ports

Fuel pump relay should already be removed from the previous test. If not:

- ☐ Remove the fuel pump relay
- ☐ Start the car
- ☐ Let the engine run until it stalls
- ☐ 1. Connect the DVOM to the battery
- 2. Place the DVOM **red test lead on the positive terminal of the battery**
- 3. Place the DVOM **black test lead on the positive cable terminal on the starter**
- ☐ Crank the engine for several seconds and record the lowest reading _____

What are the results in mV? _____

Was the reading within an acceptable range?

YES NO

- ☐ Have the second technician in your team perform the test
- ☐ Replace the fuel pump relay and start the engine, turn the engine off

Instructor's Initials _____

Engine Positive Voltage Drop

Good	0.10 – 0.30 vdc
Marginal	0.30 – 0.50 vdc
Requires Further Testing*	> 0.50 vdc



GUIDED PRACTICE 7**Charging Voltage Drop
Positive Side**

Assigned Vehicle: _____

- ☐ Place the vehicle on a lift
- ☐ Locate a DVOM and bring it to your assigned vehicle
- ☐ Set the DVOM to **DC Voltage** and place the test leads in their appropriate ports on the DVOM
- ☐ 1. Connect the DVOM
- ☐ 2. Place the DVOM **black test lead on the positive terminal of the battery**
- ☐ 3. Place the DVOM **red test lead on the B+ terminal on the generator**
- ☐ **Start the engine and raise the engine speed to 2000 RPM**

- ☐ Record the results _____

What are the results in mV? _____

- ☐ A voltage drop of greater than 0.25 volts indicates that there is unwanted resistance on the B+ side of the charging system

Was the reading within an acceptable range?

YES NO

- ☐ Turn the vehicle off
- ☐ Have the second technician in your team perform the test

Instructor's Initials _____

GUIDED PRACTICE 8

Assigned Vehicle: _____

**Charging Voltage
at Battery Posts**

- ☐ Locate a DVOM and bring it to your assigned vehicle
- ☐ Set the DVOM to **DC Voltage** and set the test leads to measure DC voltage on the DVOM
- ☐
 1. Connect the DVOM to the battery
 2. Place the DVOM **black test lead on the ground terminal of the battery**
 3. Place the DVOM **red test lead on the positive terminal**
- ☐ **Start the engine and raise the engine speed to 2,000 RPM**
- ☐ Record the results _____
- ☐ The result should be 1.0 - 2.0 volts above the Open Circuit Voltage test results
- ☐ Record the results from the Open Circuit Voltage Test that was performed earlier in this module

Was the reading within an acceptable range?

YES NO

- ☐ Have the second technician in your team perform the test

Instructor's Initials _____

GUIDED PRACTICE 9

Assigned Vehicle: _____

**Charging Voltage Drop
Ground Side**

- ☐
- Locate a DVOM and bring it to your assigned vehicle

- ☐
- Set the DVOM to
- DC Voltage**
- and place the test leads in their appropriate ports on the DVOM

- ☐
1. Connect the DVOM

- ☐
2. Place the DVOM
- black test lead on the negative terminal of the battery**

- ☐
3. Place the DVOM
- red test lead on the case of the generator**

- ☐
- Start the engine and raise the engine speed to 2,000 RPM**

- ☐
- Record the results _____

What are the results in mV? _____

- ☐
- Test results should be less than 0.15 volts

Was the reading within an acceptable range?

YES NO

- ☐
- Turn the vehicle off

- ☐
- Have the second technician in your team perform the test

Instructor's Initials _____

GUIDED PRACTICE 10**Charging Voltage
Overcharge/Undercharge**

Assigned Vehicle: _____

- ☐ Locate a DVOM and bring it to your assigned vehicle
- ☐ Set the DVOM to **DC Voltage** and place the test leads in their appropriate ports on the DVOM
- ☐ 1. Connect the DVOM
- ☐ 2. Place the DVOM **black test lead on the negative terminal of the battery**
- ☐ 3. Place the DVOM **red test lead on the positive terminal of the battery**
- ☐ **Start the engine and raise the engine speed to 2,000 RPM**
- ☐ Record the results _____
- ☐ **Turn on the vehicle's loads** such as the Defroster, Blower (set to high speed), Windshield Wipers, and High Beams
- ☐ Record the results _____
- ☐ Results should be:
 - 15 volts or lower without a load
 - Greater than 13.4 volts with a load

Was the reading within an acceptable range?

YES NO

- ☐ Turn the vehicle off
- ☐ Have the second technician in your team perform the test

Instructor's Initials _____

GUIDED PRACTICE 11**Diagnostic Charge**

Assigned Vehicle: _____

- ☐ Locate a GR8 and bring it to your assigned vehicle
- ☐ Record the following information:

Last six digits of VIN _____

CCA _____

- ☐ Turn the GR8 ON

- ☐ **Connect the GR8 Charge clamps to the battery**



CAUTION: Never connect both sets of clamps to the to the same battery at the same time. Only the charge clamps of the GR8 should be connected to the battery when performing the diagnostic charge.

- ☐ Use the **GR8 Instruction Manual** to fill in the blanks. After you have filled in the blanks, complete the test
- ☐ Select DIAGNOTIC CHARGE from the main menu
- ☐ Enter _____ VIN and press NEXT
- ☐ Enter the location of the _____ and press NEXT
- ☐ Enter the _____ and press NEXT
- ☐ Enter the _____ and select NEXT
- ☐ Use the keypad to enter the _____ and press NEXT to start the diagnostic charge
- ☐ While the test is running, record the following information:
 - Volts _____
 - Amps _____
 - Amp Hours _____
 - Max Time _____

(Max Time is the time needed to fully charge the battery)



- ☐ Press and hold the STOP button to abort the diagnostic charge
- ☐ Disconnect the charge clamps and turn the GR8 Off
- ☐ Have the second technician in your team perform the test

Instructor's Initials _____

GUIDED PRACTICE 12 Assigned Vehicle: _____

ECM Re-Flash



☐ Locate a GR8 and bring it to your assigned vehicle

☐ Turn the GR8 ON

☐ **Connect the GR8 Charge clamps to the battery**

CAUTION: Never connect both sets of clamps to the to the same battery at the same time. Only the charge clamps of the GR8 should be connected to the battery when performing the ECM re-flash.

☐ Select ECM RE-FLASH from the main menu

☐ While the ECM re-flash is running, write down the following information:

Volts _____

Amps _____

Voltage Setting _____

☐ Press and hold the STOP button to to stop the ECM re-flash

☐ Disconnect the charge clamps and turn the GR8 OFF

☐ Have the second technician in your team perform the test

Instructor's Initials _____

GUIDED PRACTICE 13 Assigned Vehicle: _____**Battery Test**

- ☐ Locate a GR8 and bring it to your assigned vehicle
- ☐ Record the following information:
Last six digits of VIN _____
CCA _____
- ☐ Make sure all vehicle accessories are off, the key is not in the ignition (if applicable) and the doors are closed
- ☐ Turn the GR8 ON
- ☐ **Connect the GR8 Test clamps to the battery**



CAUTION: Never connect both sets of clamps to the to the same battery at the same time. Only the test clamps of the GR8 should be connected to the battery when performing the battery test.

- ☐ Select BATTERY TEST from the main menu
- ☐ Enter the last six digits of the VIN and press NEXT
- ☐ Enter the location of the battery and press NEXT
- ☐ Enter the vehicle type and press NEXT
- ☐ Enter the battery type and select NEXT
- ☐ Use the keypad to enter the battery rating (CCA) and press NEXT to start the diagnostic charge
- ☐ Follow the instructions and use the GR8 control module to get the battery temperature. When the temperature is entered, press NEXT to start the test.
- ☐ After the test has completed, print out the test results
- ☐ Press END after you have printed the test results
- ☐ Disconnect the Test clamps and turn the GR8 OFF
- ☐ Have the second technician in your team perform the test

Instructor's Initials _____

GUIDED PRACTICE 14

Assigned Vehicle: _____

System Test

- ☐ Locate a GR8 and bring it to your assigned vehicle

- ☐ Record the following information:

Last six digits of VIN _____

CCA _____

- ☐ Make sure all vehicle accessories are off, the key is not in the ignition (if applicable) and the doors are closed

- ☐ Turn the GR8 ON

- ☐ **Connect the GR8 Test clamps to the battery**



CAUTION: Never connect both sets of clamps to the same battery at the same time. Only the test clamps of the GR8 should be connected to the battery when performing the system test.

- ☐ Select SYSTEM TEST from the main menu
- ☐ Enter the last six digits of the VIN and press NEXT
- ☐ Enter if you are using the Amp Clamp option of not and press NEXT
- ☐ Enter the location of the battery and press NEXT
- ☐ Enter the vehicle type and press NEXT
- ☐ Enter the battery type and select NEXT
- ☐ Use the keypad to enter the battery rating (CCA) and press NEXT to start the diagnostic charge
- ☐ Follow the instructions and use the GR8 control module to get the battery temperature. When the temperature is entered, press NEXT to start the test.
- ☐ Follow the instructions on the GR8 screen to complete the test





NOTE: If the GR8 displays "**CHARGE & RETEST**" during the system test, press the NEXT key to continue the test.

- ☐ After the system test is complete, print out the test results
- ☐ Disconnect the Test clamps and turn the GR8 OFF
- ☐ Have the second technician in your team perform the test

Instructor's Initials _____

GUIDED PRACTICE 15

Assigned Vehicle: _____

Manual Charge☐ Turn the GR8 ON☐ **Connect the GR8 Charge clamps to the battery**

CAUTION: Only the charge clamps of the GR8 should be connected to the battery when performing a manual charge.

☐ Select MANUAL from the main menu☐ Enter the battery type and select NEXT☐ Select LIMIT VOLTAGE and press the NEXT key to continue☐ Set the voltage limit to 13.5 volts☐ Set the charge time to CONTINUOUS and press NEXT to start the manual charge

Instructor's Initials _____



NOTE: Let the battery charge for the next group.

**GUIDED PRACTICE 16
(OPTIONAL)**

Load Testing a Battery

(Load Type Electrical System
Tester)



Assigned Vehicle: _____

- ☐ Locate the AVR and bring it to your workstation
- ☐ Follow the tester instructions for the battery load test, or instructions provided by your instructor
- ☐ Complete the test and answer the following questions

What is the CCA for the battery being tested?

What was the voltage reading during the test?

Did the battery pass the load test? YES NO

- ☐ Have the second technician in your team perform the test

Instructor's Initials _____

**GUIDED PRACTICE 17
(OPTIONAL)****Starter Draw Test**

(Load Type Electrical System
Tester)



Assigned Vehicle: _____

☐ Locate the AVR and bring it to your workstation

☐ Remove the fuel pump relay

☐ Start the car

☐ Let the engine run until it stalls

☐ Follow the tester instructions for the starter draw test, or instructions provided by your instructor

☐ Complete the test and answer the following questions

☐ Record your results _____

Was the reading within an acceptable range?

YES NO

☐ Replace the relay. Start the engine and idle for 60 seconds then shut OFF engine.

☐ Have the second technician in your team perform the test

Instructor's Initials _____

**GUIDED PRACTICE 18
(OPTIONAL)**

Current Output Test



Assigned Vehicle: _____

- ☐ Locate the ARV and bring it to your assigned vehicle
- ☐ Follow the tester instructions for the current output test, or instructions provided by your instructor
- ☐ Complete the test and answer the following questions
- ☐ Record the results _____

Was the reading within an acceptable range?

YES NO

- ☐ Turn the vehicle OFF
- ☐ Have the second technician in your team perform the test

Instructor's Initials _____

GUIDED PRACTICE

The following questions are designed to help you determine how you would use the tests you performed, and the results you obtained, when actually diagnosing a vehicle.

Answer the questions using:

- Information from KGIS
- Content from the theory module (AED.07)
- Your own knowledge

Your instructor will tell you which vehicle information to use when answering the questions.

Vehicle: _____

BATTERY

1. When would you perform an open circuit voltage test?

2. An open circuit voltage test result of 12.58 volts would indicate that the battery is _____ % charged.

3. When would you load test a battery?

4. According to the service manual, what are the four steps to diagnosing a battery in a 2012 Optima?

1. _____

2. _____

3. _____

4. _____



5. If the battery state of charge was too low, what would you do next if you were diagnosing a vehicle that won't start after sitting for three days?

6. According to the service manual, how long should you wait for all electrical systems to go to "sleep" before performing a parasitic draw test?

7. According to the service manual, when performing a parasitic draw test the maximum draw should be:

8. If the results were not within specifications, what would you do to diagnose the cause of the parasitic draw?

**SYSTEM GROUND VOLTAGE
DROP TEST**

9. When you perform a system ground voltage test, you are testing for _____
from the _____
to the _____.

10. What symptoms might you see if the system ground voltage drop were 0.03 volts?

STARTER

11. A battery cranking voltage test result of 8.9 volts indicates the _____ needs further testing.

12. When would you perform a starter draw test?

13. If the result of a starter draw test was 146 amps, which test would you perform next?

14. If a technician was checking the voltage between the negative battery post and engine block, what would they be testing?

15. An engine positive cranking voltage drop test result of 0.94 volts would indicate:

CHARGING SYSTEM

16. What is a quick way to determine if a charging system voltage is normal?

17. The proper way to perform a current output test is to test the output with no load, then turn on the lights, blower and A/C to provide a load and recheck the output.

True False (circle one)

18. If the charging system had high voltage drop when testing the ground side, what would you test next?

19. According to the service manual, the standard voltage drop for the alternator output wire is:

20. What would be the most likely cause of charging system overcharge?

GR8 21. Which GR8 test provides the most thorough battery test?

22. The GR8 Battery Test should be used to confirm that a battery that is suspected to be _____ is actually _____.

23. What systems does the GR8 System Test test?

24. If you get a REPLACE BATTERY result from the system check, what do you need to do before replacing the battery?

25. The GR8 Manual Charge will charge the battery based on:

[illegible]